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# वार्षिक प्रतिवेदन Annual Report 2023



भा.कृ.अ.प.-केन्द्रीय रोपण फसल अनुसंधान संस्थान  
कासरगोड़, केरल, भारत

ICAR-CENTRAL PLANTATION CROPS RESEARCH INSTITUTE  
KASARAGOD 671 124, KERALA, INDIA





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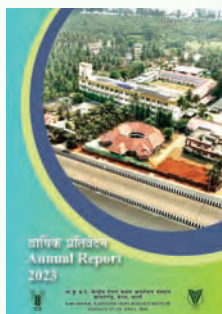
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## I. प्रस्तावना

### पिछलें साल का पुनरावलोकन



मुझे भा.कृ.अनु.प.-केंद्रीय रोपण फसल अनुसंधान संस्थान (भा.कृ.अनु.प.-के.रो.फ.अ.सं.) का वार्षिक प्रतिवेदन २०२३ प्रस्तुत करते हुए गर्व हो रहा है। 'जनवरी २०२३ में, संस्थान ने १०७ वर्ष पूरे किए और इस लंबी यात्रा के दौरान हितधारकों के आर्थिक और सामाजिक कल्याण के लिए फसल सुधार, प्रबंधन, प्रसंस्करण और प्रसार प्रौद्योगिकियों के सभी पहलुओं में उन्नत अनुसंधान के दायित्व का लगातार प्रयास जारी है। संस्थान ने बुनियादी ज्ञान की खोज और कृषि व्यवसाय और सामाजिक लाभ के लिए अपने शोध को लागू करने के बीच संतुलन बनाने के लिए सफलता के साथ प्रयास किया है। वर्ष के दौरान पामिरा (बोरेसेस फ्लैबेलिफेरा) मौजूदा नारियल, सुपारी और कोको के अलावा एक ऐसे कम उपयोग वाले ताड़ को, जिससे नीरा (फूल रस) प्राप्त किया जा सकता है, संस्थान की फसल के रूप में शामिल किया गया है।

वर्ष के दौरान जलवायु परिवर्तन के अलावा वृक्षारोपण क्षेत्र को कई अन्य चुनौतियों का सामना करना पड़ा था जैसे नारियल की कीमत में गिरावट, उच्च उत्पादन लागत, कुशल जनशक्ति की अनुपलब्धता, गैर-पारंपरिक क्षेत्रों में फसलों का प्रसार और वैश्विक प्रतिस्पर्धा। इनमें से कुछ मुद्दों को हल करने के लिए संस्थान ने निम्नलिखित विषयगत क्षेत्रों में अपने शोध को फिर से उन्मुख किया है। जलवायु-लचीला लक्षणों/जीनोटाइप और गुणवत्ता मापदंडों के लिए फिनोटाइपिंग और जीनोटाइपिंग, बड़े पैमाने पर गुणा प्रोटोकॉल का अनुकूलन, उच्च उत्पादकता और पर्यावरणीय सुरक्षा के लिए संसाधन उपयोग दक्षता, दक्षता और कम लागत के लिए एआई और आईसीटी का उपयोग, स्थायी उत्पादन के लिए एकीकृत कीट और रोग प्रबंधन, बायोएक्टिव यौगिकों का निष्कर्षण और विशेषता और मूल्य वर्धन, ताड़ और कोको में फसल प्राप्ति से पहले और बाद के मशीनीकरण, प्रौद्योगिकी मूल्यांकन, प्रभाव और उद्यमिता विकास। प्रत्येक विषयगत क्षेत्र के तहत किए गए अनुसंधान और निष्कर्षों की व्याख्या बाद के अध्यायों में की गई है। संस्थान परियोजना समन्वयक (ताड़) और विभाग और क्षेत्रीय स्टेशन प्रमुखों की नियमित रूप से नियुक्ति होने से उत्साहित हूँ। यह कृषक समुदाय के सामने आने वाली समस्याओं के अनुसंधान और समाधान के लिए अपनी प्रतिबद्धता को सशक्त बनाने में मददगार हैं। यह सभी उत्पन्न और व्यावसायिक, उच्च प्रभाव पत्रिकाओं में प्रकाशन, पुरस्कार और प्राप्त मान्यता, किसानों और गणमान्य व्यक्तियों के संस्थान और अन्य लोगों के दौरे से स्पष्ट है संस्थान हमेशा अनुसंधान और नवाचार की सीमाओं को आगे बढ़ाने के लिए प्रौद्योगिकी के क्षेत्र में स्थापित नेताओं के साथ टीम बनाने में विश्वास रखता रहा है। इसके लिए, पिछले वर्ष सहयोग स्थापित करने में विशेष रूप से फलदायी रहा है और एम्स, सीएसआईआर-सीसीएमबी, सीएसआईआर-सीएफटीआरआई, सीएसआईआर-एनबीआरआई, एस डी एम भा.कृ.अनु.प.-आईआईएसआर, केएमसी, निट्टे विश्वविद्यालय और एसडीएम विश्वविद्यालय, जैसे संस्थानों के साथ सुपारी चबाने के स्वास्थ्य पहलुओं पर वैज्ञानिक साक्ष्य विकसित करने के लिए, एक प्रस्ताव तैयार किया है। सेंसर और अन्य कृषि तकनीकों को विकसित करने के लिए इंजीनियरिंग कॉलेजों, एनआईटी के साथ अन्य सहयोग पर विचार किया गया है।

जबकि के.रो.फ.अ.सं. के संचालन के लिए सरकारी वित्त पोषण महत्वपूर्ण बना हुआ है, गैर-सरकारी स्रोतों से समर्थन अर्थात सीडीबी, डीएसडी, डीसीसीडी, आरकेवीवाई और अन्य संगठनों ने न केवल अपने बुनियादी ढांचे के आधुनिकीकरण के लिए बेहद फायदेमंद साबित किया है, बल्कि हमारे वैज्ञानिकों को जमीनी स्तर पर अनुसंधान जारी रखने में भी मदद की है।

आने वाले वर्षों में, हमेशा की तरह हम अनुसंधान में उत्कृष्टता और रोपण फसल क्षेत्र में सुधार के लिए, योगदान जारी रखेंगे। के.रो.फ.अ.सं की ओर से, मैं भा.कृ.अनु.प. के महानिदेशक और डेयर के सचिव, उप महानिदेशक (बागवाली) सहायक महानिदेशक (बागवाली) और बागवाली विभाग के सभी सदस्य, संस्थान परिषद के सदस्यों को हमारे प्रयासों का समर्थन और मार्गदर्शन के लिए धन्यवाद देता हूँ।। संपादकीय बोर्ड के भरसक प्रयासों की विशेष सराहना करता हूँ।

के.ब. देवदार

(के. बालाचंद्र हेब्बार)  
निदेशक

## I. PREFACE

### The Year in Retrospect

I am honored to present the 2023 Annual Report of the ICAR-Central Plantation Crops Research Institute (ICAR-CPCRI). In January 2023, CPCRI completed 107 years and during this long journey the Institute has steadfastly stayed the course in its promise to “provide for advanced original investigations in all aspects of crop improvement, management, processing and dissemination technologies for the economic and social welfare of stakeholders”. The Institute has also strived, with success, to strike a balance between the pursuit of basic knowledge and applying its research for agribusiness and social benefit. During the year Palmyrah (*Borassus flabelifera*) a sap yielding underutilized palm has been included as institute mandate crop in addition to the existing coconut, arecanut and cocoa.

Plantation sector in addition to climate change had faced many other challenges during the year like price crash of coconut, high production cost, non-availability of skilled manpower, spread of crops in non - traditional areas and global competition. In order to address some of these issues the institute has reoriented its research in the following thematic areas Viz. Phenotyping and genotyping for climate-resilient traits/genotypes and quality parameters, Optimization of mass multiplication protocols, Resource use efficiency for higher productivity and environmental security, Use of AI and ICT for increased efficiency and reduced costs, Integrated pest and disease management for sustainable production, Extraction and characterization of bioactive compounds and value addition, Pre- and post-harvest mechanization in palms and cocoa, Technology evaluation, impact and entrepreneurship development. The research carried out under each thematic area and the findings are enumerated in subsequent chapters. The institute is bolstered with the regular joining of project coordinator (palms) and division and regional station heads and could make good progress of its commitment to research and solve the problems faced by farming community. It is all evident from the technologies generated and commercialized, publications in high impact journals, awards and recognition received, farmers and dignitaries visit to institute and others.

The Institute has always believed in teaming up with technology leaders to push the boundaries of research and innovation. To this end, the past year has been particularly fruitful in establishing collaboration and prepares a proposal with institutions like AIIMS, CSIR-CCMB, CSIR-CIMAP, CSIR-CFTRI, CSIR-NBRI, IISc, KMC, NITTE University, and SDM University, to develop scientific evidence on health aspects of arecanut chewing. Other collaborations contemplated are with engineering colleges, NITs to develop sensors and other agro techniques.

While government funding continues to be critical for CPCRI's operations, support from non-government sources Viz. CDB, DASD, DCCD, RKVY and other organizations has proved extremely beneficial not only to modernize its infrastructure but also helped our scientists continue to pursue ground-breaking research.

In the coming years, we will persist in our pursuit of excellence in research and contribute to improvement of plantation sector, as we have always done. On behalf of CPCRI, I thank the Director General ICAR and Secretary DARE, Deputy Director General (Hort. Sci), Assistant Director General (FPC) and all the members of SMD, members of the Council of the Institute for their valuable support and guidance in all our endeavors. A special appreciation to Editorial Board for their sincere effort.



(K Balachandra Hebbar)  
Director

## II. कार्य सारांश



कोको किस्मों (वीटीएलसीएस-1 और वीटीएलसीएस-2) को उनके बेहतर प्रदर्शन के आधार पर विमोचित किया गया है।

कासरगोड़ में, जमैका लंबा नारियल किस्म, जिसे फल, कोपरा और पुष्प गुच्छ से मिलने वाली कल्प रस उपज में अपने लगातार और बेहतर प्रदर्शन के लिए उपयुक्त है, को रिलीज करने की सिफारिश की गई है। जर्मप्लाज्म संग्रह के तहत, 28 नए नारियल संग्रह, पतली भूसी, बड़ी फल की मात्रा, उच्च कर्नेल और ज्यादा नारियल पानी जैसी बेहतर विशेषताओं के गुण के कारण, नान कौड़ी, एक्स द्वीप निकोबार और उत्तर और मध्य अंडमान से प्राप्त किए गए।

इस अवधि के दौरान, अनुसंधान केंद्र, किडु में द्वीप (अंडमान एवं निकोबार तथा लक्षद्वीप) तथा तटीय (केरल) पारिस्थितिक प्रदेशों के 31 नारियल अभिगमों की विशेषता रही है। इनमें से, उच्च फल और कोपरा उपज क्षमता वाले चैंपिन माइक्रो टॉल को आगे के अध्ययन के लिए पहचाना गया है। विभिन्न स्थानों पर मूल्यांकन परीक्षणों ने विशिष्ट नारियल जर्मप्लाज्म को चुनायित किया है, जैसे कि बेहतर नारियल पानी की लक्षणों के साथ PHOT(मोहितनगर में) और केर संकर हाइब्रिड उच्च उपज क्षमता (असम में) के साथ। कासरगोड़ अंडमान ग्रीन बौना, किंग नारियल और समोन येलो ड्वार्फ के मूल्यांकन परीक्षणों में अच्छा प्रदर्शन किया है। इस अवधि के दौरान इक्कीसविशेषता-विशिष्ट नारियल जर्मप्लाज्म की पहचान और वर्गीकरण किया गया है। नई मीठी - कर्नेल नारियल प्रकार (रत्नागिरि के मोहचो नरेल मीठी फल) की संख्या बढ़ाने के प्रयास किए गए हैं।

### अनुवांशिक संसाधन प्रबंधन और उपयोग

भा.कृ.अनु.प.-के.रो.फ.अ.सं., कासरगोड़ ने नारियल जर्मप्लाज्म के दुनिया के सबसे व्यापक संग्रह की मेजबानी कर रहा है, जिसमें शामिल 27 देशों से प्राप्त 455 प्रजातियाँ, कासरगोड़ में और दक्षिण एशिया एवं मध्य पूर्व के लिए अंतर्राष्ट्रीय नारियल जीन बैंक, भा.कृ.अनु.प.-के.रो.फ.अ.सं. अनुसंधान केंद्र, किडु में संरक्षित हैं। तीन अधिदेश फसलों के संग्रह में शामिल नारियल की 455 प्रजातियाँ, सुपारी की 172 प्रजातियाँ, कोको की 531 प्रजातियाँ, इन फसलों में अधिकतम विविधता को संरक्षित करना, फसल सुधार कार्यक्रमों में संरक्षण, विशेषता और बाद में उपयोग की सुविधा के लिए उपलब्ध है। क्षेत्रीय फ्रील्ड जीन बैंक से परे, जाइगोटिक भ्रूण, पराग और डीएनए के क्रायोप्रिजर्वेशन के माध्यम से महत्वपूर्ण प्रजातियों को भी संरक्षित किया गया है।

एक बेहतर 'इंटर से' नारियल हाइब्रिड 'कल्प वज्र' एक लंबी विविधता, जड़ (मुरझा) रोग से प्रभावित क्षेत्रों में खेती के लिए उपयुक्त है, और नारियल की किस्म 'कल्प सुवर्णा' एक बेहतर बौनी, दोहरी उद्देश्य किस्म, नारियल पानी और कोपरा के लिए उपयुक्त है, जिस को इस साल विमोचित किया गया है। इसके अतिरिक्त, उच्च घनत्व वाले रोपण प्रणालियों के लिए उपयुक्त दो उच्च उपज देने वाले कोको हाइब्रिड (वीटीएलसीएस-1 और वीटीएलसीएस-2) और दो

भा.कृ.अनु.प.-के.रो.फ.अ.सं. प्रादेशिक केंद्र विट्टल में, 25 विशेषताओं और 69 चुनिंदा किस्मों के आधार पर कोको के लिए डीयूएस दिशानिर्देश स्थापित किए गए हैं। 21 हाइब्रिड और 10 एलाइट क्लोनों सहित 31 कोको जीनोटाइप का आकलन किया गया और पत्ती, फूल, फल और बीन लक्षणों के साथ-साथ वसा सामग्री (मल्टीस्केल स्कोर का उपयोग करके) के आधार पर वर्गीकृत किया गया। इस अवधि के दौरान, भा.कृ.अनु.प.-सीआईएआरआई, पोर्ट ब्लेयर और भा.कृ.अनु.प.-के.रो.फ.अ.सं., कासरगोड़ द्वारा विकसित बौने सुपारी ताड़ को भा.कृ.अनु.प. के पीजीआरसी द्वारा पंजीकृत किया गया है।

नारियल में टेस्टा तेल की जैव रासायनिक संरचना पर अध्ययनों से पता चला कि कल्पा में 88.8% की सबसे अधिक वसा सामग्री है। विशेष रूप से, सुपारी जर्मप्लाज्म में क्रमशः सीएल-32 और एससीआरडीटीसी-18 में 13.6% और 13.45% की उच्च वसा सामग्री देखी गई है। नीरा उपज के लिए नारियल जीनोटाइप का मूल्यांकन करने में, नाइजीरियाई टॉल ने 48.03 लीटर प्रति पुष्पगुच्छ की उच्चतम उपज का प्रदर्शन किया है, इसके बाद फिलीपींस कलाम्भीम टॉल (36.10 लीटर/ पुष्पगुच्छ) हैं। "(i)" नारियल पराग के फाइटोकेमिकल विश्लेषण ने पराग को मानव पोषण के लिए एक उत्कृष्ट पूरक के रूप में दिखाया है, जिसमें 39% तक की कच्ची प्रोटीन सामग्री है।

नारियल में सूखा सहिष्णुता के संकेतक के रूप में अपने ओस्मोटिक समायोजन क्षमता के लिए पराग की स्क्रीनिंग का पता लगाने के प्रयास किए गए हैं। विशेष रूप से ताड़ के कैबेज के लिए नारियल के बीज उत्पादन पर प्रारंभिक अध्ययनों ने आशाजनक परिणाम दिखाए हैं। सूखा सहिष्णुता के लिए नारियल संकरों की स्क्रीनिंग पर अध्ययनों में, CODxADOT, MYDxWCT और CODxLCT ने सूखे की स्थिति के तहत अच्छा प्रदर्शन दिखाया है।

वर्ष के दौरान रोपण सामग्री उत्पादन के संदर्भ में, 144873 नारियल, 588970 सुपारी और 48893 कोको रोपण सामग्री का उत्पादन और वितरण किया गया है।

### जैव प्रौद्योगिकी

नारियल ऊतक संवर्धन में कैलस प्रेषण, भ्रूणजनक कैलस गठन और सोमैटिक भ्रूण विकास को बढ़ाने में नूतन विकास नियामकों के उपयोग के माध्यम से महत्वपूर्ण प्रगति हुई है। अपरिपक्व पुष्पगुच्छ एक्सप्लान्ट संवर्धन की शुरुआत के लिए सबसे उपयुक्त 2, 4-डी (1 एमजी एल -1) के साथ Y3 को पहचाना गया है। पुष्पगुच्छ के आदर्श आकार पर अध्ययनों में, 5-10 सेमी और 10.5-15 सेमी लंबे पुष्पगुच्छ से एक्सप्लान्ट में इष्टतम शूट रीजनरेशन प्राप्त किया गया है। ऊतक संवर्धन की शुरुआत के दौरान बढ़ी हुई वनस्पति ऊतक को टीडीजेड (5 एमजी एल -1) के साथ अकेले या एनएए के साथ संयुक्त पूरक वुडी प्लांट माध्यम का उपयोग करके पाया गया है। मीठे कर्नेल जीनोटाइप के भ्रूण बचाव में, लगभग 30 प्लांटलेट पोट किए गए हैं और 60 प्लांटलेट इन विट्रो संवर्धन के अंतिम चरण में हैं। हिरेहल्ली बौने, बौने सुपारी संकर और वाईएलडी-प्रतिरोधी ताड़ों के इन विट्रो गुणन की दिशा में प्रयास किए गए हैं। इन लक्षणों को प्रभावित करने वाले संभावित चुनिंदा जीन की पहचान के साथ-साथ मिश्रित रैखिक मॉडल (एमएलएम) विश्लेषण का उपयोग करके उपज योगदान लक्षणों से जुड़े एसएनपी के पहचान की गई थी। नारियल और सुपारी के लिए वी-क्रायो मेश और वी-क्रायो प्लेट विधि के सत्यापन पर क्रायो संरक्षण अध्ययनों के परिणामस्वरूप वी-क्रायो मेश (73% और 33%) के साथ अधिकतम प्लूम्यूल और प्लांटलेट रिकवरी हुई, इसके बाद क्रमशः वी-क्रायो प्लेट (56% और 23%) और एनकैप्सुलेशनविट्रीफिकेशन विधियों (46% और 20%) का उपयोग किया गया।

कल्प जीनोम रिसोर्स' बौनी नारियल किस्म चौघाट ग्रीन ड्वार्फ (सीजीडी) के लिए एक व्यापकजीनोमिक्स प्लेटफार्म, ट्रांसक्रिप्टोम और फिंगरप्रिंटिंग डेटा को एकीकृत करके प्रयोक्ताओं को जीनोम और ट्रांसक्रिप्टोम संसाधनों की पहचान की प्रणाली खोजी गई है। राइनोसेरोस बीटल पर एक कीट बायोएस्से अध्ययन, एसआईआरएनए को मूल्यांकन के लिए संकलन के माध्यम से प्रशासित किया गया था। गैर-लक्षित आईएमपीआई डीएसआरएनए की तुलना में डीएसआरएनए के प्रयोग के बाद तीसरे दिन से शुरू होकर 50% की काफी बढ़ी हुई मृत्यु दर नोट की गई थी। नारियल गैंडे बीटल के विभिन्न विकासात्मक चरणों में अभिव्यक्ति अध्ययन के लिए उपयुक्त संदर्भ जीन (ओं) की पहचान पर अध्ययन ने एनएडीएच/ईएफ1ए को सबसे विश्वसनीय संदर्भ जीन संयोजन के रूप में पहचाना, जो जीन अभिव्यक्ति अध्ययनों में आरटी-क्यूपीसीआर डेटा का मजबूत सामान्यीकरण प्रदान कर सकता है।

### उत्पादकता और पर्यावरणीय सुरक्षा के लिए इनपुट उपयोग क्षमता

कासरगोड़ में एक हेक्टेयर में नारियल आधारित फार्मिंग सिस्टम मॉडल जिसमें नारियल ताड़, नारियल ताड़ के अंतस्थ फसल, भूमि की सीमा में केला, चारा ज्वार (सीओ 31- मल्टी कट चारा ज्वार) बकरी इकाई और पोल्ट्री शामिल हैं, जिस में से, रुपये 5,43,682 की शुद्ध वापसी हासिल की गई और 884.9 के.जे.प्रति हैं के उत्पादन के साथ उच्च ऊर्जा दक्षता का प्रदर्शन किया गया। 100% जैविक पोषण के साथ नारियल के बागान में अंतस्थ फसल मल्टी-कट ज्वार (सीओ 31) से 79.9 टन प्रति हैं हरा चारा प्राप्त हुआ। नारियल आधारित उच्च घनत्व वाली बहु प्रजाति फसल प्रणाली, जिसमें नारियल, काली मिर्च, केला, रोबस्टा कॉफी, दालचीनी और जायफल शामिल हैं, पोषक तत्व प्रबंधन प्रथाओं और बायोमास पुनर्चक्रण के साथ, नारियल मोनोक्रोप की तुलना में 4.9 गुना अधिक रु. 4,81,416 शुद्ध आय प्राप्त हुआ, जिसकी राशि नारियल उद्यान में इंटरक्रॉपिंग दालचीनी रोपण की पेंटागोनल विधि (प्रति गड्डे में 5 पौधों के साथ 0.6 मीटर = 1.2 मीटर की दूरी पर) के बाद 631.92 ग्राम हर पेड़ और 979.0 किलोप्रति हेक्टेयर की काफी अधिक शुष्क दालचीनी छाल की उपज दर्ज की गई। काहिकुची में, संसाधन पुनर्चक्रण और जीवों का अनुप्रयोग और उच्च घनत्व बहु प्रजाति फसल प्रणाली में सुपारी के लिए 3/4 वें आरडीएफ सुपारी में अधिकतम उपज दे सकते हैं, जिससे 25% आरडीएफ की बचत हो सकती है। सर्दियों की सब्जियों के बीच सुपारी में गोभी 13.17 टन प्रति हेक्टेयर की उपज के साथ 2.04 का उच्च बी:सी अनुपात वाली सबसे अच्छी इंटरक्रॉप के रूप में उभरी।

वाईएलडी-प्रभावित सुपारी उद्यानों में मानसून के दौरान प्लास्टिक मल्टिंग से वाईएलडी-लक्षण अभिव्यक्ति में देरी दिखाई दी, उपज में वृद्धि हुई और मिट्टी के भौतिक गुणों में सुधार हुआ। सुपारी कोको फसल प्रणालियों में कार्बन सीक्वेस्ट्रेशन का अनुमान 167 टन CO<sub>2</sub> प्रति हेक्टेयर पर लगाया गया था, जो जलवायु परिवर्तन से निपटने में महत्वपूर्ण योगदान देता है। 19-71 जीएन, 2.5-9.4 जीपी, 27-118 जीके, 5.5-21.1 जीसीए, 4.3-15.6 जीएमजी, 71-336 एमजी एफईई, 182-1304 एमजी एमएन, 29-152 एमजी जेडएन, 68-255 एमजी सीयू, और 37-153 एमजी प्रति पेड़ प्रति वर्ष के साथ 20 कोको जीनोटाइप परिवर्तनशीलता देखी गई, जो पोषक तत्वों की भारी मात्रा को हटाने, विशेष रूप से पोटेशियम को दर्शाता है। नाइट्रोजन, जिंक और पोटेशियम उपयोग कुशल कोको किस्मों की पहचान की गई थी। हाइब्रिड और बौने नारियल ने ड्रिप फर्टिगेशन (200% आरडीएफ) के माध्यम से उच्च उर्वरक स्तरों पर अच्छी प्रतिक्रिया दी, जिसके परिणामस्वरूप कल्प शंकर और सीओडी, और टीएसएस और सीओडी में पानी की मात्रा में वृद्धि हुई। कार्बनिक पोषक तत्व प्रबंधन प्रथाओं के तहत नारियल की खेती ने अन्य उपचारों पर कोपरा वजन के 174 ग्राम के साथ 102.4 फल/ताड़/वर्ष की उच्च उपज दर्ज की। फार्म यार्ड खाद (50 किग्रा/ ताड़) के साथ संयुक्त ह्यूमिक एसिड (10% और 20%) ने मिट्टी के कार्बनिक कार्बन को बढ़ाया और पोषक आयन की धीमी रिहाई की सुविधा प्रदान की। सांख्यिकीय मॉडल का उपयोग करके नारियल में क्षेत्र-विशिष्ट पोषक तत्वों की सिफारिशों ने संकेत दिया कि रैंडम वन मॉडल ने 77.08% सटीकता दर, और मिट्टी पीएच, कार्बनिक कार्बन और मिट्टी उपलब्ध पोषक तत्वों (K, Cu, Mn) और पत्ती पोषक तत्व (P, Ca और Mg) के साथ नारियल की उपज पर महत्वपूर्ण प्रभाव के साथ बेहतर प्रदर्शन किया। वाईएलडी

स्थानिक राइजोस्फीयर मिट्टी में जीवाणु माइक्रोबायोम को मेटाजेनोमिक्स के माध्यम से विशेषता दी गई थी, जो सुपारी राइजोस्फीयर में कीमोलिथोट्रोफिक संभावित माइक्रोबायोम की उच्च प्रचुरता को प्रकट करता है और वे वाईएलडी स्थानिक राइजोस्फीयर में N, Fe, S और P के माइक्रोबायोम-मध्यस्थ पोषक तत्व परिवर्तनों के प्रमुख खिलाड़ी हैं।

ताड़ और कोको के संयुक्त बायोमास अवशेषों को वर्मीकम्पोस्ट में बदलने के लिए एक प्रौद्योगिकी कोसीपीसीआरआई केंचुए, यूट्रिलस प्रजाति का उपयोग करके मानकीकृत किया है, जिसमें नारियल की भूसी बयोचार एक नवीनतम घटक के रूप में है। अंतिम उत्पाद में नारियल पत्ती वर्मीकम्पोस्ट की तुलना में वृद्धिशील पीएच, विद्युत चालकता, %P, %K, और डिहाइड्रोजेन गतिविधि है। पौधे रोगजनक और शिकारी सूत्रकृमि अनुपस्थित थे। कोको भूसी बयोचार कणों के 0.5-1.0 मिमी आकार में तेजी से रिलीज के साथ अधिकतम उपलब्ध पोषक तत्व देखा गया था और इसलिए, इसके कुशल उपयोग के लिए 1 मिमी आकार से कम के बयोचार की सिफारिश की जा सकती है। परिपक्व नारियल पानी (तरल एंडोस्पर्म) को बदलने की प्रौद्योगिकी, जो नारियल गुठली प्रसंस्करण उद्योगों से सिरका, बड़ी मात्रा में बर्बाद हो जाती है, 'नाटा' बैक्टीरियल सेलूलोज (बीसी) तक को नारियल सिरका से अलग स्वदेशी बैक्टीरिया का उपयोग करके मानकीकृत किया गया है। अधिकतम बीसी उत्पादन बीसी 4 को छोड़कर सभी अलग के लिए 4-5 की पीएच रेंज में प्राप्त किया गया था, जिसमें अधिकतम बीसी उत्पादन 5-10 की व्यापक पीएच रेंज में प्राप्त किया गया था। नारियल बीज की बुवाई के समय और जड़ उभरने के समय माइक्रोबियल अध्ययनों ने संकेत दिया कि, राइजोस्फीयर और एंडोस्फीयर माइक्रोबियल आबादी बीज की बुवाई के 5 से 6 महीने बाद स्थिर आबादी तक पहुंच गई। जड़ मुरझा रोग ग्रस्त इलाकों के लिए मानक सिफारिशों के अनुसार उर्वरक और कार्बनिक खाद (लोबिअ के साथ हरी पत्ती खाद और बेसिन प्रबंधन सहित) के साथ संयुक्त पीजीपीआर उपभेदों और मिश्रण के साथ बायोप्रिमिंग ने जड़ मुरझा रोग अनुक्रमण विधि के अनुसार पौध को 'स्पष्ट रूप से स्वस्थ' रखा। सुपारी की भूसी पर स्वाभाविक रूप से बढ़ते मशरूम की पहचान को प्रिनोप्सिस सिनेरिया सीसीबीवीएच1 के रूप में की गई थी।

### एकीकृत कीट और रोग प्रबंधन

कासरगोड़ जिले के कुछ हिस्सों में लेसियोडिफ्लोडिया थियोब्रोमा के फलों की सड़न के साथ संबंध की सूचना दी गई थी। 'रोग प्रतिरोधी सुपारी जर्मप्लाज्म/हाइब्रिड की पहचान में तेजी लाने के लिए अत्यधिक विषाक्त फाइटोथोरा मेडाई आइसोलेट (पी19) के जूसपोर निलंबन का उपयोग करके 'डिटैच लीफ असे' नामक एक लागत प्रभावी, सरल और तेजी से पहचान तकनीक विकसित की गई। स्थानिक रूप से स्पष्ट जोखिम अनुमान और उत्पन्न संभावना वितरण मानचित्र भारत के दक्षिणी इलाकों में बेसल स्टेम रॉटरोग के लिए प्रबंधन रणनीतियों के विकास में मदद करेंगे। यूएवी स्प्रेयर का उपयोग करके नारियल के ताड़ के लिए मानक संचालन प्रक्रिया (एसओपी) को अनुकूलित किया गया था। भारत में पहली बार सुपारी के रिंग स्पॉट रोग के साथ वायरस के संबंध की सूचना मिली थी। तनाव चक्का 1 ने कोको में पीजीपीआर बैसिलस प्रजाति बीज उपचार और मिट्टी भिगोना दोनों में अधिकतम पौधे की ऊंचाई और बायोमास दर्ज किया।

भारत में सीआरबी-जी हैप्लोटाइप की अनुपस्थिति को दर्शाने वाले विभिन्न क्षेत्रों से प्राप्त नारियल गैंडे बीटल नमूनों में कोई ए से जी संक्रमण नहीं देखा गया। विविध सफेद मक्खियों में, पैरालीरॉड्स बोंडारी, इसके बाद एलोरोटिकस रुजियोपरकुलेटस और एलोरोटिकेल्स एट्रेटस द्वारा वर्तमान में एडिस्पर्सस के पूर्ण विस्थापन के साथ-साथ नारियल प्रणाली में पी मिनेई के साथ नारियल की ताड़ पर सीमित रूप से दर्ज किया गया था। पाम व्हाइटफ्लाई एल्यूरोटिकेल्स एट्रेटस के विभिन्न चरणों के मॉर्फोमेट्रिक वर्ण और इसकी आणविक पहचान स्थापित की गई थी। कायमकुलम से एकत्र किए गए सीओआई जीन का उपयोग करके कोरीड बग, पैराडासिनेस रोस्ट्रेटस का आणविक लक्षण वर्णन किया गया और कोरीडी प्रजातियों के साथ इसका संबंध स्थापित किया गया। कोरीड बग के लिए लगभग 11 नए होस्ट दर्ज किए गए और फलों का उपयोग करके प्रयोगालय पालन प्रोटोकॉल विकसित किया गया। स्थानिक क्षेत्र में आरपीडब्ल्यू के लिए परिष्कृत एकीकृत प्रबंधन कार्यनीति का वैधीकरण किया गया। ब्यूवेरिया बेसियाना के रूप में पहचाने गए संक्रमित लाल पाम वीविल से अलग एक एंटोमोपैथोजेनिक कवक को आरपीडब्ल्यू के जैव-दमन का वादा करते हुए पाया गया और इस बायोकंट्रोल एजेंट का एक ठोसपदार्थ बायोफार्मूलेशन विकसित किया गया। नारियल गैंडा भृंग के क्षेत्र स्तर प्रतिकर्षण के लिए एक कीट प्रतिकर्षक 'कनेल पेलेट' विकसित किया गया था। संरक्षण जैविक नियंत्रण रूगोज सर्पिलिंग व्हाइटफ्लाई के खिलाफ समान रूप से प्रभावी था क्योंकि ताड़ का इसरिया फुमोसोरोसी के साथ-साथ नीम तेल के साथ छिड़काव किया गया था। एक एंटोमोपैथोजेनिक कवक (ईपीएफ), सिम्पलिलियम लैनोसोनिवियम (कॉर्डिसिपिटिसिया: हाइपोक्रिलेस) रूगोज स्पिरालिंग व्हाइटफ्लाई के निम्फाल कैडवर्स से अलग आरएसडब्ल्यू के सभी विकासात्मक चरणों के लिए विषाक्त और रोगजनक पाया गया और लागत प्रभावी बायोफार्मूलेशन विकसित किए गए। नारियल / सुपारी प्रणाली में कीट प्रबंधन के लिए ईपीएन के गैलेरिया कैडवर आधारित पाउच निर्माण का विकास किया गया। नारियल प्रणाली में स्ट ग्रब और कटू भृंग के प्रबंधन के लिए 'कल्प ईपीएन' का फील्ड उपचार प्रभावी पाया गया। अमरूद में रूट-नॉट नेमाटोड (मेलोइडोगाइन एंटरोलोबी) के प्रारंभिक चरण के संक्रमण के लिए फील्ड स्तर प्रबंधन रणनीतियों को विकसित किया गया था। ट्राइकोडर्मा समृद्ध नीम केक प्रौद्योगिकी को नारियल / सुपारी प्रणाली में इंटरक्रॉप की गई काली मिर्च में रूट-नॉट नेमाटोड के स्थायी प्रबंधन के लिए प्रभावी पाया गया था। परीक्षण किए गए छह ईपीएन आइसोलेट्स में स्टीनेरनेमा कार्पोकैप्सा, हेटरोहाब्डिटिस इंडिका और मेटरहब्डिटिस सरैनाई दीमक के प्रबंधन का वादा कर रहे थे। ब्रोइंग नेमाटोड, रोडोफोलस सिमिलिस की उपस्थिति 9 से 70 नेमाटोड/बीजों की आबादी सीमा के साथ मूल्यांकन किए गए सभी बीजों में देखी गई थी। भा.कृ.अनु.प.-के.रो.फ.अ.सं.फार्म, कासरगोड़ में नारंगी बौनी विविधता पर आरएसडब्ल्यू के खिलाफ मूल्यांकन किए गए सात आवश्यक तेल रिपेलेंट की फील्ड प्रभावकारिता, नियंत्रण परीक्षण से बेहतर थी। आरपीडब्ल्यू डिटेक्टर ने हार्डवेयर और सॉफ्टवेयर दोनों मोर्चों पर और सुधार किया है। कासरगोड़ में नारियल की नारंगी बौनी विविधता पर आम के कीट के बग, रेस्ट्रोकोकस इसरियोइडस (ग्रीन) (हेमिप्टेरा: होमोप्टेरा: स्यूडोकोसिडाई) के नए वितरण रिकॉर्ड को प्रलेखित किया। मेटारैजियम एनिसोपलिया सुपारी में पुष्पगुच्छ कैटरपिलर के प्रबंधन का वादा कर रहा था। प्रयोगशाला स्थितियों के तहत रेड माइट पर कोक्सिनेलिड भक्षक स्टीथोरस केरलिकस के बड़े

पैमाने पर गुणन के लिए एक सरल और लागत प्रभावी प्रोटोकॉल पाया गया।

### जलवायु लचीलापन और मूल्यवर्धन प्रौद्योगिकियां

भविष्य की जलवायु परिस्थितियों में नारियल की खेती के लिए पर्यावास उपयुक्तता का एक मैक्सिमम मॉडल-आधारित आकलन से पता चला है कि जैविक चरण, अर्थात् तापमान मौसम, तापमान वार्षिक रेंज और वर्षा, प्रमुख योगदानकर्ता हैं जो भारत में नारियल के लिए जलवायु उपयुक्तता निर्धारित करते हैं। अनुमानों से पता चलता है कि कुछ वर्तमान नारियल की खेती वाले क्षेत्र (दक्षिण आंतरिक कर्नाटक और तमिलनाडु के मैदान) अनुपयुक्त हो जाएंगे, जिसमें फसल परिवर्तन की आवश्यकता होगी, जबकि अन्य क्षेत्रों को जीनोटाइपिक या कृषि प्रबंधन (पूर्वी तट और दक्षिण आंतरिक मैदान) में अनुकूलन प्रक्रियाओं की आवश्यकता होगी। अन्य क्षेत्रों में, नारियल उगाने की जलवायु उपयुक्तता (पश्चिमी तट) में वृद्धि होगी। एडु बायोकेमिस्ट्री अध्ययन में, 15% दोषपूर्ण नारियल के आटे और 85% चावल के आटे की संरचना के मिश्रण ने अनुकूल स्टार्च : प्रोटीन इंटरैक्शन दिखाया, जिससे चावल के आटे में स्टार्च ग्रेन्यूल की अमूर्त संरचना को क्रिस्टलीय संरचना में बदल दिया, सकारात्मक रूप से ग्लाइसीमिक सूचकांक में गिरावट के साथ सहसंबंध पाया गया।

फसल कटाई के बाद के प्रबंधन के क्षेत्र में, एक गम अरबिका आधारित खाद्य कोटिंग ने गुणवत्ता परिवर्तनों में देरी करने और प्रशीतित स्थितियों (4-10 °C) के तहत 20 दिनों तक नारियल हॉस्टोरियम के शेल्फ जीवन को बढ़ाने में अपनी प्रभावकारिता का प्रदर्शन किया। नारियल नीरा सिरप से लिप्ट रेडी-टू-सर्वनारियल के हॉस्टोरियम के उत्पादन के लिए एक प्रोटोकॉल विकसित किया गया है। इन्फारेड-असिस्टेड हॉट एयर डिहाइड्रेटेड नारियल फ्लेक्स (IRHAD) का उपयोग करके 65 डिग्री सेल्सियस पर सूखे नारियल के लच्छों ने एक बेहतर पोषक और भौतिक प्रोफाइल का प्रदर्शन किया।

उभरती सुखाने वाली तकनीकों - इन्फारेड ड्राइंग (IRD), इन्फारेड हॉट एयर ड्राइंग (IRHAD) और पारंपरिक हॉट-एयर ड्राइंग विधि (HAD) के एक तुलनात्मक विश्लेषण से पता चला है कि नारियल चिप्स के IRHAD (65 डिग्री सेल्सियस) उपचार ने एक बेहतर सुखाने की दर, एक कम सुखाने का समय और जैव रासायनिक रूप से बेहतर नारियल चिप्स पाया गया। इसी तरह, कोपरा गुणवत्ता पर इन्फारेड और हॉट एयर ड्राइंग तकनीकों के स्वतंत्र और संयुक्त प्रभाव से पता चलता है कि 60 डिग्री सेल्सियस के तापमान पर IRHAD विधि को अपनाने से कोपरा से तेल निष्कर्षण बढ़ाने के लिए एक महत्वपूर्ण कारक वसा सामग्री (68.4%) को संरक्षित किया जा सकता है।

पौधे-आधारित स्टार्च के साथ नारियल की मृदु फल के छिलकों से एक बायोडिग्रेडेबल, इको-फ्रेंडली पत्रा विकसित किया गया था। एक्स-रे विवर्तन (एक्सआरडी) और स्कैनिंग इलेक्ट्रॉन माइक्रोस्कोपी (एसईएम) अध्ययनों ने स्टार्च के हाइड्रोक्साइल समूहों और नारियल भूसी मैट्रिक्स के सेल्यूलोज के बीच मजबूत क्रियात्मकता का प्रदर्शन किया। उत्पाद ने 20 दिनों की अवधि के भीतर एक उल्लेखनीय उपघटन दर (~70%) प्रदर्शित की।

### आउटरीच गतिविधियां, सामाजिक-आर्थिक दृष्टिकोण और सांख्यिकीय पहलू

प्रौद्योगिकी हस्तांतरण दृष्टिकोण और स्थान विशिष्ट जरूरतों के लिए अनुकूलित रणनीतियां, कीट और रोग प्रबंधन में समुदाय और क्षेत्र के व्यापक दृष्टिकोण विकसित करना और किसानों की आय को दोगुना करने, किसान उत्पादक संगठनों (एफपीओ) को बढ़ावा देने और एमजीएमजी के माध्यम से कृषि परिवारों तक पहुंचने के लिए परिदृश्य आधारित भागीदारी सामाजिक मॉडल हस्तक्षेप और कार्यक्रम हैं। इस वर्ष, संस्थान ने कुल 65 संस्थागत और ऑफ-कैंपस प्रशिक्षण कार्यक्रमों का आयोजन किया जिससे 3,648 किसान और अन्य हितधारक लाभान्वित हुए हैं। इसके अलावा, वर्ष के दौरान मुख्यालयों, क्षेत्रीय केंद्रों और संस्थान के अनुसंधान केंद्रों पर एक दिवसीय एक्सपोजर विजिट कार्यक्रम आयोजित किए गए, जिससे किसानों सहित कुल 7468 हितधारकों को लाभ हुआ। इसके अलावा, इस संस्थान द्वारा 539 उद्यमियों को लाभान्वित करने के लिए सात उद्यमिता विकास कार्यक्रम आयोजित किए गए थे। 50 ग्रामीण बेरोजगार अनुसूचित जाति/अनुसूचित जनजाति के युवाओं के लिए छह दिनों की अवधि का विशेष प्रशिक्षण कार्यक्रम आयोजित किया गया। प्लेटिनम जुबली समारोह के संबंध में सीपीसीआरआईएस कायमकुलम में एक किसान मेला आयोजित किया गया था। 2023 के दौरान विभिन्न क्षेत्रों में कुल 23 एफएलडी किए गए।

किसानों और अन्य हितधारकों को 90,81,185/- रुपये के बीज और अन्य कृषि इनपुट बेचे गए। एनईएच के तहत कार्य प्रणाली के हिस्से के रूप में, देश के उत्तर पूर्वी हिस्सों के गांवों के लिए प्रशिक्षण और इनपुट वितरण कार्यक्रम आयोजित किए गए थे। पूर्वोत्तर केवीके, वैज्ञानिकों के लिए पांच दिवसीय क्षमता निर्माण कार्यक्रम आयोजित किया गया था। एससीएसपी और टीएसपी के तहत, 20 से अधिक प्रशिक्षण और इनपुट वितरण कार्यक्रम आयोजित किए गए थे और देश के विभिन्न हिस्सों में लगभग 1500 से अधिक एससी / एसटी किसान लाभार्थियों में से थे।

भारत में कृषक उत्पादक कंपनियों का प्रोफाइल विश्लेषण के संबंध में असम में कृषक उत्पादक कंपनियों पर एक अध्ययन किया गया। विश्लेषण ने यह संकेत दिया कि अधिकांश एफपीसी सदस्यों के पास क्षेत्र में एफपीसी के सीईओ के माध्यम से बाजार तक पहुंच है। भा.कृ.अनु.प. फार्मर फर्स्ट प्रोग्राम (एफएफपी), व्यक्तिगत-आधारित फार्म मॉडल को सामाजिक स्केल किया गया है। महिला एसएचजी किसानों के साथ कुल 223 विस्तार गतिविधियां आयोजित की गईं। महिला एसएचजी किसानों की भागीदारी उनकी आय को दोगुना कर सकती है जो तिल और कंद की फसलों की क्षमता को आकर्षित करती है। संस्थान का कंटेंट मोबिलाइजेशन मुख्य रूप से यू-ट्यूब, फेस बुक, ट्वीटर(X), वाट्सप और प्रिंट मीडिया के माध्यम से होता है।

नारियल के हार्ड ही में मूल्य में आई गिरावट के आर्थिक कारणों का अध्ययन नीतिगत दृष्टिकोण से किया गया है। यह विचित्र है कि पिछले पांच वर्षों में नारियल और नारियल के तेल की कीमतों में सबसे कम मूल्य होने का अनुभव हुआ है। भारत के नारियल मूल्य वर्धित उत्पादों की निर्यात टोकरी ने 10 साल की अवधि में महत्वपूर्ण बदलाव का अनुभव किया, जिसमें सक्रिय कार्बन ने 2022 में 67% की हिस्सेदारी के साथ कुल नारियल निर्यात पर हावी किया, जबकि

2013 में हिस्सेदारी 45% थी। एक सुप्रबंधित नारियल उद्यान से प्राप्त आंकड़ों के आधार पर केरल में नारियल के उत्पादन की लागत रु. 12.10 प्रति फल होती है।

भारत के तमिलनाडु, केरल, कर्नाटक और आंध्र प्रदेश राज्यों के लिए नारियल रोगों के संभावित जोखिम क्षेत्रों का स्थानिक अध्ययन और मूल्यांकन किया गया। तमिलनाडु के तंजावुर, पुदुकोट्टई और तिरुवरूर जिलों में नारियल की गंभीरता दरों की बेसल स्टेम सड़न अधिक थी और आंध्र प्रदेश के पूर्वी गोदावरी और पश्चिम गोदावरी से सटे कुछ क्षेत्र भी थे। एक सांख्यिकीय अध्ययन से पता चला है कि मानसूनअवधि के आधार पर कवकनाशी अनुप्रयोग का समय क्षेत्र की स्थितियों के तहत फल के विकासात्मक चरणों के आधार पर अनुप्रयोग की तुलना में सुपारी के एफआरडी का बेहतर नियंत्रण प्रदान करता है। वृक्षारोपण के आयु वितरण का विश्लेषण किया गया और एक कार्यप्रणाली प्रस्तावित की गई। मशीन लर्निंग का उपयोग करके स्थान विशिष्ट दृष्टिकोण का उपयोग करके नारियल के ताड़ में फोलियर/मिड्री पोषक तत्वआधारित उपज पूर्वानुमान मॉडल विकसित किया गया था। इनके अलावा, कुछ महत्वपूर्ण सांख्यिकीय गतिविधियों में इनवेसिव सफ़ेद मक्खी की कीट गतिशीलता का स्थानिक विश्लेषण शामिल है, जिसमें परिणामों से पता चला है कि सफ़ेद मक्खी प्रजातियों ने पाम ट्री के शीर्ष श्रृंग लीफलेट को प्राथमिकता दी, नवीन दृश्य के लिए आर/पायथन में कंप्यूटर प्रोग्राम को लागू किया और जीजीई बाइप्लॉट विश्लेषण के लिए कम्प्यूटेशनल स्क्रिप्ट, बीएलयूपी अनुमान, सरिमा और एआरसीएच टाइम सीरीज मॉडल रोपण डेटा पर आधारित हैं।

डॉ. एम.आर. मणिकंदन को कृषि के क्षेत्र में एनएएस फेलो 2024 (इंजी. एवं तकनीकी) के रूप में चुना गया। डॉ. एस. बीगम को

भारतीय बागवानी विज्ञान अकादमी, नई दिल्ली की एसोसिएटशिप से सम्मानित किया। डॉ. आर. पांडीसेल्वम ने स्टैनफोर्ड विश्वविद्यालय, यूएसए और एल्सवियर द्वारा प्रकाशित विश्व शीर्ष 2% वैज्ञानिक सूची में स्थान अर्जित किया है। डॉ. टी. शिवकुमार, एसएमएस ने कासरगोड़ में आयोजित स्थापना दिवस समारोह के दौरान भा.कृ.अनु.प.-के.रो.फ.अ.सं. का सर्वश्रेष्ठ तकनीकीकर्मचारी पुरस्कार प्राप्त किया। डॉ. मधु, टी.एन. चेन्नई में 'कृषि और औद्योगिक एन्टोमोलॉजी और पर्यावरण विज्ञान में हालिया प्रगति और खाद्य पर्यावरण सुरक्षा पर उनका प्रभाव' पर पांचवें राष्ट्रीय सम्मेलन में युवा वैज्ञानिक पुरस्कार प्राप्त किया। डॉ. पांडीसेल्वम, जर्नल ऑफ फूड साइंस एंड टेक्नोलॉजी से खाद्य प्रौद्योगिकी की श्रेणी में 'बेस्ट पेपर अवार्ड' प्राप्त किया। सूखा सहिष्णुता के लिए कोको जीनोटाइप के ट्रांसक्रिप्टोमिक विश्लेषण के लिए सर्वश्रेष्ठ पोस्टर प्रस्तुति पुरस्कार से श्रीमती सुचित्रा को पुरस्कृत किया गया।

2 सितंबर, 2023 को भा.कृ.अनु.प.-के.रो.फ.अ.सं., कासरगोड़ और सीडीबी, कोच्चि द्वारा संयुक्त रूप से विश्व नारियल दिवस का आयोजन किया गया था, जिसमें माननीय कृषि और किसान कल्याण राज्य मंत्री सुश्री शोभा करंदलाजे ने इस अवसर पर भाग लिया। वर्ष के दौरान अन्य महत्वपूर्ण घटनाओं में 20-22 फरवरी 2023 के दौरान एआईसीआरपी-पीएचईटी की वार्षिक समूह बैठक और 14 नवंबर 2023 को हितधारकों की भागीदारी के साथ सुपारी और मानव स्वास्थ्य पर एक बहु-संस्थागत बैठक शामिल है। पूर्वोत्तर में सुपारी के वैज्ञानिक उत्पादन और प्रसंस्करण प्रौद्योगिकियों को अपनाने के लिए भा.कृ.अनु.प.-के.रो.फ.अ.सं., कासरगोड़ और भा.कृ.अनु.प.-एटीएआरआई, गुवाहाटी, असम के बीच 26 अप्रैल 2023 को एक महत्वपूर्ण एमओए पर हस्ताक्षर किए गए थे।

### III. EXECUTIVE SUMMARY

#### Genetic resource management and utilization

ICAR-CPCRI, Kasaragod hosts the world's most extensive collection of coconut germplasm, encompassing 455 accessions from 27 nations, conserved at Kasaragod and International Coconut Gene Bank for South Asia and the Middle East at ICAR-CPCRI Research Centre, Kidu. The collection across three mandate crops includes 455 accessions in coconut, 172 accessions in arecanut and 531 accessions in cocoa germplasm, preserving maximum diversity in these crops to facilitate conservation, characterization and subsequent utilization in crop improvement programs. Beyond the *ex-situ* field gene bank, significant number of accessions are also preserved through cryopreservation of zygotic embryos, pollens and DNA.

A superior *inter se* coconut hybrid 'Kalpa Vajra' a tall variety, suitable for cultivation in areas affected by root (wilt) disease, and a coconut variety 'Kalpa Suvarna' an improved dwarf, dual-purpose variety, suitable for tender nut and copra have been released this year. Additionally, two high-yielding cocoa hybrids (VTLCH-1 and VTLCH-2) and two cocoa varieties (VTLCS-1 and VTLCS-2) suitable for high-density planting systems have been released based on their superior performances.

At Kasaragod, the Jamaica Tall coconut variety, known for its consistent and superior performance in nut, copra and inflorescence sap yield, has been recommended for release. Under germplasm collections, 28 new coconut collections, exhibiting superior attributes like thin husk, larger fruit volume, higher kernel content and higher tender nut water, have been obtained from Nan Cowry, Aves Island of Nicobar district and North & Middle Andaman district.

During the period, 31 coconut accessions native to Island (A&N and Lakshadweep) and Coastal (Kerala) ecosystems have been characterized at Research Centre, Kidu. Among these, Champin Micro Tall, with high nut and copra yielding potential, has been identified for further studies. Evaluation trials at different locations have

highlighted specific coconut germplasms, such as PHOT with superior tender nut traits (in Mohitnagar) and Kera Sankara hybrid with higher yield potential (in Assam). In evaluation trials at Kasaragod Andaman Green Dwarf, King Coconut and Samoan Yellow Dwarf have exhibited good performance. Twenty-one trait-specific coconut germplasms have been identified and categorized during this period. Efforts to enhance the population of novel sweet-kernel coconut types (Ratnagiri's Mohacho narel sweet kernel) have been undertaken.

At ICAR-CPCRI RS, Vittal DUS guidelines for cocoa have been established, based on 25 characteristics and 69 reference varieties. 31 cocoa genotypes, including 21 hybrids and 10 elite clones, were assessed and classified based on leaf, flower, fruit and bean traits, as well as fat content (using multiscale scores). During the period the dwarf arecanut palm, developed by ICAR-CIARI, Port Blair and ICAR-CPCRI, Kasaragod has been registered by PGRC of ICAR.

Studies on the biochemical composition of Testa oil in coconut revealed that Chandra Kalpa has the highest fat content of 88.8%. Notably, in arecanut germplasm higher fat content of 13.6% and 13.45% has been observed in Cal-32 and SCRDT-18, respectively. In evaluating coconut genotypes for inflorescence sap yield, Nigerian Tall have exhibited the highest yield of 48.03 L per inflorescence, followed by Philippines Kalamthem Tall (36.10 L/inflo.). Phytochemical analysis of coconut pollens have shown pollens to be an excellent supplement for human nutrition, with a crude protein content of up to 39%.

Efforts have been made to explore the screening of pollen grains for their Osmotic Adjustment Potential as an indicator for drought tolerance in coconut. Preliminary studies on coconut seedling production exclusively for palm cabbage have shown promising results. In the studies on screening coconut hybrids for drought tolerance, CODxADOT, MYDxWCT and CODxLCT have shown good performance under drought imposed conditions.

In terms of planting material production, 144873 coconut, 588970 arecanut and 48893 cocoa planting materials have been produced and distributed during the year.

### Biotechnological investigations

Significant progress has been made in enhancing callus induction, embryogenic calli formation and somatic embryo development in coconut tissue culture through the utilization of novel growth regulators. Y3 with 2, 4-D ( $1 \text{ mg L}^{-1}$ ) is identified as the most suitable for culture initiation in immature inflorescence explants. In the studies on ideal size of inflorescence, optimal shoot regeneration in explants have been obtained from 5-10cm and 10.5-15cm long inflorescences. Increased vegetative buds during culture initiation have been found using Woody Plant medium supplemented with TDZ ( $5 \text{ mg L}^{-1}$ ), alone or combined with NAA. In the embryo rescue of sweet kernel genotypes, approximately 30 plantlets have been potted and 60 plantlets are in the final stage of in vitro culture. Efforts have been directed towards in vitro multiplication of Hirehalli Dwarf, dwarf arecanut hybrids and YLD-resistant palms. SNP's linked to yield-contributing traits were identified using mixed linear model (MLM) analysis, along with the identification of potential candidate genes influencing these traits. Cryopreservation studies on validation of V-cryo mesh and V-cryo plate method for coconut and arecanut resulted maximum plumule and plantlet recovery with V-cryo mesh (73% and 33%), followed by V-cryo plate (56% and 23%) and encapsulation vitrification methods (46% and 20%), respectively.

The 'Kalpa Genome Resource' a comprehensive genomics platform for the dwarf coconut variety 'Chowghat Green Dwarf' (CGD), integrating transcriptome and fingerprinting data has been hosted allowing users to access genome and transcriptome resources. An insect bioassay studies on the rhinoceros beetle, siRNAs were administered through injection for evaluation. A considerable significantly elevated mortality rate of  $\geq 50\%$  was noted starting from the 3<sup>rd</sup> day after administering oral dsRNA in comparison to the non-target IMPI dsRNA. The study on identification of suitable reference gene(s) for expression studies in different developmental stages of coconut rhinoceros beetle identified NADH/EF1A as the most reliable reference gene

combination, that can provide robust normalization of RT-qPCR data in gene expression studies.

### Input use efficiency for productivity and environmental security

Coconut based farming system model in one hectare at Kasaragod comprising coconut, pepper trailed on the coconut trunk, banana in the border of the plots, fodder sorghum (CO 31- multi cut fodder sorghum) in the interspaces of coconut, along with a dairy unit, goat unit and poultry, achieved a net return of Rs. 5,43,682 and demonstrated high energy efficiency with an output of  $884.9 \text{ KJ ha}^{-1}$ . Intercropping multi-cut sorghum (CO31) in coconut gardens with 100% organic nutrition yielded  $79.9 \text{ t ha}^{-1}$  of green fodder. A coconut based high density multi species cropping system, comprising coconut, pepper, banana, Robusta coffee, cinnamon and nutmeg, along with nutrient management practices and biomass recycling, resulted in 4.9 times higher net returns compared to coconut monocrop, amounting to Rs. 4,81,416. Intercropping cinnamon in the coconut garden following the pentagonal method of planting ( $0.6 \text{ m} \times 1.2 \text{ m}$  spacing with 5 plants per pit) recorded a significantly higher dry quill yield of  $631.92 \text{ g tree}^{-1}$  and  $979.0 \text{ kg ha}^{-1}$ . At Kahikuchi, resource recycling and application of organics and  $3/4^{\text{th}}$  RDF to arecanut in HDMSCS could give maximum yield in arecanut, saving 25% RDF. Cabbage emerged as the best intercrop in arecanut among winter vegetables, yielding  $13.17 \text{ t ha}^{-1}$  with a higher B:C ratio of 2.04.

Plastic mulching during the monsoon in YLD-affected arecanut gardens delayed symptom expression, increased yield and improved soil physical properties. The carbon sequestration in arecanut-cocoa cropping systems was estimated at  $167 \text{ t CO}_2 \text{ ha}^{-1}$ , offering significant contribution in combating climate change. Genotypic variability in nutrient export was observed among 20 cocoa genotypes with 19-71 g N, 2.5-9.4 g P, 27-118 g K, 5.5-21.1 g Ca, 4.3-15.6 g Mg, 71-336 mg Fe, 182-1304 mg Mn, 29-152 mg Zn, 68-255 mg Cu, and 37-153 mg per tree per year, indicating removal of huge quantities of nutrients, especially potassium. Nitrogen, zinc and potassium-use efficient cocoa varieties were identified. Hybrid and dwarf coconuts responded well to high fertilizer levels

through drip fertigation (200% RDF), resulting in increased yields in Kalpa Shankara and COD, and TSS and water content in COD. Cultivation of coconut under organic nutrient management practices recorded a higher yield of 102.4 nuts/palm/year with 174 g of copra weight/nut over other treatments. Humic acid (10% and 20%) combined with farmyard manure (50 kg/palm) enhanced the soil organic carbon and facilitated the slow release of nutrient ions. Region-specific nutrient recommendations in coconut using statistical models indicated that the Random Forest model performed better, with a 77.08% accuracy rate, and soil pH, organic carbon, and soil available nutrients (K, Cu, Mn and leaf nutrients (P, Ca, and Mg) were found to have significant effects on the yield of coconut. The bacterial microbiome in the YLD endemic rhizosphere soil was characterized through metagenomics, revealing a higher abundance of the chemolithotrophic potential microbiome in the arecanut rhizosphere and they are the key players of microbiome-mediated nutrient transformations of N, Fe, S and P in YLD endemic rhizosphere.

A technology to convert combined biomass residues of palms and cocoa to vermicompost has been standardized using CPCRI earthworm, *Eudrilus* sp., with primed coconut husk biochar as a novel component. The final product has incremental pH, electrical conductivity, P, K and dehydrogenase activity compared to coconut leaf vermicompost. Plant pathogenic and predatory nematodes were absent. Maximum available nutrient with faster release was noticed in the size 0.5-1.0 mm of cocoa husk biochar particles and hence, the biochar of less than 1mm size can be recommended for efficient utilization of the same. Technology to convert mature coconut water (liquid endosperm), which is wasted in huge quantities from coconut kernel processing industries, to vinegar, 'nata'/bacterial cellulose (BC), has been standardized using indigenous bacteria isolated from coconut vinegar. Maximum BC production was obtained in the pH range of 4-5 for all the isolates except BC4, in which maximum BC production was obtained in a wider pH range of 5-10. Microbial studies at the time of sowing coconut seed nut and at the time of root emergence indicated that, rhizosphere and endosphere microbial populations reached a stable population

at 5 to 6 months after the sowing of seed nut. Biopriming with PGPR strains and consortia combined with fertilizers and organic manures (including green leaf manure and basin management with cowpea) as per the standard recommendations for RWD tract kept the seedlings 'apparently healthy' as per the root (wilt) disease indexing method. The naturally growing mushrooms on the arecanut husk were identified as *Coprinopsis cinerea* CCBVH1.

### Integrated pest and disease management

The association of *Lasiodiplodia theobromae* with fruit rot of soursop in parts of Kasaragod district was reported. A cost effective, simple and rapid technique called "Detached Leaf Assay", using zoospore suspension of highly virulent *Phytophthora meadii* isolate (P19) was developed to accelerate identification of disease resistant arecanut germplasm/hybrids. The spatially explicit risk estimation and probability distribution maps generated will help in the development of management strategies for the basal stem rot disease in southern tracts of India. The Standard Operating Procedure (SOP) for coconut palms using UAV sprayer was optimized. The association of virus with ring spot disease of arecanut was reported for the first time in India. The PGPR *Bacillus* sp. strain SHIS11 recorded maximum plant height and biomass in both seed treatment and soil drenching in cocoa. No A to G transition observed in coconut rhinoceros beetle samples obtained from different regions indicating the absence of CRB-G haplotype in India. Among the diverse whiteflies, *Paraleyrodes. bondari*, followed by *Aleurodicus rugioperculatus* and to limited extent by *Aelurotrachelus atratus* were currently recorded on coconut palms with complete displacement of *A. dispersus* as well as *P. minei* in coconut system. Morphometric characters of different stages of palm whitefly *Aleurotrachelus atratus* and its molecular identification was established. Molecular characterization of coreid bug, *Paradasynus rostratus* using *COI* gene collected from Kayamkulam was conducted and its linkage with Coreidae species was established. About 11 new hosts for the coreid bug was recorded and the laboratory rearing protocol using fruits was developed. A refined integrated management strategy for RPW in endemic area was validated. An entomopathogenic fungus isolated from

infected red palm weevil, identified as *Beauveria bassiana*, was found promising for the bio-suppression of RPW and a solid state formulation of this biocontrol agent was developed. An insect repellent “kernel pellet” was developed for the field level repulsion of coconut rhinoceros beetle. Conservation biological control was equally effective against rugose spiralling whitefly as that of palms sprayed with *Isaria fumosorosea* as well as neem oil. An entomopathogenic fungus (EPF), *Simplicillium lanosoniveum* (Cordycipitaceae: Hypocreales) isolated from the nymphal cadavers of rugose spiralling whitefly was found virulent and pathogenic to all the developmental stages of RSW and cost-effective bioformulations were evolved. Developed *Galleria* cadaver-based sachet formulation of EPN for the insect pest management in coconut/arecanut system. Field treatment of Kalpa EPN was found effective for the management of root grub and pumpkin beetle in coconut system. Field level management strategies were developed for early stage infestation of root-knot nematode (*Meloidogyne enterolobii*) in guava. *Trichoderma* enriched neem cake technology was found effective for the sustainable management of root-knot nematodes in black pepper intercropped in coconut/arecanut system. Among six EPN isolates tested, *Steinernema carpocapsae*, *Heterorhabditis indica* and *Metarhabditi srainai* were promising for the management of termites. The presence of burrowing nematode, *Radopholus similis* was observed in all the seedlings evaluated with a population range of 9 to 70 nematodes/seedlings. Field efficacy of seven essential oil repellents evaluated against RSW on orange dwarf variety of coconut at ICAR-CPCRI farm, Kasaragod was superior to control. The RPW detector has undergone refinements and improvements both on the hardware and software fronts. Documented the new distribution record of mango mealy bug, *Rastrococcus iceryoides* (Green) (Hemiptera: Homoptera: pseudococcidae) on orange dwarf variety of coconut in Kasaragod. *Metarhizium anisopliae* was promising for the management of inflorescence caterpillar in arecanut. A simple and cost-effective protocol for the mass multiplication of coccinellid predator, *Stethorus keralicus* on red mite under laboratory conditions.

### Climate resilience and value addition technologies

A MaxEnt model-based assessment of habitat

suitability for coconut cultivation under future climatic conditions revealed that the bioclimatic variables, namely temperature seasonality, temperature annual range and precipitation, are the major contributors that determine climate suitability for coconuts in India. Projections suggest that some current coconut cultivation regions will become unsuitable (plains of South interior Karnataka and Tamil Nadu), requiring a crop change, while other areas will require adaptations in genotypic or agronomic management (east coast and the south interior plains). In other regions, the climatic suitability for growing coconut will increase (west coast).

In a biochemistry study, a mixture of 15% defatted coconut flour and 85% rice flour composition showed favorable starch: protein interactions, causing the conversion of the amorphous structure of starch granules in the rice flour to a crystalline structure, positively correlating with a decline in the glycemic index.

In the field of post-harvest management, a gum Arabic-based edible coating demonstrated its efficacy in delaying quality changes and extending the shelf life of the coconut haustorium for up to 20 days under refrigerated conditions (4-10 °C). A protocol for the production of ready-to-serve coconut flakes infused with coconut neera syrup has been developed. Coconut flakes dried at 65°C using infrared-assisted hot air dehydrated coconut flakes (IRHAD) exhibited a superior nutritional and physical profile.

A comparative analysis of emerging drying techniques—Infrared Drying (IRD), Infrared-Assisted Hot Air Drying (IRAHAD) and the conventional Hot-Air Drying method (HAD)—revealed that IRAHAD (at 65°C) treatment of coconut chips exhibited a superior drying rate, a shorter drying time and biochemically superior coconut chips. Similarly, independent and combined effects of infrared and hot air drying techniques on copra quality suggest that the adoption of the IRAHAD method at a temperature of 60°C preserved the fat content (68.4%), a crucial factor for enhanced oil extraction from copra.

A biodegradable, eco-friendly composite paper was developed from tender coconut husk with plant-based starch. X-ray diffraction (XRD) and scanning electron microscopy (SEM) studies

demonstrated robust interactions between the hydroxyl groups of starch and the cellulose of the coconut husk matrix. The product displayed a remarkable degradation rate (~70%) within a period of 20 days.

### **Outreach activities, socio-economic perspectives and statistical aspects**

Technology transfer approaches and strategies customized to location specific needs, evolving community and area wide approaches in pest and disease management and landscape based participatory social models for doubling farmers' income, promotion of Farmer Producer Organizations (FPO) and reaching out to farm households through MGMG constitute the interventions and programs. This year, institute has organized a total of 65 institutional and off-campus training programmes benefitting 3,648 farmers and other stakeholders. Besides, one day exposure visit programmes were conducted during the year at headquarters, regional stations and research centers of the institute benefitting a total of 7468 stakeholders including farmers. Also, seven Entrepreneurship Development Programmes were organized by this institute benefitting 539 entrepreneurs. Special training programme of six days duration for 50 rural unemployed SC/ST youths were conducted. A Kisan Mela was organized at CPCRI RS Kayamkulam in connection with the Platinum Jubilee celebrations. A total of 23 FLDs were carried out in different regions during 2023.

Seedlings and other farm inputs worth Rs 90,81,185/- were sold to farmers and other stakeholders. As part of the interventions under the NEH, training and input distribution programmes were conducted for villages of north eastern parts of the country. A five day capacity building programme for North – East KVK, Scientists was organized. Under the SCSP and TSP, more than 20 training and input distribution programmes were organized and around more than 1500 SC and/or, ST farmers were among the beneficiaries across the various part of the country. A study regarding 'Profile Analysis of Farmer Producer Companies in India' was conducted on the Farmer Producer Companies in Assam. Analysis also indicated that most of the FPC members have access to the market through the CEOs of FPCs in the region. The ICAR Farmer

FIRST Programme (FFP) social, individual-based farm models are out scaled. A total of 223 extension activities with women SHG farmers were organized. The participation of women SHG farmers could double their income drawing the potential of sesame and tuber crops. The content mobilization of the institute is mainly through YouTube, Facebook, WhatsApp and print media.

The economic causes of the recent price crash regime of the coconut have been studied from a policy perspective. It is striking that coconut and coconut oil prices have experienced the lowest price regime in the last five years. India's export basket of coconut value-added products experienced a radical shift over a 10-year period, wherein activated carbon dominated the total coconut exports with a share of 67% in 2022, while the share was 45% in 2013. The cost of production of coconut in Kerala based on data from a well-managed coconut garden is calculated to be Rs. 12.10 per nut.

Spatial studies and assessment of potential risk regions to coconut diseases were conducted for Tamil Nadu, Kerala, Karnataka and Andhra Pradesh states of India. The Basal stem rot of coconut severity rates were higher in Thanjavur, Pudukkottai and Thiruvavur districts of Tamil Nadu and also some areas adjacent to the East Godavari and West Godavari of Andhra Pradesh. A statistical study revealed that the timing of fungicidal application based on the monsoon period provides better control of FRD of arecanut than an application based on the developmental stages of fruit under field conditions. Age distribution of plantations was analyzed and a methodology was proposed. Foliar/ soil nutrient based yield prediction model in coconut palms was developed using location specific approach using a machine learning approaches. Apart from these, some important statistical activities includes that spatial analysis of pest dynamics of invasive whiteflies, in which the results revealed that whiteflies species preferred tip leaflets of the palm tree, implemented computer programs in R/ python for innovative visualization and also the computational script for GGE biplot analysis, BLUP estimation, SARIMA and ARCH time series models based on plantation data.

Dr. M.R. Manikantan was elected as NAAS Fellow



2024 in the field of Agrl. Engg. & Tech. Dr. S Beegum awarded Associateship of Indian Academy of Horticultural Sciences, New Delhi. Dr. R. Pandiselvam featured in world top 2% scientist list published by Stanford University, USA and Elsevier. Dr. T. Sivakumar, SMS received the Best Technical Staff Award of ICAR-CPCRI during Foundation Day Celebration held at Kasaragod. Madhu, T.N. received Young Scientist Award at 5th national conference on "Recent Advances in Agricultural and Industrial Entomology and Environmental sciences and their impact on food environmental security" at Chennai. Dr. Pandiselvam, R. *et. al.*, received "Best Paper Award" in category of Food Technology from Journal of Food Science and Technology. Best poster presentation award for Transcriptomic analysis of cocoa genotypes for drought tolerance

was presented to Suchithra *et al.*

World Coconut Day was organised jointly by ICAR-CPCRI, Kasaragod and CDB, Kochi on 2nd September, 2023 in which Sushri Shobha Karandlaje, Hon'ble Minister of State for Agriculture and Farmers Welfare graced the occasion. Other important events during the year include the Annual Group Meeting of AICRP-PHET during 20th-22nd February 2023 and a multi-institutional meeting on arecanut and human health with the involvement of stakeholders on 14th November 2023. An important MoA for enhancing adoption of scientific production and processing technologies of arecanut in NE was signed between ICAR-CPCRI, Kasaragod and ICAR-ATARI, Guwahati, Assam on 26th April 2023.

## IV. VISION, MISSION AND MANDATE

### **Vision**

To develop CPCRI as a technology generation and repository centre, wherein the Institute strives to showcase, demonstrate and compare world-wide technologies in the commodity chains of coconut, arecanut, cocoa and palmyrah to make India the global leader.

### **Mission**

To develop technologies that enhance resource use efficiency, profitability and livelihood security of people who depend on plantation crops.

### **Mandate**

- Basic, strategic and applied research to enhance sustainable productivity, quality and utilization of coconut, arecanut, cocoa and palmyrah,
- Repository of plantation crops genetic resources and scientific information,
- Transfer of technology, capacity building and impact assessment of technologies, and
- Coordinate research and validation of technologies on plantation crops through AICRP on Palms.

## V. INSTITUTE PROFILE

ICAR-Central Plantation Crops Research Institute (ICAR-CPCRI), the premier research institution in the National Agricultural Research System of India, is presently mandated to conduct research on plantation crops (coconut, arecanut, cocoa and palmyrah). It had a modest beginning as Coconut Research Station in 1916 under the erstwhile Madras Presidency. Since its inception, it has served the farming community with a distinction through exemplary research, generation of appropriate technologies and development of the skilled human resource.

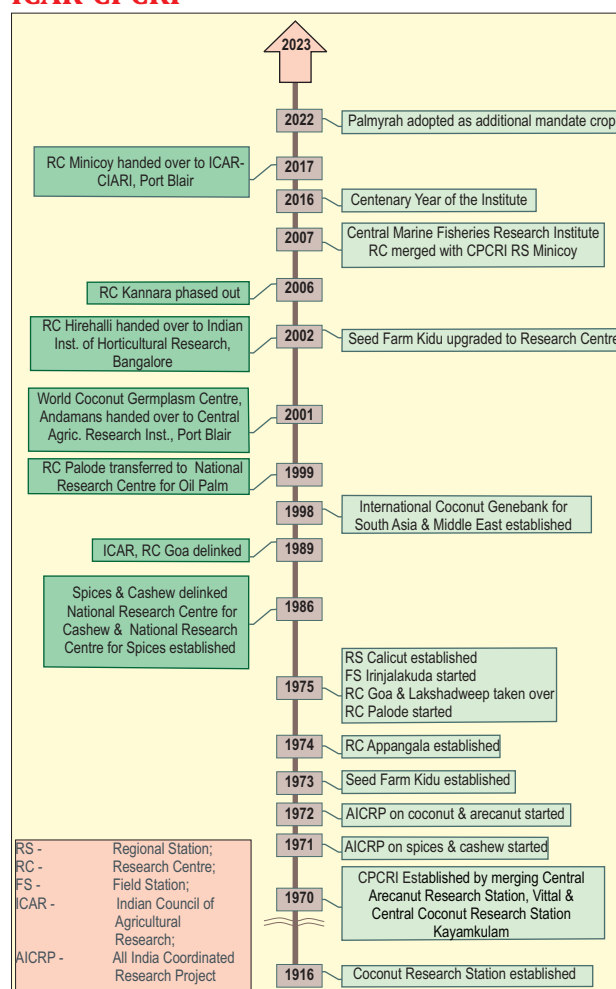
### Historical Perspective

The Coconut Research Station at Kudlu (Kasaragod) was taken over by the Indian Central Coconut Committee and established the Central Coconut Research Station (CCRS), Kasaragod in 1947 and the Central Coconut Research Station (CCRS) at Kayamkulam was also established exclusively for tackling diseases in coconut. Coconut research became an integral part of the national agricultural research system in 1966 when the Indian Central Coconut Committee was abolished and the coconut research was taken over directly by the Indian Council of Agricultural Research. In 1970, the Central Plantation Crops Research Institute was established with the headquarters at Kasaragod, by merging the Central Coconut Research Stations at Kasaragod and Kayamkulam and the Central Arecanut Research Station at Vittal along with its five substations at Kannara, Mohitnagar, Kahikuchi, Hirehalli and Palode.

Since 1986, crops like spices, cashew, and oil palm were taken out of the purview of the institute with the formation of dedicated research institutions like Indian Institute of Spices Research, Kozhikode, Directorate of Cashew Research, Puttur and Indian Institute of Oil Palm Research, Pedvegi. Some of the erstwhile Research Centres at Hirehalli, Palode, Appangala, Kannara, Port Blair and Minicoy were either handed over to sister ICAR institutions or phased out. At present, the mandated crops are limited to coconut, arecanut, cocoa and palmyrah and the research and frontline extension aspects of these crops are

undertaken under five divisions *viz.*, Crop Improvement, Crop Production, Crop Protection, Physiology, Biochemistry and Post-Harvest Technology and Social Sciences at the Institute. The Regional Station at Kayamkulam (Kerala) is mandated to work on pests and disease problems in coconut, while the Regional Station at Vittal (Karnataka) caters to research and extension in arecanut and cocoa. The Research Centres at Kahikuchi (Assam) and Mohitnagar (West Bengal) undertake location-specific research in these crops, while the Research Centre at Kidu (Karnataka) hosts the International Coconut Gene Bank for South-Asia and the Middle East (ICG-SAME) and also caters to the large-scale production of quality planting materials in the mandate crops. Besides, there are two KVKs (at Kasaragod and Kayamkulam) functioning under the Institute.

### Milestones in the timeline of ICAR-CPCRI





involving talls and dwarfs as parents have been released for commercial cultivation. The high yielding varieties are capable of yielding 3.12 to 6.28 tonnes of copra ha<sup>-1</sup> annually, as compared to 2.96 t copra ha<sup>-1</sup> in West Coast Tall local. Kalpa Vajra a tall *inter se* progeny of coconut developed by hybridization (WCT x WCT) was recommended specifically for the root (wilt) disease prevalent tract. It has an average yield of 91 nuts/palm/year compared to WCT with a yield of 69 nuts/palm/year in the region. Kalpa Vajra is also suitable for tender nut, copra and inflorescence sap production. A dwarf selection of coconut was also released in the name of Kalpa Suvarna, suitable for copra and tendernut production, which gives an average yield of 97 nuts/ palm/ year. Quantity of tender nut water is around 431 ml and very good in taste with TSS of 5.8 °Brix. The variety is recommended for the states of Karnataka and Kerala. Eleven improved varieties of arecanut, including nine selections and two dwarf hybrids, have been released till date. The improved varieties with annual average yield of 2.54 to 4.15 kg dry kernel palm<sup>-1</sup> year<sup>-1</sup> and higher dry kernel recovery, in comparison to South Kanara Local (2 kg dry kernel palm<sup>-1</sup> year<sup>-1</sup>), have significantly improved arecanut productivity in the country. In cocoa, eight high yielding varieties have been released from the Institute, which include three elite clones and five hybrids, which yield up to 3.0 kg dry bean tree<sup>-1</sup> year<sup>-1</sup> with varying processing qualities, as

Through intensive breeding and evaluation, 23 improved coconut varieties including six hybrids

compared to 1.0 kg dry bean tree<sup>-1</sup> year<sup>-1</sup> in existing cocoa plantations.

The Institute has been producing quality planting materials annually in coconut, arecanut and cocoa to the tune of 1.2 lakhs, 5 lakhs and 0.5 lakhs respectively, for distribution to farmers and other stakeholders. Seed gardens of improved varieties have been established in the Institute as well as in farmer's fields to augment planting material production. ICAR-CPCRI nurseries at Kasaragod, Kidu, Kayamkulam and Vittal were graded with 'four-star' status in the five star scale by National Horticultural Board. DUS guidelines for coconut, arecanut and cocoa were published through PPV&FRA, New Delhi.

### Biotechnology and Bioinformatics

Achievements under biotechnology include standardization of embryo culture protocol for germplasm exchange, standardization of regeneration protocol for inflorescence tissues of arecanut and cryopreservation of coconut DNA, embryo and pollen. In arecanut, the protocol developed for somatic embryogenesis and plantlet regeneration from immature inflorescence explants has been commercialized. A simple and easy vitrification protocol has been developed for cryopreservation of coconut zygotic embryos from both tall and dwarf accessions. The protocol developed for cryopreservation of coconut pollen for the first time by the Institute, has been commercialized; this would be instrumental in enhancing hybrid seed production as it facilitates year round availability of coconut pollen for all stakeholders across the coconut growing states of India. The safe movement of coconut germplasm through embryo cultures, instead of seed nuts, is recommended by FAO/ IPGRI.

Sequence characterized amplified regions (SCAR) markers have been developed for confirming the hybridity at seedling level in both coconut and arecanut. A panel of SSR markers has been identified for confirming the hybridity of D x T hybrids (CGD x WCT) which will ensure supply of genuine hybrid material to farmers. In coconut, DNA fingerprint profiles have been developed for 25 each of tall and dwarf accessions using polymorphic SSR and EST-SSR markers. Transcriptome analysis of response of coconut to root (wilt) disease and somatic embryogenesis

have been undertaken using RNA-Seq and transcripts up/ down-regulated have been identified. Many of transcripts down-regulated in root (wilt) diseased palms were primarily involved in defense responses, signaling pathways, cellular transport and other metabolic processes. Transcriptome analysis of coconut embryogenic calli, derived from plumular explants of West Coast Tall, resulted in the identification of 14 genes with important roles in somatic embryogenesis. Work on deciphering the genome sequence of Chowghat Green Dwarf has been initiated.

The Institute has developed various tools and databases, which include MAPS (Microsatellite Analysis and Prediction Software), stand alone EST-SSR analysis pipeline (SEMAT), prediction tools for resistant gene analogues and enzymes in gibberellic acid biosynthesis using machine learning algorithms, prediction of miRNAs in date palm, coconut and *Phytophthora* spp. and transcriptome based reconstruction of carotenoid biosynthetic pathway in cocoa and gibberellic acid biosynthetic pathway in coconut.

### Cropping and Farming Systems

Coconut based farming system comprising coconut, pepper trailing on the coconut trunk, banana in the border of the plots, fodder grass (Hybrid Bajra Napier Co 5) in the interspaces of coconut, dairy unit and poultry could fetch net returns of Rs. 6,53,853/- per year. Groundnut intercropping under coconut by following integrated nutrient management and application of farmyard manure recorded a higher pod yield of 2189 and 1932 kg of Girnar-2 and Girnar-3 varieties per hectare of coconut garden, respectively. Compared to coconut monocropping, the cropping system can provide more than three times farm income. In the case of arecanut, the income enhancement through intercropping would be 75% to 130%.

Drip irrigation in arecanut, coconut and cocoa has reduced the use of water to the extent of 35-40 per cent, with increase in yield by 30-40 per cent. Drip fertigation in these crops has reduced the use of chemical fertilizer from 50 to 75 per cent, with increase in yield by 35-40 per cent. *In situ* soil and water conservation techniques such as, half-moon bund reinforced with pineapple planting, trench filled with coconut husk and bund reinforced with

pineapple planting and providing catch pits help in augmenting the soil moisture availability in coconut plantations having mild slope and could enhance coconut yield up to 60%. This could reduce soil erosion from 2.73 t ha<sup>-1</sup> to 0.02 t ha<sup>-1</sup> and consequent reduction of nutrient loss due to soil erosion (N from 7.98 to 0.36 kg ha<sup>-1</sup>, P from 12.52 to 0.9 kg ha<sup>-1</sup> and K from 28.5 to 1.1 kg ha<sup>-1</sup>).

The productivity of coconut in coastal sandy soil, which is made of 99% sand, is very low (30 nuts palm<sup>-1</sup> year<sup>-1</sup>) due to the porous nature and low fertility. Incorporation of coconut husk in the interspaces of the coconut garden and growing various intercrops like vegetables, flowers, grasses and pineapple and fertigation along with mulching to coconut has increased the yield of coconut to 140 nuts palm<sup>-1</sup> year<sup>-1</sup>. The intercrops generated an additional income of Rs. 2.5 to 3.5 lakh ha<sup>-1</sup> of coconut garden.

### Bioresources Utilization

Recycling crop wastes in coconut, arecanut and cocoa through vermicomposting and mushroom production helps in disposing of wastes, improving soil fertility, reduction in use of chemical fertilizers and sustaining the yield besides enhancing nutritional security. Coconut gardens of one hectare area can generate up to eight tonnes of leaf biomass residues every year. Technology has been developed to utilize these wastes for production of vermicompost, vermiwash, compost and mushrooms. From about eight tonnes of leaf residues, 3-4 tonnes of vermicompost could be produced annually using the local isolate of *Eudrilus* sp. The leaf residues can also be used for growing oyster mushroom to the tune of 1,660 kg of fresh mushroom. The coconut leaf vermicompost can also meet 50% of the nitrogen requirement of coconut palms grown in one hectare area saving expenditure on inorganic fertilizer. A 'push-pull' strategy was successfully adopted to harvest earthworms from vermicompost heaps through the use of behaviour-modifying stimuli. Vermiwash, produced from coconut leaf vermicomposting unit, is a good liquid fertilizer for organic farming. On farm coir pith composting technology has been developed to produce organic input to the plantation as well as for soil-less medium to produce quality planting material.

Arecanut and cocoa gardens generate biomass of 4-5 and 0.7-0.8 million tonnes ha<sup>-1</sup> respectively and these wastes could be effectively utilized for production of oyster mushroom and livestock feed, in addition to vermicompost. Recyclable biomass in arecanut supplies approximately 95 g N, 10 g P<sub>2</sub>O<sub>5</sub> and 110 g K<sub>2</sub>O palm<sup>-1</sup> year<sup>-1</sup> that has the potential to meet nitrogen and phosphorus requirements of arecanut, which can save the cultivation cost to the extent of Rs. 5,200 ha<sup>-1</sup>. Arecanut leaf sheath and bunch waste from one hectare garden can be used in production of 643 kg fresh mushroom.

Cocoa pod husk biochar (CPHB) with about 60% of the total potassium content and around 7.5% exchangeable K, can be used as an amendment to improve soil potassium. Biochar production units by gasification and pyrolysis has been developed.

In the area of microbial bioresources, plant growth promoting rhizobacteria (PGPR) based bioinoculant products, 'Kera Probio<sup>®</sup>' containing *Bacillus megaterium* and 'Cocoa Probio<sup>®</sup>' containing *Pseudomonas putida* have been released for production of healthy and vigorous coconut and cocoa seedlings. The genes involved in the plant growth promoting properties and other important metabolic functions of three PGPRs, one each from coconut, arecanut and cocoa, have been identified through whole genome sequencing. An efficient zinc solubilizer has been identified from alkaline soil which could increase availability of soluble zinc in soil. This bioresource could prove to be useful in regions where zinc availability is a problem.

### Reducing Crop Losses

Bud rot, stem bleeding, basal stem rot and root (wilt) of coconut; fruit rot, inflorescence die back and yellow leaf disease of arecanut and black pod and stem canker in cocoa are the major diseases that cause substantial crop losses. Integrated disease management strategies developed for the major diseases over the years has resulted in saving of thousands of coconut and arecanut palms and reduced the loss due to black pod diseases in cocoa. Most importantly, the disease management strategies are being continuously refined based on the change in pathogen population, soil and climatic factors and screening of new and native

bioagents or fungicides or host plant resistance. The role of slug *Deroceros* sp. in spreading of bud rot has been confirmed by observing the presence of sporangia of *P. palmivora* in faecal matter of the slugs collected from bud rot affected garden and proving its pathogenicity on coconut. Prophylactic treatments of Bordeaux mixture (1%) or placement of two perforated sachets containing chlorothalonil 78 WP (3g) or *Trichoderma* coir pith cake in the innermost leaf axil of coconut with the onset of monsoon (first week of June) can prevent the appearance of bud rot in disease endemic areas. Basal stem rot disease caused by *Ganoderma lucidum* is another major disease of coconut and soil application of *Trichoderma* enriched neem cake (5 kg palm<sup>-1</sup>) at quarterly interval was very effective in reducing the disease incidence. Identified and characterized a native plant growth promoting isolate of *Trichoderma asperellum* (isolate AT172) having antagonistic activity against arecanut basal stem rot pathogen *G. lucidum*. Stem bleeding disease of coconut could be effectively controlled using fungicides (Hexaconazole 5EC and Propiconazole 25 EC).

Root (wilt) disease of coconut caused by phytoplasma is another major disease and efforts were made to improve the PCR-based diagnostic techniques for reliable early detection of phytoplasma. A package of practice for the disease affected regions had been standardized for improving the yield.

Spraying of Bordeaux mixture (1%) or mandipropamid (0.5%) was effective in reducing the fruit rot disease of arecanut. Among the foliar fungal diseases, inflorescence die back of arecanut caused by *Colletotrichum* spp. and die back of cocoa caused by *Lasiodiplodia* spp. were the major diseases observed. Inflorescence dieback of arecanut could be effectively managed using Propiconazole 25% EC @ 3 ml per litre. *Lasiodiplodia theobromae* causing cocoa die back disease was prevalent in Andhra Pradesh, which could be controlled by removal of symptomatic branches and applying 10% Bordeaux paste on cut branches.

Clean and green innovative pest management technologies have been developed and field validated for the bio-suppression of rhinoceros beetle, red palm weevil, leaf eating caterpillar and

eriophyid mite infesting coconut. IPM module for the management of rhinoceros beetle through integration of biocontrol agents viz., *Oryctes rhinoceros* Nudivirus (OrNV), Green Muscardine Fungus (GMF), *Metarhizium anisopliae*, botanicals (leaf axil filling with neem/ marotti/ pongamia cake @ 250g mixed with equal volume of sand) and aggregation pheromone embedded nanomatrix trap @ 1trap ha<sup>-1</sup> has been developed. Area-wide (1575 ha) farmer-participatory experiments undertaken at Krishnapuram (Kerala), Semanampathy (Tamil Nadu), Voodimudi (Andhra Pradesh) and Doddenhally (Karnataka) significantly reduced the spear leaf and inflorescence damage to an extent of 81.2%. Recently, an agro-ecosystem based pest regression strategy through ecological bio-engineering has been designed for managing rhinoceros beetle, exploiting the interplay of mixed-volatile cues of crop plurality of coconut with spices and fruit trees.

Integrated management technologies involving complete destruction of infested palm, close monitoring and sustained surveillance for early diagnosis, leaf axil filling of chlorantraniliprole sachet, curative management with imidacloprid (0.02%) and pheromone trap @ 1 trap ha<sup>-1</sup> were effective in the management of red palm weevil. Community level technology convergence and large-area adoption of IPM technologies conducted in 2150 ha in Bharanikavu (Kerala), Palladam (Tamil Nadu), Ambajipet (Andhra Pradesh) and Bidramamandi (Karnataka) could reduce the pest incidence to 56.8%. An acoustics-sensor based red palm weevil detector in coconut was developed with 80% accuracy.

Rugose Spiralling Whitefly (RSW) (*Aleurodicus rugioperculatus* Martin) is invading coconut palms in all South Indian States. Aphelinid parasitoid, *Encarsia guadeloupae* Viggiani as natural enemy was successful in controlling the larvae of white flies; conservatory bio-suppression was successful by pesticide holiday. Subsequent development of sooty mould was completely devoured by *Leiochrinus nilgiranus* Kaszab beetles.

For the bio-suppression of leaf eating caterpillar, augmentative release of stage-specific parasitoids viz., *Goniozus nephantidis* and *Bracon brevicornis* @ 20 parasitoids per palm, removal of

heavily damaged outer three leaves and improving soil and palm health of infested garden reduced the leaf damage to 95.3% in a period of 12-15 months. Area-wide field validation and demonstration experiments conducted at Kasaragod (Kerala), Sethumada (Tamil Nadu), Matlapalem (Andhra Pradesh) and Arsikere (Karnataka) in an area of 550 ha recorded a minimal pest incidence of 2.4% from an initial damage level of about 73.4% indicating the success of the technology.

IPM technologies for the suppression of eriophyid mite developed by the Institute involving 2% neem oil-garlic emulsion spray, root feeding of azadirachtin 10000 ppm @ 10 ml + 10 ml water and soil and palm health management practices reduced pest incidence to the tune of 71.4%. From an initial pest incidence of 58.6% observed in Krishnapuram (Kerala), Kottur (Tamil Nadu), Ambajipet (Andhra Pradesh) and Boranakoppalu (Karnataka), the pest incidence was reduced to 16.3% in a period of two years indicating the success of the technology at national level.

Integrated pest management strategies involving soil application of neem cake (2 kg palm<sup>-1</sup>), drenching the root zone with chlorpyrifos 20 EC @ 2.5 ml L<sup>-1</sup> or imidacloprid 17.8 SL @ 675 ml ha<sup>-1</sup> or bifenthrin 10 EC @ 20 litre ha<sup>-1</sup> and entomopathogenic nematodes (EPN), *Steinernema carpocapsae* @ 1.5 lakh IJ ha<sup>-1</sup> during May-June and September- October reduced the arecanut white grub population significantly. Placement of the neonicotinoid, thiamethoxam (2g) in perforated poly sachets in the innermost two leaf axils of areca palms during April-May safeguarded arecanut palms from spindle bug damage. Spraying of neem oil emulsion (0.5%) has been effective in controlling sporadic pests viz., phytophagous mites and pentatomid bugs on arecanut.

Standard operational procedure for drone based spray using specially designed nozzles was developed.

### Climate Resilient Technologies

Coconut, arecanut and cocoa are highly sensitive to climate change variables like high temperature and water deficit stress. The impact, adaptive strategies and the mitigation potential of the above crops were studied to develop climate resilient

technologies. The impact of climate change variables, elevated carbon dioxide [ECO<sub>2</sub>] and elevated temperature [ET], on coconut seedlings was studied in an open top chamber. The study indicated that the present level of biomass could be produced in future climate with less expense of water due to high water use efficiency observed under [ECO<sub>2</sub>]; however, at high temperature biomass production would be less. As an adaptive strategy, coconut genotypes were phenotyped for water deficit and high temperature stress. At 100% Field capacity (FC), tall genotypes exhibited high WUE (3.5 g biomass L<sup>-1</sup> water), while at 25% FC, dwarf genotypes had high WUE (3.8). Tall genotypes had highly sensitive stomata while, dwarfs exhibited better root growth under stress. Furthermore, studies on leaf epicuticular wax content revealed that tall cultivars (Kalpa Pratibha and Kalpatharu) showed relatively high wax content than dwarf varieties.

Chemometric approach to detect adulteration in virgin coconut oil and coconut oil divulge that among the various parameters examined, total polyphenol content (TPC) and fatty acids profile of the oils are excellent candidate markers.

At the reproductive phase, pollen germination was found to be very sensitive to high temperature. It was 63% at 30°C and got drastically reduced to 14% at 45°C. A clear contrast was observed between tall and dwarfs in terms of pollen germination at high temperatures, which can be an important selection criterion in evolving varieties with tolerance to high temperature.

As a measure of water conservation, institute has developed hydraulically efficient, environmentally compatible and cost effective filtration systems and structures for roof water harvesting, run-off collection, storage and percolation tanks. Low-cost water harvesting structures like check dam, sub surface dam, vented cross bars, storage structures using ferro-cement technology could augment surface/ sub surface water resources.

### Product Diversification, Value Addition and Mechanization

Value addition and product diversification can ensure the sustainable livelihood of plantation farmers and entrepreneurs. In this context, 'coco-sap chiller' technology developed for collecting

fresh, hygienic and unfermented coconut inflorescence sap (Kalparasa) is very useful. Other value added products like virgin coconut oil, coconut chips could improve the profitability and employment generation in coconut sector. In an effort towards product diversification and value addition, collection of fresh coconut inflorescence sap ('Kalparasa®') using a simple coco sap chiller paved the way for preparation of coconut palm sugar from the inflorescence sap and preparation of confectioneries. Coconut milk residue and jackfruit seed powder enriched biscuits, coconut milk residue based extrudate, 'Kalpa Krunch', coconut sugar based dark chocolate 'Kalpa Bar', 'Kalpa Drinking Chocolate' and coconut milk based 'Frozen coconut delicacy' using tender coconut water and pulp also have been developed. Bean-to-bite cocoa dark chocolate using coconut palm sugar could grab the attention of entrepreneurs.

Farm mechanization and various processing machineries developed at the institute could contribute substantially in reducing the production cost, increased labour efficiency and enhanced product output and quality. The safety attachment incorporated by the Institute to Chamberi Joseph model of climbing device has become an effective solution since it could be operated even by women with proper training. This gives much required confidence to the climbers, especially the beginners. Apart from this, machineries and gadgets developed for labour saving and gender main streaming viz., power operated tender coconut cutter, power operated coconut and arecanut husking machines, coconut de-shelling and shell removing machines for copra making and wet processing respectively, tender coconut punch and cutter, copra and coconut chips dryers of varying capacities and using different fuel sources, testa remover, manual and power operated coconut slicing machines, coconut milk expellers of various capacities, VCO cookers, VCO fermentation tank and copra moisture meter are the other major contributions from the institute. A recent addition to this impressive array of gadgets is the gender-friendly self-loading arecanut dehusking device (with dust control) along with the arecanut grading attachment. So far the institute could obtain national patents for seven of its devices.

## Capacity Building Programmes

For technology transfer, efforts have been made to adequately promote the mandate crops of the institute through effective extension activities including trainings, farmer participatory approaches in technology development and dissemination, participation in exhibitions and conducting Kisan Melas, and production and distribution of planting materials of mandate crops. Training and frontline demonstrations on selected technologies, on campus and off campus training programmes for extension personnel and research-extension-farmer interface programmes have been conducted. Besides, the institute has participated in exhibitions, radio talks, television interviews, phone-in programme and press meets. Mega Expo and Kisan Mela were organized in addition to release of various publications and documentation of farmers' experiences and felicitation of the innovative farmers across the country.

ICT tools like videoconferencing to conduct trainings, conferences and interaction workshops with various stakeholders was utilised. Statistical Databases created, technical bulletins, CD ROMs, extension pamphlets, information brochures published. Krishi Vigyan Kendras under the institute catered to the training needs of farmers of Kasaragod and Alappuzha Districts in Kerala State. Cyber extension programmes were further strengthened with the addition of mobile video conferencing unit for facilitating the Research-Extension-Farmer interfaces. The Institute website (<https://cpcri.icar.gov.in>) is being updated regularly with latest information. Besides, several innovative steps were taken to meaningfully engage the visual and print media for disseminating the research accomplishments to the farming community.

## Socio-Economic Studies and Policy Interventions

The impact of changing trade policy environment (domestic / international) on mandate crops in terms of prices (cointegration also) and demand-supply equations was studied and continuously monitored. Consultancy briefs (yearly basis) on production and trade aspects of the coconut sector were submitted to CACP as inputs to facilitate the fixation of minimum support prices of copra. Policy

brief on minimum support price for arecanut was also submitted. In a policy note submitted to CACP on raw coconut procurement, it was suggested to establish panchayat level hubs with forward and backward integration along with unit level collection centers under the supervision of CPS networks for efficient raw coconut procurement. Minimum import price of arecanut was arrived at for providing better market price for indigenous growers.

### **Statistical Models to Improve Field Experiments**

Analysis of covariance technique in field experiments is made more robust/flexible by taking the relationship between the response variable and covariate as non-parametric instead of linear. Semi-parametric additive regression model has been proposed to estimate/eliminate the positional effect in field experiments, when the number of experimental units is comparatively small. Crop production model in arecanut was developed based on the semi parametric regression technique. A data driven technique was developed to estimate the trend and relative growth rate of time series data. The method was

extended for handling sudden shifts or changes in the trend or growth rate functions by adding dummy variables for the jumps. It has been applied to estimate trend and growth rate of area, production and yield of major crops in India. Robust spatial smoothing technique was developed to estimate the spatial effect of a field in the presence of outliers or extreme observations. It is based on fitting M-type robust nonparametric spatial regression following iterative kernel weighted local regression surface technique. Yield prediction in cocoa was done using biometrical/partial harvest data. Besides, weather based crop yield modelling was carried out in mandate crops. Pest and disease incidence and severity were regularly assessed employing appropriate sampling strategies in Kerala and Karnataka.

### **Technology Commercialization**

The Institute Technology Management Committee is responsible for protection of IP assets and commercialization. Till now, the Institute has commercialized more than 55 technologies, signed 375 MoAs for transfer of technology know-how and realized a revenue of Rs. 1.045 crore.

## CROPS, AREA, ALTITUDE AND RESEARCH UNDERTAKEN AT DIFFERENT LOCATIONS

### Headquarters

**KASARAGOD** (Estd.: 1916), Crops: Coconut and Cocoa, Area 78 ha; 10.7m MSL

Priority areas of research: Genetic resources management, breeding, biotechnology, water and nutrient management, organic cultivation, cropping/ farming system, microbiology, pests and diseases management, physiology and biochemistry, value addition and farm mechanisation, economics, statistics and transfer of technology. Various activities are envisaged under five divisions viz., Crop Improvement, Crop Production, Crop Protection, Physiology, Biochemistry and Post Harvest Technology and Social Sciences.



### Regional Stations

**KAYAMKULAM** (Estd.: 1947), Crops: Coconut and palmyrah, Area 24.17ha, 3 m MSL

Priority areas of research: Palmyrah collections, etiology and management of root (wilt) and other diseases, pests and nematodes management.



**VITTAL** (Estd.: 1956), Crops: Arecanut and Cocoa, Area 68.34 ha; 58 m MSL

Priority areas of research: Genetic resources management, breeding, production and protection, cropping systems and drought tolerance.



### Research Centres

**KAHIKUCHI** (Estd.: 1958), Crops: Arecanut and Cocoa, Area 15.76 ha; 48 m MSL

Priority areas of research: Cropping system, crop protection and production of quality planting materials.



**KIDU** (Estd.: 1972), Crops: Coconut, Arecanut and Cocoa, Area 120 ha; 281m MSL

Priority areas of research: National coconut gene bank, International Coconut Gene bank for South Asia (ICGSA), soil and water conservation, quality planting material production.

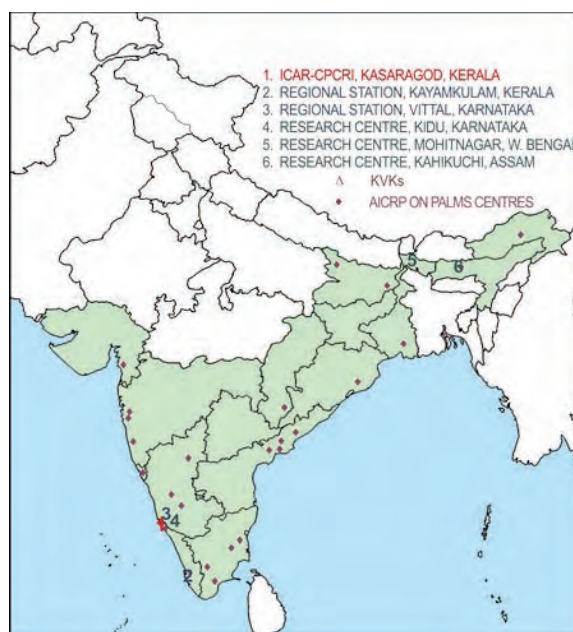


**MOHITNAGAR** (Estd.: 1958), Crops: Coconut and Arecanut, Area 25.99 ha; 91.3 m MSL

Priority areas of research: Genetic resources management, cropping system, soil, water and nutrient management.



## LOCATION OF STATIONS / CENTRES UNDER CPCRI



### Staff strength as on 31-12-2023

#### ICAR-CPCRI KASARGOD

| Category       | Sanctioned | In position | Vacant |
|----------------|------------|-------------|--------|
| Scientific     | 65         | 61          | 4      |
| Technical      | 75         | 53          | 22     |
| Administrative | 50         | 43          | 7      |
| Supporting     | 128        | 60          | 68     |
| Total          | 318        | 217         | 101    |

#### ICAR-KVK, ALAPPUZHA

| Category       | Sanctioned | In position | Vacant |
|----------------|------------|-------------|--------|
| Scientific     | 1          | 1           | -      |
| Technical      | 11         | 10          | 1      |
| Administrative | 2          | 0           | 2      |
| Supporting     | 2          | 2           | 0      |
| Total          | 16         | 3           | 3      |

#### ICAR-KVK, KASARGOD

| Category       | Sanctioned | In position | Vacant |
|----------------|------------|-------------|--------|
| Scientific     | 1          | 1           | -      |
| Technical      | 11         | 6           | 5      |
| Administrative | 2          | 1           | 1      |
| Supporting     | 2          | 1           | 1      |
| Total          | 16         | 8           | 8      |

### Budget and Expenditure 2023-24 (Rs. in lakhs)

| Head               | Allocation | Expenditure |
|--------------------|------------|-------------|
| Budget             | 1580       | 1580        |
| Revenue generation | 412.86     |             |



## Germplasm Conservation

### Coconut

ICAR-CPCRI plays a vital role in conserving coconut accessions, which encompass germplasm originating from 27 nations. These collections are stored in *ex situ* gene banks located at Kasaragod and Kidu, together accounting 455 coconut accessions. Additionally, selected duplicates are held at ICAR-CPCRI, Research Centres in Mohitnagar and Kahikuchi. Notably, the ICAR-CPCRI, Research Centre in Kidu serves as the host for the International Coconut Gene Bank for South Asia and the Middle East (ICG-SAME). Complementary cryo-conservation of zygotic embryos (30 accessions), pollen (20 accessions) and DNA (16 accessions) is done in collaboration with ICAR-NBPGR, New Delhi. ICAR-CIARI, a collaborating partner, has conducted explorations in coconut grown areas in the Nancowry Islands (Nicobar district) and Aves Island (North and Middle Andaman districts) to document the diversity of coconut populations. In this endeavour, 28 seed nuts were collected to conserve traits such as large fruit volume, higher kernel content, more nut water content, and thin husk.



Fig. 1.1. Coconut collections from Nicobar Islands

### Arecanut

ICAR-CPCRI is conserving 172 arecanut accessions within its field gene banks at Vittal, with additional safety duplicates conserved in alternate field gene banks located at Kahikuchi (Assam) and Mohitnagar (West Bengal).

### Cocoa

The field gene bank of CPCRI, Regional Station, Vittal, Karnataka, presently conserves 531 cocoa

collections, consisting of 475 exotic and 56 indigenous types. To ensure the safety of core germplasm, duplication efforts are carried out in alternate gene banks located in Karnataka (CPCRI, Research Centre, Kidu), Tamil Nadu (TNAU), and Andhra Pradesh (YSRHU). Additionally, a cocoa type from Kottayam, featuring a 2 g bean size (Fig. 1.2) was recently acquired and incorporated into the germplasm holdings within the field gene bank.

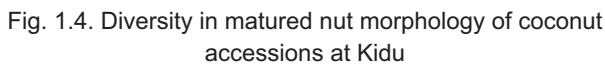
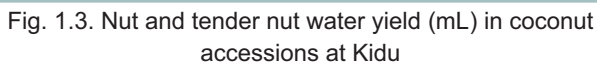


Fig. 1.2. Cocoa collections from Kottayam, Kerala

## Germplasm characterization and evaluation

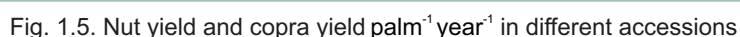
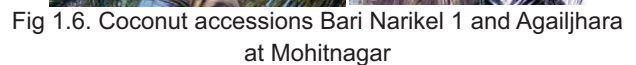
### Coconut

At ICAR-CPCRI, Research Centre, Kidu, a study was conducted to evaluate the genetic diversity of 31 coconut accessions from island and coastal ecosystems of India. Among these, Champin Micro Tall stood out for its remarkable nut production ( $196 \text{ palm}^{-1}$ ) and copra output ( $27 \text{ kg palm}^{-1}$ ). Noteworthy selections for tender nut traits included Kodiaghat Brown Tall, Dugong Creek Tall, and Nicobar Tall Auckchung (Fig. 1.3), while for matured nut traits, Nicobar Tall Tamaloo, Nicobar Tall Auckchung, Dugong Creek Tall, and Kodiaghat Brown Tall displayed superior characteristics (Fig. 1.4). Biochemical analysis of copra oil revealed that Chengara Tall had superior values for fatty acids such as lauric acid, capric acid, and caprylic acid, while Laccadive Small Tall Agathi had the highest myristic acid content. These findings earmarked them as promising candidates for breeding programs aimed at enhancing quality traits.



exhibited the highest yield ( $59.4 \text{ nuts palm}^{-1} \text{ year}^{-1}$ ), followed by Chandra Sankara ( $56.6 \text{ nuts palm}^{-1} \text{ year}^{-1}$ ). Notably, in the dwarf category, MYD displayed the highest yield of  $71.7 \text{ nuts palm}^{-1} \text{ year}^{-1}$ , distinguishing itself from other dwarfs. In terms of copra yield among tall accessions, Agailjhara stood out with  $24.63 \text{ kg palm}^{-1} \text{ year}^{-1}$ , followed by Bari Narikel 1 ( $17.47 \text{ kg palm}^{-1} \text{ year}^{-1}$ ). Among hybrids, Kera Sankara recorded the highest copra yield of  $12.1 \text{ kg palm}^{-1} \text{ year}^{-1}$ , while among dwarfs, MYD produced the highest copra yield of  $9.68 \text{ kg palm}^{-1} \text{ year}^{-1}$  (Fig. 1.5).

In Mohitnagar, tender nut traits of 17 coconut accessions were studied. PHOT with bigger sized nuts had the highest tender nut weight of 3.3 kg, followed by Agailjhara Tall with 2.09 kg (Fig. 1.6), while the lowest was observed in Chandra Shankara (1.8 kg). In the study, tender nut water content ranged between 306.7 mL to 758.3 mL, with PHOT recording the highest water content. Agailjhara had the highest Brix value for tender water (5.37 °Brix), followed by AGT (5.1°Brix), while ADOT had the lowest (3.13 °Brix). Fruit component traits varied notably among the accessions, with Chinnaasukhan Tall displaying the highest fruit length (24.03cm), and PHOT with the highest fruit circumference of 58.67 cm.



At Mohitnagar, in the study on evaluation of 21 coconut accessions at juvenile phase, ECT had the largest trunk circumference at the base (107.6 cm) and highest trunk height (134.7cm). Among the accessions, PALT, ECT, and SSGT demonstrated the maximum number of leaves palm<sup>-1</sup> (14.3). Notably, WCT displayed the longest leaf (4.9 m) as well as longest leaflet-bearing portion (3.7m). In terms of leaflet length, MVT, NLAD, and NIGD stood out as noteworthy varieties. Additionally, at Mohitnagar 15 local coconut accessions are being screened for cold tolerance. Alongside other phenological traits, such as trunk height (Lataguri-II; 3.5 m), base circumference (GB II; 149 cm), inflorescence number (MNR-III; 9.33) and nuts per inflorescence (Mal-III; 13.67), cold injury symptoms chiefly leaf scorching has been recorded among the lines, with intensity ranging between 0% to 5%.

Similarly, in Assam, 15 coconut varieties were evaluated, Kera Sankara exhibited the highest fruit yield of 132 fruits palm<sup>-1</sup> year<sup>-1</sup>, while among local accessions, KKHC 4 had the highest yield of 38 fruits palm<sup>-1</sup> year<sup>-1</sup>. For reproductive traits, Laksha Ganga displayed longer inflorescences (120.25cm), while, Assam Green Tall recorded maximum female flowers with population mean of 36.67, during the period from November to February. Chandra Sankara, Kera Sankara, and Gangabondam Green Dwarf varieties, each produced one inflorescence per month. In Kasaragod, several accessions, including Benaullim Tall, Lifou Tall, and Kappadam Tall, recorded higher fruit yield than the local control. Jamaica Tall, with consistently high nut and copra yield and a high inflorescence sap yield, is recommended for release. Comparative evaluation of 29 dwarfs at Kasaragod revealed variations in palm morphology. Andaman Green Dwarf and Malayan Green Dwarf had higher plant height and longer leaves, while Samoan Yellow Dwarf had a slender trunk and shorter palm height. Andaman Green Dwarf, King Coconut, Samoan Yellow Dwarf, and others recorded higher nut yield than the local check COD. Fruit traits showed variations in color, weight (250-2500 g), tender nut water content (37-603 mL), and copra content (44-173 g).

All nine palm progenies of Ratnagiri's Mohacho Narel, sweet kernel coconut population, yielded sweet kernel fruits (5-24%, fig. 1.7). Selected

palms were *inter-se* mated for population improvement, with resulting seed nuts sown for germination. Germination time ranged from 155-225 days, and specific cross combinations showed 50-100% germination (Fig. 1.8). About 45 *inter-se* mated seedlings were field-planted. Embryo culture of 230 sweet kernel fruit embryos produced 30 seedlings, while 60 plantlets are under *in vitro* culture.

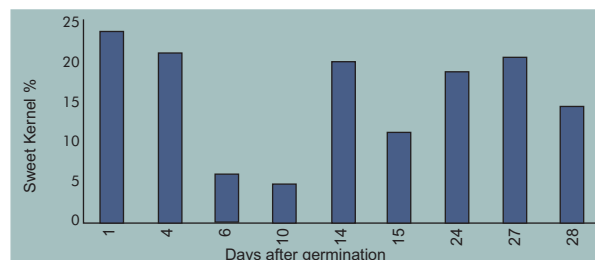


Fig. 1.7. Kernel sweetness in Mohacho Narel progenies

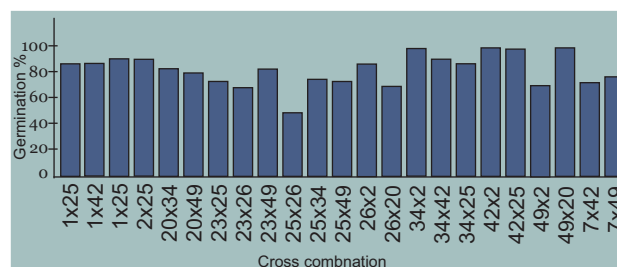


Fig. 1.8. Germination percentage in cross combinations  
Trait-specific coconut germplasm identified are:

- Lesser fruit weight - GDD, LMT, AYKT, NKOD
- Higher endosperm content - SNRT, Guam III, Nuwallis, Tamaloo, Auckchung
- Higher shell thickness, shell weight - Philippines Kalambhahim Tall
- Higher oil content - LCT, LMT, Guam III, Kodiaghat Brown Tall
- Low fruit to husk ratio - LCOD
- Higher female flower production - LMT, Spicata, SDKT, ADOT Rotuma Tall, Ghaigaita Tall, Karkar Tall, Niu Guinea Tall
- Higher testa oil content - LCT, WCT, TPT, PHOT

### Evaluation of coconut accessions at WCGC, Sipighat

ICAR-CIARI monitored 30 conserved coconut accessions at WCGC, Sipighat, including 24 from Pacific Ocean region and six from Nicobar Islands. Performance evaluations by ICAR-CIARI focused on palm morphology, inflorescence traits, and fruit yield in specific varieties like Rennel Tall, Katchal Tall, AuckChung Tall, Tahiti Tall, Tamaloo Tall, and Pao Pao Tall, which are identified for crop improvement. *Inter se* mating and selfing were undertaken for germplasm multiplication and

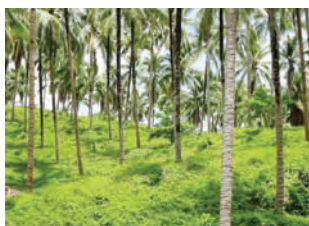


Fig. 1.9. Coconut germplasm at WCGC, Sipighat



Fig. 1.10 High yielding palm of WCGC Accession 21

crossing programs in Acc. 1, 2, 5, 8, 10, 11, 14, 18, 19, 21, 22, 24, 27 and 28. Fatty acid profile analysis across the 30 WCGC accessions showed variation in lauric acid proportion among individual palms, with non-significant population-wise differences. Continuous observations on growth and morphology were made in selected mother palms in the Niu Leka Green Dwarf coconut population. Niu Leka Dwarf with orange-coloured fruits (5/39) and higher tender nut water content (470 mL) was identified for the development of a homogeneous population.

### ***Evaluation of germplasm for testa oil characteristics***

The biochemical composition of testa oil of 20 coconut varieties (4 dwarfs, 11 Talls, 5 hybrids) revealed varied fat contents ranging from 64.85% (Chandra Sankara) to 88.80% (Chandra Kalpa). Testa oil had 2.95- 5.58 mg GAE 100g<sup>-1</sup> of total polyphenol content, flavonoids 1-4.09 mg QE 100g<sup>-1</sup>,  $\beta$ - carotene 0.1-0.4 ppm and chlorophyll 0.95-3.37 mg kg<sup>-1</sup>. The antioxidant capacity DPPH (% scavenging activity) and FRAP (mg trolox 100g<sup>-1</sup> of oil) in testa oil ranged from 49-85.41% and 3.21-6.86 mg trolox 100g<sup>-1</sup> of oil, respectively. Crude protein content varied between 8.56-11.39 g 100g<sup>-1</sup> dry weight of testa. Ash content varied from 1.8-2.45% and contained substantially more nitrogen and potassium.

### ***Inflorescence sap yield of coconut genotypes***

Among the coconut genotypes evaluated for inflorescence sap yield viz., Tiptur Tall (TT), Fiji Longtongwan Tall (FJLT), Nigerian Tall (NIT), Philippines Kalambahim Tall (PKBT), Gudanjali Dwarf (GDD) and Chowghat Orange Dwarf (COD) highest yield of 36.10 L inflorescence<sup>-1</sup> was recorded in PKBT (Philippines Kalambahim Tall) followed by NIT(Nigerian Tall) (48.03 L inflorescence<sup>-1</sup>). Mean sap yield per day ranged from 177.86 to 1069.27 mL. Evaluation of another eight accessions for production of inflorescence

sap, indicated higher average yield (around 2 L day<sup>-1</sup>) in Laguna Tall followed by Kappadam Tall (Fig. 1.11).

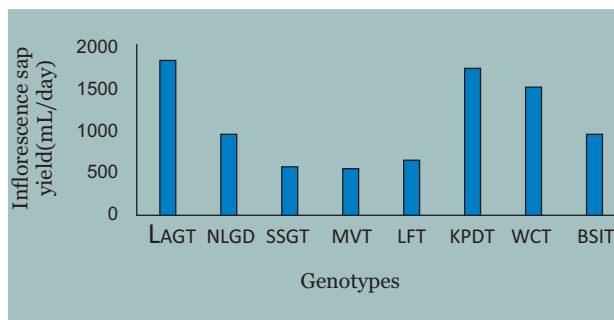


Fig. 1.11. Inflorescence sap yield form different coconut accessions

### ***Coconut pollen for human nutrition***

Phytochemical properties of coconut pollen from 12 accessions (six tall, three dwarfs, and three hybrids) were studied to explore its potential as a nutritive food ingredient. WCT exhibited the highest pollen yield per inflorescence (15.9g), followed by MYD x WCT, while MYD recorded a lower yield (6.8g). Biochemical analysis revealed variations in total sugars (4.22%-6.86%), reducing sugars (0.84%-2.05%), starch (7.58%-13.31%), crude protein (31.73%-38.88%), crude fat (6%-10%) and ascorbic acid (0.065%-1.05%). The highest energy value was in the hybrid MYD x WCT (271.20 kcal 100g<sup>-1</sup>), followed by tall WCT (250.84 kcal 100g<sup>-1</sup>). Hybrid COD x WCT recorded the highest total polyphenol content (4.63 mg GAE g<sup>-1</sup>), and DPPH assay showed higher scavenging activity in COD x WCT (36.2%). Antioxidant activity significantly correlated with total polyphenol content. Liquid chromatography-mass spectrometry revealed fructose as the major sugar, and amino acid profiling showed the presence of all essential amino acids. Essential amino acid content was highest in dwarf COD (93.67mg g<sup>-1</sup>), followed by MGD and SNRT. Phenolic acid profiling identified 18 phenolic acids, with higher total phenolic acid content in FJT, MGD, WCT, MYD, SNRT, and COD. The study highlights the diverse phytochemical composition of coconut pollen, showcasing its potential as a valuable nutritional resource.

### ***Coconut seedlings for palm cabbage***

Eight coconut accessions, including six tall (Andaman Giant Tall, Java Tall, Philippines Lono Tall, Strait Settlement Apricot Tall, Bari Narikel Tall 1, and Klapawangi Tall) and two dwarfs

(Malayan Green Dwarf and Niu Leka Green Dwarf), were evaluated for cabbage (Heart of palm) yield and quality (Fig. 1.12). Vegetative characters such as total number of leaves and collar girth of seedlings correlated positively with coconut cabbage yield. The maximum yield of cabbage core portion in the tall accession Strait Settlement Apricot Tall was 202 g, followed by Philippines Lono Tall (181g). Quality parameters, including total moisture content (ranged between 82.5% to 87.25%), cabbage pH (5.7 to 6.5), and various nutritional components, were estimated. Total sugar content ranged from 6.68% (Niu Leka Green Dwarf) to 10.81% (Malayan Green Dwarf), reducing sugar from 2.07% (Strait Settlement Apricot Tall) to 3.71% (Klapawangi Tall), starch from 15.85% (Malayan Green Dwarf) to 21.96% (Niu Leka Green Dwarf), total protein from 10.07% (Philippines Lono Tall) to 13.14% (Klapawangi Tall), free amino acid content from 0.63% (Niu Leka Green Dwarf) to 1.3% (Java Tall), crude fat from 3.5% (Malayan Green Dwarf) to 4.9% (Philippines Lono Tall), total polyphenol content from 0.49 (Java Tall) to 0.69 (Niu Leka Green Dwarf) mg GAE g<sup>-1</sup>, and ascorbic acid in the range of 0.065 mg g<sup>-1</sup> (Niu Leka Green Dwarf) to 0.01 mg g<sup>-1</sup> (Strait Settlement Apricot Tall). The energy value was maximum in Bari Narikel Tall 1 with 1.25 kcal g<sup>-1</sup>, followed by Malayan Green Dwarf (1.21 kcal g<sup>-1</sup>), and the minimum was recorded in Niu Leka Green Dwarf (1.14 kcal g<sup>-1</sup>).



Fig. 1.12. Harvested coconut cabbage or heart of coconut

### ***Drought tolerance in coconut: Osmotic potential adjustment in pollen grains***

Pollen response to osmotic stress was assessed by immersing fresh pollen grains from eight Dwarf x Tall coconut hybrids (COD x ADOT, COD x WCT, COD x LCT, COD x WAT, MYD x ADOT, MYD x WCT, MYD x LCT, and MYD x WAT) in 40% and 50% PEG 6000 solutions, with and without KCl. This allowed estimation of osmotic adjustment (OA) based on quantitative measurements of pollen cytoplasm area. Significant differences were noted between hybrids for both Intrinsic OA (effect of PEG solutions without KCl) and Induced OA (effect of KCl on PEG solutions). The projected area of pollen cytoplasm decreased after immersion in PEG solutions but increased substantially after adding KCl. The intrinsic osmotic adjustment value (B/A) ranged from 0.957 to 1.177, with MYD x LCT exhibiting the highest value. This hybrid also recorded the highest total osmotic adjustment value. The induced osmotic adjustment value ranged from 1.010 (COD x LCT) to 1.127 (MYD x ADOT), with an average of 1.053 (Fig. 1.13).

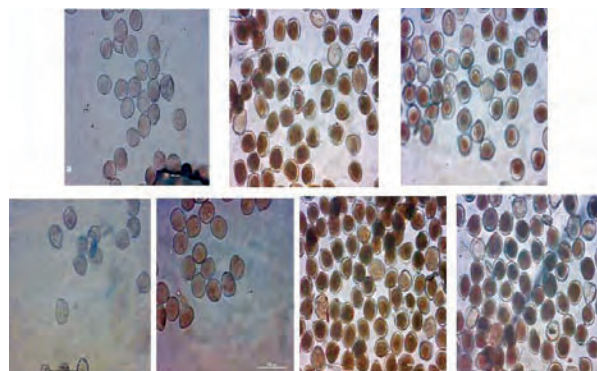


Fig. 1.13. Effect of osmotic stress and osmolyte on size and shape of pollen grains (a, Control; b, Pollen grains 40% PEG; c, Pollen grains 50% PEG; d, Pollen grains 40% PEG with 10mM KCl, e, Pollen grains 40% PEG with 20mM KCl, f, Pollen grains 50% PEG with 10mM KCl and g, Pollen grains 50% PEG with 20mMKCl).

### ***Screening for single temperature stress tolerance***

The impact of temperature (15°C to 50°C) on coconut hybrid pollen germination and tube growth was studied. Optimal conditions were at 25°C, yielding the highest germination (38%) and longest tube length (773µm). Across eight hybrids, cardinal temperatures (T<sub>min.</sub>, T<sub>opt.</sub>, and T<sub>max.</sub>) varied. T<sub>max.</sub> for germination ranged from 46.1°C to 50.5°C, and T<sub>opt.</sub> for tube length varied from 20.4°C to 26.9°C. The hybrid COD x ADOT

exhibited greater tolerance to high temperatures, with higher cardinal temperatures for both germination and tube length.

### Pre-treatment effect on pollen germination at different temperature

Freshly extracted pollens were subjected to treatments at temperatures 50°C and -20°C for 5 and 10 minutes, respectively. The *in vitro* pollen germination was observed for both treatments across different temperature range (15°C, 20°C, 25°C, 30°C, 35°C, 40°C, 45°C) to assess their impact on germination percentage and tube lengths compared to untreated pollens. The highest germination and tube length were noted at 25°C, outperforming the other tested temperatures. Both heat and cold-treated pollens exhibited better germination and tube length across all temperature range compared to untreated pollens. Notably, among the two treatments, cold-treated pollens demonstrated germination (8.25%) even at 40°C, while the control and heat treatments resulted in nil germination at 40°C (Fig. 1.14).

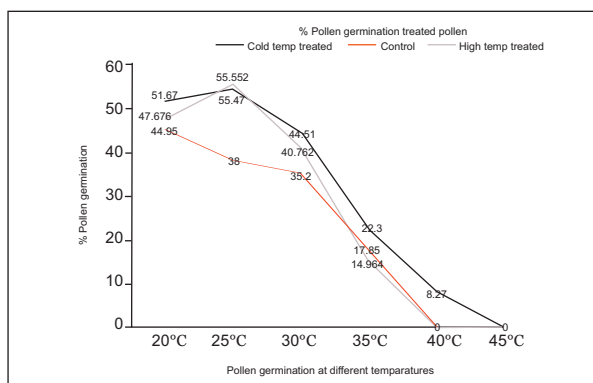


Fig. 1.14. Pollen germination at different temperatures

## Germplasm Utilization

### Genetical investigations and breeding for improved varieties in coconut

In a hybrid evaluation trial at ICAR-CPCRI Research Centre, Kidu, planted in 1996, WAT x NAT and WAT x RET performed well for bunch and nut yield, while PHOT x GBGD excelled in tender nut and matured nut traits, achieving highest tender nut water yield (553 mL nut<sup>-1</sup>), maximum husked fruit weight (849 g nut<sup>-1</sup>) as well as copra output of 206 g nut<sup>-1</sup>. Similarly, in 1998 trial, CGD x LCT displayed the highest mean number of female flowers per inflorescence (51), followed by CRD x WCT (33). WCT x CGD showed a higher mean number of bunches per palm (13.42), followed by CRD x WCT (12.60). PHOT x

CGD had larger nuts, achieving the highest tender nut water (577 mL), maximum husked fruit weight (749 g), and highest copra output of 188 g (Fig. 1.15).



Fig. 1.15. Nut profiles of improved coconut hybrids

D x T hybrid evaluation trial with 26 combinations, planted during 2014, MYD x SNRT led in female flowers per inflorescence (27.88), followed by CRD x SNRT (24.50). MGD x CCNT produced the most bunches on the crown (14.88), followed by CGD x SNRT (14). MOD x CCNT had larger nuts, averaging a tender nut weight of 3335 g and the highest tender nut water of 877 mL, followed by GBGD x CCNT (827 mL) and COD x PHLT (821 mL) (Fig. 1.16).

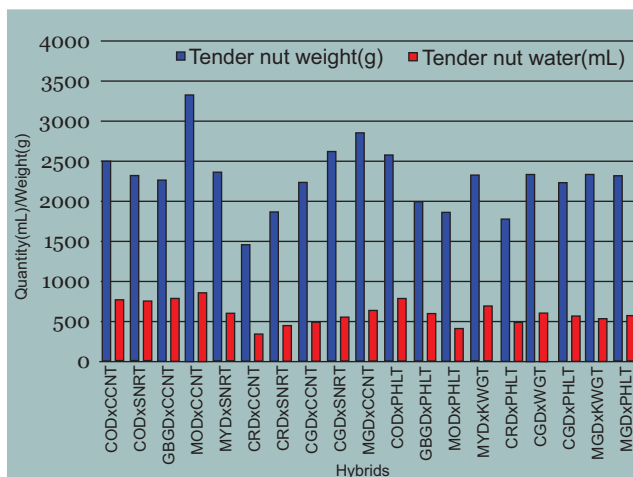


Fig. 1.16 Mean performance of D x T hybrid combinations planted during 2015-16 (RC Kidu)

In D x T hybrid trials at Kasaragod (2013 & 2014 planting), combinations involving CCNT showed higher fruit yield. In the Dwarf x Dwarf trial at Kidu (2003 planting), MOD x NL had the highest inflorescence characteristics, and MYD x NL displayed the highest mean number of female flowers per inflorescence. Among 14 Dwarf x Dwarf hybrids planted at Kasaragod in 2016, positive heterosis for fruit yield was observed, with CGD x CRD, GBGD x CRD, MOD x NLAD, CGD x MOD, COD x GBGD, CRD x CGD, CRD x GBGD, MYD x NLAD, and MYD x CGD showing higher yield potential. Additionally, GBGD x COD, MOD x NLAD, and MYD x NLAD recorded higher tender nut water content (>450 mL).

## Coconut hybrids and drought tolerance

Coconut hybrids were screened for stomatal traits, the key determinants of water use efficiency. Variability in stomatal conductance was noted among hybrids. COD x WCT, MYD x ADOT, and WCT exhibited higher conductance, suggesting better drought adaptation. Chlorophyll fluorescence could be a criterion for high yield under drought, with COD x WAT showing the highest fluorescence. Antioxidant enzyme activities (SOD, PPO, POD) increased during summer, notably in COD x WAT (SOD) and WCT (PPO, POD).

Cluster analysis grouped hybrids into three clusters (Fig. 1.17). Cluster II, including COD x ADOT, MYD x WCT, and COD x LCT, performed better under drought, based on the screening parameters. However, since drought tolerance is a complex phenomenon, further screening for additional parameters is recommended.

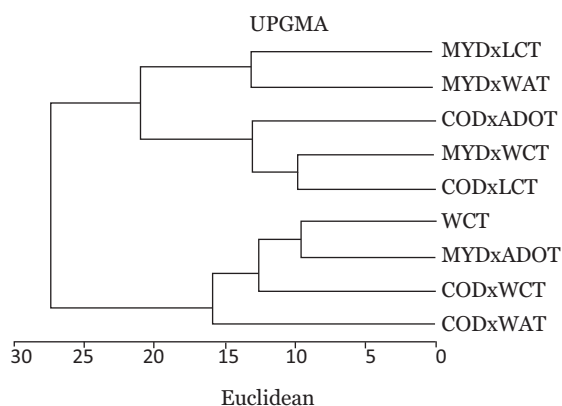


Fig. 1.17. UPGMA cluster analysis of coconut hybrids

## Molecular markers-based diversity analysis

Among the 70 SSR primers, reported in the Tropgene database for coconut (CIRAD), 30 primers were utilized for parental polymorphism screening. Fourteen SSR markers displaying polymorphism were chosen for the characterization of 96 palms representing 16 coconut germplasms. Consequently, a total of 49 primers exhibited polymorphism, with 35 previously screened and selected SSR primers retaining their polymorphic nature.

## Novel trait: Dwarfness

A unique dwarf palm, distinguished among seedlings from the Lakshadweep population, has been incorporated into the breeding program. Despite flower initiation, the palm failed to

produce fruit. Consequently, pollen from this palm was used to fertilize a COD palm, and the offspring, consisting of 54 seedlings, were subsequently planted in the Kasaragod field in 2018. The juvenile palms' heights ranged from 197 to 526 cm. Presently, three palms resulting from the cross between the distinct dwarf palm and COD have begun to flower. The seedlings produced from this cross have been classified into three groups based on their height (as shown in Table 1).

Table 1. Morphological traits of the juvenile palms

| Group      | Height (cm) | Petiole length(cm) | Leaf length(cm) | Leaflet length(cm) | Leaflet breadth(cm) |
|------------|-------------|--------------------|-----------------|--------------------|---------------------|
| 1          | 275.0       | 73.3               | 222.0           | 53.9               | 3.5                 |
| 2          | 370.7       | 85.1               | 275.1           | 74.8               | 4.0                 |
| 3          | 462.8       | 114.7              | 329.5           | 84.0               | 4.4                 |
| Population | 366.9       | 88.2               | 273.6           | 71.8               | 4.0                 |
| Mean       |             |                    |                 |                    |                     |

## Development of coconut inbreds

In the inbred development programme in WCT, out of 182 WCT S3 seedlings planted at Kidu, 96 seedlings have started flowering and among the yielding ones, palm number 144 has recorded highest tender nut water of 820 mL with good taste (5° Brix) followed by palm number 169 and 171 both yielding 740 mL per fruit.

## Coconut accessions for resistance /tolerance to root (wilt) disease

In an evaluation trial involving thirteen tall accessions planted during 2014, root (wilt) disease incidence was lowest in St. Vincent Tall (4.17%) followed by Andaman Giant Tall (4.76%). Highest root (wilt) disease was recorded in Andaman Ordinary Tall (38.09%) followed by Federated Malay States Tall (35.7%). The average nut yield was highest in Kalpa Haritha (72.04 nuts palm<sup>-1</sup> year<sup>-1</sup>) followed by St. Vincent Tall (71.9 nuts palm<sup>-1</sup> year<sup>-1</sup>).

In an evaluation trial involving six green dwarfs planted in 2013, the highest root (wilt) incidence after ten years of planting was in Niu Leka Dwarf (50 %) followed by Gangabondam Green Dwarf (42.10 %). Kalpasree remained disease-free even after ten years of planting. During the reporting period, the mean yield was highest in Kalparaksha (86.6 nuts palm<sup>-1</sup> year<sup>-1</sup>) followed by Kalpasree (77.6 nuts palm<sup>-1</sup> year<sup>-1</sup>).

## Production of D x D / D x T progenies

Artificial pollination was conducted on 27 CGD

palms to produce hybrid combinations, including CGD x GBGD, CGD x CCNT, CGD x AGT, CGD x STVT, and CGD x TT. A total of 67 inflorescences, comprising 1601 female flowers, were involved in the pollination process for the production of these five hybrid combinations.

Artificial pollination using ground pollinator was also carried out on 169 dwarf parental palms involving 711 inflorescences and 14858 female flowers. Collected 34,979 seed nuts of different coconut varieties including 5420 artificially pollinated (CGD x WCT) seed nuts.

## Germplasm utilization in cocoa

### Performance of cocoa genotypes in CPCRI centres

At Kahikuchi, among 13 cocoa genotypes evaluated over 10 years VTLC19, VTLC2 (fig. 1.18), VTLC4 and VTLC11 produced higher pod yield of 64.73, 62.41, 59.84 and 53.26 respectively with single dry bean weight of 0.93, 1.06, 0.93, 1.03 g respectively, which can be recommended for cultivation in the region.



Fig. 1.18. Performance of VTLC2 at Kahikuchi

At Mohitnagar, under arecanut, VTLC-5 yielded maximum number of pods (105.3) followed by VTLC-1 (99.3). Dry bean yield of 3.5 kg was obtained in VTLC-5 followed by VTLC-1 (3 kg). In coconut garden, maximum number of pods (91) was recorded in VTLC-4 followed by VTLC-11 (71). The bean weight was maximum (2.88 kg) in VTLC-4 followed by VTLC-5 (2.37 kg).

To multiply the best performing genotypes in the region, grafting trial was taken up in six accessions during August - September with bulk Forastero root stocks. Maximum survival percentage was recorded in VTLC-23 (73.8%) followed by VTLC-42 (53.4%) during September month (Table 2).

Table 2. Survival percentage of grafting of different cocoa accessions at Mohitnagar

| Sl. No. | Accessions | Month of grafting | No. seedling grafted | Survival after 20 days | Survival % after 20 days | Survival after 4 months | % of after survival 120 days |
|---------|------------|-------------------|----------------------|------------------------|--------------------------|-------------------------|------------------------------|
| 1       | VTLC-5     | August            | 58                   | 12                     | 25                       | 11                      | 19.0                         |
| 2       | VTLC-66    | August            | 47                   | 16                     | 34                       | 16                      | 34.0                         |
| 3       | VTLC-5     | September         | 51                   | 26                     | 50.1                     | 17                      | 33.3                         |
| 4       | VTLC-61    | September         | 52                   | 18                     | 34.6                     | 15                      | 29.0                         |
| 5       | VTLC-23    | September         | 107                  | 99                     | 92.8                     | 79                      | 73.8                         |
| 6       | VTLC-42    | September         | 58                   | 31                     | 53.4                     | 31                      | 53.4                         |

At Kayamkulam, among the twenty five promising accessions planted in 2017, highest pod yield was recorded in VTLC-21 (72 pods tree<sup>-1</sup>) followed by VTLC-20 (68 pods tree<sup>-1</sup>). Single bean weight was the highest in VTLC-11 (1.08 g) followed by VTLC-6 (1.04 g).

### Evaluation of cocoa clones:

- Among Ghana clones, VTLC-554 and VTLC-556 had 63 and 60 pods in 18 m<sup>2</sup> canopy with 2.5 and 2.4 kg dry beans tree<sup>-1</sup>year<sup>-1</sup>, respectively (Table 3).
- Evaluation of old cocoa collections at Kahikuchi indicated that highest number of pod per tree (110 no.) was recorded in KHIC-26 (Fig. 1.19). Pod weight ranged from 208 g to 600 g per pod where KHIC-2 had the maximum pod weight with higher husk thickness (1.5 cm). Single dry bean weight ranged from 0.60 g to 1.40 g per bean, where KHIC-12 had maximum single dry bean weight. Dry bean yield ranged from 0.75 kg to 2.98 kg tree<sup>-1</sup>. Highest dry bean yield was recorded in KHIC-21 (2.98 kg tree<sup>-1</sup> year<sup>-1</sup>) followed by KHIC-13 (2.91 kg tree<sup>-1</sup> year<sup>-1</sup>).

Table 3. Best performing Ghana cocoa clones

| Clones   | Canopy area (m <sup>2</sup> ) | Pod yield tree <sup>-1</sup> year <sup>-1</sup> | No. of beans pod <sup>-1</sup> | Single dry bean wt. (g) | Dry bean yield (kg tree <sup>-1</sup> year <sup>-1</sup> ) | Shelling (%) | Nib recovery (%) | Fat (%) |
|----------|-------------------------------|-------------------------------------------------|--------------------------------|-------------------------|------------------------------------------------------------|--------------|------------------|---------|
| VTLC-554 | 17.0                          | 63                                              | 42                             | 1.11                    | 2.5                                                        | 14.1         | 86.7             | 50.0    |
| VTLC-556 | 16.4                          | 60                                              | 40                             | 1.10                    | 2.4                                                        | 11.7         | 88.6             | 50.5    |
| VTLC-557 | 13.9                          | 56                                              | 42                             | 1.10                    | 2.2                                                        | 15.4         | 85.7             | 50.3    |
| VTLC-549 | 16.7                          | 52                                              | 42                             | 1.08                    | 2.0                                                        | 15.3         | 85.5             | 51.2    |



Fig. 1.19. Best performing cocoa genotypes at Kahikuchi

## Arecanut

### *Arecanut germplasm registration*

Dwarf arecanut palm developed by ICAR-CIARI, Port Blair, Andaman and Nicobar Islands and ICAR-CPCRI, Kasaragod, Kerala, has been registered by Plant Germplasm Registration Committee (PGRC) of ICAR.

### *Tender nut processing in arecanut germplasm*

Tender nut processing of various arecanut germplasm including Sagar x HD, HD x Sagar, Ratnagiri x HD, HD x Ratnagiri, Thirthahalli x HD, HD x Rangron, Sirsi x HD, HD, Sagar, Ratnagiri, Thirthahalli, Sirsi, Cal-10, Cal-32, Dangapara, Kamalpur, and Thargira was performed (Fig. 1.20). Percentage recovery of dry processed tender nut from fresh kernel ranged from 11.46% to 31.97%. Thirthahalli x HD showed the highest recovery at 31.97%, followed by Sirsi (25.95%), while the lowest recovery of 11.46% was observed in HD x Rangron.

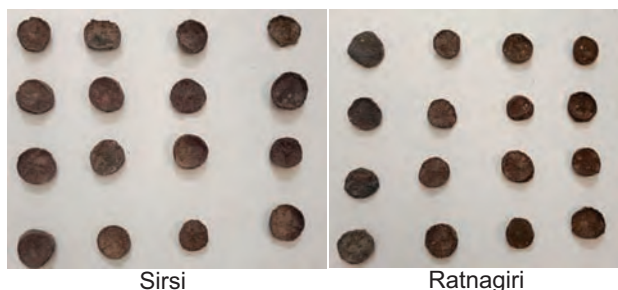


Fig. 1.20. Processed dry tender nuts

### *Fat content of arecanut germplasm*

Estimated fat content from dry kernels of various arecanut germplasm including Cal-2, Cal-4, Cal-5, Cal-6, Cal-7, Cal-10, Cal-17, Cal-32, Cal-33, Cal-35, SCDRTC-18, SCDRTC-92, Thargira, Nalbari, Berubari, Panicha, K & J hills, Saragoan 1, Saragoan 2, Kahikuchi, Borihat, Chayagoan 2, Shellshella, Aunihati 2, Dangapara, Aunihati 1, Moralpara, Kamalpur, and Bongera. The fat content ranged from 6.18% to 13.6%. Cal-32

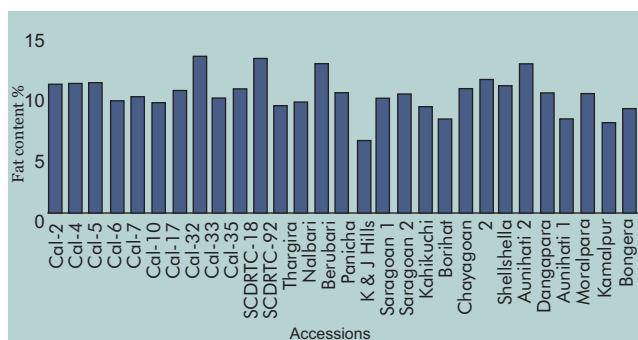


Fig. 1.21. Fat content in different arecanut germplasm

exhibited the highest fat content at 13.6%, followed by SCDRTC-18 (13.45%) and Aunihati-2 (12.92%). The lowest fat content, 6.18%, was observed in the arecanut germplasm Panicha.

### *Evaluation of arecanut germplasm at Vittal and Mohitnagar*

The ongoing evaluation of arecanut germplasm at ICAR-CPCRI involves different batches planted in various locations. In Vittal, Halsoor from Karnataka, Maharashtra, and Gujarat recorded the highest chali yield, followed by Thirthahalli. Nalbari showed the highest chali yield among North Eastern collections, while Rajarhat and Boragar exhibited promising yields in another batch. From Andaman and Nicobar Islands, Cal-32 had the highest chali yield, and among Konkan collections, Murud displayed high yielding tendencies, followed by Vanand and M. Raigad.

At Mohitnagar Center, the 1988 batch planting showed the maximum dry kernel or chali yield of 3.723 kg palm<sup>-1</sup> in Mohitnagar, followed by VTL-5 (2.129 kg palm<sup>-1</sup>). In the 1990 batch, VTL-29 recorded the highest chali yield (1.318 kg palm<sup>-1</sup>), followed by Kamrup (1.533 kg palm<sup>-1</sup>) and Sweet Arecanut (1.33 kg palm<sup>-1</sup>). VTL-27 in the 1991 batch exhibited a high yielding tendency with a dry kernel yield of 4.092 kg palm<sup>-1</sup>. In the 1992 batch, VTL18 (c) produced the maximum chali yield of 3.075 kg palm<sup>-1</sup>. Among the North Eastern collections planted in 1994, Nalbari produced the highest chali yield (3.555 kg palm<sup>-1</sup>), followed by Kahikuchi (3.453 kg palm<sup>-1</sup>), while Haldibari had the lowest yield of 1.214 kg palm<sup>-1</sup>. Similarly in 1997 batch, Calicut-20, Calicut-27, and Calicut-10 recorded the maximum chali yield 3.002 kg, 2.938 kg and 2.678 kg palm<sup>-1</sup>, respectively, while minimum yield was observed in Calicut-7 (1.32 kg palm<sup>-1</sup> year<sup>-1</sup>).

### *Evaluation of arecanut dwarf hybrids at Vittal, Mohitnagar and Kahikuchi*

At Vittal, the cross between Sumangala x HD exhibited maximum dry kernel yield (2.12 kg palm<sup>-1</sup> year<sup>-1</sup>) followed by HD x Mohitnagar (1.37 kg palm<sup>-1</sup> year<sup>-1</sup>). At Mohitnagar Center, among the eight arecanut hybrids, Sumangala x HD produced the maximum inflorescence per palm (3.1), followed by HD x Mohitnagar (2.9) and HD x

Mangala (2.8). However, in terms of dry kernel yield, Mohitnagar x HD had highest yield (2.203 kg palm<sup>-1</sup> year<sup>-1</sup>) followed by HD x Mohitnagar (2.079 kg palm<sup>-1</sup> year<sup>-1</sup>). Over seven years, Mohitnagar x HD consistently produced the maximum chali yield (2.190 kg palm<sup>-1</sup> year<sup>-1</sup>) followed by HD x Mohitnagar (VTLAH-2) with 1.930 kg and HD x Sumangala (VTLAH-1) with 1.138 kg chali palm<sup>-1</sup>.

### **Inter-specific hybridization for fruit rot and YLD screening**

Inter-specific hybridization between *Areca triandra* x *A. catechu* (Shatamangala) and *Areca catechu* (Shatamangala) x *Areca triandra* have been done for screening against *Phytophthora meadii* (fruit rot pathogen), along with seedlings of the parental plants (Shatamangala and *Areca triandra*). About 1566 and 1651 female flowers have been pollinated to obtain inter-specific hybrids from each of the *A. triandra* x *A. catechu* (Shatamangala) and *A. catechu* (Shatamangala) x *A. triandra* combination, respectively. The detached leaf assay technique was employed to identify inter-specific hybrid seedlings displaying resistance reactions. The resultant resistant seedlings will further be planted in the field for evaluation.

## **Varietal Development**

### **Coconut**

#### **Kalpa Vajra**

'Kalpa Vajra', an improved tall coconut variety developed through crossing high-yielding and root (wilt) disease-free WCT palms, received approval for release from the Central Sub-Committee on Crop Standards, Notification, and Release of Varieties for Horticultural Crops on May 4, 2023. The average yield of Kalpa Vajra over four years at ICAR-CPCRI campus was 80 nuts per palm per year. Performance trials at CDB Farm, Neriamangalam, showed a five-year average yield (2015-20) of 94 nuts per palm per year, outperforming WCT (OP) with 69.4 nuts per palm per year. Kalpa Vajra planted at Neriamangalam exhibited root (wilt) disease incidence of less than 20%. Given its superior performance, Kalpa Vajra is recommended for cultivation in areas with root (wilt) disease.

#### **Kalpa Suvarna**

A high-yielding dual-purpose (tender nut and copra) dwarf coconut variety, was recommended for release by the 32<sup>nd</sup> AICRPP workshop held during September 13 to 15, 2023. The variety is

well-suited for cultivation in the Western Ghats and coastal plains of Karnataka and Kerala. The variety features large green fruits with an attractive oblong shape. Palms yield an average of 108 nuts palm<sup>-1</sup> year<sup>-1</sup>, producing 20.1 kg of copra palm<sup>-1</sup> with good-quality tender nut water (TSS of 5.8° Brix and an average yield of 431 mL nut<sup>-1</sup>) (Fig. 1.22). These dwarf palms, 5.62 m in height at the 50th year, are regular bearers, flower early in 32 months, and have a short canopy, making them suitable for high-density planting (7.0m x 7.0m).



Fig. 1.22. Kalpa Suvarna variety palm with tender and matured nut

### **Cocoa**

Two cocoa varieties (Fig. 1.23) were released through the Central Sub-Committee on Crops Standards, Notification and Release of Varieties for Agricultural Crops.

**VTLCS 1 (Vittal Cocoa Selection 1):** It is a selection from Malaysian Trinitario type cocoa, red to orange in colour having smooth surface and shallow furrows. The trees with medium canopy (15-20 m<sup>2</sup>) yielded on an average 55 pods tree<sup>-1</sup> year<sup>-1</sup>, both as clones and seedlings, under arecanut and coconut. Pods are of 360 g weight with 41 beans per pod. With a single dry bean weight of 1.13 g this clone recorded the highest yield of 2.55 kg tree<sup>-1</sup> year<sup>-1</sup> (1700 kg ha<sup>-1</sup>). This is suitable for chocolate industry with a shelling percentage of 11%, nib recovery of 89% and 52% fat content. It withstands major pests, diseases, water limited conditions and recommended for all cocoa growing regions both in traditional and non-traditional belts.

**VTLCS 2 (Vittal Cocoa Selection 2):** It is a selection from an Upper Amazon Forastero type of cocoa, green to yellow in colour having smooth surface and shallow furrows. The trees with medium canopy (15 m<sup>2</sup>) yielded on an average 54

pods tree<sup>-1</sup> year<sup>-1</sup>, both as seedlings and clones, under arecanut and coconut. Pods weigh 400 g with 41 beans pod<sup>-1</sup>. Beans are very bold and big with 3 cm length, 1.5 cm width, 1.21 g weight, 15% shell, 58% nib recovery and 53% fat content. The total yield is 2.7 kg tree<sup>-1</sup>year<sup>-1</sup> (1840 kg ha<sup>-1</sup>). It is tolerant to major biotic and abiotic stress and suitable for all cocoa growing regions.



Fig. 1.23. High yielding cocoa varieties recommended for release

Following cocoa hybrids (Fig. 1.24) were recommended for release during 32<sup>nd</sup> AICRP on Palms workshop.

**VTLCH-1 (Vittal Cocoa Hybrid 1)** is an early-bearing, high-yielding cocoa variety suitable for cultivation under arecanut and coconut shades. Whether grown from seedlings or clones, it thrives well in high-density planting conditions with trees reaching 3.5-4 m in height and an optimal canopy size of 15-18 m<sup>2</sup>. It yields on an average 45-50 pods/tree/year, each pod weighing 350-400 g with 35-44 beans pod<sup>-1</sup>. The individual dry bean weight is 1-1.10 g, resulting in an average yield of 1.5-2.5 kg tree<sup>-1</sup> in various locations, translating to 750-1250 kg ha<sup>-1</sup> with 500 trees. The beans meet international standards (>1 g size) and are suitable for the chocolate industry, with a shelling percentage of 13%, nib recovery of 87%, >50% fat, and 1% free fatty acids. Additionally, it is rich in Fe and Zn contents. This versatile variety is recommended for cultivation in Karnataka, Kerala, and Tamil Nadu.

**VTLCH-2 (Vittal Cocoa Hybrid 2)** is also an early-bearing and high-yielding hybrid, suitable under arecanut and coconut shades, as seedlings and clones, and is adaptable to high-density planting. The trees, with a height of 3.5 m and an optimal canopy size of 14-20 m<sup>2</sup>, yield on an

average 40-50 pods/tree/year, each weighing 350-400 g with 35-50 beans pod<sup>-1</sup>. The individual dry bean weight is 1 to 1.20 g, resulting in an average yield of 1.5-2.5 kg tree<sup>-1</sup>. The beans, meeting international standards (>1 g size), are ideal for the chocolate industry, exhibiting a shelling of 13%, nib recovery of 87%, >50% fat, and 1% free fatty acids. Rich in iron (Fe) and zinc (Zn) contents, this variety is specifically tolerant to black pod rot, tea mosquito bug, and low moisture stress. It is well-suited for cultivation in four states: Karnataka, Kerala, Andhra Pradesh, and Gujarat.



Fig. 1.24. High yielding cocoa varieties recommended for release

### DUS Centre for Coconut

Candidate variety (REG/2015/415) under DUS testing at ICAR-CPCRI, Kasaragod, recorded higher palm height, longer leaves and leaflets in comparison to the reference varieties. Flower initiation was observed in 65% of palms in the candidate variety, whereas 100% flowering was recorded in reference 1 and reference 2 and 25% flowering in reference 3 varieties. Inflorescence characteristics were recorded during September-November. Higher inflorescence length, medium stalk girth, medium number of spikelets and absence of intra spadix overlapping was observed under DUS testing of candidate variety. On site observations pertaining to inflorescence, tender nut and fruit characteristics were under taken on adult palms of this candidate variety at RCRS, Bhatye. Inflorescence length ranged from 90-130cm, stalk girth from 9-11cm, number of spikelet varied from 30-48 and female flowers from 12-26. Volume of tender nut water varied from 210-490 mL.



Fig. 1.25. Tender nut of REG/2015/415

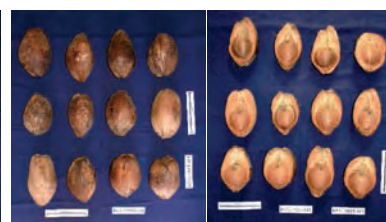


Fig. 1.26. Fruits of REG/2015/415

On site recording of DUS descriptor data of the farmer's variety (Reg/2014/1949) of Edava long fibre coconut from Thiruvanthapuram was continued. Inflorescences were observed to have higher girth with length ranging from 80-120 cm, number of spikelets ranging from 30-58 and female flowers from 11-30. The volume of tender water recorded varied from high to very high. Mature fruits were harvested for fruit characters (Fig. 1.27).



Fig. 1.27. Edava long fibre coconut

Growth characteristics were recorded on extant/reference/example varieties of coconut planted in the DUS plot. In the current year, significantly higher trunk length, leaf length, leaflet length and leaf production were observed in Chandra Kalpa, whereas broader leaflet was observed in Kalpa Dhenu under 4m x 4m spacing. Higher fruit yield was recorded from 6m x 6m spacing, while both inflorescence production and fruit set was very poor at close spacing of 4m x 4m as recommended in DUS test. Based on the data recorded for DUS descriptor traits in varieties/germplasm and considering the data range actually observed, reclassification of states is proposed for the characteristics, namely plant: collar girth (cm), plant: number of leaves produced in one year, leaflet: length (cm) (+), leaflet: breadth (cm); plant petiole color and fruit color.

### DUS Centre on Cocoa

DUS guidelines for cocoa were developed based on 25 characteristics. Thirtyone cocoa genotypes (including 21 hybrids and 10 elite clones) were classified based on three each of leaf and flower characters, twelve fruit and six bean traits and fat content as qualitative parameter in the form of multiscale scores. Colour in immature pod, pod shape, basal constriction, apex form and bean shape were used as grouping characters (Fig. 1.28). Apart from DUS characters, some noticeable variations were observed for types of staminodes in flowers, sweetness of pulp and fresh bean colour, which can act as morphological and qualitative markers for specific traits.

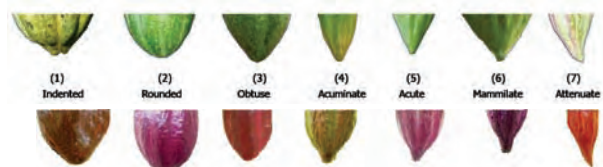


Fig. 1.28. Different fruit characters in cocoa

### Planting Material Production

At Kasaragod, 23138 hybrid coconut seed nuts were produced and sown in the nursery from the flowers pollinated last year. This year 114197 female flowers from 480 WCT and 20 LCT palms were pollinated. About 29451 seed nuts of other coconut varieties were produced and sown in the nursery for seedling production. Breeder seed production was undertaken in released varieties like Kera Chandra, Kalpa Prathiba, Kalpa Raksha, Kerakeralam, Kalpa Tharu, Kalpa Ratna, Kalpa Haritha, Kalpa Surya and Kalpa Jyothi both at Kasaragod and Kidu. Seedling observations on characters of polybag and nursery raised seedlings have been recorded to develop seedling standards for six, nine and twelve month old seedlings.

At Kayamkulam, a total of 34,979 coconut seedlings including 5420 hybrid seedlings have been produced during the period (Table 4).

At Research Centre Kidu, 30,700 hybrid coconut seed nuts consisting of 28,943 T x D hybrid seed nuts (17785 WCT x COD, 3018 WCT x MYD, 1791 LCT x COD and 1719 LCT x GBGD) and 1757 D x T hybrid seed nuts (895 MYD x TPT, 141 COD x WCT and 862 CGD x WCT) were produced and sown for hybrid seedling production. Under assisted pollination, about 4489 (D x T) nuts have been produced from the emasculated COD palms and sown for seedling production. About 26,081 seed nuts of tall and dwarf coconut varieties have been produced of which about 12,320 seed nuts have been distributed to farmers and rest (11,520 nos.) have been sown for raising seedlings. Inter-se mating for seed nut production in released coconut varieties have been done for gap filling. An average of 24% of nuts have been harvested out of the total pollination done during the previous year. In Arecanut, about 73,000 arecanut seedlings, 3.12 lakhs arecanut seed nuts and 352 VTLAH-2 hybrid seedlings have been produced. During the period, 17 inflorescences in selected Hirehalli dwarf palms have been pollinated using Sumangala and Mohitnagar male parents for VTLAH-1 and VTLAH-2 seed nut production, respectively.

Table 4. Planting materials production in Coconut, Arecanut and Cocoa

| Centre     | Coconut |        |        | Arecanut |          |        | Cocoa |                 |       | Total  |
|------------|---------|--------|--------|----------|----------|--------|-------|-----------------|-------|--------|
|            | Variety | Hybrid | Total  | Seed nut | Seedling | Total  | Pod   | Seedling/grafts | Total |        |
| Kasaragod  | 29451   | 23138  | 52589  | -        | -        | -      | -     | -               | -     | 52589  |
| Vittal     | 120     | -      | 120    | 60197    | -        | 60197  | 366   | 48563           | 48893 | 109210 |
| Kayamkulam | 29559   | 5420   | 34979  | -        | -        | -      | -     | -               | -     | 34979  |
| Kidu       | 26081   | 30700  | 56781  | 312192   | 73371    | 385563 | -     | -               | -     | 442344 |
| Kahikuchi  | -       | -      | -      | -        | 51500    | 51500  | -     | -               | -     | 51500  |
| Mohitnagar | 404     | -      | 404    | 83157    | 8500     | 91657  | -     | -               | -     | 92061  |
| Total      | 85615   | 59258  | 144873 | 455546   | 133371   | 588917 | 366   | 48563           | 48893 | 782683 |

At Regional Station, Vittal, 60197 planting materials of arecanut were produced and distributed during the period. Similarly, in cocoa, a total of 113 seed pods; 30456 seedlings and 959 grafts were sold and a revenue of Rs. 3,37,285/- has been generated. During the period, 84 coconut seedlings were supplied and a revenue of Rs.17,640 have been obtained.

At Research Centre, Mohitnagar, 83157 seed nuts and 8500 seedlings of arecanut, 404 seedlings of coconut, 2775 black pepper cuttings and 200 acid lime seedlings were produced and distributed.

At Research Centre, Kahikuchi, 51,500 arecanut seed nuts and 2,562 cocoa seeds have been sown for production of quality planting materials.

### Biodegradable eco grow bags for raising arecanut seedlings

Eco grow bags were prepared using arecanut leaf sheath. Technological improvements were made to eco grow bags to prevent rotting of bags during rainy season. Eco grow bags (Fig. 1.29 a & b) were treated with 1% Bordeaux mixture and 10% Bordeaux paste for increased shelf life. For further increasing the shelf life, bags were coated with eco-friendly resins/gums/wax.



Fig. 1.29 a. Arecanut seedlings in leaf sheath bags



Arecanut leaf sheath bags coated with eco-friendly wax

Fig. 1.29 b. Biodegradable eco pots for planting material production

Cocoa is a recalcitrant seed, loses its viability after extraction from pods. To extend its germinability and to encourage long distance transportation, cocoa seed balls (Fig. 1.30) were prepared with clay loam soil mixed with coir compost and little sand on top at the point of sowing and with alternate days of watering. Normally, cocoa seeds germinate in 7 days and in these seed balls cocoa seeds took 15-18 days for germination.



Fig. 1.30 Cocoa seed balls

Table 5. Revenue from planting material production (Rs in Lakhs)

| Centre     | Coconut | Arecanut | Cocoa | Others | Total   |
|------------|---------|----------|-------|--------|---------|
| Kasaragod  | 31.290  | -        | -     | -      | 31.290  |
| Vittal     | -       | 10.955   | 4.984 | -      | 15.939  |
| Kayamkulam | 43.382  | -        | -     | -      | 43.382  |
| Kidu       | 55.794  | 90.931   | -     | -      | 146.725 |
| Mohitnagar | 0.522   | 2.446    | -     | 0.805  | 3.773   |
| Kahikuchi  | -       | 12.308   | -     | -      | 12.308  |
| Total      | 130.988 | 116.640  | 4.984 | 0.805  | 253.417 |

### Coconut Tissue Culture

#### Callus induction, formation of embryogenic calli and somatic embryos using novel growth regulators

Plumular explants inoculated in Y3 medium with picloram (200 or 100  $\mu\text{M}$ ) and meta-topolin (5  $\mu\text{M}$ ) responded differently. Proliferation of callus, formation of embryogenic calli, per cent germination of somatic embryos and shoot regeneration were improved in Y3 medium supplemented with 200  $\mu\text{M}$  picloram and 5  $\mu\text{M}$  meta-topolin compared with Y3 medium supplemented with 2,4-D and TDZ.



Fig. 2.1. Callogenesis (a) and shoot regeneration (b & c) from plumular explants of coconut in medium supplemented with meta-topolin

#### Axillary *in vitro* shoots in coconut

Procedure for obtaining axillary *in vitro* shoots in coconut was attempted in indigenous accessions. Zygotic embryos from mature coconut (11-12 months) were cultured in a germination medium with 60 g  $\text{L}^{-1}$  sucrose. After attaining 3-4 cm (~3 months after inoculation), plantlets were trimmed near the meristematic region, and a transverse cut was made. Segments were cultured in Y3 medium with TDZ (0.2 mg  $\text{L}^{-1}$ ). Cultures were maintained under 16h light conditions. The cut procedure was repeated until a proliferating mass appeared in the meristematic region. Proliferating mass (PM) was scooped and transferred to dark condition for multiplication. About 25% of the cultures produced proliferating mass. Four plantlets obtained through the regeneration of proliferating mass were shifted to light and are now in the rooting stage. An attempt was made to grow the proliferating mass in liquid. A clump of proliferating mass was cultured in a 100 mL conical flask with 6 mL of Y3 media supplemented with 0.2 mg  $\text{L}^{-1}$  of TDZ. Cultures were maintained under dark conditions in an orbital shaker. The relative growth rate in PM in liquid was 0.016 g. g $^{-1}$  day $^{-1}$ . Results indicate the possibility of maintaining proliferating mass in liquid cultures.

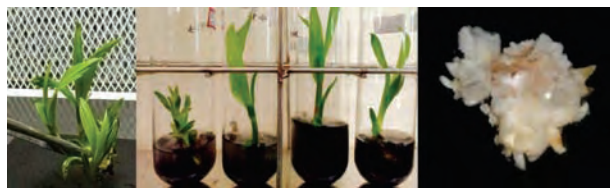


Fig. 2.2. Initiation of proliferating mass and regeneration of *in vitro* axillary cultures in coconut

#### Methylation/ metabolome status in somatic embryogenesis

In Y3 medium, the impact of three concentrations of the demethylating agent 5-azacytidine (5-azaC) - 10, 15, and 20  $\mu\text{M}$  - on callus induction from plumular explants revealed that highest callus induction percentage was observed with 15  $\mu\text{M}$  5-azaC. After 12 weeks, the fresh weight of the callus in the 15  $\mu\text{M}$  5-azaC treatment (0.6 g) surpassed that of the control (0.4 g). Histological studies revealed multiple meristematic areas in embryogenic calli from the 15  $\mu\text{M}$  5-azaC treatment, suggesting potential for multiple somatic embryos. In addition, the combination of picloram (75  $\mu\text{M}$ ) and 5-azaC (15  $\mu\text{M}$ ) exhibited an enhanced response for both callus induction and somatic embryogenesis.



Fig. 2.3. Initiation of callus and regeneration of *in vitro* cultures from thin layer sections of coconut zygotic embryos

#### Initiating cell suspension at various stages of development of embryogenic calli

Embryogenic callus derived from plumular explants was used to initiate cell suspension culture. After one month, cell division occurred in both the Y3 medium supplemented with picloram (0.2 mg  $\text{L}^{-1}$ ) and the Y3 medium supplemented with 2,4-D (0.2 mg  $\text{L}^{-1}$ ). Clumps of cells were observed rather than single cells. Elongated and round-shaped cells were observed. Cells with dense cytoplasm and small vacuoles, and cells with large vacuoles were also present. After 50 days, 2 mL of suspension from the cultures was added to fresh culture media. Cell aggregates were observed after 10 days of subculture.

### Immature inflorescence culture

Immature inflorescences (with outer spathe length of 2-15 cm) were collected from 25 to 30 year old West Coast Tall palms. Rachillae bits of 0.5-1 mm size were inoculated in various media-hormone combinations. The explants in media containing Y3 with 2,4-D ( $1 \text{ mg L}^{-1}$ ) swelled rapidly, whereas those in Y3 with picloram turned brown. Browning of explants in media containing MS with 2,4-D ( $100 \text{ mg L}^{-1}$ ) and 2iP ( $3 \text{ mg L}^{-1}$ ) was 52% at the end of six weeks and, after the first subculture, turned to 75 per cent. The cultures in the woody plant medium (WPM) remained fresh without browning even after eight months. The results indicate that, using immature inflorescences explants, the Y3 with 2,4-D ( $1 \text{ mg L}^{-1}$ ) is best suited for culture initiation.

For studying the effect of the size of inflorescences, inflorescences were grouped into 4 categories, i.e. those with outer spathe lengths of <5 cm, 5 cm – 10 cm, 10.5 cm - 15 cm and >15 cm and the rachillae bits were inoculated in Y3 with 2,4-D and incubated in dark condition. Maximum shoot regeneration was recorded when explants were taken from 5-10 cm inflorescence, followed by 10.5 to 15 cm. There was no shoot regeneration in explants from the larger inflorescence of more than 15 cm. Rachillae bits from smaller inflorescence produced calli. In contrast, larger inflorescences formed floral structures and were confirmed by histological studies. This indicates that the right developmental stage of inflorescence is important for the success of direct organogenesis.

For culture initiation using immature inflorescence, WPM supplemented with TDZ ( $5 \text{ mg L}^{-1}$ ) alone or in combination with NAA resulted in more vegetative buds. The varietal response on culture initiation was evaluated using inflorescences collected from Kalpasree, Gangabondam Green Dwarf, Niu Leka Dwarf, Kalpa Sankara and WCT. There was no variation in the initial media, but varietal and palm-to-palm variations were noticed in subsequent subcultures. For multiple shoot induction,  $1/2 \text{ MS}$  supplemented with  $1 \text{ mg L}^{-1}$  each of NAA, BAP and 2iP resulted in more multiple shoots compared to 2iP-free media.

### Embryo rescue of sweet kernel genotypes in coconut

Plantlets developed from embryos of *Mohachao Narel* (sweet kernel genotypes from Maharashtra)

were grown in a solid Y3 medium supplemented with BAP, IAA and IBA, followed by liquid Y3 media of the same composition. The plantlets developed were potted in the potting mixture containing coco peat and vermiculite (ratio 3:1). About 30 plantlets are potted, and 60 plantlets are in the final stage of *in vitro* culture.

### Arecanut Tissue Culture

#### *In vitro* multiplication of arecanut palms

Immature inflorescence from arecanut varieties [VTLAH-1, VTLAH-2, Hirehalli Dwarf (HD), Shatamangala, tissue cultured Yellow Leaf Disease resistant palm (STC1 and STC2)] and YLD disease free palms were collected from ICAR-CPCRI Regional station Vittal and Sampaje-Sullia region of Karnataka. Also, inflorescence explants were collected from YLD tolerant palm in Sringeri, Karnataka. The explants were inoculated for callus induction and somatic embryogenesis. The immature inflorescence collected by non-destructive sampling was inoculated in Y3 media supplemented with a high concentration of auxin (Picloram). After two weeks, they are sub-cultured at monthly intervals to serially reduce concentration of picloram, i.e., P100, P50 and P25. Callus initiation percentages in previous year (2021) cultures were 25% (VTLAH-1), 60% (Shatamangala), 56% (S-18), 34% (S-34), 4% (S-42), 50% (HD) and 60% (VTLAH-2). Approximately 100 somatic embryos of Shatamangala, 480 SE of YLD-resistant palms, 40 SE in VTLAH1 and 10 in HD varieties were generated from the previous year's culture. 24 plantlets obtained from S18 YLD tolerant palm are in different growth and developmental stages under light.

From the previous year's (2020) cultures, around 1000 somatic embryos of VTLAH-2 were separated from embryogenic calli cultures. Around 300 plantlets are in different growth stages, and 27 plants were subjected to *in-vitro* hardening.

To attain the synchronous maturity in somatic embryos, an experiment was carried out by using ABA (@1, 2.5, 5, 10, 15 and 20  $\mu\text{M}$ ) on somatic embryos generated from immature inflorescence culture of VTLAH-2. Hormone-free Y3 media served as control. Results obtained from the experiment indicate that ABA @ 1-5  $\mu\text{M}$  is good for the synchronous maturity of somatic embryos and for reducing secondary somatic embryogenesis. In contrast, higher concentrations were found to inhibit/delay the somatic embryo growth and development.

## Cryopreservation Studies

### Coconut and arecanut cryopreservation

Different cryopreservation techniques, such as encapsulation vitrification of plumules, V-cryomesh and V-cryoplate techniques, were validated to identify the best suitable method for the long-term conservation of coconut plumules. Plumule (post-thaw recovery after one month) and plantlet recovery in different methods ranged from 46-73% and 16-33%, respectively. Maximum plumule and plantlet recovery were achieved with V-cryomesh (73% and 33%), followed by V-cryoplate (56% and 23%) and encapsulation vitrification methods (46% and 20%), respectively.

## Molecular Studies

*Agrobacterium tumefaciens* strain EHA105 strain carrying pBI121 was utilised for the coconut transformation using various explants, such as zygotic embryos and embryogenic calli derived from plumular explants and axillary shoots. The explants were wounded by pin-pricking followed by vacuum infiltration with the *Agrobacterium* culture. The explants were inoculated in the Y3 medium, and co-cultivation was performed in the co-cultivation medium. The explants were then cultured on the selection medium (containing 150 mg L<sup>-1</sup> kanamycin and 500 mg L<sup>-1</sup> cefotaxime) to select transformants. After incubation at 28°C for three weeks, the tissues were subjected to *GUS* staining.

### Pseudomolecule level genome assembly of CGD using Hi-C

Previously, the genome of Chowghat Green Dwarf (CGD) could be assembled into 26855 scaffolds totalling ~1.93 Gb genome size, which is ~ 70% of the reported coconut genome size of ~2.7 Gb. Since the genome assembly lacks assembly contiguity, it was decided to use short/long read platforms in combination with Hi-C technology to upgrade the existing genome sequence of coconut. High-quality gDNA was isolated from the spindle leaf tissue of CGD. A short read sequencing library was prepared using Illumina protocol, quality checked, and sequenced using a 2 x 150 bp Paired-End (PE) module on the Illumina NovaSeq6000 platform. Data amounting to 90 Gb was generated. Long-read sequencing was performed using Oxford Nanopore Technology (ONT) MK1C sequencing platform. A total of ~40 Gb of raw data at ~15X depth was generated using three flow cells.

Further, the Hi-C library was prepared and sequenced using the Illumina NovaSeq6000 platform to generate 120 GB of raw data; at an estimated depth of 40X.

The incorporation of HiC data helped in the generation of contiguous assembly. The obtained assembly was of size 2.68 Gb with a total of 1930 scaffolds and N50 of 174 Mb; this was a significant improvement from the previous N50 of 128 Kb. The assembly has been resolved in 16 pseudo-chromosomes, comprising of 2.63 Gb and 1914 singletons comprising 56 Mb.

### Association mapping in coconut using SNPs

An association mapping panel comprising 96 coconut palms was evaluated for yield contributing traits. Genome-wide association studies (GWAS) were conducted based on two years of phenotype data and a set of 83,000 single nucleotide polymorphisms (SNPs) derived by genotyping by sequencing. SNPs significantly associated with these traits were identified using mixed linear model (MLM) analysis. Putative candidate genes that contribute to these traits were also identified.

### ‘Kalpa Genome Resource’

The ‘Kalpa Genome Resource’ is a broad and interactive genomics platform for the dwarf coconut variety ‘Chowghat Green Dwarf’ (CGD). The portal can be utilised to access the genome and transcriptome resources on the platform in list view or as a visualisation track. The platform consists of many user-friendly tabs. The ‘Annotation’ tab offers various options like searching for ‘simple sequence repeats’, ‘repeat elements’, ‘genes of interest’ and ‘locus identifier’. The ontology classification, attributes, NCBI accession number with a hyperlink to the NCBI portal, and the transcript expression levels under various life cycle stages and stress conditions can be viewed. The Gene Ontology (GO) with locus identifier and KEGG Orthologs containing the annotated sequence can also be viewed. Using the BLAST search, inputting the query sequence or uploading the query file, the percentage identity, E-Value, Bitscore, and query coverage can be obtained. JBrowse can be accessed through the CGD Coconut option in the dropdown menu. The downloadable files for the complete genome, transcriptome, proteins, genes, gene annotations,

repeat elements, SSR, and transcript expression data can be found in the download tab.

### **In silico analysis of odorant binding proteins (OBPs) of rhinoceros beetle homology modelling and docking studies**

A total of 20 OBP genes were identified from the in-house generated draft genome of the coconut rhinoceros beetle. The predicted protein sequences of these OBPs were taken for homology-based modeling. Major components of eight essential oils were screened for their binding to the modelled rhinoceros beetle OBPs by *in-silico* docking studies. Cymene gave the best binding with OBP, followed by cinnamate and thymol.

### **Genome sequencing and analysis of *Phytophthora meadii***

Short read sequencing library was prepared using Illumina protocol, quality checked, and sequenced using a 2 x 150 bp Paired-End (PE) module on an Illumina NovaSeq6000 platform. A total of 80 million reads, amounting to 12 Gb, was generated at an estimated depth of 200x. In addition, long-read sequencing was performed using Oxford Nanopore Technology (ONT) and a total of 1.4 million reads was generated, amounting to 2.04 Gb of raw data. Hybrid whole genome assembly was completed using Nanopore long and Illumina short reads. Primary assembly was performed using the MASURCA tool. A total ungapped genome size of 58701594 base pairs was assembled. The final assembly contains 877 contigs, with an N50 size of 158923 base pairs. Assembly completeness was estimated using BUSCO using the Stramenopiles database. The genome assembly recovered 100% of the total 100 BUSCO gene groups, with 98% complete single copy ortholog and 2% duplicate ortholog gene groups.

### **Identification of novel targets from genome resources of rhinoceros beetle**

Four novel targets (*viz.*, transcription elongation factor's *spt 5* and *spt 6*; DEAH/H-box ATP-dependent RNA helicase *prp16*; and RNA polymerase II subunit *rpII-330*) were mined from the in-house generated genome and transcriptome data of rhinoceros beetle.

### **Insect bioassay studies in rhinoceros beetle by injecting siRNAs**

Oral feeding of siRNA was carried out in second instar rhinoceros beetle larvae. Negative controls

were injected with an equivalent amount of water or dsRNA corresponding to the *Galleria mellonella* metalloproteinase inhibitor (*IMPI*) gene. Survival rates were monitored every day for 10 days. Significant mortality of  $\geq 50\%$  was observed from the 3<sup>rd</sup> day after oral feeding of dsRNA. The detected mortality rates were significantly higher compared to the non-target *IMPI* dsRNA.

### **Identification of suitable reference gene(s) of coconut rhinoceros beetle**

To identify suitable reference gene(s) possessing stability, six prospective genes (*viz.*, *NADH*, *ACTIN*, *EF1A*, *RPL3*, *SDHA*, and *ARF6*) were selected and evaluated for their potential use as reference gene(s) across different developmental stages of coconut rhinoceros beetle. A comprehensive approach was utilised based on five statistical models, *viz.*, GeNorm, BestKeeper, NormFinder, RefFinder and the  $\Delta C_t$  value, and based on the obtained stability values of candidate genes, a consensus ranking was generated. The expression levels of *NADH*, *EF1A* and *RPL3* were the most stable across the developmental stages with significant statistical reliability. Further, this study identified *NADH/EF1A* as the most reliable reference gene combination, which could provide robust normalisation of RT-qPCR data in gene expression studies in coconut rhinoceros beetle.

### **Expression profiling of specific miRNAs and their targets of interest**

The expression patterns of 6 miRNAs (*cnu-miR166a*, *cnu-miR167c*, *cnu-miR169a*, *cnu-miR171a*, *cnu-miR397* and *cnu-miR535a*) were studied in different stages of zygotic and somatic embryogenesis by stem-loop qRT-PCR. The expression level of these miRNAs was studied in the eight months old, ten months old and twelve months old zygotic embryos and calli and somatic embryos. The corresponding targets studied for expression levels are as follows: glutathione S-transferase U8-like, ribonuclease 3-like protein 3 and homeobox-leucine zipper protein HOX32 for *cnu-miR166a*; eIF-2- $\alpha$  kinase activator GCN1 for *cnu-miR167c*; nuclear transcription factor Y subunit A-7 protein and NEDD1-like protein for *cnu-miR169a*; scarecrow-like protein 22 and scarecrow-like protein 27 for *cnu-miR171a*, laccase-17, laccase-22 and laccase-24 for *cnu-miR397*; cadmium/zinc-transporting ATPase HMA2-like for *cnu-miR535a*.

## Management of coconut based farming systems for sustainable productivity

One hectare of coconut based farming system model comprising coconut, pepper trailed on the coconut trunk, banana in the border of the plots, fodder sorghum (CO 31- multi cut fodder sorghum) in the interspaces of coconut, dairy unit and poultry yielded 18190 coconuts, 17486 litres of cow milk, 105.3 kg of live weight of poultry birds, 2877kg of banana and 327.6 kg of pepper and 61.25 kg of fish. The system was economical with net returns Rs. 543682/- and energy efficient with higher energy output (884.9 KJ ha<sup>-1</sup>). Coconut monocropping recorded lower energy output (259.8 KJ ha<sup>-1</sup>). A similar trend was noticed for net energy, energy use efficiency, energy productivity, and energy profitability. Multicut Sorghum - CO31 fodder grass intercropped in coconut garden under different nutrient management practices, recorded 79.9 t ha<sup>-1</sup> green fodder yield and 55.1 t ha<sup>-1</sup> dry fodder yield with 100% organics (sum of five harvests in a year) (Fig. 3.1 & 3.2).

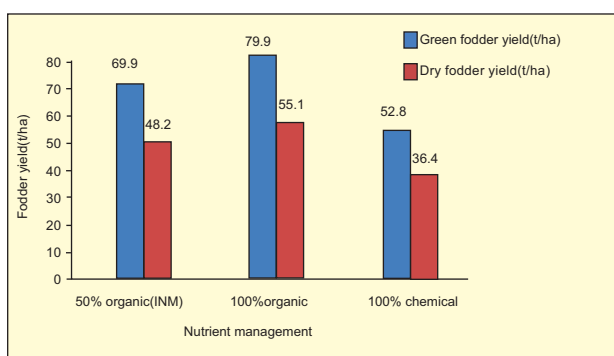


Fig. 3.1. Green and dry fodder yield of multi cut fodder sorghum (CO31) in coconut based farming system



Fig. 3.2. Cultivation of multi cut fodder sorghum (Co31) in coconut based farming system

## Evaluation of coconut based high density multi species cropping system under organic and integrated management

Coconut based high density multi species cropping system initiated in 2007 (Fig. 3.3) comprising coconut, pepper (trailed on the coconut trunk), banana cv. Kadali, Robusta (inter-row space of palms), cinnamon (inter-row space of palm) and nutmeg (between 4 coconut palms) under different nutrient management practices (two-third, one-third and non-application of RDF) along with recycling of biomass, biofertilizer application and green manuring resulted in non-significant difference among the treatments with respect to nut and copra yield (128-136 nuts palm<sup>-1</sup> year<sup>-1</sup> with 3132 to 3606 kg ha<sup>-1</sup> of copra yield). Similarly, there was no significant yield difference among treatments for Banana cv. Kadali (8.8 – 9.2 kg bunch<sup>-1</sup>), Banana cv. Robusta (11.4 to 13.7 kg bunch<sup>-1</sup>), Cinnamon (61.5 to 67.2 kg ha<sup>-1</sup>) and pepper (323.0 to 390.5 kg ha<sup>-1</sup>). By generating additional returns from the component crops and achieving a higher nut yield with reduced external inputs, the high density multispecies cropping system (HDMSCS) proves economically advantageous, boasting net returns of Rs. 4,81,416. This amount is 4.9 times greater than that obtained from monocrop coconut cultivation.



Fig. 3.3. Coconut based multi species cropping system

## Performance of dwarf jack in coconut system

Performance evaluation of four varieties of dwarf jack (Sindoor Varikka, Thailand Red, Thailand Pink, and Gumless) has been taken up since November 2021 in the interspaces of coconut var. Kalpasankara planted at 9 m x 9 m spacing. Among the varieties, the first fruiting was recorded nine

months after planting in Var. Thailand Pink (Fig. 3.4). The fruiting was 100% in this variety during the second season, whereas other varieties are yet to start yielding. The intercultural operation such as pruning is not required for this variety as it has a unique pyramidal type of canopy structure. All the other varieties were pruned at the 9th and 15th months after planting to maintain their canopy within the coconut ecosystem. During the first season (June- July 2023) and second season (Dec 23 – Jan 24) Thailand Pink produced 56 and 123 bulbs, respectively. The average weight of fruits is 4.25 kg. The average weight of a bulb is 35 g with a seed weight of 10 g.



Fig. 3.4. Performace of Dwarf Jack Var. Thailand Pink in coconut system

### Demonstration of cinnamon intercropping in coconut garden

An experiment on cinnamon intercropping in coconut with the pentagonal method of planting (5 plants per pit) in different spacing revealed that spacing significantly influenced the growth and yield of cinnamon (Fig. 3.5). During the second harvest, two shoots were retained per plant. With regard to individual plant quill yield, higher individual plant quill yield was recorded in widely spaced (2.0 x 2.0 m) trees (229.3 and 263.8 g plant<sup>-1</sup>) during the first and second harvest over other treatments. However, with regard to per hectare yield, closer planting of cinnamon (0.6 x 1.2 m) recorded significantly higher dry quill yield of 631.92 g tree<sup>-1</sup> and 979.0 kg ha<sup>-1</sup> over other spacing treatments.



Fig. 3.5. Demonstration of intercropping cinnamon in coconut garden

### Arecanut based high density multispecies cropping system (HDMSCS) at Kahikuchi

The experiment conducted at Kahikuchi on the arecanut based high density multispecies cropping system (HDMSCS) with various crop components (pineapple, turmeric, Assam lemon and banana) and both organic and integrated nutrient management revealed that applying 2/3<sup>rd</sup> recommended dose of fertilizer (RDF) along with biomass recycling resulted in a higher number of bunches (1.70), number of nuts (73.60), and fresh nut weight per palm per year (3.28 kg). In the intercropped turmeric, the same treatment recorded 1209.57 kg of fresh rhizome per hectare.

### Intercropping of seasonal horticultural crops under arecanut at Kahikuchi

Based on climate feasibility and the local demand, seasonal horticultural crops were screened as intercrop under arecanut at Kahikuchi (Fig. 3.6). Among the winter vegetables viz., cabbage (var. Green Express), knol khol (local var.), and brinjal (var. Rani), as intercrops, cabbage was the best intercrop, with a yield of 13.17 t ha<sup>-1</sup> and a higher B:C ratio of 2.04.



Fig. 3.6. Intercropping of vegetables under Arecanut

### Management strategies for production, health and nutrition of arecanut and cocoa

#### Management of YLD in arecanut

A field experiment was initiated in 5 farmers' gardens in Karnataka in 2016/2017 to study the effect of plastic mulching on the yellow leaf disease of arecanut. Plastic mulching during monsoon (June-October) was found to reduce the disease index in all the management trial plots having an initial disease index of <10% during the experimental period (2017-2023) compared to the pre-experimental data (2016). The disease index during 2023 was 1.6 – 4.3%, whereas it was 2.9–5.8% in 2016 in different plots. Importantly, the

cumulative dry kernel yield  $\text{ha}^{-1}$  was higher than the control (no mulching) in different gardens. It also improved the soil physical parameters like porosity, bulk density, and water-holding capacity. The result indicates that plastic mulching during monsoon could delay the symptom expression and increase the yield in the YLD-affected garden (especially when the initial disease index is low).

### **Identification of disease-free arecanut palms in YLD hotspot area**

A field survey was conducted in Sampaje and Koppa, the hotspot areas for yellow leaf disease of arecanut. A total of 90 arecanut palms having no symptoms of yellow leaf disease were identified and marked. Among them, 15 palms were tested for the disease, and 10 palms were disease-free. Multiplication of selected elite palms through tissue culture is in progress.

### **Carbon stock in arecanut - cocoa cropping system:**

Arecanut and cocoa are perennial crops with enormous potential to sequester atmospheric carbon. The carbon stock in the arecanut - cocoa cropping system was estimated as  $45.52 \text{ t ha}^{-1}$ . The contribution of twenty-year old arecanut to the total carbon stock in the system was 84.25% ( $36.35 \text{ t C ha}^{-1}$ ), whereas, ten-year old cocoa trees accumulated  $9.17 \text{ t carbon ha}^{-1}$ . Carbon sequestration in the arecanut-cocoa cropping system was estimated at  $167 \text{ t CO}_2 \text{ ha}^{-1}$ . The net carbon accumulation in adult palm $^{-1}$  was  $0.83 \text{ t ha}^{-1}$  in arecanut, whereas it was  $1.11 \text{ t ha}^{-1}$  in cocoa. The biomass partitioning varied among different yield levels. The biomass partitioning towards yield was 9.5 - 9.8 % in arecanut and cocoa. The result indicates that the arecanut-cocoa cropping system can considerably accumulate atmospheric carbon, and this has a greater significance in the era of climate change.

### **Genetic variability in cocoa for nutrient use efficiency, water use efficiency, partitioning of micronutrients towards nib, and nutrient removal**

A hydroponic experiment was conducted with eight released cocoa varieties and three different nitrogen (N) levels (low, medium and high) to study the water and nutrient use efficiencies of released cocoa varieties/hybrids. The dry biomass significantly increased with an increase in N

concentration in the growing medium. Increasing N concentration in the nutrient solution also led to a significant increase in water use efficiency (WUE). Nitrogen uptake positively influenced the WUE ( $r=0.55^*$ ), implying the role of N in water use efficiency. The dry weight of fine roots produced by the cocoa plants was positively correlated with N uptake ( $r=0.64^*$ ) and WUE ( $r=0.49^*$ ). A significant difference was observed for WUE among eight released varieties /hybrids. Maximum mean WUE was recorded in VTLC-1 ( $2.56 \text{ g L}^{-1}$ ), which was on par with VTLC-1 ( $2.51 \text{ g L}^{-1}$ ) and VTLC-1 ( $2.49 \text{ g L}^{-1}$ ). These varieties were also nitrogen-use-efficient.

Micronutrients in the nibs of 20 cocoa genotypes was studied. Among them, the Upper Amazon genotype, VTLC 151, was found to have multiple special features. It was potassium (K) use efficient, K responsive, partitioned significantly more zinc (45.5 % higher than mean value) and boron (141 % higher than mean value) towards the nib. The iron concentration in the bean was also high (16.3 % higher than the mean value). This genotype produced a good yield ( $1.28 \text{ kg tree}^{-1}$ ) with significantly bolder beans ( $1.1 \text{ g single bean weight}$ ).

The estimation of nutrient flows and stocks is needed for the development of sustainable land use systems. Nutrient removal was studied in twenty different cocoa genotypes. The nutrient removal through yield  $\text{tree}^{-1}$  varied significantly among cocoa genotypes. It was estimated at 19-71 g N, 2.5-9.4 g P, 27-118 g K, 5.5-21.1 g Ca, 4.3-15.6 g Mg, 71-336 mg Fe, 182-1304 mg Mn, 29-152 mg Zn, 68-255 mg Cu, and 37-153 mg B. This indicates that, cocoa removes huge quantities of nutrients, especially potassium. Understanding the nutrient removal potential of different genotypes is needed to optimize the input-use in cocoa.

### **Fertigation studies in hybrid coconut (Kalpa Sankara)**

A field experiment is in progress with different nutrient levels (5 treatments ranging from 50-200% of soil test based nutrient values) through drip fertigation and with basin application of nutrients under drip irrigation in hybrid coconut (Kalpa Sankara). The treatments were applied from July 2022 to May 2023. Notably, T4 (fertigation with 200% nutrient application)

showed the highest total nut yield of 191 nuts, followed by T<sub>2</sub> (fertigation with recommended dose of nutrient application) of 172 nuts, while T<sub>5</sub> (basin application of fertilizers with drip irrigation) exhibited the lowest yield of 88 nuts. Overall, the fertigation treatments yielded significantly higher than the surface-applied treatment (T<sub>5</sub>). Leaf sample analysis revealed that Leaf K content was sufficient (>1%) in the 200% and 100% soil application treatments, slightly lower at 0.98 in the 150% treatment. 100% fertigation, recorded the second-highest yield but showed a depletion of leaf K (0.8%). All fertigation treatments demonstrated a B:C ratio above 2. In contrast, the soil application with drip irrigation treatment recorded a B:C ratio of 1.26. The result indicates that coconut hybrid Kalpa Sankara responds well to high levels of fertilizers given through drip fertigation.

### Fertigation schedule for tender nut production in dwarf coconut

The study has been taken up since October 2019 to standardize fertigation scheduling for tender nut production in sandy loam soils of root (wilt) disease-prone areas using COD as test variety (Fig. .7). The treatments comprised of 133%, 166% and 200% of STBNR (530 N:150 P:1200 K) and three levels of irrigation (66%, 100% and 133% PE) along with 100% STBNR as soil application in four split doses + 66% PE drip irrigation. The study revealed that irrigation at 100% PE and 200% nutrients resulted in early flowering and harvest of tender nuts. It also produced on an average of 133 tender nuts palms<sup>-1</sup> during the first year of harvest, with an average production of 12 bunches. The tender nuts recorded good nut water content (336 mL nut<sup>-1</sup>) with higher potassium (K) content (2689.5 ppm) and TSS (5.2 °Brix). At the end of four years, these palms recorded 11.9 kg palm<sup>-1</sup> carbon stock and carbon sequestration of 43.8 t CO<sub>2</sub> ha<sup>-1</sup>. Leaf K and zinc (Zn) content showed a positive correlation with the number of leaves, average number of nuts bunches<sup>-1</sup>, and tender nut weight. Potassium content in the soil declined with a higher level of irrigation (166% PE). The population of nitrogen-fixing microbes also showed a reduction with increased levels of irrigation. The result indicates that the coconut dwarf variety (COD) responds to increased levels of fertilizers given through drip and produces tender nuts with higher K content and TSS.



Fig. 3.7. High yielding COD variety of coconut

### Evaluation of humic acid on soil quality enhancement

A laboratory incubation study was conducted to understand the cumulative effect of adding humic acid and farmyard manure on the soil health attributes in sandy soils. There were five levels of humic acid {H<sub>0</sub> (0), H<sub>1</sub> (1%), H<sub>2</sub> (5%), H<sub>3</sub> (10%) and H<sub>4</sub> (20%)} and three levels of farmyard manure {F<sub>0</sub> (0 kg/palm), F<sub>1</sub> (25 kg/palm) and F<sub>2</sub> (50 kg/palm)} with 15 treatment combinations and three replications. The results of the study indicated a gradual increase in organic carbon content (OC) till the 60<sup>th</sup> DOI. In F<sub>2</sub>H<sub>3</sub>, OC content increased from the initial value of 0.505% to 0.860% at the 105<sup>th</sup> DOI. The highest value at the 45<sup>th</sup> DOI was 0.974%. For all the treatments, pH declined after the 30<sup>th</sup> DOI. In F<sub>2</sub>H<sub>4</sub>, the initial pH of 6.47 declined to 6.15 at the 120<sup>th</sup> DOI. A significant effect of the combination on the availability of calcium (Ca), magnesium (Mg), and K was observed. Available K for F<sub>0</sub>H<sub>0</sub> was 103.98 ppm at the initial stage, which remained steady at 100.52 ppm until the 4<sup>th</sup> month of incubation. F<sub>2</sub> level of FYM, in combination with the H<sub>1</sub>, H<sub>2</sub>, H<sub>3</sub>, and H<sub>4</sub> level of humic acid, showed the highest availability of nutrients at all the stages of incubation. F<sub>2</sub>H<sub>4</sub> at the 75<sup>th</sup> DOI showed the highest value of exchangeable Mg (148.85 ppm). The study suggests that combining humic acid and farmyard manure can enhance nutrient availability and the

capacity of the soil to retain nutrients, which can be attributed to the synergistic effect of humic acid and farmyard manure. It was observed that, 50 kgs of FYM, in combination with 10% and 20% of humic acid, can facilitate the slow release of nutrient ions, with the maximum release occurring between the 60th and 75th DoI, thereafter showing a decline. The combined effect of these two inputs could enhance the active pool of organic fractions in the soil.

### **Kalpa Poshak and Kalpa Vardhini for the growth and productivity of juvenile and adult coconut palms**

Nutrient mixtures Kalpa Poshak and Kalpa Vardhini were assessed for their effect on the growth and productivity of juvenile and adult coconut palms cultivated in acidic soils. Kalpa Poshak, specifically formulated and field-validated for enhancing the growth of juvenile palms, was applied ten days after administering the recommended dose of major nutrients. Among the Kalpa Sankara hybrid palms treated with Kalpa Poshak, 60% exhibited flowering at the 26<sup>th</sup> month post-planting. Observations on nut setting in 25-year-old WCT palms revealed a notable increase in the number of nuts, rising from 11 (2018-2019) to 34 (2022-2023) in the tagged inflorescence. In the Onattukara sandy tract, with the Kalpa Vardhini application, the average annual nut yield reached 73 per palm, with a 36% improvement over the pre-treatment (Fig. 3.8). The average values for pH, organic carbon, available P, K, Ca, and Mg at 25 cm soil depth were 5.68, 0.39%, 31.99 ppm, 101.88 ppm, 258.4 ppm, and 95.94 ppm, respectively. At 50cm, the average values were 4.76, 0.36%, 18.48 ppm, 77.13 ppm, 217 ppm, and 80.03 ppm, respectively. The total content of P, K, Ca, Mg, Zn, and Cu in the Kalpa Vardhini treated palms were 0.12%, 1.54%, 0.39%, 0.17%, 17.87 mg kg<sup>-1</sup>, and 7.77 mg kg<sup>-1</sup>, respectively. To enhance the nutrient use efficiency of micronutrients like boron (B), chemical complex formation methods using green approaches were developed and confirmed through FTIR spectroscopy.

### **Developing fertilizer recommendations for different yield levels**

In order to predict region-specific nutrient recommendations based on soil and leaf nutrient status, five gardens each of three yield levels viz.,

low yield (LY) with <60 nuts palm<sup>-1</sup> year<sup>-1</sup>, medium yield (MY) with 60-100 nuts palm<sup>-1</sup> year<sup>-1</sup> and high yield (HY) with >100 nuts palm<sup>-1</sup> year<sup>-1</sup> were selected for soil and leaf nutrient analysis from Dindigul district of Tamil Nadu. The average K concentration from the leaf samples of the three categories of palms was 0.88%, 1.03%, and 1.05%, respectively, for the low, medium, and high yielding palms, whereas the calcium content was 0.45%, 0.47%, and 0.57%, respectively, for the three categories. The bacterial population was generally higher in LY soils, whereas the population of actinomycetes and fungi was higher in MY soils. An increasing trend of fluorescent pseudomonads' population from the LY to HY soil category was noticed. Culture-dependent analysis did not reveal the presence of P solubilizers, whereas free-living N-fixing microorganisms were predominant in coconut soils, particularly in the MY category. The major contributions of different nutrients on yield were assessed using regression and correlation approaches. Further, four statistical models, namely Decision Tree, Random Forest, Artificial Neural Network, and Support Vector Machine with Linear Kernel Model, were attempted to predict region-specific nutrient recommendations in coconut. The Random Forest model performed better, with a 77.08% accuracy rate in comparison to others. Based on the Random Forest model, yield prediction equations considering the soil and leaf nutrient status were developed. From the resultant prediction equation, it was found that soil pH, organic carbon, and soil available nutrients (K, Cu, Mn leaf nutrients (P, Ca, and Mg) have a significant effect on the yield of coconut. The results of the investigation form the baseline for evolving nutrient recommendations for the coconut palms under different yield levels.



Fig. 3.8. Kalpa Vardhini treated palms

## Physico-chemical and biological studies of YLD affected soil

The bacterial microbiome in the YLD endemic rhizosphere soil characterized through metagenomics, revealed a higher abundance of the chemolithotrophic potential microbiome in the arecanut rhizosphere, and they are the key players of microbiome-mediated nutrient transformations of N, Fe, S, and P in YLD endemic rhizosphere. Iron (Fe) and sulphur (S) are the key elements in chemolithotrophic metabolism in rhizosphere microorganisms. The structural composition of the rhizosphere microbiome of arecanut palms from YLD endemic regions has been interpreted in terms of the over represented microbiome and their functional traits, which mediate soil- plant-microbe and microbe-microbe interactions, and nutrition acquisition, and we postulated the possible association of defective Rhizosphere Nitrogen Transformation Cycle (RNTC) with disease severity in YLD affected palm. About 30 sulphur oxidizing rhizobacteria has been isolated and studied for their S oxidation potential, PGP activities and Al tolerance. The liquid formulation of the elite PGPR isolates *Burkholderia* sp., *Acinetobacter* sp., *Bacillus* sp., *Pseudomonas* sp. have been developed and studied for its shelf life (Fig. 3.9). Studies on the *in vivo* growth promoting potential of the elite plant growth promoting rhizobacteria *viz.*, *Burkholderia*, *Acinetobacter*, *Pseudomonas*, *Bacillus* in arecanut nursery is in progress.

## Integrated nutrient management in arecanut at Kahikuchi

A trial was taken up in Kahikuchi, Assam, with the organic and chemical sources of nutrients and their combinations to study the impact of the source of nutrients on areca nut. Among the different treatment combinations, application of vermicompost (2/3<sup>rd</sup>) + fertilizers (1/3<sup>rd</sup>) recorded the maximum *chali* yield of 2.37 kg per palm per year.

## Vermicomposting of combined palm and cocoa wastes and quality assessment

A technology to convert combined biomass residues of palms and cocoa, including coconut leaves, cocoa leaves, cocoa pod husks, and arecanut husks, to vermicompost has been standardized using CPCRI earthworm, *Eudrilus* sp. with primed biochar as a novel component. The final product has incremental pH, electrical conductivity, % P, % K and dehydrogenase activity compared to coconut leaf vermicompost. The analysis of the nematode population indicated the absence of plant pathogenic and predatory nematodes. A significantly higher number of free-living bacterivorous nematodes belonging to the genus *Acrobeles* sp., *Cephalobus* sp., and *Mesorhabditis* sp. were recorded. Earlier analysis had also shown the absence of plant pathogenic nematodes but recorded a higher count of fungal and bacterial feeding free-living (beneficial) nematode communities. The technology will be useful for farmers cultivating coconut+cocoa and/or

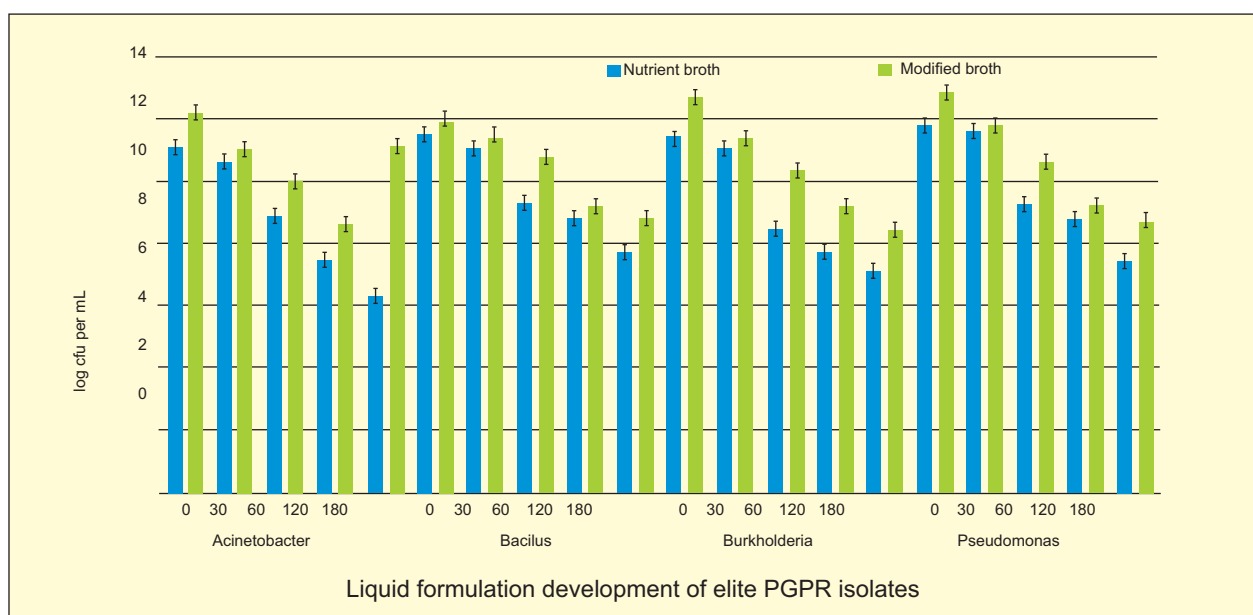


Fig. 3.9. Characteristics of PGPR formulations

arecanut+cocoa to recycle their biomass residues to good quality organic manure.

### Studies on biochar as a bioinoculant carrier

Biochar produced from cocoa pod husk wastes was tested for its ability to be an alternative carrier material of plant growth promoting rhizobacteria (PGPR). Two efficient PGPR of coconut and cocoa, *Bacillus megaterium* (Kera Probio) and *Pseudomonas putida* (Cocoa Probio), were tested for their survival when formulated with cocoa pod husk biochar as a carrier material. It was observed that the PGPRs were able to maintain population levels till one month of storage. Studies are in progress to determine their shelf life in biochar medium.

### Effect of particle size of cocoa pod husk biochar on soil nutrient availability

To study the effect of particle size of cocoa pod husk biochar on the soil nutrient availability, cocoa pod husk biochar was partitioned into three particle sizes, viz. 0.5-1.0 mm (small) (S1), 1-2 mm (medium) (S2) and 2-4.75 mm (large) (S3) and mixed with soil @ 1% (B1) and 2 % (B2), and then incubated at 50% field capacity for a period of 90 days in the dark. Nutrient analyses after the 1<sup>st</sup>, 15<sup>th</sup>, 30<sup>th</sup>, 45<sup>th</sup>, and 90<sup>th</sup> days after incubation revealed that available nutrients like P, K, Ca, Mg, S, Zn, and copper (Cu) were more in the smallest sized biochar applied soils (Fig 3.10 & 3.11). But, micronutrients like Fe were found to decrease with the incubation period, irrespective of the biochar particle size, to the point of falling below 'control' levels (Fig 3.12). Soil organic carbon was higher on the 1<sup>st</sup> day after the addition of biochar, and the smallest-sized biochar addition treatment recorded the highest values. As the study proceeded, there was a gradual decrease in organic carbon and reached a stable level, which was greater than control. But as the quantity of biochar increased, there was a corresponding increase in nutrient release, especially for P, K, Mg, S, Zn and Cu, and organic carbon. However, the quantity of biochar had no effect on the concentration of Fe. The study showed that the size and quantity of biochar have an influence on controlling the pH of the amended soil. From this study, it was concluded that amongst the particle sizes, the maximum value of available nutrients recorded in the 0.5-1.0 mm sized biochar particle treatment,

and hence, application of cocoa pod husk biochar of less than 1mm can be recommended for better results.

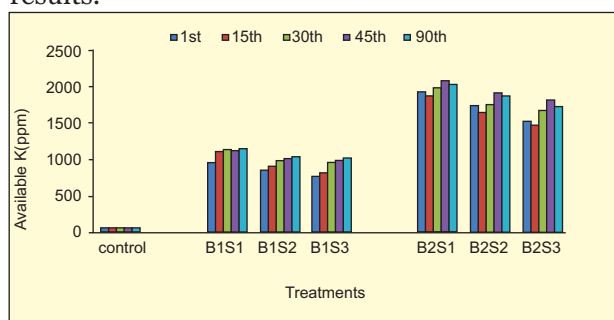


Fig 3.10. Effect of Particle Size of CPHB on Available K

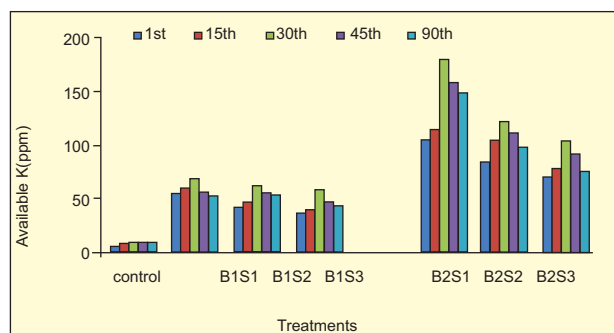


Fig 3.11. Effect of Particle Size of CPHB on Available S

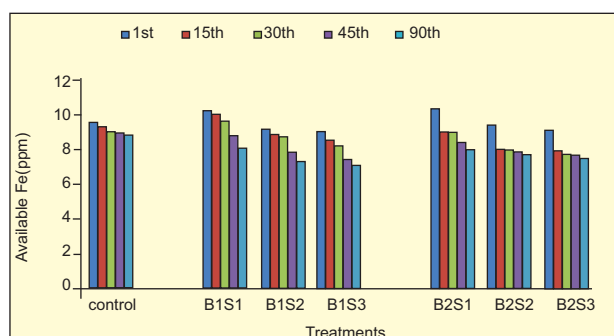


Fig 3.12. Effect of Particle Size of CPHB on Available Fe

### Cocoa pod husk biochar particle size on potassium dynamics and availability

To study the K dynamics in tropical sandy loam soil, cocoa pod husk biochar was partitioned into 0.5-1.0 mm (small), 1-2 mm (medium) and 2-4.75 mm (large) and was mixed with soil @ 1% and 2%, and incubated for a period of 90 days at room temperature keeping 50% field-capacity. Different K fractions, including water extractable, exchangeable, available, non-exchangeable, HNO<sub>3</sub> extractable, lattice (fixed), and total K, were analyzed on the 1<sup>st</sup>, 15<sup>th</sup>, 30<sup>th</sup>, 45<sup>th</sup>, and 90<sup>th</sup> day after the incubation (Fig 3.13, 3.14 & 3.15). The addition of cocoa pod husk biochar had a direct relationship with the water extractable, exchangeable, available, non-exchangeable, HNO<sub>3</sub> extractable, lattice (fixed), and total K. The particle size had a reverse relation with the potassium fractions after

each interval of incubation, except the lattice K. Lattice K had an inverse trend only on the first day after incubation, and afterward, its relation was directly proportional to the size of the biochar particle. The non-exchangeable fraction of K had a continuous improvement throughout the incubation period, irrespective of the rate of biochar for up to 90 days. Also, with the increase in biochar rate, the percentage of each fraction to the total also improved, except in the case of lattice K. The per cent of lattice K to total K was found to decrease with an increase in the addition of cocoa pod husk biochar. Hence, to get the full benefits of the addition of CPHB for K availability, it is advised to use the cocoa pod husk biochar in a powdered form of less than 1 mm particle size.

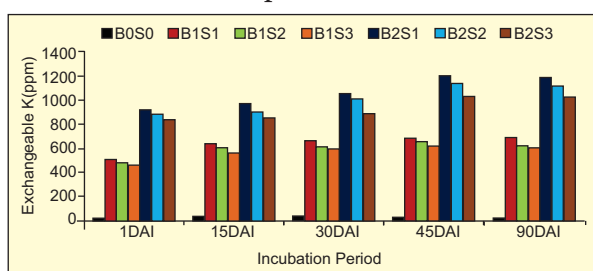


Fig. 3.13. Effect of particle size of CPHB on exchangeable K

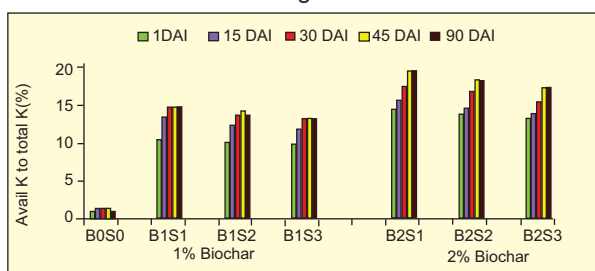


Fig. 3.14. Effect of particle size of CPHB on per cent fraction of available K to total K

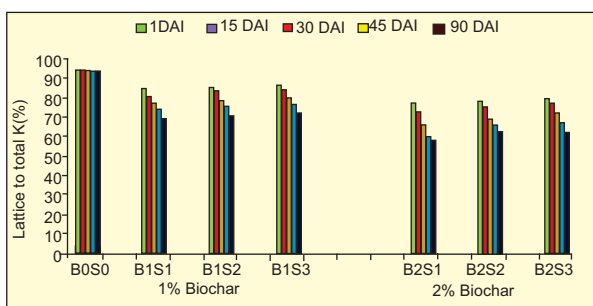


Fig 3.15. Effect of particle size of CPHB on per cent fraction of lattice K to total K

### Alternative microbiological uses for waste mature coconut water

Mature coconut water (liquid endosperm), which is wasted in huge quantities from coconut kernel processing industries, has immense potential for valorisation through the microbial process for the production of vinegar, 'nata'/bacterial cellulose

(BC), probiotic beverages, etc. A few selected indigenous isolates of acetic acid bacteria isolated from coconut sap vinegar were taken to study the effect of culture conditions like pH, carbon and nitrogen sources on BC production in mature coconut water medium. Fresh mature coconut water (MCW) has a pH range of 5.4-5.8 and contains most nutrients, thus, an ideal substrate for BC synthesis (Fig 3.16). The microplate-based rapid assay method was developed for optimization studies to select the critical cultural conditions for bacterial cellulose production in the four selected isolates. Working volume and incubation time were considerably reduced in rapid assay protocol while rapidly generating measurable BC yield. BC synthesis occurred in a wider pH range (4-11) in MCW. Maximum BC production was obtained in the pH range of 4-5 for all the isolates except BC4 in which maximum BC production was obtained in a wider pH range of 5-10.

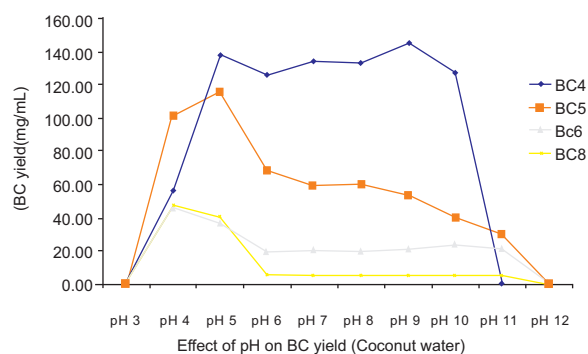


Fig. 3.16. Effect of pH on BC production in mature coconut water

BC8 showed a higher tolerance to acetic acid (up to 4%) in MCW (Fig. 3.17). Isolates utilized a wide range of carbon sources, including polysaccharides and polyols, and among the ten carbon sources tested, malic acid was found to be inhibitory for BC production.

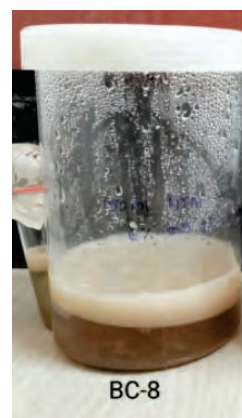


Fig. 3.17. Reaction mixture of BC8

Among the organic nitrogen (N) sources, beef extract and peptone were found to increase BC yield in most isolates. Among the inorganic N sources, urea produced a better yield, whereas the nitrate salts were inhibitory for BC production in most isolates. Optimal fermentation conditions to enhance cellulose synthesis will be explored further to develop suitable bioprocessing methods for efficient conversion of low-cost agro residues to high-value industrial products.

### **Microbiota studies in germinating coconut seedlings**

Microbial studies at the time of sowing coconut seed nut and at the time of root emergence indicated a spurt in the microbial population in the soil at the later event. The root emergence occurred from 45 to 90 DAS in different coconut varieties. Root endophyte population also changed dynamically with root elongation. Rhizosphere and endosphere microbial populations reached a stable population 5 to 6 months after the sowing of seed nut. This information will be useful for properly timing bioinoculant application for producing healthy coconut seedlings.

### **Studies on microbial community in healthy and root (wilt) affected coconut palms**

#### ***Biopriming effects of PGPR strains and consortia on coconut seedlings***

Biopriming effects of PGPR strains and consortia on biotic stress (RWD) tolerance are being studied in field-planted coconut seedlings treated/bioprimed with different PGPR strains and consortium (planted during March 2021). Bioinoculant treatments included T1-*Pseudomonas* sp.- KH3PSB2 (multiple nutrient solubilizing rhizobacteria); T2-*Bacillus* sp.- CRE-9 (Root endophyte); T3- *Margalitia* sp.- CRE-15 (Root endophyte); T4-*Azospirillum* sp.-AzoL8; T5-Consortium (of above 4 strains), T6-Control. Fertilizers and organic manures (including green leaf manure and basin management with cowpea) were applied as per the standard recommendations for the RWD tract. Soil analyses indicated sufficient levels of major and micronutrients. Deficits in secondary nutrients such as Ca and Mg were taken care of by lime and MgSO<sub>4</sub> application at recommended doses. Soil pH (~5.5) and organic carbon (0.5-0.6%) showed improvement in comparison to the initial status. Leaf nutrient status recorded during this period

indicated a sufficient level of nutrients in the crop. The microbial population of heterotrophic bacteria and filamentous fungi were significantly higher in T3. Basal girth, number of leaflets, and leaf width were higher in the treatment with bacterial consortium than in the control. Symptoms of RWD, such as leaf flaccidity and leaf rot (10% and 4%, respectively) have been noticed among the 2-year-old seedlings, and the progression of RWD will be monitored with respect to different treatments. Evaluation of PGPRs and their consortium in farmers' fields have been initiated in Pathiyoor and Devikulangara Panchayats of Alappuzha district under the ICAR-FFP programme.

#### ***Bioprospecting of actinobacteria for bioactive metabolites including plant growth/health promoting traits and hydrolytic enzymes***

Actinobacteria include a diverse group of filamentous, spore-forming bacteria that are isolated from a diverse range of ecological niches. Soil actinobacteria are reported as prolific producers of bioactive secondary metabolites, including plant growth-promoting metabolites, hydrolytic enzymes, and antimicrobial compounds suitable for agricultural applications. The present investigation has attempted to explore the diversity of the culturable actinobacterial population in soils amended with coconut palm residues and their ability to produce bioactive metabolites related to their interaction with the plant host and the microbiome in the environment. Sixteen actinobacterial isolates with distinct morphological characters were selected for screening studies. Most of the isolates (93%) secreted plant phytohormone Indole Acetic Acid (IAA). Of the 16 actinobacterial isolates, eleven solubilized tricalcium phosphate and 13 recorded zinc solubilization activity. About 37% of isolates recorded positive for all five hydrolytic enzymes: cellulase, amylase, protease, lipase, and chitinase. Two isolates showed antagonistic activity against the coconut leaf rot pathogen *Colletotrichum* sp. on a dual culture assay plate. Preliminary studies indicated that the soils amended with coconut palm agro-residues are good reservoirs of versatile and metabolically active actinobacteria, which can be a promising source for bioprospecting microbial inoculants to reclaim and sustain soil and plant health.

## Evaluation of organic cultivation of coconut-based farming system under coastal agro-ecosystem (Kasaragod)

Experiment on the cultivation of coconut under organic nutrient management practices revealed that the treatment - *In situ* organic matter (frond, leaf, inflorescence waste, husk) recycling in trenches made in the interspaces of 6 coconut palms (15 m length, 1.2 m width and 60 cm depth) + *in situ* green manuring in the basin + PGPR consortia in the basin + cowdung + 50% recommended  $K_2O$  using sulphate of potash (T4) recorded higher yield of 124.0 nuts palm<sup>-1</sup> year<sup>-1</sup> with 191.0 g of copra weight nut<sup>-1</sup> over other treatments. In cocoa intercropped with coconut, the above treatment recorded significantly higher single bean weight (1.08 g) and dry bean yield (947.3 kg ha<sup>-1</sup>). Economic analysis of the system revealed that treatment T4 recorded higher net returns of Rs. 2,67,366 ha<sup>-1</sup> (on matured nut basis) over other treatments.

## Natural Resources Management

### Impact of water/ soil moisture conservation to enhance production of coconut under rainfed and deficit irrigated farms

Kasaragod district of Kerala (No.12- western coastal plains and ghats), and Tirupur and Coimbatore district of Tamil Nadu (No.10 Southern plateau and hills) were surveyed and 26 and 14 farmers were selected respectively from Tamil Nadu and Kerala. Baseline data on techno-socio-economic variables of farmers revealed that the average size of coconut holdings was 1-2 ha. About 87% of farmers from Tamil Nadu depended on coconut farming as the main source of income for their livelihood, while only 40% of farmers from Kerala consider coconut farming as the main source of income. The awareness and knowledge level of farmers on recommended soil and water conservation techniques for coconut was not satisfactory. Analysis of the extent of soil and water conservation techniques in coconut holdings revealed that most of the growers did not adopt the recommended practices except mulching.

Mulching coconut basins with biomaterials were taken up in 596 basins in Kerala and 1755 basins in Tamil Nadu (Fig. 3.18, 3.19, 3.20, 3.21 & 3.22). On farm recycling of bio resources was taken up in Kerala (433 basins) and Tamil Nadu (3280 basins).



Fig. 3.18. Half moon bund



Fig. 3.19. Mulching the basin with coconut leaves



Fig. 3.20. Trench is filled with coconut leaves



Fig. 3.21. Trench filled with coconut husk



Fig. 3.22. General view of demonstration plot

### Natural occurrence of *Coprinopsis cinerea* on arecanut husk

Some arecanut farmers in the Dakshina Kannada district of Karnataka reported naturally growing mushroom on arecanut husk during monsoon, which was consumed by people. However, the exact species of this mushroom was unknown. To explore the bioprospecting potential of the naturally growing mushroom, a study was initiated at ICAR-CPCRI to identify the mushroom. Based on the morphological descriptions conducted with the help of Calicut University and molecular amplification of nuclear internal transcribed spacer (nrITS) and large subunit ribosomal RNA (nrLSU), the mushroom under study is identified as *Coprinopsis cinerea* CCBVH1 and a reference strain is being maintained at Plant Pathology Section, ICAR-

CPCRI, RS, Vittal (Fig. 3.23 & 3.24). The ITS gene sequence has been deposited in NCBI with the Gen Bank Accession(s): OR733603-605. The *Coprinopsis* mushroom has a very short growth cycle. The development of basidiocarp was noticed within 7-10 days and produced a typical fruiting body (mushroom). It is highly perishable and must be consumed, processed, or frozen/iced within 2–4 hr of collection.



Fig. 3.23. Areca husk mushroom  
In vivo/natural culturing and Sampling



Fig. 3.24. In vitro culturing of  
*Coprinopsis* sp.

### NCBI GenBank

Paulraj, S., Rajesh, M.K., Pandian, R.T.P., Prathibha, V.H., Sabana, A., Daliyamol, M., Subramanian, P. and Hebbar, K.B. 2023. GenBank: OR733603 (*Coprinopsis* sp. isolate AcHMF1), OR733605 (*Coprinopsis* sp. isolate AcHMF6), OR733604 (*Coprinopsis* sp. isolate AcHMF4) a small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence.

### Establishment of new laboratory at ICAR-CPCRI, Regional Station, Kayamkulam

A new laboratory for physiological studies was established at ICAR-CPCRI, RS, Kayamkulam, with the funding of RKVY with an outlay of Rs. 54.40 L (Fig. 3.25). The assets procured under the project includes; a Portable Photosynthesis System, a handheld Chlorophyll Fluorometer, a Digital Pressure Chamber, a Leaf Porometer, a Plant Canopy Analyzer, a Hot air oven and a double distillation unit. The instruments purchased under the scheme help in the analysis of the photosynthetic parameters and available natural

resources in the coconut-based cropping system of main and subsidiary crops at different growth stages and seasons. Proper diagnosis of the deficiencies and stress factors helps in adopting time-bound remedial measures, which is helpful in giving technical advice to progressive farmers and future cropping system research. The facilities can also be used in the diagnosis of the physiological parameters under various biotic and abiotic stresses resulting in yield reduction so as to devise strategies to maintain the health of palms.



Fig. 3.25. A view of the new analytical laboratory at  
Kayamkulam

### Horticulture based crop cafeteria for nutritional and livelihood security of marginal farmers

Established horticulture based crop cafeteria in Block II and Block VII of ICAR-CPCRI, Regional Station, Kayamkulam, Alappuzha District with indigenous and exotic fruit crops, medicinal plants, exotic ornamental cut foliage, spices, and medicinal plants as component crops in coconut gardens planted with Kalpa Sankara variety at 9 m x 9 m spacing. Demonstrated the concept of vertical farming in the border area for efficient usage of available land in a cropping system. Restructured the existing facility for demonstrating vermicomposting techniques with special emphasis on demonstrating nutrient recycling in a coconut-based cropping system. Installed micro irrigation systems, including sprinkler, drip, and mist irrigation systems, for demonstrating water use efficiency in a system approach. Established naturally ventilated greenhouse for the production of quality planting materials.

## Rapid and cost-effective technique for screening arecanut against *Phytophthora meadii*

*Phytophthora meadii*, which causes fruit rot disease in arecanut, is a major yield limiting factor causing yield losses of up to 90%. Bud rot and crown rot diseases occur frequently in fruit rot affected palms leading to death of the palms. Host plant resistance is considered as eco-friendly, economical and sustainable option to mitigate arecanut yield losses and also to save palms from *Phytophthora* diseases. A cost effective, simple and rapid technique called 'Detached Leaf Assay', was developed and validated using zoospore suspension of highly virulent *P. meadii* isolate (P19) to accelerate identification of disease resistant arecanut germplasm/hybrids (Fig. 4.1). Typical disease symptoms, including water-soaked lesions, were recorded three days after inoculation, lesion size increased to 1-2cm, and the leaves were covered with mycelial growth within seven days of incubation. The pathogen was re-isolated and confirmed with the original culture. Among the screened arecanut accessions, varieties and dwarf mutant only wild species showed resistance reaction against *P. meadii*.

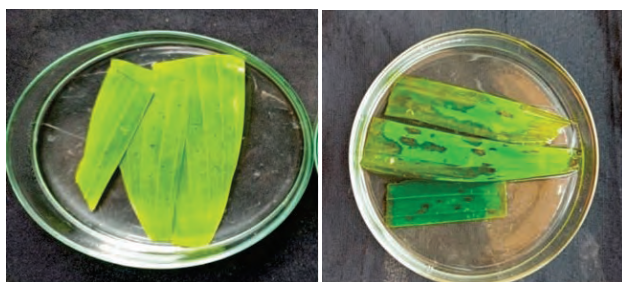


Fig. 4.1. *Phytophthora* disease test results in (a) resistant - *Areca triandra* and (b) susceptible arecanut hybrid

## Spatial studies and assessment of potential risk regions to coconut basal stem rot disease in southern tracts of India

An extensive roving survey was conducted during 2022-2023 in four different coconut growing states viz., Tamil Nadu, Kerala, Karnataka, Andhra Pradesh where Basal stem rot disease has become a major problem in the recent years. A total sample of 439 GPS locations covering four hotspot districts in each state were exclusively selected for the survey. Geospatial approaches such as inverse distance weighting (IDW) and ordinary kriging

(OK) are used to interpolate and predict the pattern in terms of the percent BSR severity cases distribution across the districts. Moran's *I* Index (Moran's *I*) statistics was used to evaluate autocorrelation in BSR spatial distribution and tested how taluks were clustered or dispersed in

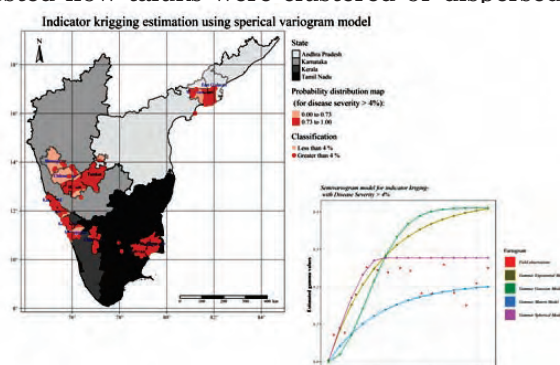


Fig. 4.2. Probability distribution map of BSR using IK.

A color-coded map (red to green) portrays the probability (high to low) of BSR risk-prone areas space. Indicator kriging (IK) was used to identify the hotspot and cold spot areas within the study area.

IDW and OK generated identical maps, whereby the BSR severity rates were higher in Thanjavur, Pudukkottai and Thiruvavur districts of Tamil Nadu and also some areas adjacent to the East Godavari and West Godavari of Andhra Pradesh. The BSR severity rates were observed in moderate status at Coimbatore of Tamil Nadu, Kozhikode of Kerala and some parts of Tumkur and Hassan of Karnataka (Fig. 4.2). Additionally, IK was used to identify hotspot disease-prone (> 4% severity) areas in Tamil Nadu, Kerala, Karnataka and Andhra Pradesh using spherical variogram model. After fitting the semi variograms with different models, the Spherical model showed the best fit with the semi variogram. The identified BSR risk areas in our study, which showed higher disease probability rates (>4%) exceeding the threshold level, need to be monitored with utmost care to contain and reduce further spread of the disease in southern tracts of India. The spatially explicit risk estimation and probability distribution maps generated in the current study will help in the development of management strategies for the disease in southern tracts of India. Additionally, this information could be useful for farmers and extension officers in setting up control measures and preventing further

invasion of BSR into neighboring districts, regions, and states in future.

- Efficacy of Oomycete-specific fungicide application against fruit rot disease in arecanut at different timings of sprays for which generalized linear mixed models (GLMMs) were employed. Study revealed better control of FRD of arecanut.
- Age distribution of plantations were analyzed to estimate the proportion of the area under each age category and yearly planting material requirement based on secondary data.
- Foliar/ soil nutrient based yield prediction model in coconut palms was developed using machine learning approach.
- Spatial analysis of pest dynamics of invasive whiteflies, Rugose spiralling whitefly (*Aleurodicus rugioperculatus*) and Bondar nesting whitefly (*Paraleyrodes bondari*) was carried out based on visual counts of different life stages at three positions on the frond (base, middle and tip). Inverse distance weighting (IDW) kriging was adopted to generate spatial surface maps and aggregation indices using the 'gstat' package in R revealed that both species preferred tip leaflets.

### Optimization of the Standard Operating Procedure (SOP) for spraying with UAV in coconut

The E610P hexacopter plant-protection UAV sprayer (M/s. EFT Electronic Technology Co., Ltd., Hefei City, China) with a payload capacity and take-off load capacity of 10 L and 25 kg respectively was used in the study. Four T-Jet nozzles are fixed under the rotor wings and maximum spray pressure generated by the system was 0.39 mpa. During the experiment, the parameters like spray droplet size, spray coverage and droplet deposition in the different layer of tree (spindle, middle and bottom) and ground were recorded. Nine different treatments with different combinations of spray height and spray time were taken up for the study. These include, T1: spray height of 1 m and spray time of 5 s; T2: spray height of 1 m and spray time of 8 s; T3: spray height of 1 m and spray time of 11 s; T4: spray height of 2 m and spray time of 5 s; T5: spray height of 2 m and spray time of 8 s; T6: spray height of 2 m and spray time of 11 s; T7: spray height of 3 m and spray time of 5 s; T8: spray height of 3 m and spray time of 8 s and T9: spray height of 3 m and spray time of 11 s.

Water-sensitive paper (WSP, Syngenta Crop Protection AG, Basel, Switzerland) of size 50 mm × 50 mm was selected as the droplet collector. An ANOVA and Duncan test was carried out to evaluate the spray droplet size, spray coverage and deposition data at different layers of coconut tree canopy with three spray height and three spray time, at a significance level ( $p > 0.05$ ). The results indicated that spray height of 2m and spray time of 8s is optimum for spraying chemicals to coconut canopy. The mean droplet size, spray coverage, and spray deposition was in acceptable range for effective control of pest and diseases and penetration efficiency was also better in this treatment (Fig.4.3).

### Field evaluation of novel fungicides against emerging arecanut leaf spot disease

Second year field evaluation trial of newer fungicides viz., Hexaconazole 5% EC (0.1%), Propiconazole 25% EC (0.1%), Tebuconazole 38.9% SC (0.1%), Carbendazim 12% + Mancozeb 63% WP (0.2%), Copper Sulphate 47.15% + Mancozeb 30% WDG (0.2%) and Mancozeb 75% WP (0.3%) were conducted against arecanut leaf spot disease in farmer plot, Swarga, Kasaragod District, Kerala during post-monsoon (September-October, 2023). Two rounds of spraying of fungicides at 25 days intervals (first round during September and Second during October) were followed. Pre- and post-treatment observations on disease intensity (%) were recorded and statistical analysis of the observations are under progress.

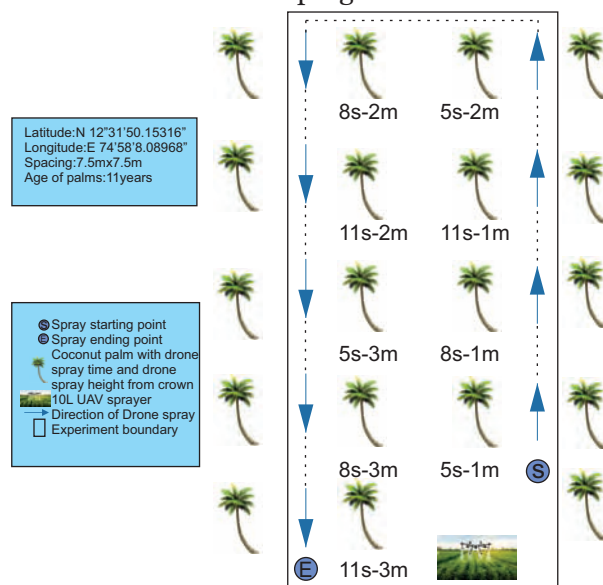


Fig. 4.3. Experimental plot layout for UAV spray test



### Field survey on emerging leaf spot disease in arecanut gardens

In order to determine the prevalence of emerging leaf spot disease in arecanut, a field survey was conducted in disease affected areas in Dakshina Kannada, Chikkamangaluru, Udupi, Kodagu, Shivamogga and Hassan districts of Karnataka as well as Kasaragod district of Kerala. Out of 75 gardens surveyed, the disease incidence was recorded as high, moderate and low in 36, 21 and 18 arecanut gardens respectively. The details are given in table 8.

Table 8. Incidence of leaf spot in different regions

| Sl. No. | Disease incidence category | Emerging leaf spot affected taluks in Karnataka and Kerala                                                                                            |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | High                       | Kalasa (Chikkamangaluru); Elaneer, Belthangady (Dakshina Kannada); Thirthahalli, Hosanagara, Agumbe (Shivamogga); Hanbal Hobli, Sakaleshpura (Hassan) |
| 2       | Moderate to High           | Koppa and N. R. Pura (Chikkamangaluru); Markanja, Sullia (Dakshina Kannada); Kasaragod and Kanhangad (Kasaragod)                                      |
| 3       | Moderate                   | Madikeri (Kodagu)                                                                                                                                     |
| 4       | Low to Moderate            | Kundapura (Udupi)                                                                                                                                     |
| 5       | Low                        | Puttur, Kadaba and Moodabidiri (Dakshina Kannada); Belur (Hassan)                                                                                     |

### Characterization of pathogen associated with ring spot disease of arecanut

Arecanut palms showing chlorotic and necrotic ring spots were observed in Karnataka and Assam during 2023 (Fig. 4.5). The symptoms were similar to that of viral ring spot disease and therefore the samples were tested for the presence of viral pathogens. Symptomatic samples showed positive amplification in PCR using coat protein primers for *Arepa virus* reported from China. The amplicons were sequenced and the sequences showed >85% similarity to coat protein gene sequences of Chinese isolates of Areca Palm Necrotic Ring Spot.

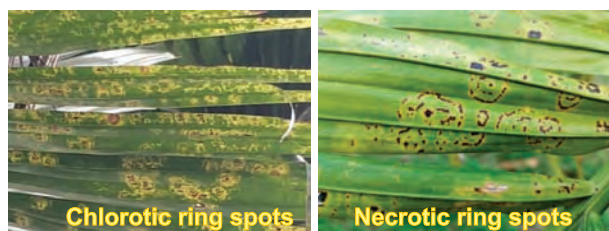


Fig. 4.5. Characteristically different appearance of disease in arecanut leaf

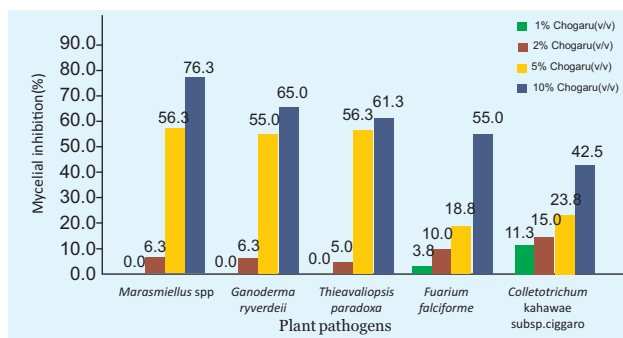
This forms the first report on the association of virus with ring spot disease of arecanut in India. Detection of such symptoms on arecanut palms from down South and North-East warrants strict surveillance and monitoring and a possible eradication considering the spread of the pathogen.

### Bio-efficacy evaluation of 'Chogaru' against major plant pathogens in arecanut

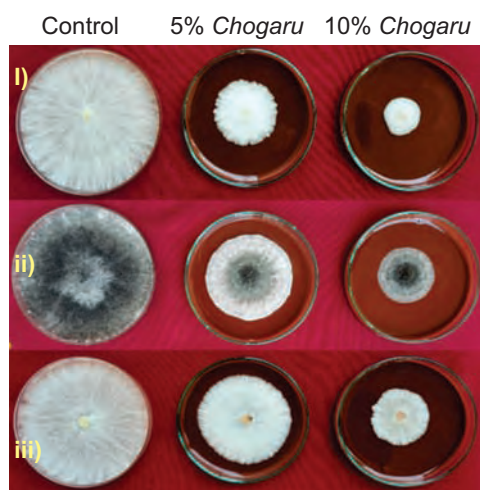
Tannin (Chogaru), a by-product of arecanut obtained during the processing of tendernut was evaluated for their antimicrobial properties against the major arecanut plant pathogens viz., *Ganoderma ryverdinii*, *Thieavaliopsis paradoxa*, *Fusarium falciforme*, *Marasmiellus* spp. and *Colletotrichum kahawae* subsp. *ciggaro* with various concentrations (v/v) viz., 1%, 2%, 5%, and 10% using poisoned food technique under in-vitro conditions. The results revealed that none of the pathogens were restricted at 1%, and 2% concentration. Whereas, 5% exhibited slight mycelial restriction and 10% showed higher reduction of mycelial growth as compared to control (Fig. 4.6).

### Evaluation of plant growth promotion activity of selected PGPRs

Four different bacterial species (*Rhizobium* sp. S8, *Bacillus* sp. SHIS10, *Priestia megaterium* SHIS11, *Bacillus licheniformis* PALEB3.) and two bacterial consortia (Consortium I: *Bacillus licheniformis* PALEB3 + *Priestia megaterium* SHIS11 + *Bacillus* sp. SHIS10 + *Rhizobium* sp. S8) and Consortium II: (*Priestia megaterium* SHIS11, *Bacillus* sp. SHIS10 and *Rhizobium* sp. S8) were evaluated for their plant growth promotion activity on cocoa by seed treatment and soil drenching. The PGPR *Bacillus* sp. strain SHIS11 recorded maximum plant height and biomass in both seed treatment and soil drenching (Fig. 4.7).



A



B

Fig. 4.6. A) Mycelial inhibition of LSD microorganism (%) over control in 'Chogaru' treatments; B) mycelial growth of i) *Marasmiellus* spp. ii) *Thieavaliopsis paradoxa* iii) *Ganoderma ryverdinii* in 5% & 10% 'Chogaru' treatments



Fig. 4.7. Growth promotion in cocoa by *Bacillus* sp. strain SHIS11 (a) Seed treatment (b) soil drench

### Fruit rot disease in Soursop caused by *Lasiodiplodia theobromae*

Fruit rot disease was observed in soursop (cv. Brazil White) grown in coconut-based cropping system in Kasaragod district of Kerala during April

to May 2022 with 20 to 25% incidence. Symptoms were visualized as dark brown color lesions either on the distal end or in middle surface of fruit (Fig. 4.8). The fungal isolates were characterized as whitish grey color colonies which turned to dark grey with wooly aerial mycelium after five days of incubation and reverse colony was dark black in color (Fig. 4.9). Based on these morphological traits, the fungus was tentatively identified as *Lasiodiplodia theobromae*. To confirm the identification, internal transcribed spacer (ITS) rDNA, elongation factor 1 $\alpha$  and  $\beta$ -tubulin gene regions were amplified and sequenced. Sequences were deposited in GenBank with accession numbers OR936549, OR957987, OR957988 for ITS, elongation factor 1 $\alpha$  and  $\beta$ -tubulin genes respectively. BLAST analysis of the sequences showed 99 to 100% identity with available sequences of *Lasiodiplodia theobromae* (MT372632, MN921238 and JX462278). The combined multi-locus sequence analyses, coupled with a critical examination of the phenotypic characters and pathogenicity, revealed the association of *Lasiodiplodia theobromae* with fruit rot of soursop. Identification was also satisfied with Koch postulates.

Dark brown lesion on fruits

Internal rotting



Fig. 4.8. Rot disease of soursop

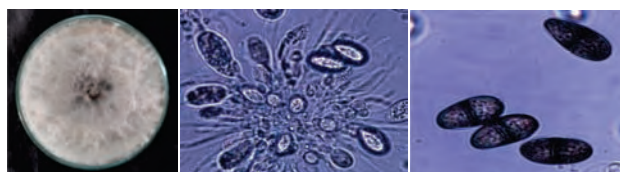


Fig. 4.9. (A) *Lasiodiplodia theobromae* colony on PDA, (B) Conidiogenous cells and (C) matured conidia

### By-product utilization from cocoa for multiplication of *Trichoderma*

Cocoa juice from sweet mucilaginous pulp of wet beans, cakes and balls made out of cocoa husk + coir compost were used for multiplication of *Trichoderma* and utilized in nursery and plots as biocontrol agents against black pod rot disease of cocoa.

## COCONUT

### Coconut rhinoceros beetle

#### Coconut rhinoceros beetle-Guam strain (CRB-G)

CRB-G haplotype which is insensitive to *Oryctes rhinoceros* nudiviruses has been causing serious damage to coconut palms in South-East Asia inflicting huge economic loss. Identification of CRB-G haplotype by A to G transition at the nucleotide position 288 on a *MseI* restriction site of *COI* gene was undertaken. No A to G transition observed in coconut rhinoceros beetle samples obtained from Kayamkulam (Kerala), Ratnagiri (Maharashtra), Pollachi (Tamil Nadu), Ambajipeta (Andhra Pradesh) indicating the absence of CRB-G haplotype in India based on the samples studied (Fig. 5.1). More than 95% infection by the grubs of coconut rhinoceros beetle to *Oryctes rhinoceros* nudivirus was observed in the laboratory inoculation studies.



Fig. 5.2. Kernel pellets

| Species/Abbrv                                                    | 288                                                                                       |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1. KY313846.1 <i>Oryctes rhinoceros</i> Guam-1                   | C T T C C T C C C T C T T G A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 2. RB17_ <i>Oryctes rhinoceros</i> _ KYLM_ OrNvescape            | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 3. RB18_ <i>Oryctes rhinoceros</i> _ KYLM_ OrNvescape            | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 4. RB19_ <i>Oryctes rhinoceros</i> _ KYLM_ OrNvescape            | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 5. RB20_ <i>Oryctes rhinoceros</i> _ KYLM_ OrNvescape            | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 6. RB21_ <i>Oryctes rhinoceros</i> _ KYLM_ OrNvescape            | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 7. RBD11_ <i>Oryctes rhinoceros</i> _ KYLM_ fieldcollected       | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 8. RBD12_ <i>Oryctes rhinoceros</i> _ KYLM_ fieldcollected       | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 9. RBD13_ <i>Oryctes rhinoceros</i> _ KYLM_ breedingsitelog1     | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 10. RBD14_ <i>Oryctes rhinoceros</i> _ KYLM_ breedingsitelog1A   | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 11. RBD15_ <i>Oryctes rhinoceros</i> _ KYLM_ breedingsitelog2    | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 12. RBD16_ <i>Oryctes rhinoceros</i> _ KYLM_ breedingsitelog2A   | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 13. RB24_ <i>Oryctes rhinoceros</i> _ RATNAGIRI_ MAHARASHTRA     | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 14. RB32_ <i>Oryctes rhinoceros</i> _ POLLACHI_ TAMILNADU        | C T T C C T C C C T C T T A A C T C T A C T T C T A G C A A G A G A C T A G T A G A A A A |
| 15. RB36_ <i>Oryctes rhinoceros</i> _ AMBAJIPETTA_ ANDHRAPRADESH | C T T C C T C C C T C T T A A C T C T T C T T C T A G C A A G A G A C T A G T A G A A A A |
| 16. RB37_ <i>Oryctes rhinoceros</i> _ AMBAJIPETTA_ ANDHRAPRADESH | C T T C C T C C C T C T T A A C T C T T C T T C T A G C A A G A G A C T A G T A G A A A A |

Fig. 5.1. Nucleotide sequence of coconut rhinoceros beetle

### Development of Kernel pellets (Neem: Karanja (1:1 w/v)) against Rhinoceros beetle and its field efficacy

An insect repellent “Kernel pellets” was prepared by mixing Neem and Pongamia seed kernel oilcake in equal proportion and tested for its efficacy on COD palms to safeguard against rhinoceros beetle under field conditions during May – Dec. 2023 (Fig. 5.2). Leaf axil filling of 150 gms of kernel pellets recorded 62% reduction in percent damaged leaves, which was effective when compared to 50 and 100 gms of kernel pellets tested. The powdered neem cake when applied @ 250 gms palm<sup>-1</sup> recorded 56.8 % reduction in damaged leaves. However, the control palms had rhinoceros beetle induced leaf damage up to 75%.

### Red palm weevil

#### Integrated Management of red palm weevil

A fatal incidence (6.1%) of red palm weevil through crown region was observed in 2019 at Bayar, Kasaragod. About 17.1% palms recovered after curative crown treatment with insecticide and recorded no fresh incidence. Though the field was absolutely free from RPW in 2020, there was an increased incidence (12.21%) in 2021 through the bole region. Curative trunk injection with imidacloprid and basin drenching with chlorpyrifos led to the recovery of 87% palms. In 2022, the crown and bole entry of RPW was reduced to 0.002% and 0.006%, respectively. During 2023 the field was absolutely free from RPW attack (Fig. 5.3). Based on the practical

learning, a refined integrated management strategy for RPW in endemic area was formulated as given below.

1. Adopt prophylactic leaf axil filling with the following repellents sequentially at monthly intervals in endemic areas
  - a. Crown pouring with imidacloprid 17.8 SL @ 1.0 mL<sup>-1</sup>
  - b. Leaf axil filling with 1:1 mixture of sand and neem cake @ 500 g / palm
  - c. Leaf axil filling with naphthalene balls @ 12 – 15 g palm<sup>-1</sup> and cover it with sand
  - d. Leaf axil filling with 1: 40 mixture of chlorantraniliprole (0.4 G) and sand @ 250g palm<sup>-1</sup>.
2. Provide prophylactic bole protection simultaneously
  - a. Drenching (by wetting the basal portion of trunk with chlorpyrifos @ 5ml L<sup>-1</sup> (One round)
  - b. Sprinkling neem cake around the collar region
3. Mulching has to be done 5-10 cm away from the collar region, so that collar region is clearly monitorable

Prophylactic leaf axil filling intervals can be widened to 45 days - 60 days, once the pest incidence reduced to less than 1%.

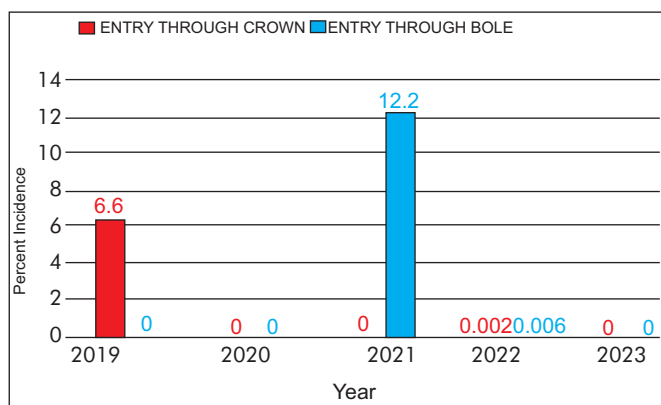


Fig. 5.3. Red palm weevil incidence in coconut farm

### Efficacy of native isolate of entomopathogenic fungus, *Beauveria bassiana* on Red palm weevil management

An entomopathogenic fungus isolated from infected red palm weevil was identified as *B. bassiana* based on morphological characters and by sequencing ITS region. Five different spore concentrations ranging from  $1 \times 10^4$  to  $1 \times 10^8$  spores mL<sup>-1</sup> were tested against grubs of RPW. Results

revealed that the susceptibility of grubs of *R. ferrugineus* to *B. bassiana* at the concentration  $1 \times 10^7$  –  $1 \times 10^8$  spores mL<sup>-1</sup> caused 100% larval mortality within 7-8 days (Fig. 5.4, 5.5, & 5.6).



Fig. 5.4. White colony, powdery texture on upper surface

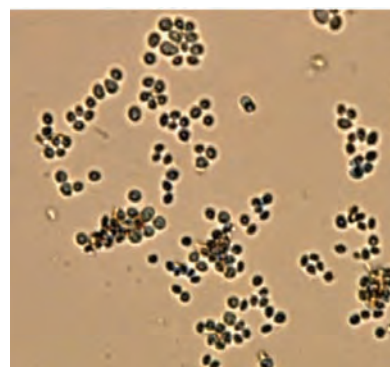


Fig. 5.5. Conidia – round, tend to oval, hyaline



Fig. 5.6. *B. bassiana* infected grubs

### *B. bassiana* sachets and its efficacy against RPW

A solid state formulation of *B. bassiana* (BB) was developed and was tested against the grubs of red palm weevil through pot studies. Coconut palm trunk of 1.5 m height was taken and larval entry holes were manually made and grubs were released for feeding. The sachet containing BB spores (5 gm or  $5 \times 10^8$  cfu gm<sup>-1</sup>) were dissolved in water (100 mL) and poured onto the trunk with grubs. Subsequent observation indicated that the released larva had in contact with the EPF suspensions and exhibited

mycosis symptom within 7- 8 days (Fig. 5.7 & 5.8). In control, larvae were active, feeding and no mortality was observed.



Fig. 5.7. BB sachets Fig. 5.8. Palm trunk showing BB infected grubs of RPW

### Electrophysiological response of red palm weevil against essential oil

Ten essential oils were tested for the antennal response of red palm weevil electrophysiologically. It indicated no significant difference in antennal response with respect to essential oils tested. However, rosemary oil, citriodora oil and bergamont oil induced more response in male RPW. Similar trend was observed in antennal response of female beetle with respect to different essential oils. However, female antenna was more responsive when compared to male antenna (Fig. 5.9(a,b), 5.10(a,b), 5.11).

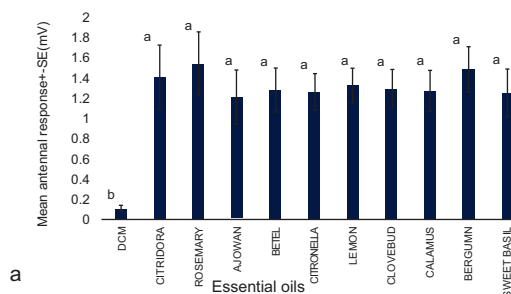


Fig. 5.9. a Mean antennal response of female red palm weevil antenna against essential oils

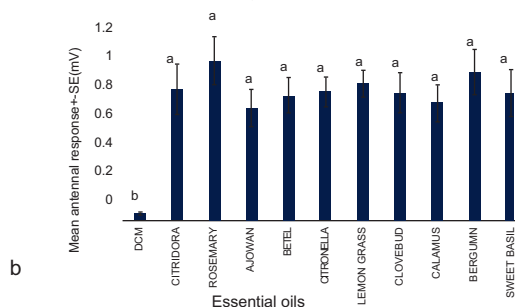


Fig. 5.9.b Mean antennal response of male red palm weevil antenna against essential oils

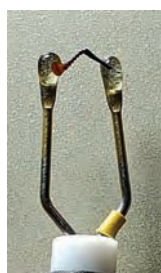


Fig. 5.11. RPW antenna mounted on electrode

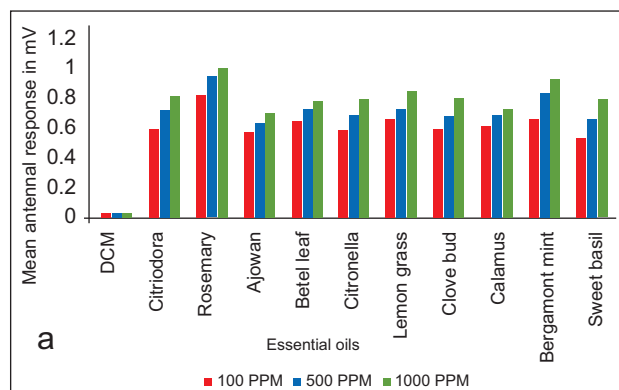


Fig. 5.10 a. Dose-Response of Male RPW antenna to essential oils

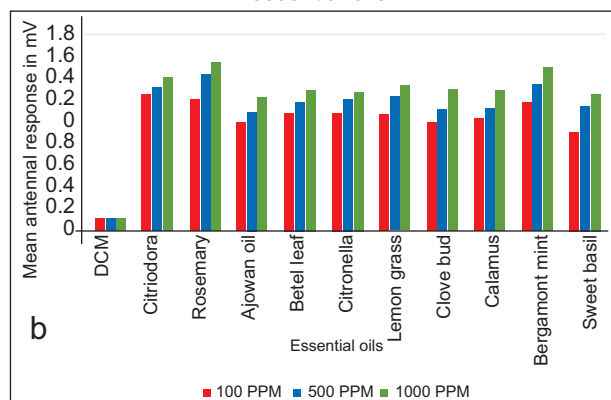


Fig. 5.10 b. Dose-Response of Female RPW antenna to essential oils

### Red Palm weevil detector

The RPW detector has undergone refinements and improvements both on the hardware and software fronts. On the hardware side, the detector probe was refined to provide better endurance in the application environment. On the software front, the detection algorithm underwent refinements to improve accuracy and reduce the processing time. At present the team is working towards a cloud-based solution to reduce the cost of the hardware and to make it more affordable for the users.

### Rugose spiralling whitefly

#### Molecular lineage of exotic whiteflies in India

Morphological and molecular identification of exotic whiteflies are very important due to the close similarity of these whiteflies as well as their co-existence in the same niche. From the phylogenetic tree (Fig. 5.12), it was revealed that the five exotic whiteflies viz., *Aleurodicus dispersus*, *Aleurodicus rugioperculatus*, *Paraleyrodes bondari*, *Paraleyrodes minei* and *Aleurotrachelus atratus* infesting coconut belonged to two taxonomically distinct subfamilies, Aleurodicinae (*Aleurodicus* spp., *Paraleyrodes* spp.) and Aleyrodinae

(*Aleurotrachelus*) and are cohabiting on the same host plant in South Indian conditions. Currently coconut palms are mostly infested by *P. bondari*, followed by *A. rugioperculatus* and to limited extent by *A. atratus*. Complete displacement of *A. dispersus* as well as *P. minei* are noticed at this point in time.

Molecular phylogenetic analysis also indicated the relative affinities of the five whitefly species and their inter-relationship. There are three groups in which the invasive whitefly species infesting coconut could converge viz., *Aleurodicus*, *Paraleyrodes* and *Aleurotrachelus* (*Aleurothrixus*) groups. The first two groups are intrinsically more closely related than that of *Aleurotrachelus* (*Aleurothrixus*) group, which is distantly placed. The native *Aleurocanthus arecae* was found associated with *Aleurotrachelus* (*Aleurothrixus*) group but forms a sub-clade with other *Aleurocanthus* spp. viz., *Aleurocanthus woglumi*, *Aleurocanthus spiniferus* and *Aleurocanthus camelliae*. Furthermore, the solanum whitefly, *Aleurothrixus trachoides* was more related to *Aleurotrachelus atratus* than *Aleurothrixus floccosus* indicating a change in the

generic name of solanum whitefly as *Aleurotrachelus trachoides*. *Bemisia tabaci* was more related with *Aleurotrachelus* (*Aleurothrixus*) group than to that of *Paraleyrodes* and *Aleurodicus* group and also formed sister clade with *Aleurolobus* sp. *Paraleyrodes pseudonaranjiae* was more associated with *Paraleyrodes bondari* than *Paraleyrodes minei* and distinct from that of *Aleurodicus* group with more affiliation toward *Paraleyrodes* group.

### Morphometry of palm whitefly, *Aleurotrachelus atratus*

Morphometric characters of different stages of palm whitefly (Fig. 5.13) were recorded through a research microscope. The eggs are black in colour and nymphs have eight white puffs on their dorsal surface. Pupae are black with puparium length of 1065.54 $\mu$  and width of 631.13 $\mu$ . They have unilobed

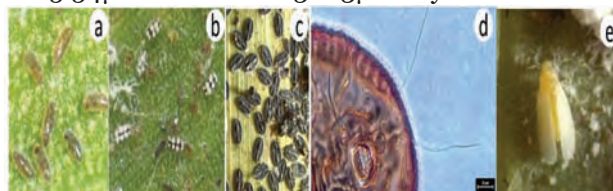


Fig. 5.13. Different stages of palm whitefly a) eggs b) nymphs c) pupae d) unilobed lingula and caudal setae e) adult

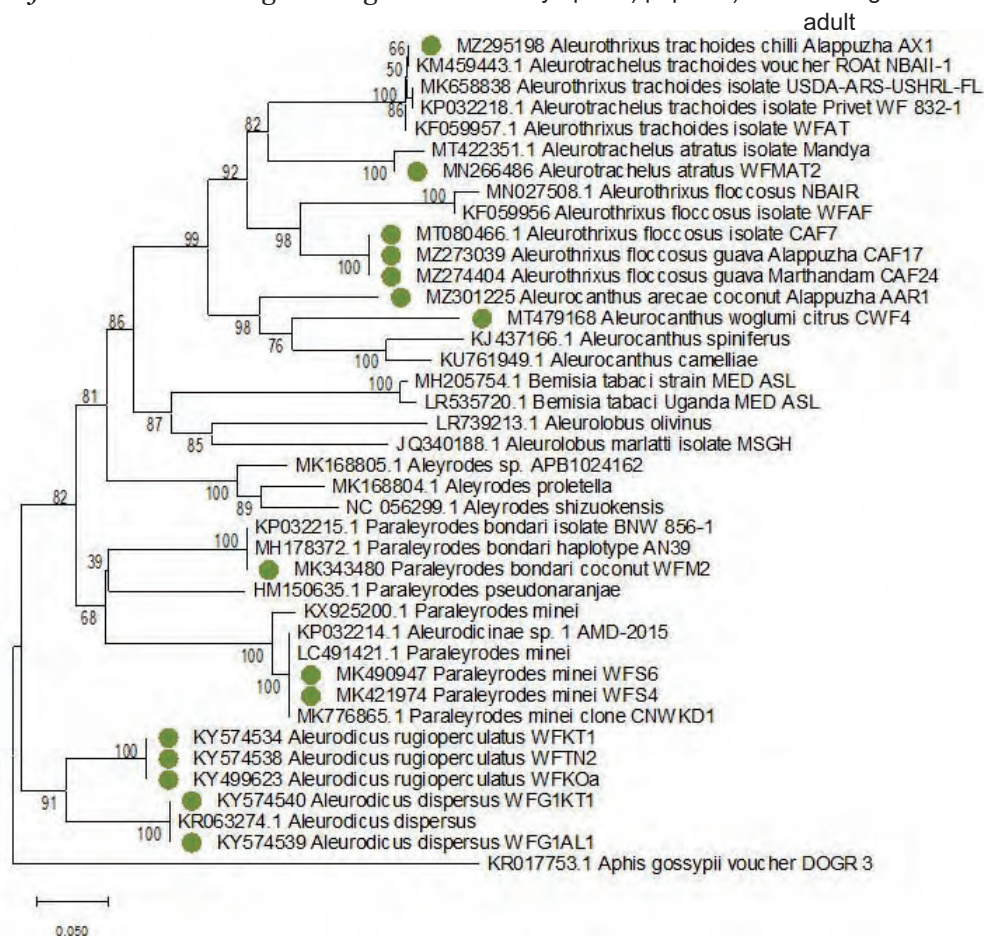


Fig. 5.12. Phylogenetic tree of exotic whiteflies

lingula, 2 caudal setae, long A8 setae and submarginal wax globules.

### Biological control of rugose spiralling whitefly in coconut

Bio-suppression of the exotic rugose spiralling whitefly (*Aleurodicus rugioperculatus*) using the entomopathogen, *Isaria fumosorosea*, neem oil and conservation biological control is presented in Table 9

Table 9. Biological suppression of rugose spiralling whitefly

| Treatments                               | RSW population (No. of colonies / leaflet)* |               |               |               |               | Parasitism (%) |
|------------------------------------------|---------------------------------------------|---------------|---------------|---------------|---------------|----------------|
|                                          | Pre-treatment                               | After 15 days | After 45 days | After 60 days | Reduction (%) |                |
| <i>Isaria fumosorosea</i> (5g per litre) | 0.57 (0.69)                                 | 0.30 (0.25)   | 0.05 (0.10)   | 1.53 (0.72)   | -168.33       | 21.29          |
| Neem oil (0.5%)                          | 0.54 (0.73)                                 | 0.50 (0.32)   | 0.50 (0.32)   | 0.57 (0.24)   | -6.67         | 27.17          |
| Conservation biological control          | 0.29 (0.39)                                 | 0.96 (0.52)   | 0.61 (0.38)   | 0.64 (0.19)   | -122.73       | 25.49          |
| CD (P=0.05)                              | ns                                          | ns            | ns            | ns            |               |                |

The rugose spiralling whitefly (RSW) population was very low (0.29 to 0.57 live colonies per leaflet) as pre-treatment count (Table 9). All the treatments were effective in pest regulation and are non-significant. Conservation biological control was equally effective as that of palms sprayed with *I. fumosorosea* as well as neem oil. There was an increase in pest population after 60 days of treatment application, and it was least observed in palms sprayed with neem oil. Parasitism by *Encarsia guadeloupae* was ranging from 21% to 27% in all the treated palms

### Development of bio - formulations of entomopathogenic fungi, *Simplicillium lanosoniveum* for coconut whitefly management

An entomopathogenic fungus (EPF), *Simplicillium lanosoniveum* (Cordycipitaceae: Hypocreales) isolated from the nymphal cadavers of rugose



Fig. 5.14. Conidiated rice formulation

spiralling whitefly (RSW), *Aleurodicus rugioperculatus* was found virulent and pathogenic to all the developmental stages of RSW. Mass production techniques of *S. lanosoniveum* on cost – effective solid and liquid substrates were standardized. Different bio – formulations of *S. lanosoniveum* viz., rice grain, talc and oil based formulations (Fig. 5.14, 5.15 & 5.16) were developed with enhanced bio - efficacy and longer shelf life. Among them, oil based formulation was found as best one with good spore viability up to six months.



Fig. 5.15. Oil based formulation of *S. lanosoniveum*

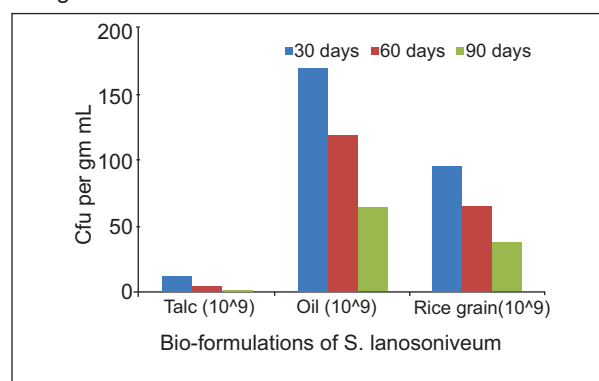


Fig. 5.16. Colony forming units (cfu) per mL or gm in bio-formulations of EPF at monthly intervals.

### Field demonstration of Oil based formulation of EPF, *Simplicillium lanosoniveum* in the whitefly infested coconut gardens at Pollachi, Coimbatore

Large scale demonstration on the efficacy of EPF, *S. lanosoniveum* along with yellow sticky traps was carried out in five different farmers' gardens covering 5 acres at Pollachi block, Coimbatore District, Tamil Nadu during 2023 (Fig. 5.17 & 5.18). Two sprays of foliar application of *S. lanosoniveum* (CPCRI SL 951 -  $1 \times 10^8$  conidia  $\text{mL}^{-1}$ ) @ 5 ml per litre of water at fortnightly intervals were carried out. Post treatment observation showed that whitefly population was reduced by 68%.



Fig. 5.17. Spraying oil based formulation of *S. lanosoniveum* in the demonstration plot



Fig. 5.18. Demonstration plot with yellow sticky traps

### Field bio-efficacy of essential oil repellants

Field efficacy of seven essential oil repellents were evaluated against RSW on orange dwarf variety of coconut at ICAR-CPCRI farm, Kasaragod. All the essential oils tested were found superior to control. Per cent reduction in RSW colony at 14 days post treatment was in the following decreasing order *viz.*, citriodora oil (58.38%) > black pepper oil (53.13%) > citronella oil (42.22%) > Jasmine oil > Neem oil > Turmeric oil. Per cent reduction in pupal population of RSW was highest on palms treated with citronella oil (75.99 %) followed by black pepper oil (74.41%) > Citriodora oil (71.84%). The effect of oil treatment on pupal population persisted up to 21 days. Mean per cent pupal parasitism by *E. guadeloupae* on RSW pupae was 68.7±12.6. Parasitism was not affected by citriodora oil, black pepper oil and citronella oil treatments in the field. However, in neem oil and nutmeg oil treated palms the per cent parasitism was reduced from 73.33% to 52.66% and 63.24% to 41.8%, respectively at three weeks post treatment. Among the essential oil repellents tested, citriodora, black pepper, and citronella oils were superior in reducing the number of viable colonies

of RSW and immature stages as manifested in the reduction of pupal populations. The essential oils tested deterred the orientation of RSW adults from the source. Eight essential attractants were tested for their attractancy by swabbing them individually (0.05 % in castor oil) on yellow corrugated sheet (50 cm x 30 cm size) (Fig. 5.19). None of the tested oils exhibited attractancy towards the source.



Fig. 5.19. Yellow corrugated sheets swabbed with 0.05 % EO + castor oil

### Coreid bug

#### Molecular characterization of coreid bug

Molecular characterization of coreid bug collected from Kayamkulam was carried out by amplification of mitochondrial cytochrome oxidase I gene (*COI* gene). There is no other nucleotide sequence information about *P. rostratus* in the NCBI database. In the phylogenetic tree, the coreid bug sequence aligned next to the *Coreidae* sp. (Fig. 5.21).

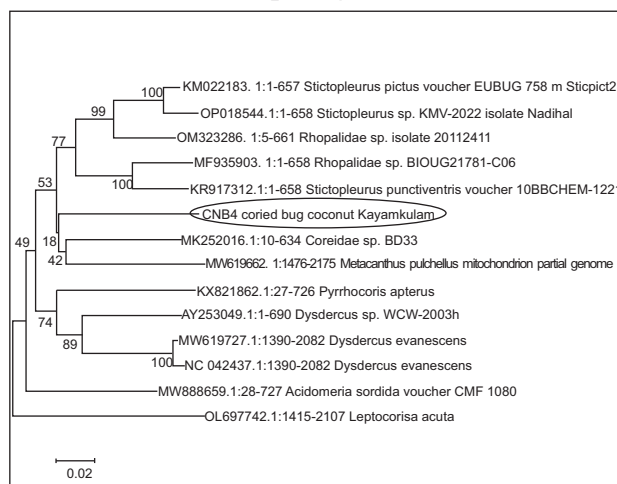


Fig. 5.20 Phylogenetic tree of coreid bug

### Host range of coreid bug, *Paradasynus rostratus* Distant

Nut crinkler or coreid bug *Paradasynus rostratus* Distant (Hemiptera: Coreidae) is an emerging as well as endemic pest of coconut in Southern Kerala. Besides coconut, a few other hosts *viz.*, guava, neem, cashew, cocoa, tamarind, para rubber tree, mango, tapioca, passion fruit, black pepper and gamboge tree were already been reported. Eleven

new hosts, belonging to ten botanical families are recorded as hosts of coreid bug during current study (Table 10). New hosts included Java apple (*Syzygium samarangense*), soursop (*Annona muricata*), custard apple (*Annona squamosa*), nutmeg (*Myristica fragrans*), mangosteen (*Garcinia mangostana*), tomato (*Lycopersicon esculentum*), West Indian cherry (*Malpighia marginata*), Rambutan (*Nephelium lappaceum*), loquat (*Eriobotrya japonica*), coral vine (*Antigonon leptopus*) and sapota (*Manilkara zapota*) resulting in wider adaptability and sustenance. Adult coreid bugs either laid eggs or fed on these hosts and laid on an average  $35.89 \pm 16.68$  eggs per cluster. A total of 23 hosts were recorded for egg laying or feeding including the new hosts in the pest endemic area (Fig. 5.21).

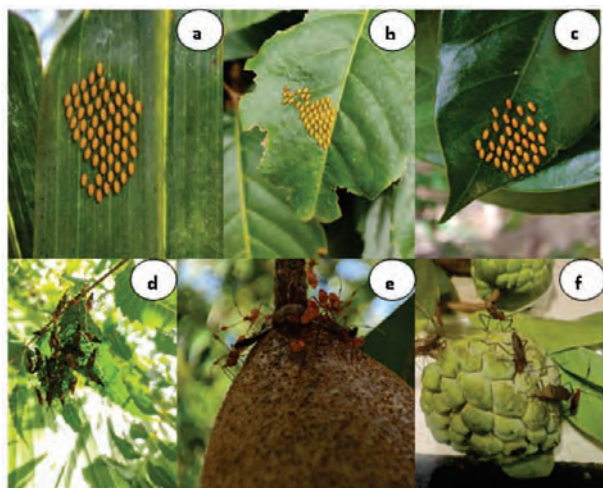


Fig. 5.21. Eggs of coreid bug on a) coconut, b) rambutan, c) nutmeg; Nymphs of coreid bug on d) neem, e) sapota, f) custard apple

### Laboratory rearing of coreid bug

Laboratory rearing protocol for coreid bug, *P. rostratus* was standardized (Fig. 5.22). First two nymphal instars were fed with neem twigs. From 3<sup>rd</sup> instar onwards the nymphs and adults were maintained on ripe tomato and tomato was replaced at 5 days interval. The life cycle comprised of 8-10 days of egg period, 4-5, 5-7, 4-6, 4-7 and 8-10 days respectively of five nymphal stages as well as 45-110 days of adult period. The fecundity of female insects ranged from 10-71 with an average of 40 eggs per cluster.



Fig. 5.22. Laboratory rearing of coreid bug on neem and tomato

### Mango mealy bug

Documented the new distribution record of mango mealy bug, *Rastrococcus iceryoides* (Green) (Hemiptera: Homoptera: pseudococcidae) on orange dwarf variety of coconut in Kasaragod (Fig. 5.23). The mealy bugs were found infesting tender nuts and rachillae of inflorescence. It was found associated with red ant *Oecophylla smaragdina* and predatory butterfly, *Spalgisepius* larva.

Table 10. Newly recorded hosts of coreid bug

| Sl. No. | Host               | Scientific name                                         | Family        | Pest stage observed |
|---------|--------------------|---------------------------------------------------------|---------------|---------------------|
| 1       | Java apple         | <i>Syzygium samarangense</i> (Blume) Merr. & L.M. Perry | Myrtaceae     | Egg, nymph, adult   |
| 2       | Soursop            | <i>Annona muricata</i> Lin.                             | Annonaceae    | Egg, adult          |
| 3       | Nutmeg             | <i>Myristica fragrans</i> (Houtt.)                      | Myristicaceae | Egg                 |
| 4       | Mangosteen         | <i>Garcinia mangostana</i> L.                           | Clusiaceae    | Egg                 |
| 5       | Rambutan           | <i>Nephelium lappaceum</i> L.                           | Sapindaceae   | Egg                 |
| 6       | Loquat             | <i>Eriobotrya japonica</i> Lindl.                       | Rosaceae      | Egg                 |
| 7       | Coral vine         | <i>Antigonon leptopus</i> Hook. & Arn.                  | Polygonaceae  | Egg                 |
| 8       | Tomato             | <i>Solanum lycopersicum</i> L.                          | Solanaceae    | Nymph, adult        |
| 9       | West Indian Cherry | <i>Malpighia emarginata</i> DC.                         | Malpighiaceae | Nymph, adult        |
| 10      | Custard apple      | <i>Annona squamosa</i> L.                               | Annonaceae    | Nymph, adult        |
| 11      | Sapota             | <i>Manilkara zapota</i> (L.) P. Royen                   | Sapotaceae    | Nymph               |



Fig. 5.23. Mango mealy bug on coconut

## ARECANUT

### Inflorescence caterpillar

#### Evaluation of *Metarhizium anisopliae* against inflorescence caterpillar

Pathogenicity evaluation of *Metarhizium anisopliae* TMBMA1 (Fig. 5.24) was carried out against arecanut inflorescence caterpillar (*Tirathaba mundella*) under *in-vitro* conditions. Mature larvae weighing 0.044 to 0.064 g were used in the study. The spore concentration of the EPF stock solution was  $2.64 \times 10^7$  cfu mL<sup>-1</sup>. The larvae were immersed in three different dilutions *viz.*,  $10^{-1}$ ,  $10^{-2}$  and  $10^{-3}$  and mortality was recorded at daily interval. Significant difference was observed in the larval mortality of treated and control batches. It is observed that there was considerable increase in the mortality rate with progression in time. At the highest tested dilution ( $10^{-1}$ ), the mortality was recorded to be 60% at 3 days after treatment (DAT). Maximum mortality of 100% was found at 10 DAT. The maximum mortality recorded in lowest tested dilution ( $10^{-3}$ ) was 30% at 10 DAT. There was considerable reduction in the feeding as well as mobility when the larvae were exposed to the fungal treatment. The larval cadaver was initially covered by white mycelial growth and later on greenish powdery growth was evident as fungus sporulates (Fig. 5.25). The results demonstrated that *M. anisopliae* TMBMA1 is highly virulent to *T. mundella* under laboratory conditions.



Fig. 5.24. *Metarhizium anisopliae* TMBMA1

A) culture; B) Microscopic image of spores

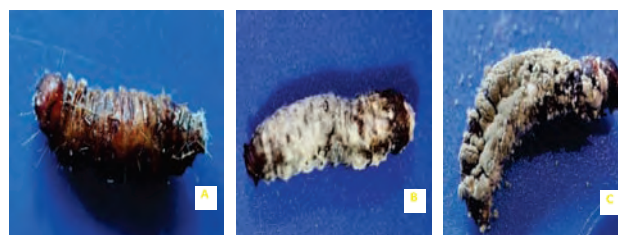


Fig. 5.25. Progression in the infection cycle in *T. mundella* larva on treatment with *Metarhizium anisopliae* TMBMA1A) Died larva at 3 DAT; B) Mycelial growth observed on larva at 5 DAT; C) Sporulation on larva at 7 DAT

## Red mite

### Simple and cost effective methodology for mass multiplication of coccinellid predator, *Stethorus keralicus*

Red mite (*Raoilla indica*) is an important non-insect pest on arecanut and causes significant damage during summer season. The coccinellid predator, *S. keralicus* (Coleoptera: Coccinellidae) is a key predator on *R. indica*. Both grubs and adults feed on all stages of mites to keep the mite population at lower level. However, intervention of pesticides can cause significant damage to their establishment under field level. In this regard, we have developed a simple and cost effective protocol for mass multiplication of *S. keralicus* on red mite under laboratory conditions (Fig. 5.26 & 5.27)



Fig.5.26. Mass production of *S. keralicus* under laboratory conditions

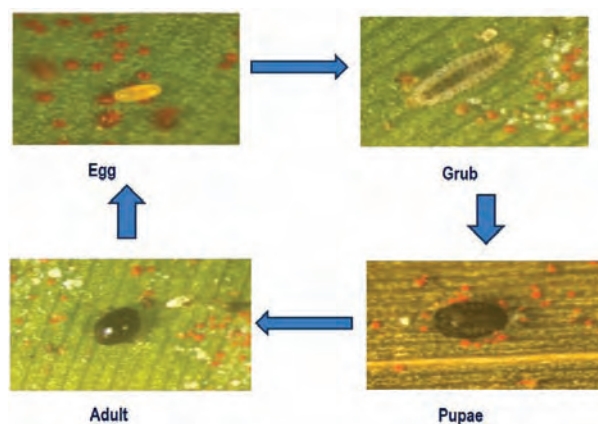


Fig. 5.27. Different stages of *S. keralicus*

## Spindle bug

### Laboratory efficacy of EPN against spindle bug in arecanut

The bio-efficacy evaluation of EPN, *S. carpocapsae* (CPCRI-SC1), *S. abbasi* (CPCRI-SA1) and *H. indica* (CPCRI-HI1) against arecanut spindle bugs in laboratory conditions evinced 80% - 100% mortality of spindle bugs within three days of treatment application (Fig. 5.28).



Fig. 5.28. Vials with different EPN treatments

## COCOA

### Mealy bugs

#### Infestation of mealy bug complex in cocoa

Mealy bug infestation was observed in both cocoa nursery and garden with an incidence level of 6.8 %. Two different species was associated with cocoa viz., *Planococcus* sp., *Ferrisia* sp. (Fig. 5.29). Mealy bugs were found colonizing on different plant parts and inflict considerable damage. Mealy bug colonizes the tender parts of the cocoa plant such as growing tips, terminal bud, flower cushions, young cherelles and mature pods. Damage to tender apical shoots leads to deformation into slender brush like process. Colonization of flower cushion results in cushion abortion, withering and drying up (Fig. 5.30). Feeding on pods results in irregular cracks and pitting on pods.



Fig. 5.29. Various species of mealy bugs infesting cocoa  
A) *Planococcus* ; B) *Ferrisia*



Fig. 5.30. Damage symptoms on cocoa due to mealy bug infestation A) Drying of apical shoot; B) Damage to flower cushion; C) Pod damage

## Entomopathogenic nematodes

### EPN infected *Galleria* cadaver based sachet formulation for pest management in coconut/arecanut system supports sustainable organic farming

Developed native strain of entomopathogenic nematode, *Steinernema carpocapsae* (CPCRI-SC1) and *Heterorhabditis indica* (CPCRI-HI1) infected *Galleria mellonella* as cadaver-based sachet formulation with enhanced bio-efficacy and shelf life of up to twenty five days at ambient temperature of  $28 \pm 2^\circ \text{C}$  (Fig. 5.31). This cadaver-based formulation made delivery of EPN easier by farmers on to the regular leaf whorl/ foliar spot application (@ 1 sachet 5 litre<sup>-1</sup> water) as prophylactic and curative measures for the bio-suppression of leaf eating caterpillars in various vegetables, maize, fruits and spices intercropped in coconut/arecanut based cropping system as well as homestead coconut farming (Fig. 5.32). The developed formulation is cost effective and support sustainable organic farming. The infective juveniles released from the cadaver-based sachet formulation recorded 100% survival and high vigor as compared to aqueous suspension due to more available space as well as better oxygen supply to the emerging IJs.



Fig 5.31. Emerged IJs from EPN infected *G. mellonella* cadaver in covered sachet, ready to spray



Fig 5.32. Demonstration of use of cadaver based EPN formulation in a farmer's coconut garden for pest management

### Field efficacy of EPN against pumpkin beetle in watermelon

Release of infective juveniles of Kalpa EPN, *S. carpocapsae* (CPCRI-SC1) @ 1500 mL solution (1 crore IJs) per square meter through drip irrigation at 15 days interval significantly reduced pumpkin beetle damage on watermelon in coconut eco-system at Kanhangad village Kerala (Fig. 5.33).



5.33. Plant protection with EPN

### Bio-suppression of *M. enterolobii* in Guava seedlings under coconut plantation

Planting of nematode free seedlings, uprooting and burning of severely infested seedlings, exposure of tree basin to sunlight together with basin free from weeds, application of neem cake enriched

*Trichoderma harzianum* @ 1kg tree<sup>-1</sup> and planting of marigold seedlings of 4-6 numbers around the tree were observed to reduce the incidence of yellowing in seedlings due to suppression of root-knot nematode (*Meloidogyne enterolobii*) in guava.

### Field demonstration of EPN in coconut for sustainable management of root grubs

Organized field demonstration of aqua suspension of EPN, *S. carpocapsae* (CPCRI-SC1) and *H. indica* (CPCRI-HI1) as soil drenching @ 20 lakh IJs palm<sup>-1</sup> twice in a year followed by application of neem cake @ 5kg palm<sup>-1</sup>, reduced coconut root grub population to the tune of 55% and 51% respectively in farmers garden at Kundapur and Puttur Taluk of Karnataka.

### Trichoderma enriched neem cake for management of root-knot nematodes in black pepper intercropped in coconut/arecanut system

Successfully demonstrated the application of *Trichoderma harzianum* (CPTD-28) enriched neem cake @ 1.0 kg vine<sup>-1</sup> for three years (2021 - 2023) in the farmer garden at Karkala Taluk of Karnataka. The application during pre and post monsoon season significantly suppresses the buildup of root-knot nematode (*Meloidogyne incognita*) population to the tune of 62% in treated pepper vines trailed on coconut and arecanut palms (Fig. 5.34). The regular application of *T. harzianum* enriched neem cake showed consistent improvement in overall health of vines and increased the yield up to 13.7% by reducing the damage caused by *M. incognita*. Further, the buildup of *Trichoderma* population in treated soil had an inhibitory effect on incidence of nematode and fungal disease complex in coconut/arecanut based system.



Fig. 5.34 a) *Trichoderma* enriched neem cake effective for biosuppression of *M. incognita* population in coconut interspaces, b) black pepper roots showing knots/galls, c) second stage male juvenile of *M. incognita*

## Entomopathogenic nematodes against termites

The efficacy of entomopathogenic nematodes (EPN) against termites in coconut system was evaluated under laboratory condition. Among six EPN isolates tested, *Steinernema carpocapsae*, *Heterorhabditis indica* CPCRIH0703 and *Metarhabditi srainai* CPCRIIM2021 are promising by inducing 100% mortality within 48 hours of treatment application. The emergence of infective juveniles was also observed from 80, 70 and 60% cadavers killed by these three isolates, respectively (Table 11).

Table 11. Efficacy of EPN against termites

| EPN Species/Isolate                | % Mortality in 48 h | % Emergence |
|------------------------------------|---------------------|-------------|
| <i>Steinernema carpocapsae</i>     | 100                 | 80          |
| <i>S. abbasi</i>                   | 80                  | 20          |
| <i>S. hermaphroditum</i>           | 50                  | 0           |
| <i>Steinernema</i> sp. S0804       | 60                  | 0           |
| <i>Heterorhabditis indica</i> H12H | 100                 | 60          |
| <i>Metarhabdus rainai</i>          | 100                 | 70          |
| Control                            | 10                  | 0           |

## Field evaluation of capsule formulation of EPN against red palm weevil

Evaluated the efficacy of capsule formulation of entomopathogenic nematode (EPN) as a prophylactic application against red palm weevil (RPW) in the farmers' fields in Pathiyoor Panchayath. A total of 40 palms were treated with EPN capsule @ 4 capsules per palm. No red palm weevil infestation was observed in any of the palms treated up to six months of treatment application.

## Recovery of new EPN isolates

Recovered four new EPN isolates from Pathiyoor, Mavelikkara and Palode area of Alappuzha and Thiruvananthapuram districts, Kerala. Based on the colour of the wax moth cadaver, two of the isolates were found as *Steinernema* spp. and the other two were identified as *Heterorhabditis* spp. Both the steinernematids belong to *feltiae* group which is identified based on the size of infective juveniles.

## Association of plant parasitic nematodes in coconut seedlings

Root samples were collected from 20 coconut seedlings in the nursery of the institute research farm. The presence of burrowing nematode, *Radopholus similis* was observed from all the seedlings evaluated with a range of 9 to 70 nematodes seedlings<sup>-1</sup> from the complete root system of these seedlings (Fig. 5.35). Soil sample was also collected from the root zone of a patch of stunted seedlings and higher population of *R. similis* (12 nematodes g<sup>-1</sup> soil) and *Pratylenchus* sp. (18 nematodes g<sup>-1</sup> soil) was observed. In addition, presence of *Xiphinema* sp., *Tylenchorhynchus* sp. and *Hoplolaimus* sp. was also observed from the root zone. The feeding impact of these nematodes on the seedling growth is being ascertained.



Fig. 5.35. Damage by burrowing nematode in tender creamy white roots of seedlings

## Plant parasitic nematode

### Bio-efficacy of tender nut husk extract against *Menoidogyne incognita* juveniles

Three different concentrations (25, 50, 75 and 100%) of the supernatant filtrate collected from the tender nut extract incubated at 28±1°C, inhibited the egg hatching and induced up to 72 - 100% juvenile mortality in *M. incognita* (Fig. 5.36).

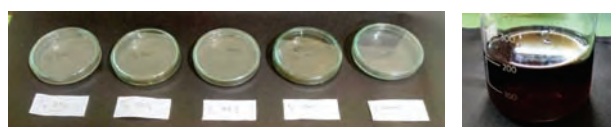


Fig. 5.36. Bio-efficacy test of supernatant solution of tender nut husk extract

## Current and future climate suitability for coconut (*Cocos nucifera* L.) in India

Coconut, an important plantation crop grown in coastal belts and plains, is highly vulnerable to climate change. Urgent measures are imperative for its appropriate protection and management. Climate change and climate variability are projected to alter the geographic suitability of lands for crop cultivation. Early awareness regarding the future climatic scenarios in the current cultivation zones is essential for the adaptation and sustainable cultivation of this perennial tree crop in vulnerable regions. MaxEnt model was used to evaluate the bioclimatic variables determining the suitable habitat for coconut cultivation and to forecast regions with climates potentially conducive for coconut cultivation under future conditions. Coconut's vulnerability to climate change in India, based on climate projections for the 2050s and the 2070s under two Representative Concentration Pathways (RCPs): 4.5 and 8.5, was analyzed. Based on the current cultivation regions and climate change predictions from seven ensembles of Global Circulation Models, changes in relative climatic suitability for coconut cultivation using the MaxEnt model were predicted (Fig. 6.1).

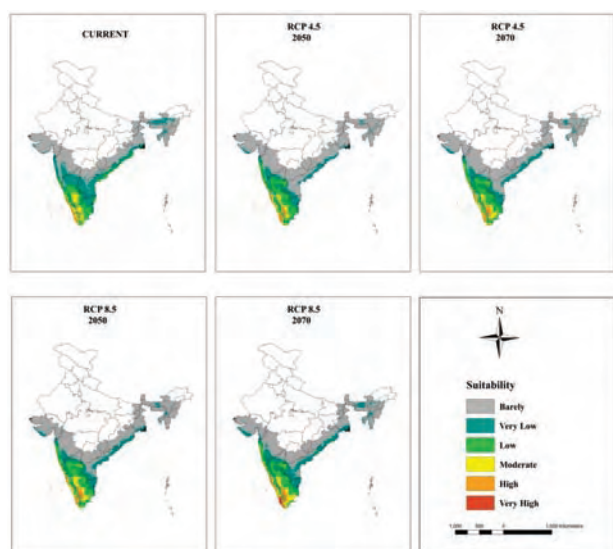


Fig. 6.1. Climatically suitable regions for coconut production in India under the current and future climates of the 2050s and 2070s for RCP 4.5 and RCP 8.5 as modeled by MaxEnt

Bioclimatic variables Bio 4 (temperature seasonality, 34.4%) and Bio 7 (temperature annual range, 28.7%) together contribute 63.1%, which along with Bio 15 (precipitation seasonality, 8.6%) determined 71.7% of the climate suitability for coconuts in India. The model projected that some current coconut cultivation producing areas will become unsuitable (plains of South interior Karnataka and Tamil Nadu) requiring crop change, while other areas will require adaptations in genotypic or agronomic management (east coast and the south interior plains), and yet in others, the climatic suitability for growing coconut will increase (west coast). These findings suggest the need for adaptation of strategies to ensure the sustainable cultivation of coconut, particularly in current cultivated areas.

## Effect of defatted coconut flour (DCF) on the glycemic index of rice and the underlying starch protein interactions

In recent times, there has been a growing research focus on the interaction between starch and protein in foods. This heightened interest stems from the significant role of protein in lowering the glycemic index of starchy food products. Hence, the effect of defatted coconut flour (DCF) and rice flour combination in reducing the glycemic index of the latter was evaluated. Various composition of DCF (*viz.*, 2.5%, 5%, 10%, 15%) with rice flour was prepared and the *in vitro* starch digestibility was studied. The protein content of the samples increased with the addition of DCF and the composition (DCF: 15% and 85% Rice flour) showed a protein content of  $495.5\mu\text{g mL}^{-1}$ , while, the control rice flour showed only  $150.5\mu\text{g mL}^{-1}$ . The moisture content of all the prepared samples ranged between 9.01% to 9.23%. Additionally, the analysis of bulk and the tapped densities revealed inverse relationship with an increase in DCF content. The composition (DCF: 15% and 85% rice flour) showed a low glucose release of  $0.488\mu\text{g mL}^{-1}$  when compared to the other compositions (DCF: 2.5% and 97.5% Rice flour, DCF: 5% and 95% Rice flour, DCF: 10% and 90% Rice flour). The Fourier transform infrared spectroscopy (FTIR) analysis of the DCF: rice flour compositions were performed at a base range of  $4000\text{-}400\text{cm}^{-1}$  with 63 scans (Fig.

6.2). The results suggest that 15% DCF with 85% rice flour composition showed starch : protein interaction causing the conversion of the amorphous structure of starch granules in the rice flour to a crystalline structure which is positively correlated with the decrease in the glycemic index of the starchv foods.

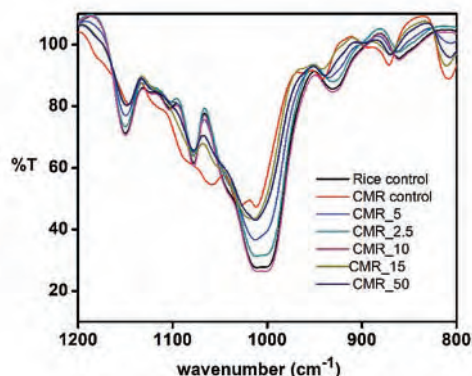


Fig. 6.2. FTIR spectra of defatted coconut flour and rice paste samples (rice, DCF/CMR, DCF: 2.5% and 97.5% Rice flour; DCF: 5% and 95% Rice flour, DCF: 10% and 90% Rice flour) having different starch:protein ratio

### Gum arabic as a novel edible coating on the biochemical attributes of coconut (*Cocos nucifera* L.) haustorium during storage

The coconut haustorium is a good source of nutritionally important bioactive compounds, encompassing soluble sugars, proteins, fats, and essential mineral nutrients such as potassium, phosphorus along with various phytochemicals boasting antioxidant potential. These bioactives make it an excellent candidate for the production of infant food supplement and the development of extruded food products. However, the inherent perishable nature of the haustorium, attributed to its bioconstituents, poses a considerable challenge. To address the challenge of extending the shelf life of coconut haustorium under refrigerated conditions, various preservation methods were explored, including coating with coconut *neera* (inflorescence sap) honey, acacia gum, vinegar and freeze drying following flash freezing in liquid nitrogen treatment. Among these methods, coating of haustorium with 15% gum arabic emerged as a promising technique. This method effectively preserved biochemical attributes such as comprehensive acid value (CAV), phenolic acids and resultant antioxidant properties, for upto 20 days under refrigerated conditions. CAV of the treated haustorium was  $6.1 \pm 0.5$  mg NaOH g<sup>-1</sup> as against the high values of  $18.6 \pm 1.57$  mg NaOH g<sup>-1</sup> in

untreated control. Phytochemical profiling of nutritionally important phenolics acids ( $\mu\text{g g}^{-1}$  dry weight) in haustorium after 20 days of treatment revealed the presence of gallic acid (183.43), chlorogenic acid (8.13), caffeic acid (65.23), vanillic acid (72.69), p-coumaric acid (traces), sinapic acid (1.59), 2,4 DHBA (2.49) and ferulic acid (1.19). In summary, the gum arabic-based edible coating demonstrated its efficacy in delaying quality changes and extending the shelf life of the haustorium for up to 20 days under refrigerated conditions (4-10 °C) (Fig. 6.3). These findings present significant opportunities for the effective post-harvest management of this nutritionally important component of the coconut.

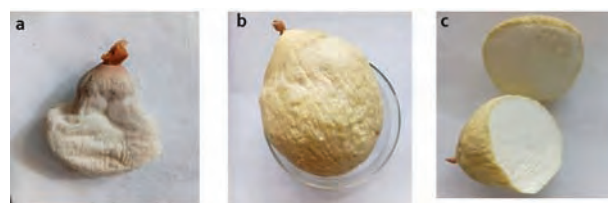


Fig. 6.3. Effect of edible coating (15% Acacia Gum) on haustorium after 15 days of refrigerated storage: b&c compared with untreated control (a)

### Neera syrup infused coconut flakes

A protocol and methodology for the production of ready-to-serve coconut flakes have been developed. This approach incorporates different drying systems and an osmotic process utilizing coconut inflorescence sap concentrate, specifically neera syrup at 50°Brix (Fig. 6.4). The infusion of neera syrup not only elevates nutritional profile of the product but also significantly contributes to its sensory attributes by enhancing taste, flavor, and color. Coconut flakes were subjected to drying using various methods at three distinct temperatures (55, 65, and 75 °C) to determine the optimal drying method and temperature. Upon evaluation of the bioactive profile, coconut flakes dried at 65° C using infrared-assisted hot air drier (IRHAD) exhibited an outstanding nutritional and physical profile. Noteworthy attributes include a Hausner ratio of 1.04 and Carrs index of 4.22, rehydration ratio of 2.37%, hygroscopicity of 1.68%, bulk density of 0.552 g mL<sup>-1</sup>, tapped density of 0.610 g mL<sup>-1</sup>, protein content of 3.92%, carbohydrate content of 33.86%, fat content of 34.29%, ash content 1.92%, total phenolic content of 1.054 mg GA g<sup>-1</sup>, DPPH activity of 88.81% and FRAP of 0.00009 mg TE g<sup>-1</sup>.



F.g. 6.4. Kalparasa infused coconut flakes

### Cocoa fermenter

Traditionally, cocoa beans undergo box fermentation and sun-drying methods, leading to challenges such as inconsistent quality, contamination risks, weather dependency, and limited control over processing conditions. The CPCRI fermenter unit could process high quality beans in terms of bulk density, percentage shell mass and nib recovery rate. Biochemical analyses, encompassing fat, protein, carbohydrate, ash, moisture, total phenol, and antioxidant activity, were conducted on both fermented and dried cocoa bean samples. Cocoa beans from the CPCRI fermenter unit displayed elevated values for carbohydrate (28.94%), ash (3.64%), moisture (7.17%), total phenol content (0.047 mg GAE g<sup>-1</sup>) and FRAP (0.0323 mg TE g<sup>-1</sup>). In sensory evaluation, chocolates crafted from cocoa beans processed in the CPCRI fermenter unit garnered higher preference compared to those from the traditional box fermentation method.

### Infrared assisted hot air drying of coconut chips

A comparative analysis was conducted to assess the emerging drying techniques like infrared drying (IRD), infrared-assisted hot air drying (IRAHAD), and the conventional hot-air drying method (HAD) focusing on drying rate and the biochemical profile of coconut (Fig. 6.5). The mass reduction during drying was monitored at 15-minute intervals, and the drying curve was constructed to analyze drying kinetics and its modeling for each technique. Notably, IRAHAD at 65°C exhibited a superior drying rate and a shorter drying time compared to infrared drying and hot air drying. Drying kinetics and modeling revealed that infrared-assisted hot air drying at 65°C was the most effective drying technique for producing high-quality coconut chips (Fig. 6.6). For 0.5 mm slice thickness, the diffusion approximation was identified as the best modeling for IRD, IRAHAD, and HAD. The initial moisture content of the slices was 39.68% (w.b.), reducing to a final moisture content of 2.88% (w.b.). The biochemical properties of the resultant chips included significant values for moisture content (2.4±0.33%), ash (1.29±0.27%), total carbohydrate (0.43±0.02 g glucose g<sup>-1</sup>), protein (4.87±0.10%), fat (36.20±0.90%), total phenol (0.039±0.009g gallic acid eq g<sup>-1</sup>), DPPH (34.51±3.76%) and FRAP (0.0097±0.0006 trolox eq g<sup>-1</sup>).

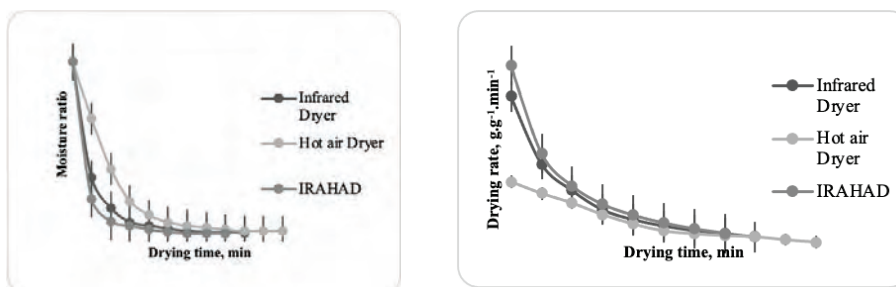


Fig. 6.5. Effect of drying time on moisture ratio and drying rate of coconut chips for IRD, HAD, IRAHAD at 65°C



Fig. 6.6. Coconut chips prepared with (a) hot air drying, (b) infrared drying and (c) infrared assisted hot air drying at 65°C

## Gelatinized Starch Doped Microcellulose Paper from Tender Coconut Husk

Cellulose-starch based composite papers have emerged as promising packaging materials due to their biodegradable and renewable properties. In this study, a biodegradable composite paper was developed from tender coconut husk with starch playing a key role as an additional component. Potato and corn starch were incorporated into the matrix at varying ratios of 5%, 10%, and 15%, and the resulting biocomposite papers were assessed for mechanical, physical, and structural properties, along with their biodegradability. The tensile strength and elongation percentage of the developed biocomposite paper ranged from  $12.45 \pm 1.69$  MPa to  $9.52 \pm 4.30$  MPa and  $9.76 \pm 0.99\%$  to  $15.52 \pm 3.27\%$ , respectively. The results revealed a decreasing trend in tensile strength with an increase in starch concentration, attributed to a reduction in hydrogen bonding density. Moisture content analysis indicated no significant difference between the types of starch used. The composition of the paper was thoroughly analyzed using Fourier transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), and scanning electron microscopy (SEM) as presented in Fig.6.7. These analyses demonstrated robust interactions between the hydroxyl groups of starch and the cellulose of the coconut husk matrix (Fig. 6.8). Furthermore, the material exhibited a notable degradation rate of approximately 70% within a 20-day period, underscoring its suitability for the production of biodegradable materials. The study concludes that tender coconut husk holds the potential as a promising raw material for the production of paper intended for eco-friendly packaging applications.

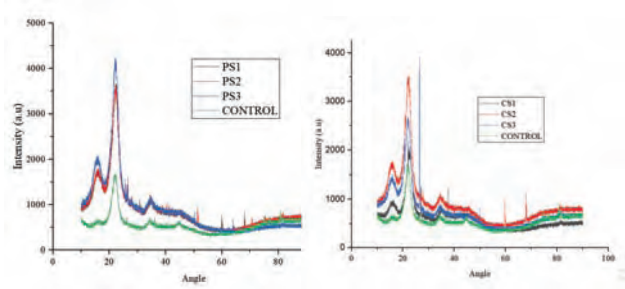


Fig. 6.7. XRD Patterns of samples coconut husk-based paper with (a) corn starch and (b) potato starch as additives

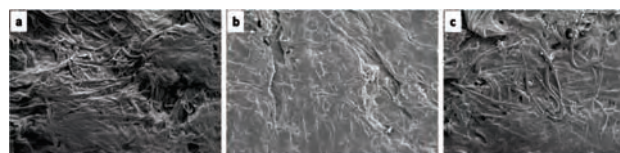


Fig. 6.8. SEM images of the developed packaging material, (a)control, (b) corn starch as additive (15% incorporation), and (c) potato starch as additive (15% incorporation)

## Drying kinetics of coconut for copra production under different drying system

Copra, the dried form of coconut, plays a vital role as a primary ingredient in oil production. The initial moisture content of coconut kernels stands at 50-55% (wb), which is significantly reduced to 6-8% (wb) during the drying process. Traditional methods, including sun, solar, and hot air drying are commonly employed to produce copra. However, improper or insufficient drying may lead to moisture retention, resulting in fungal contamination and the production of discolored copra. Hence, to address this, the study examined the independent and combined effects of infrared and hot air drying techniques on enhancing the copra quality. Three drying methods were employed: infrared drying (IFD), hot air drying (HAD), and infrared-assisted hot air drying (IRAHAD). Coconut pieces were exposed to varying drying temperatures ( $50^{\circ}\text{C}$ ,  $60^{\circ}\text{C}$ , and  $70^{\circ}\text{C}$ ) with a consistent air speed of  $2\text{ m s}^{-1}$ . Optimal results were obtained in IRAHAD method at a temperature of  $60^{\circ}\text{C}$ , preserving a crucial fat content of 68.4%, essential for enhanced oil extraction from copra. Drying rates in IRAHAD were twice as high as those in IFD and HAD. Based on the highest  $R^2$  value and the lowest error values, the page model emerged as the most suitable technique for characterizing the drying kinetics of copra. At a drying temperature of  $60^{\circ}\text{C}$ , the logarithmic model and the diffusion approximation model were deemed the best fit for HAD and IRAHAD, respectively.

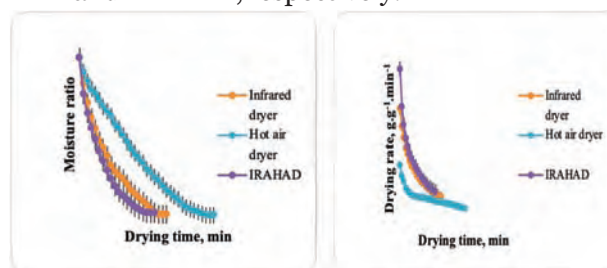


Fig. 6.9. Effect of drying time on the moisture ratio and drying rate of copra for IFD, HAD, IRAHAD

## Technology transfer and co-learning action research approaches

The stakeholder base of the coconut sector encompasses farmers, women SHGs, youth, extension officials, NGOS, entrepreneurs, startups, national and overseas organizations, policy and decision making bodies, and the public. Technology transfer approaches and strategies customized to location specific needs, evolving community and area wide approaches in pest and disease management and landscape based participatory social models for doubling farmers' income, promotion of Farmer Producer Organizations (FPO), and reaching out to farm households through *Mera Gaon Mera Gaurav* constitute the interventions and programmes. The activities in this regard and the research components are summarized and presented as below:

### Capacity building programmes

During the year, various capacity building programmes were conducted by the institute for farmers, extension personnel and other stakeholders as detailed below:

### Farmers' training programmes

A total of 61 training programmes on various thematic areas pertaining to coconut, arecanut and cocoa were organized at headquarters (Fig. 7.1), regional stations and research centres of the institute to benefit farming community (Table 12).

Table 12. Farmers' training programmes conducted during 2023

| Place        | No. of training programmes | No. of farmers benefitted |
|--------------|----------------------------|---------------------------|
| Kasaragod    | 27                         | 671                       |
| Kayamkulam   | 10                         | 808                       |
| Vittal       | 03                         | 800                       |
| Kahikuchi    | 01                         | 17                        |
| Mohitnagar   | 10                         | 357                       |
| Kidu         | 10                         | 892                       |
| <b>Total</b> | <b>61</b>                  | <b>3545</b>               |



Fig. 7.1. Shri Inbasekhar IAS., Collector Kasaragod district inaugurating the training programme for farmers at Kasaragod

### Training programmes for extension personnel

Four training programmes were organized for 103 extension personnel (Table 13) at Kasaragod and Kayamkulam.

Table 13. Training programmes for extension personnel during 2023

| Place        | Topic of training                                                            | No of extension personnel |
|--------------|------------------------------------------------------------------------------|---------------------------|
| Kasaragod    | Scientific cultivation of coconut, arecanut and cocoa                        | 20                        |
|              | Crop protection in coconut and arecanut – pre- monsoon prophylactic measures | 30                        |
| Kayamkulam   | Biodiversity, biocontrol and biosecurity issues in coconut                   | 27                        |
|              | Invasive coconut whitefly identification and their management                | 26                        |
| <b>Total</b> |                                                                              | <b>103</b>                |



Fig. 7.2. Exposure visit of farmers in Kayamkulam-  
Interactive discussion with Scientists

### One day exposure visit programmes

One day exposure visit programmes were conducted during the year at headquarters, regional stations (Fig. 7.2) and research centers of the institute benefitting a total of 7468 stakeholders. The visitors were enlightened upon various extension activities and research achievements of the institute.

### Entrepreneurship Development Programmes

Seven Entrepreneurship Development Programmes were organized by this institute benefitting 539 entrepreneurs.

### Other technology transfer initiatives

#### Kasaragod

A training and Scientist-Farmer interface programme on 'Soil and water conservation and technologies for planting material production in coconut and horticulture crops' was organized in collaboration with Centre for Research and Development, Nileswar under the NABARD assisted project at Thayannur on 11<sup>th</sup> May 2023 to benefit 40 selected tribal farmers including 22 women (Fig. 7.3).



Fig. 7.3. Scientists-farmer interface at Thayannur,  
Kasaragod

#### Kahikuchi

Team of scientists from Kasaragod and Kahikuchi surveyed the disease infected arecanut gardens at Laitumsaw village of East Khasi Hills, Meghalaya during 24-25<sup>th</sup> April 2023 and conducted capacity development programme on 'Scientific cultivation of arecanut and integrated disease management' for arecanut farmers and extension personnel (Fig. 7.4).



Fig. 7.4. Arecanut survey team with the farmers at  
Meghalaya

#### Kidu

Special training programme of six days duration for 25 rural unemployed scheduled caste and scheduled tribe youths on 'Handicrafts preparation by using coconut shell and coconut palm climbing' was conducted at research centre Kidu in two batches.

Table 14. One day exposure visit programmes

| Sl. No.      | Place      | Category of stakeholders |                     |             |             | Total       |
|--------------|------------|--------------------------|---------------------|-------------|-------------|-------------|
|              |            | Farmers                  | Extension personnel | Students    | Others      |             |
| 1            | Kasaragod  | 656                      | 138                 | 1327        | 2059        | 4180        |
| 2            | Kayamkulam | 268                      | 62                  | 1357        | 31          | 1718        |
| 3            | Vittal     | 220                      | 155                 | 455         | 0           | 830         |
| 4            | Kahikuchi  | 23                       | 15                  | 45          | 12          | 95          |
| 5            | Mohitnagar | 55                       | 143                 | 146         | 38          | 382         |
| 6            | Kidu       | 178                      | 0                   | 85          | 0           | 263         |
| <b>Total</b> |            | <b>1180</b>              | <b>358</b>          | <b>2960</b> | <b>2140</b> | <b>7468</b> |

A PPV & FRA sponsored one day training programme on “Awareness on protection of plant varieties and farmers rights” was conducted at ICAR-CPCRI, Research Centre, Kidu on 23.11.2022. A total of 127 participants (77 - male; 50 - female) participated in the programme (Fig. 7.5).



Fig. 7.5. Training programme on protection of plant varieties and farmer's rights at RC Kidu

### Kayamkulam

Two days training on 'Hybridization techniques with emphasis on modified ground pollination technique in coconut' was conducted at Elavally (Thrissur) for officials of Dept. of Agriculture, Govt. of Kerala during 29-30th November 2022. A total of 130 participants attended the training.



Fig. 7.6. Farmers from Thiruvananthapuram District visiting fertigation plot of RS, Kayamkulam on 22.12.2023

A training programme was organized for 100 participants in the categories of agricultural officials, progressive farmers, and final year students from BSc Agriculture and VHSE during 15-26 May 2023 (Fig. 7.7). The training session covered the production and protection aspects of plantation crops, spices, medicinal plants, ornamentals, and indigenous and exotic fruit crops.



Fig. 7.7. Participants of the training programme with experts

## Frontline demonstrations

### Kasaragod

Seventeen front line demonstrations on soil and water conservation technologies including trenches filled with coconut husk and mulching of coconut palm basins were conducted in 11 farmers' garden in Tamil Nadu and six farmers' gardens in Kerala state (Fig.7.8)



Fig. 7.8. Soil and water conservation pit

### Kidu

Front line demonstrations on 'Performance of cocoa clones' were conducted in two farmers' gardens covering a total of 0.5 ha.

### Kayamkulam

Front line demonstrations on integrated root (wilt) disease management' and 'intercropping of millets, ginger and groundnut in coconut garden were conducted in farmers' field covering 437 farmers and 120.9 ha.

### Vittal

Front line demonstrations on 'Integrated management of inflorescence dieback disease in arecanut' were conducted in three farmers' gardens covering a total area of 3 ha.

## Kisan Mela

A Kisan Mela was organized at CPCRI RS Kayamkulam on 13.05.2023 in connection with the Platinum Jubilee celebrations. Dr. A.K. Singh, DDG (Hort.) inaugurated the kisanmela. About

1350 farmers attended the kisanmela. Various events were held as part of kisanmela including a seminar on 'Potential of plantation sector in farmers' development', planting of Kalpavajra block to mark 75 years of regional station, release of coconut variety 'KalpaVajra', exhibition of technologies, and exchange of MoUs on transfer of institute technologies and release of publications.

### Exhibitions

The institute participated in 31 exhibitions conducted at various localities of the country.

### Farm Advisory Services through ATIC

Various services including exposure visits, replying queries on farming related topics and sales of planting material and other inputs were offered to farmers and other stakeholders through Agricultural Technology Information Centre (ATIC) at this institute. Seedlings and other farm inputs worth Rs 90.81 lakhs were sold.

### Activities under Schedule Caste Sub Plan (SCSP) and Tribal Sub Plan (TSP)

#### Kasaragod

A seven days Off-campus training programme on 'Preparation of bee colonies for honey production-cum field demonstration of Honey frame (super) construction' was organize for Scheduled Caste farmers from 01.02.2023 to 07.02.2023 at Pillankatta village in Badiadaka Panchyath of Kasaragod (Fig. 7.9).



Fig. 7.9. View of SC farmers learning about honey frame (super) construction

Twenty one un-employed youths belonging to SC and ST communities were trained in computer applications and scientific cultivation of coconut, nursery raising techniques, identification of pest and disease incidence and its effective management in the field (Fig. 7.10 & 7.11). Imparted hands on training on production of biological agents such as *Trichoderma* coir pith cake, *Bracon* and *Goniozous*, EPN and *Metarhizium* talc formulation.



Fig. 7.10. Launch of skill development training to unemployed SC and ST youths



Fig. 7.11. SCSP trainees with Shushri Shobha Karandlaje, Union Minister of State for Agriculture and Farmer Welfare

Three day Bee Keeping training for scheduled caste farmers was organized from 29.12.2023 at ICAR-CPCRI, Kasaragod, training covered theory class followed by field level demonstration of honey bee rearing techniques in detail. Bee keeping kits were distributed to the 22 farmers, who participated in the training (Fig. 7.12).



Fig. 7.12. Beekeeping kit distribution

ICAR-CPCRI in collaboration with ICAR–KVK, Dharwad and ICAR- KVK, Hulikoti, Karnataka organized training programme on Coconut production technologies, along with seedling distribution during 10- 11 October 2023 at KVK, Dharwad and KVK, Hulkoti, Gadag for the SC farmers under SCSP. Capacity building programme on coconut cultivation was organized at ICAR-KVK, Gadag on 10<sup>th</sup> October 2023, wherein about 200 farmers participated

Around 442 coconut hybrid seedlings of Kera Sankara (T x D) were distributed to 85 SC farmers of Madhur, Badiadka, Puthige, Mogral Puttur Panchayaths as part of the training programme on the occasion of Kisan Diwas.

### Vittal

A training programme on 'Basic and advanced nursery techniques in arecanut' under Scheduled Caste Sub Plan, was organized at ICAR-CPCRI, Regional Station, Vittal, for the period from 01.02.2023 to 31.03.2023.

### Kayamkulam

Various activities under the Tribal Sub Plan (TSP) scheme were conducted in three tribal settlements namely Pottamavu, Thiruvananthapuram district as well as Kocharippa and Idappana, Kollam District. About 250 families are living altogether in these three settlements.

### Mohitnagar

Two Seedling distributions cum skill development training programmes were organized in Lataguri, Jalpaiguri district by the Research Centre, Mohitnagar during December, 2023 under SCSP and TSP Plans. A total of 100 SC/ ST farmers were benefitted by the programme.

Altogether 22 Capacity development interventions/training programmes were organized under SCSP, that include skill development programmes, hands on trainings, entrepreneurship development programmes, and refresher courses benefitting 704 participants across the country (Fig. 7.13).

Similarly under Tribal Sub plan scheme 23 training and skill development programmes were conducted for a total of 582 participants. Furthermore, seven demonstration programmes were conducted for the benefit of 90 respondents



Fig. 7.13. Glimpses of SCSP/ TSP events organized by this institute during 2023

Table 15. Inputs/equipments/machinery distributed under SCSP

| Sl. No. | Item                                                     | Quantity distributed                                                                 | Amount spent | No of beneficiaries |
|---------|----------------------------------------------------------|--------------------------------------------------------------------------------------|--------------|---------------------|
| 1       | Coconut seedlings                                        | 8088                                                                                 | 10.375 lakh  | 424                 |
| 2       | Bee keeping kit                                          | Bee hives with bees -350<br>stands -350, Smokers -35, caps -35 & honey extractor -35 | 8.00 lakh    | 35                  |
| 3       | Arecanut seedlings                                       | 825                                                                                  | 1.28295 lakh | 75                  |
| 4       | Black pepper rooted cuttings                             | 85                                                                                   | 0.255 lakh   | 157                 |
| 5       | Bay leaf rooted layers                                   | 276                                                                                  | 0.828 lakh   | 157                 |
| 6       | Acid lime rooted cuttings                                | 274                                                                                  | 0.822 lakh   | 157                 |
| 7       | Vegetable kits and grow bags                             | 19 kits                                                                              | 0.092 lakh   | 19                  |
| 8       | Mushroom spawn                                           | 19 packets                                                                           | 0.009 lakh   | 19                  |
| 9       | Farm implements (Spade, rose can, hoe, climbing machine) | 19 each                                                                              | 0.553 lakh   | 19                  |
| 10      | Fertilizer                                               | Urea-60 kg,<br>Rajphos-90 kg,<br>Potash-120 kg<br>Kalpa poshak-800g                  | 0.0605 lakh  | 19                  |
| 11      | Organic manures                                          | Neem cake- 380 kg<br>Kera Probio-19 kg<br>Vermicompost-380 kg                        | 0.133 lakh   | 19                  |
| 12      | Utilities                                                | (pen, bag, uniform, certificate folder, etc.)                                        | 0.159 lakh   | 19                  |

Table 16. Inputs/equipments/machinery distributed under TSP

| Sl.No. | Item                                                                                                                 | Quantity distributed | No. of beneficiaries |
|--------|----------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|
| 1.     | Distribution of planting material/ seeds of horticultural crops                                                      |                      |                      |
|        | No. of plants/cuttings                                                                                               | 5760                 | 705                  |
|        | Seed (kg)                                                                                                            | 300                  | 250                  |
| 2.     | Distributions of agricultural inputs like pesticides, fungicides, fertilizers, traps and small farm implements, etc. |                      |                      |
|        | Implements (Nos.)                                                                                                    | 32 climbing devices  | 32                   |

### Profile analysis of farmer producer companies

As part of the ICAR-NAHEP-KAU-CAAST sub-project on 'Profile Analysis of Farmer Producer Companies in India', a study was conducted on the Farmer Producer Companies in Assam representing the north-east region of the country. In Assam, a large proportion of FPCs (94.44%) provide training and extension services to their members. Analysis of gender representation in governance of FPCs indicated that only 23% of the Board of Directors of FPCs were women. The analysis indicated that financial networks of FPCs must integrate risk mitigation and resilience strategies.

As part of the project, workshops were conducted in collaboration with Assam Agricultural

University, KrishiVigyan Kendra- Darrang district and NABARD on "Farmer Producer Companies in Assam – Status and strategies' at Mangaldoi, Darrang district, on 09.03.2023 and at Kahikuchi on 21.11.2023 (Fig. 7.14 & 7.15).



Fig. 7.14. Interaction with FPC stakeholders of Assam



Fig. 7.15. Meetings held with FPOs in Assam

### Community Farm School (CFS) approach for coconut root (wilt) disease affected areas

Community Farm School approach aims for a paradigm shift in conserving common resources like healthy high yielding mother palms of coconut, integration of traditional and experiential knowledge/skill, and significant improvement in knowledge and adoption. Five community nurseries produced 10,000 poly-bag bio-primed coconut seedlings in five panchayats covering 5,500-hectare area involving 1100 coconut farmers.

### ICAR Farmer FIRST Programme (FFP)

Participatory technology integration to empower and ensure livelihood security of farmers in Alappuzha district under the Farmer FIRST Project is being carried out in Pathiyoor panchayat of Alappuzha district since 2016-17, and has extended to Devikulangara panchayat from 2021 onwards. Presently, the FFP social, individual-based farm models are out scaled to eight panchayats of Muthukulam block panchayat. The intervention modules during 2023 were Crop (13 technologies, 930 farmers), Horticulture (6 technologies, 348 farmers), Livestock and Poultry (5 technologies, 215 farmers, 96 animals & 3400 birds), Natural Resource Management (3 technologies, 90 farm families), Value addition and EDP (11 technologies, 53 units) and Integrated Farming Systems (IFS) (1 technology, 141 farm families). Thus, a total of 39 technologies from various sources in six modules benefiting 1647 farmers were implemented covering 1600-hectare area.

### Extension activities/Capacity building programs in FFP

A total of 223 extension activities including on-farm training programs, method and result

demonstrations, diagnostic field visits, interventions appraisal visits, and focus group discussion (FGD) on outcome assessment and convergence with women SHG farmers were organized, benefiting 1222 male farmers and 1672 female farmers (Fig. 7.16).

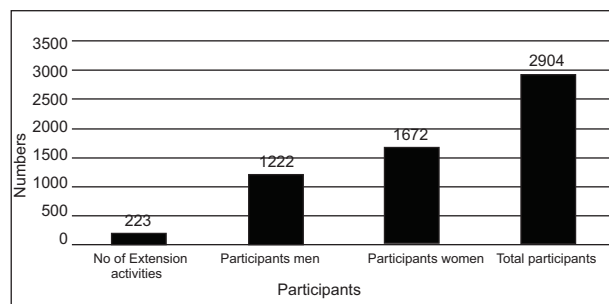


Fig. 7.16. Farmers participation in FFP extension activities

A total of 29 Capacity building programs on various thematic areas were conducted benefiting 1059 participants in FFP locations (319 men and 740 women).

### Integrated coconut community programme for doubling productivity

This innovative extension intervention intends to augment coconut technology dissemination, adoption, value addition, and project investment from agencies/institutions, for doubling the income of homestead coconut farmers of root (wilt) affected areas. The interventions in three FFP locations were: Participatory coconut seedling production by documenting and marking root (wilt) disease-free mother palms, Authentication of seed nuts collected in farmers' fields through QR code affixing, Community Nursery establishment (Fig. 7.17), Production and distribution of 2000 Bio primed polybag seedlings in each FFP locations, and Demonstration of scientific management practices for seedling management.



Fig. 7.17. Participatory community coconut nursery

## Impact of FFP technological interventions (Pre and Post)

In order to understand the significant difference between pre and post interventions, a t-test was conducted and the results are furnished in the Table 17.

Table 17. Out scaling of Technologies and Adoption

|                            |                                                               |          |
|----------------------------|---------------------------------------------------------------|----------|
| 1                          | Cropping intensity in coconut gardens (%)                     | 0.59**   |
| 2                          | Organic Manures (t/ha)                                        | 0.22**   |
| 3                          | Chemical Fertilizers (kg/ha)                                  | 29.4***  |
| 4                          | Use of Pesticide (chemical & organic/bio agents) (kg or l/ha) | 31.00**  |
| 5                          | Barren land changed to crop cultivation (%)                   | 33.0***  |
| 6                          | Subsistence crops to commercial crops                         | 29.60*** |
| 7                          | Labour use (man-days/year)                                    | 28.20**  |
| ** $p=0.05$ , *** $p=0.01$ |                                                               |          |

The FFP panchayat (Pathiyoor) was declared as fallow-free by Haritha Mission, Govt of Kerala, and increased coconut-based cropping intensity (intercrops) cultivating HYV of 17 crops including commercial crops like sesame, pulses, groundnut, turmeric, millets and enterprises in coconut processing.

## Outscaling of technologies and adoption

The high-yielding varieties of tubers, millets, pulses, maize, soybean, sunflower, ginger, and



Fig. 7.18. Little millet (Var. DHLM 36-3) in coconut interspaces

Table 18.. Impact of FFP interventions in improving income of women farmer SHGs

| Sl. No. | Crop       | Self Help Groups |         | No. of SHG members |         | Gross income (Rs lakhs) |         | Net income (Rs lakhs) |         |
|---------|------------|------------------|---------|--------------------|---------|-------------------------|---------|-----------------------|---------|
|         |            | 2018             | 2022-23 | 2018               | 2022-23 | 2018                    | 2022-23 | 2018                  | 2022-23 |
| 1       | Sesame     | 15               | 11      | 83                 | 74      | 7.97                    | 12.12   | 6.38                  | 8.99    |
| 2       | Tubers     | 12               | 14      | 78                 | 51      | 1.98                    | 1.94    | 1.35                  | 1.56    |
| 3       | Vegetables | 11               | 11      | 92                 | 82      | 0.66                    | 1.83    | 0.43                  | 1.54    |
| 4       | Millets    | 6                | 8       | 43                 | 24      | 0.58                    | 0.91    | 0.66                  | 0.50    |
| 5       | Fodder     | 6                | 5       | 30                 | 17      | 0.18                    | 0.32    | 0.15                  | 0.26    |

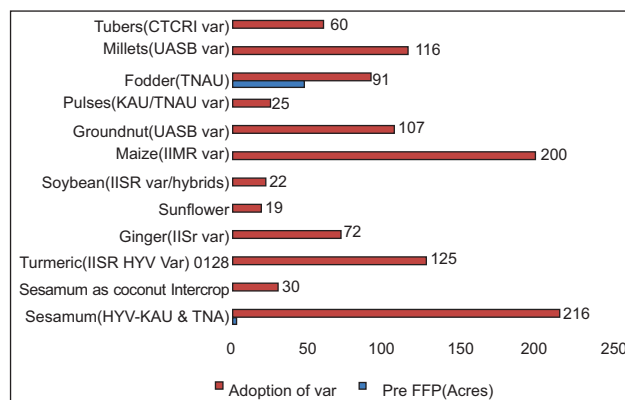


Fig. 7.19 Adoption of HYV of ICAR/Agri. Universities(acres)

turmeric were introduced in the FFP locations and continued adoption was documented. HYV of fodder and sesame only were reported before the FFP program (Table 18).

## FFP impact on cropping/farming intensity

The shift in farming combinations among FFP farmers indicated the adoption of intensification with appropriate combinations (Fig. 7.20). Poultry and fisheries were the choice added in homesteads. The involvement of women farmers was more in poultry and fisheries.

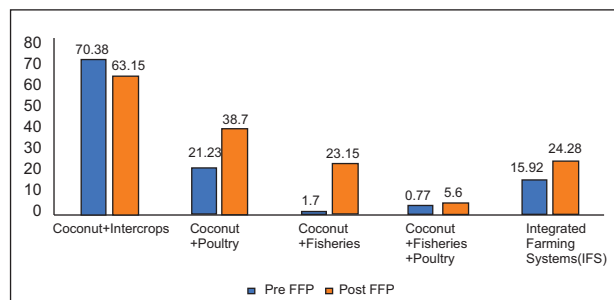


Fig. 7.20. Impact of FFP activities on shift in farming systems

## Impact of FFP interventions in improving income of women farmer SHGs

The participation of women SHG farmers could double their income through exploiting the potential of sesame and tuber crops, which are most suited for the Onattukara region where FFPs are located (Tabl 18).

### FFP Content Mobilization in social media

The content mobilization is mainly through YouTube, Facebook, WhatsApp and print media. Success stories of farmers are uploaded on the YouTube channel and daily activities on the Facebook page.

- CPCRI FFP YouTube channel; - 456 subscribers, 24 videos, 18.3k views
- FFP CPCRI Facebook page - 772 followers, 502 posts
- FFP farmers WhatsApp groups-12 groups, 851 members
- Print/TV/Radio - 18 news coverage, 3 TV programs, 2 radio programs

### Farmer Producer Organization (FPO)- Odanadu farmer producer Company Ltd.,(OFPC) - ICAR CPCRI FFP as POPI and Funded by NABARD

The OFPC is graded as 'A' by NABARD signifying a greater achievement within 3 years of initiation, with 500 shareholders. In 2023, it produced and marketed 4.1t and 1.3t of KalpaVardhini and Keraprobio respectively enabling adoption in 4100 bearing palms, as responsible technology promoters.

### Agri-business incubation

Incubation services, chiefly technology backstopping, were provided to 10 registered entrepreneurs. Eight incubatees registered in the previous years were graduated and six incubatees started business using Institute technologies. Three EDPs were conducted: one on Kalparasa (Fig. 7.22) and two on coconut value addition. Nine training programmes/technology awareness camps were organized in 2023.

A certificate programme (online) aiming agribusiness initiation viz., Kalpa Startup Master Class Series, was conducted during 15<sup>th</sup> Nov to 4<sup>th</sup> Dec 2023. It had a registration of over 260 people and nearly 100 attended every day. Certificates were issued to those who attended the classes regularly. A boot camp was conducted on 19 December 2023 at ICAR-CPCRI for the participants.

Smt. Shobha Karandlaje, Hon'ble Minister of State for Agriculture and Farmers' Welfare, Government of India felicitated eight

entrepreneurs who are successful in agri-business with Institute technologies on 2 September 2023 in a function held at ICAR-CPCRI in connection with the world coconut day (Fig. 7.23).



Fig. 7.22. Participants of EDP on Kalparasa



Fig. 7.23. Sushri. Shobha Karandlaje, Hon'ble Minister of State for Agriculture and Farmers' Welfare felicitating entrepreneurs

### Price and policy issues: Coconut sector

The economic causes of the recent price crash regime of the coconut have been studied from a policy perspective. Through the selective interviews of the stakeholders and the critical analysis of the secondary sources, a pragmatic cause-and-effect relationship has been elucidated. It is striking that coconut and coconut oil prices have experienced the lowest price regime in the last five years. The price wedge between domestic and international prices has narrowed down, and the trade sensitivity of the exports to the prices is well depicted in (Fig. 7.24 & 7.25), wherein the exports picked up drastically whenever the domestic prices were on a par or lower than the world prices of the coconut oil. Factors such as the over-supply of copra in the market in the recent period (more than 15%), diminishing local demand for coconut oil, and the availability of other edible oils at cheaper rates have significantly influenced the price crash of coconuts and coconut oil. Moreover, the industry lacks strong corporate demand and bulk procurement, which should have otherwise elevated the prices in the short term. It is

noteworthy that, due to the supply deficit in edible oil, allowing duty-free imports of 4 million tons of soybean oil, and slashing the base import prices of crude and refined palm oil have also detrimentally affected coconut oil prices.

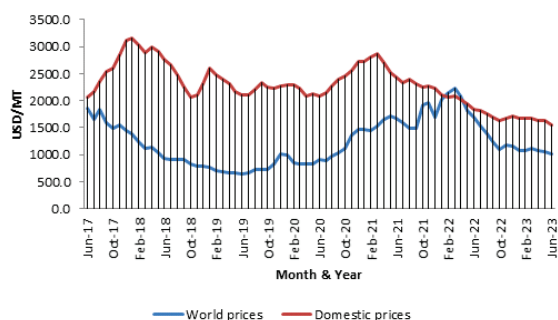


Fig. 7.24. Price comparison of coconut oil: Domestic & World

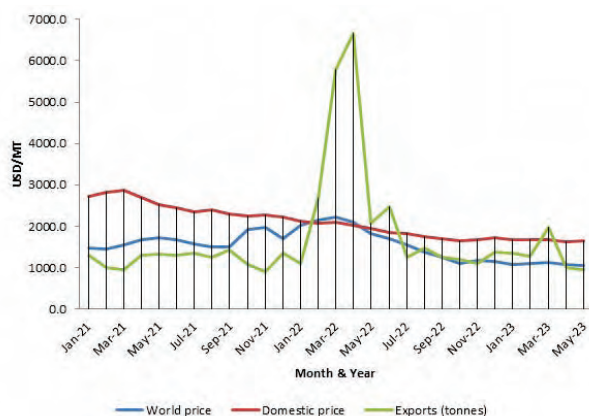


Fig. 7.25. Export sensitivity of coconut oil to the prices

### International trade dynamics

India's export basket of coconut value-added products experienced a radical shift over a 10-year period, wherein activated carbon dominated the total coconut exports with a share of 67% in 2022, while the share was 45% in 2013 (Table 19). The drastic reduction of copra exports is also noteworthy during the same period (a reduction from 20% to 2%). It is striking that, in other high-value products of coconut, the country has not made any significant strides. Having said that, the recent export figures of coconut products provide an optimistic outlook on the trade front, where the total export revenue (2022-23) is estimated at Rs. 3237 Cr, which shows a remarkable 41% increase from the export revenue in the year 2021-22. India should prioritize export-oriented product lines based on its comparative advantage and refine such product chains, especially in the Desiccated Coconuts, Activated Carbon, and Coir sectors. Alongside, we should also attempt to increase our market access (in a phased manner) of high-value products such as coconut milk/cream, and VCO.

Table 19. India's coconut export basket (% share in value terms)

| Product            | 2013 | 2022 |
|--------------------|------|------|
| Activated Carbon   | 45   | 67   |
| Copra              | 20   | 2    |
| Coconut Oil        | 12   | 8    |
| Shell charcoal     | 6.5  | 2    |
| Fresh coconut      | 5.2  | 8.5  |
| Dried coconut      | 3.6  | 7    |
| Desiccated Coconut | 1.5  | 1    |
| Others             | 6.2  | 4.5  |

### Production economics of mandate crops

The cost of production of coconut in Kerala based on data from a well-managed coconut garden, is calculated to be Rs. 12.10 per nut. In this scenario, about 60 percent of the total cost is incurred due to labour charges. This shows the higher per-unit labour charges prevailing in Kerala, which can be attributed to higher labour demand and higher cost of labour in Kerala. In addition, the lack of availability of sufficient skilled labourers for the harvesting of coconut leads to a higher cost of cultivation of coconut in Kerala. Currently, the wage rate prevailing is around Rs 800 per day, which is one of the highest costs prevailing for agricultural labour in India. The total cost of cultivation per hectare is Rs 1,90,599, with an average productivity of 90 nuts per palm per year, following the standard cultivation practices. The cost of production per kg of arecanut (chali) in a well-maintained garden on average would be Rs. 280.05/-.

### Concept note on revamping the Coconut FPOs

Based on a multi-stakeholder interactive mode, a mentoring workshop was organized at ICAR-CPCRI, RS Kayamkulam. A strategic framework has been developed for revamping the FPOs to a sustainable pavement, based on the critical analysis of policy issues, challenges, and ongoing conflicts. A policy paper in this regard was developed and circulated among the stakeholders. Policy issues of the coconut-based Farmer Producer Organizations were studied, and strategies for the sustainable way ahead have been suggested. It is obvious that the bottom line of any FPO is the capability of continuous profit generation. Nevertheless, in the case of FPOs in coconuts, it is pertinent to consider other



parameters besides profit. In this context, specific support from the government could turn out to be of great economic significance. It was found that the running of an FPO is overly influenced by the dominant member (in terms of production, political orientation, etc.), rendering the majority passive members. Therefore, conscious efforts are to be taken to keep the FPOs democratic so that the interests of most of the members are taken care of. Further, the technologies need to be suitable to the

economic conditions and farming practices of smallholder farmers. The organizational, financial, and commercial sustainability of FPOs can be further enhanced by leveraging their strengths and addressing their constraints through pragmatic policies, and there is a need for government support to create a FPO nurturing ecosystem for revitalizing such organizations in the coconut sector.



Protection. The notable achievements of these projects include:

- VPM 6 (IC 599264), a selection from the ECT population located in Adirampattinam village on the East Coast of Tamil Nadu underwent evaluation at the Coconut Research Station, Veppankulam, starting from 2006. Subsequently, it has been recommended for release (Fig. 8.2). VPM 6 recorded a maximum yield of 166 nuts in a single year, with a mean yield of 107 nuts palm<sup>-1</sup> year<sup>-1</sup>. Additional metrics include Copra content (137.40 g nut<sup>-1</sup>); -copra yield (18.00 kg palm<sup>-1</sup> year<sup>-1</sup>); oil content (67.94 %) and oil yield (2.67 tonnes ha<sup>-1</sup>). The selection exhibited adaptability to East Coast region of Tamil Nadu and displayed resilience in various stress conditions such as under planting, drought and cyclone due to its quick recovery habit.

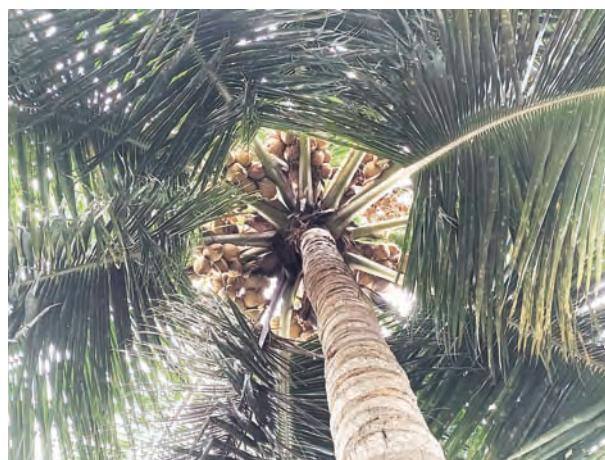


Fig. 8.2. Crown view of VPM 6

### Hybrids for Andhra Pradesh:

## Dwarf x Dwarf hybrids for Maharashtra and Tamil Nadu

- Among the Dwarf x Dwarf combinations assessed at Ratnagiri Center since 2011, the

## Coconut

There are currently 30 ongoing projects in coconut spread across various centers, focusing on Crop Improvement, Crop Production, and Crop

hybrid GBGD x MOD has been identified as a promising cross for earliness. The hybrid COD x MYD recorded the highest tender nut yield (76.7 nuts palm<sup>-1</sup> year<sup>-1</sup>), followed by the hybrid GBGD x MOD (68.3 nuts palm<sup>-1</sup> year<sup>-1</sup>) within the entire Dwarf x Dwarf coconut hybrids. Furthermore, the hybrid COD x MYD recorded the maximum volume of tender nut water (602.7 mL/nut), while the hybrid GBGD x MOD recorded the maximum TSS with a score of 5.7° Brix.

- At the Veppankulam Centre, there was considerable variation in nut yield, ranging from 55.6 to 97.0 nuts in COD and GBGD x MOD, respectively. The setting percentage exhibited a range from 21.59% (COD) to 33.78% (GBGD x MOD). The whole fruit weight displayed variability, spanning from 766.00 g to 1180.33 g, with an average of 928.85 g. Notably, GBGD x MOD showed a higher tender nut volume at 603.83 mL nut<sup>-1</sup>, while COD had a lower volume at 326.40 mL nut<sup>-1</sup>.
- Tall x Tall hybrids: At Aliyarnagar, among the five Tall x Tall cross combinations evaluated since 2011, BGR x ADOT was the highest nut yielder (78 nuts palm<sup>-1</sup> year<sup>-1</sup>), which was on a par with WCT x TPT (61 nuts palm<sup>-1</sup> year<sup>-1</sup>).
- **Coconut based multispecies cropping systems under coastal littoral sandy soil:**

Evaluation at Ratnagiri Center indicated maximum nut yield in Coconut + *Garcinia indica* + Pineapple cropping system under the recommended nutrient application. The vegetable (snake guard) yield was maximum in the coconut + *Garcinia indica* + Vegetable Crops cropping system with green manuring + biofertilizers + organic recycling +100% Recommended Dose of Fertilizer (RDF). The maximum pineapple (var. kew) yield was recorded in Coconut + *Garcinia indica* + pineapple cropping system with green manuring + biofertilizers + organic recycling + soil test based nutrient application. Maximum height and girth of *Garcinia indica* were recorded in Coconut + *Garcinia indica* + Vegetable Crops cropping system with green manuring + biofertilizers + organic recycling +100% RDF.

- **Coconut based farming systems for Tamil Nadu:**

Integration of coconut with pasture crops (Cumbu Napier hybrid + *Desmanthus*), fodder trees (*Sesbania grandiflora* + *Leucaena leucocephala* + *Glyricidia*), and the Tellicherry breed of goats recorded a net income of Rs. 2,54,206/- ha<sup>-1</sup> with a BC ratio of 3.16 as compared to monocrop of coconut at Aliyarnagar center yielding Rs. 1,51,312/- ha<sup>-1</sup> with a BC ratio of 2.25.

- **Management of leaf blight disease in coconut:**

Root feeding with propiconazole at a concentration of 5 mL in 100 mL of water, administered at three-month intervals during January, April, July, and October, led to a reduction in leaf blight incidence by 27.0% after 36 months of treatment. This particular approach also yielded the highest coconut production, with an output of 138 nuts palm<sup>-1</sup> year<sup>-1</sup>, and achieved a favorable Benefit-to-Cost (B:C) ratio of 3.7. In comparison, the untreated control exhibited a lower yield of 97 nuts palm<sup>-1</sup> year<sup>-1</sup>.

- **Initiatives for Pest Management:**

A total of 6600 numbers of *Bracon hebetor*, 1,77,050 numbers of *G. nephantidis*, 17100 numbers of *P. imbrues*, 242 Tricho cards, and 32,61,100 numbers of *P. astur* eggs were supplied to the farmers of East Godavari, West Godavari, Visakhapatnam, and Srikakulam districts of Andhra Pradesh, Bhadrachalam, Kothagudem, Medchal, and Khammam districts of Telangana from Ambajipeta center. Coconut fronds or leaflets containing parasitized puparia were collected from the affected ecosystem and released in newer areas of infestation. At Aliyarnagar Center, a total of 12,561 packets of *Encarsia* parasitoid were distributed to about 4000 farmers.

## Oil Palm

- Oil palm research within the All India Coordinated Research Project on Palms is actively underway at seven centers, focusing on aspects related to crop improvement and crop production. Thirty eight (38) hybrids are under evaluation in four trials. The salient achievements are as follows:

- **Better Performing hybrids:** Among the 10 hybrids evaluated at Pattukkottai for growth and yield parameters, the hybrid NRCOP 4 recorded a significantly higher palm height (5.27 m), while the number of leaves produced year<sup>-1</sup> (25.60) was highest in NRCOP 2. The highest bunch weight of 173.25 kg palm<sup>-1</sup> ha<sup>-1</sup> yield of 24.78 t ha<sup>-1</sup> was recorded in hybrid NRCOP 9.

## Cocoa

- **Better performing cocoa hybrids and clones :** Cocoa genotypes are under evaluation in 7 AICRP (Palms) centers covering the west coast, east coast, and NE regions and different cropping systems, including arecanut, coconut, and oil palm gardens. In the 12 year old trial at Kasaragod (Kerala), Ambajipeta (Andhra Pradesh), and Veppankulam (Tamil Nadu), VTLCH-2 and VTLCH-1 were identified as the best performers, respectively, in the regions under coconut. From the initial years of evaluation, it was observed that among 8 year old trees in Aliyarnagar (Tamil Nadu), Ratnagiri (Maharashtra) and Kahikuchi (Assam), VTLC-16, VTLC-17, and VTLC-20 were best performing, respectively, whereas VTLC-57 was high yielding at Vijayarai (Andhra Pradesh) under oil palm.

## Palmyrah

Palmyrah research under AICRP on palms is in progress at four centers on crop improvement and post-harvest technology. So far, 272 germplasm at HRS Pandirimamidi and 265 germplasm at AC & RI, Killikulam, have been conserved. The salient achievements under post-harvest technology are as follows:

- **Technology for sap powder:**

Freeze-drying has successfully yielded sap powder characterized by excellent solubility and easy rehydration. This product possesses features that inhibit chemical and enzymatic reactions, due to low temperature processing. Importantly, the sap powder meets the prevailing legislative standards for fruits and vegetable products having a moisture content of 5.6%

- **Technology for dehydrated ice apple:** The optimal conditions for osmotic dehydration of ice apple (palmyrah tender fruit

endosperm) in sucrose solutions are achieved at specific parameters. These include a temperature of 50°C, a sugar syrup concentration of 50% w/w, and an immersion time of 120 minutes. These conditions have been identified to maximize water loss while minimizing the incorporation of solids. This approach aims to produce a stable processed product with desirable characteristics.

## Meetings Held

The 32<sup>nd</sup> Annual Group Meeting of ICAR-AICRP on Palms was held at the AICRPP Centre, Horticultural Research Station, Kahikuchi, Assam Agricultural University, Jorhat, on September 13–15, 2023 (Fig. 8.3). Dr. V. B. Patel, Assistant Director of General (Fruits and Plantation Crops), Horticulture Science Division, ICAR, New Delhi, presided over the inaugural function, wherein Dr. N. K. Mohan, Hon'ble Member, Agricultural Commission, Govt. of Assam, was the Chief Guest. Dr. K. B. Hebbar, Director, ICAR-CPCRI, Kasaragod; Dr. K. Suresh, Director, ICAR-IIOPR, Pedavegi; Dr. R. K. Mathur, Director, ICAR-IIOR, Hyderabad; Dr. V. Venkatasubramanian, Director, ICAR-ATARI, Bangalore; Dr. G. Kathirvel, Director, ICAR-ATARI, Guwahati; and Dr. D. Prasath, Project Coordinator (Spices), ICAR-IISR, Kozhikode, were present for the inauguration. Dr. K.U.K. Namboothiri and Dr. P. Chowdappa, former directors of ICAR-CPCRI, participated in the inaugural session as ICAR-invited experts.

Dr. B. Augustine Jerard, Project Coordinator (Palms), ICAR-AICRP on Palms, ICAR-CPCRI, Kasaragod, presented progress report. Dr. V. B. Patel, Assistant Director General (F&PC), in his Presidential Address, congratulated the group of Palm Scientists for doing exemplary work in the sector. He also enumerated the vision of the Government of India to minimize imports and maximize the export potential of plantation crops and products. Dr. K. B. Hebbar, Director, Central Plantation Crops Research Institute, Kasaragod, expressed that the centers of ICAR-AICRP on Palms are well positioned in different agro-climatic regions of India to address location-specific problems and highlighted the existing scope for multi-disciplinary and multi-location research. The Chief Guest, Dr. N. K. Mohan, in his Special Address underlined the rich biodiversity of the

horticultural sector in the Northeastern Region and the role of Britishers in establishing the first tea garden in the country and uplifting the economy of the region.

Aliyarnagar Centre, TNAU, Tamil Nadu, was adjudged the Best AICRP (Palms) Centre for the year 2022–23.



Fig. 8.3. Release of Publications during 32<sup>nd</sup> Annual Group Meeting of ICAR- All India Coordinated Research Project (AICRP) on Palms



### CPCRI-KASARAGOD

The ICAR- Krishi Vigyan Kendra, Kasaragod has initiated 5 OFTs, 14 frontline demonstrations two EDP programmes, one nutrigarden programme, SCSP and various extension activities during 2023. The major achievements of these programmes are given below:

#### Salient achievements under OFT

##### Assessment of foam mat drying for mango pulp

Foam mat drying offers a number of advantages over traditional sun drying including uniform drying, high efficiency, cost effectiveness and improved quality. Foam mat drying for mango pulp was assessed by using natural emulsifiers and stabilizers and dried as mat in hot air dryer at 60°C for 5-6 hours. The foaming agent Tween- 80 gave good result and during reconstitution, it formed fruit juice without lumps and was having less soluble fiber content.

##### Assessment of black pepper varieties

The Black pepper varieties Arka Coorg Excel, Vijay and P-5 were planted during 2020 in 6 farmer's fields in Munnad, Parappa and Sheni villages. The Arka Coorg Excel vines have started yielding from 2<sup>nd</sup> year onwards with around 60 spikes and yielded dry pepper of 3.2 kg standard<sup>-1</sup> during the 3<sup>rd</sup> year. The farmers observed that it has bold berries and is easy to pluck the pepper spikes.

##### Assessment of different varieties of amaranthus

Performance evaluation of two varieties of Amaranthus namely Vaika and Vlathankara were carried out in 5 farmers field of Kasaragod and



Fig. 9.1. OFT on Amaranthus cultivation

Kanhangad block of Kasaragod. Both the varieties have significant vegetative growth than the local varieties. Due to the bright red colour and good taste, Vlathankara variety got more market demand and market value. It attained an average height of 43.6 cm during harvest after 27th days of sowing and yielded 115 kg cent<sup>-1</sup> of land (Fig. 9.1).

#### Assessment of different HYV of Okra.

Assessment of two okra high yielding and yellow vein resistant varieties of Okra (*Abelmoschus esculentus* L. Moench) Anjitha and Phule Vimkuthi were conducted in 5 farmers field of Kasaragod and Kanhangad block of Kasaragod. The improved varieties recorded significant improvement in growth and yield parameters over the local adopted check variety. Both the varieties are free from yellow vein mosaic disease under the field condition.

#### Salient achievements under FLD

##### Demonstration of bud rot management in coconut with *Trichoderma* coir pith cakes

Demonstrations were carried out in Kasaragod and Kanhangad block of Kasaragod. First application was done during the first fortnight of June after cleaning the crown from the dried spathe and other debris and *Trichoderma* coir pith cake (2no) was placed in the innermost leaf axils. The second application was done during the first fortnight of September.

##### Demonstration of Ganoderma wilt management in coconut

Demonstration was carried out in Kasaragod and Kanhangad block of Kasaragod. Basal drenching of the symptomatic palms and surrounding four palms with 0.2 % Hexaconazole 5 SC (40 litres palm<sup>-1</sup>) were done. After fifteen days of drenching, applied 5 kg neem cake enriched with *Trichoderma harzianum* palm<sup>-1</sup> at monthly intervals for six months. It was also recommended to maintain optimum moisture in the garden for the better proliferation of the fungus by mulching the basin with dried leaves during summer. It was recommended to follow this practice for at least 3 years to make the garden free from Basal Stem Rot/Ganoderma wilt.

### Demonstration of Nauroji Novel Organic liquid nutrients in vegetables

A banana pseudostem based organic liquid nutrients invented by scientists of NAU, Navsari, known as "Nauroji Novel Organic Liquid Nutrient" along with natural plant extracts were demonstrated in two crops Okra and Amaranthus as per the recommendations in 10 farmers fields (0.4 ha) of Kasaragod and Kanhangad block of Kasaragod. It was observed that the Nauroji Novel Organic Liquid Fertilizer increased the yield and net return in Okra and Amaranthus.

### Demonstration through millet cafeteria

Frontline demonstration of eight millets was conducted in 4 farmers fields in Manjeshwar and Kanhangad block of Kasaragod and one in KVK farm. Sowing of 8 different millets were completed in the month of December.

### Demonstration of drone technology in rice

A frontline demonstration on application of micronutrients in paddy using drone was undertaken at Bambrana Padasekhram in Kumbala Panchayat during October 2023 with the participation of 34 farmers (Fig. 9.2). Significant increase in yield was recorded in demonstration plots which yielded 2.7 tons hectare<sup>-1</sup> in comparison to 2.33 tons ha<sup>-1</sup> in control which is an increase of 16.3 per cent.



Fig. 9.2. Drone micro nutrient application in paddy field

### Demonstration of nutri garden in homesteads

Nutri Garden demonstration was carried out in 25 households (20-SC, 1-ST) selected from Chattanchal, Kudlu, Badiadka and Meenja villages and training was imparted to 28 beneficiaries on 02.02.2023. Vegetable seed kits consisting of Amaranthus, Chilli, Local cucumber, Ashgourd, Pumpkin, Okra, Cowpea, Bitter gourd, Passion fruit

seedling and inputs such as Neem oil, *Pseudomonas*, Coir pith, Dolomite were distributed to each households.

### FLD on baked coconut chips

Frontline Demonstration on Baked coconut chips was conducted on October 20, 2023 at CPCRI Kasaragod with a participation of 53 potential entrepreneurs. The technology is taken up by 3 units with brand name RAYA, Bullarkod, Ajja Farms, Kudlu (Mrs Prathima Naik), Zorras Foods, Uduma (Mrs Suharabi Fayaz) and Hitahara, Farm products, Kuruvail, Kudlu.

### Introduction of mosaic virus resistant cassava

The mosaic disease in cassava is a major threat in cassava cultivation and it causes 25-80% yield reduction. Sree Raksha, a high yielding mosaic resistant variety released and ICAR-CTCRI was introduced in this programme. The FLD was laid out in 4 acre of land belonging to 10 farmers in Kodom-Belur panchayath. The introduced variety is not affected by mosaic virus disease and an average yield of 41.75 t ha<sup>-1</sup> was recorded in this demonstration.

### EDP on Millets

An EDP training on millet value addition was imparted to 20 entrepreneurs between July 11 - Dec 18, 2023. The products marketed under this brand name include pre processed ragi powder, ragi malts, multi grain poshak powder, multi millet health mix, millet pastas, amylase rich sprouted ragi powder, millet milk shake mix, etc (Fig. 9.3).

### Capacity Development Programmes

A total of 58 training programmes (12 off campus, 46 on campus) were conducted with a participation of 1300 farmers /Farm women ( Male, 427 No.; Female, 873 No).



Fig. 9.3. Trainees displaying value added products from millets

## Extension activities

### Seminar on 'Enhancing productivity and value addition in millets'

The ICAR-KVK, CPCRI Kasaragod, organized a seminar on 'Enhancing Productivity and Value Addition in Millets' at ICAR- CPCRI on March 18, 2023. More than 60 various food products and value added products made out of Millets were displayed during the event by the KVK trained SHG groups. A total of 52 farmers participated in the meeting and 9 officials from CPCRI and KVK also participated.

### International Women's Day

On the occasion of International Women's Day on March 8, 2023 at ICAR-CPCRI, various millet products and traditional ITK based value added products of four women entrepreneurs trained by KVK were launched by Dr. K.B. Hebbar, Director, ICAR-CPCRI, Kasaragod. The Value added products launched were ragi malt and amylase rich ragi mix by Mrs. Rathi Mohan, Bekal Muliya Homestay under the brand name "RaMo", Poshak powder, Coconut milk residue fortified sprouted ragi pasta by Mrs. Usha Nair, Periya under the brand name "Usha's Nature's Guide", sprouted ragi and foxtail millet based healthy snack foods (Murukku), coconut milk residue fortified Tavara (*Cassia tora*) pasta by Mrs. Suharabi Fayaz, Uduma by brand name "Zorra's foods" and coconut oil based herbal hair oil by Ms. Remya K.V, Maipady by brand name 'Pulury Food Products'. The programme was attended by around 200 participants including KVK entrepreneurs, Staffs of CPCRI and school children.

### PM Kisan Sammelan

The ICAR-Krishi Vigyan Kendra, CPCRI, Kasaragod has organised the PM Kisan Sammelan on July 27, 2023 in CPCRI (Fig. 9.4) The webcasting of the Honourable Prime Minister's programme was telecasted to the farmers of the Kasaragod district for creating awareness on various programmes launched by the Government of India. The meeting was attended by 54 farmers and 20 extension functionaries.

### Workshop on water budgeting

Organised a workshop on "Importance of water budgeting and watershed management in local self government institutions on July 26, 2023 at ICAR-



Fig. 9.4 Live broadcast of PM Kisan Sammelan

CPCRI Kasaragod. The workshop was organized in collaboration with KSCSTE-CWRDM, Kozhikode. The event was inaugurated by Smt. Baby Balakrishnan, Hon'ble President, Kasaragod District Panchayat and was presided over by Dr. K Balachandra Hebbar, Director, ICAR-CPRI, Kasaragod. Dr. A C Mathew, Principal Scientist, ICAR-CPCRI, Kasaragod delivered the keynote address. The meeting was attended by around 150 officials, Presidents and secretaries from various Panchayats, Agricultural Officers, ATMA and NGO's staff members.

### Promotion of coir pith compost production technology

Trained farmers for production of coir pith compost by forming Joint Liability Groups (JLG) with farmers which was named "Sukrutham" JLG group. The group erected compost production unit for the commercial production of compost (Fig. 9.5). Around 30 tons of organic manure was produced and sold at a rate of Rs.15/kg.



Fig. 9.5. Demonstration of on-farm coir pith composting

### ICAR 95<sup>th</sup> Foundation and technology days celebrated by ICAR-KVK Kasaragod

Organized a one day work shop on Millets on July 18, 2023. An exhibition of value added products from millets prepared by self help groups was also

arranged along with the workshop. The workshop was attended by 112 participants including farmers, entrepreneurs and extension personnel.

### **KVK's Participation in “Viksit Bharat Sankalp Yatra”**

Viksit Bharat Sankalp Yatra is a nationwide campaign to raise awareness through outreach activities to achieve saturation of schemes of Govt. of India across the country covering all Gram Panchayats, Nagar Panchayats and Urban Local Bodies. Kasaragod KVK is also actively participating in the Viksit Bharat Sankalp Yatra from November 27, 2023, the official launch in the district. Thirty Eight Grama panchayats of six blocks of Kasaragod district with an average daily participation of 175-200 citizens was completed.

### **Farmer Scientist Interface**

Organized a Farmer-Scientist interface programme at Friend's Club, Kayyar, Jodukallu, Magalpady Panchayat of Manjeshwar Block on August 4, 2023. The programme was attended by 56 progressive farmers of the region.

### **Kisan Diwas**

Kisan Diwas 2023 was celebrated from December 21 to 23, 2023. Around 442 Hybrid coconut seedlings allocated to SC farmers/farm women were distributed to around 85 SC beneficiaries as part of the programme. Swachhta Pledge was taken by the participants of the programme. A total of 70 farmers belonging to the Scheduled caste community participated in the programme.

### **SAC meeting**

The SAC meeting of KVK Kasaragod was organized on February 17, 2023 at CPCRI Kasaragod. Dr. K.B. Hebbar, Director, ICAR-CPCRI chaired the meeting and the meeting was attended by 31 officials representing various departments, ATARI representative and two farmer representatives.

### **National Nutrition Month**

Two events highlighting the significance of the theme “Healthy food, affordable for all” were organized as on and off campus which includes holistic nutrition for upscaling nutrition status of the community (September 26, 2023) and significance of moringa cultivation for nutrition and health security at Chenekode SC colony (September 15, 2023).

### **Programmes under International Year of Millets**

Highlighting the significance of millets, seven programmes including six training programmes including one for extension functionaries, 1 millet workshop was organized and conducted with a participation of 282 women beneficiaries from Kutumbashree, ICDS officials and entrepreneurs. Two millet exhibitions were organized wherein more than 40 Millet items prepared by women were displayed. Display of millet value added products of successful entrepreneurs of KVK was organized during the millet workshop conducted on July 18, 2023.

### **SCSP Programme**

A total of ten programmes have been conducted under the approved action plan of SCSP for 2023-24 under ICAR- KVK, CPCRI covering more than 200 farm families of SC colonies coordinated by Dr. Saritha Hegde, SMS, Community Science and nodal officer, SCSP. Activities conducted include “Demonstration of Carbon fibre pole harvester cum sprayer at Kubanoor, Mangalpady (July 3, 2023) with the participation of 46 farmers (Fig. 9.6).



Fig. 9.6. Carbon fibre pole

Awareness programme on “Significance of Moringa cultivation for Nutritional security” marking the event of National Nutrition Month (September 23, 2023) at Chenekcode, SC colony and distribution of 250 moringa seedlings (PKM-1) to around 45 families at Chenekcode and Majal SC colonies of Madhur Panchayath; distribution of around 1025 arecanut seedlings (Sree Mangala) to SC farmers/farm women; skill oriented training for 8 days (September 20- 30, 2023) on use of brush cutter for creating self employment amongst un employed youths; skill oriented training on areca leaf sheath plate making (September 22, 2023) and training on organic vegetable cultivation (October 11, 2023) conducted with a participation of 25 participants wherein IIHR vegetable kits and 50 hybrid coconut seedlings were distributed to the participants (Fig.9.7) in

coordination with SCSP programme of ICAR-CPCRI, Kasaragod; training on nutritional significance of papaya and its cultivation in homesteads and distribution of 250 Papaya seedlings (Red Lady var.) to the beneficiaries at Goliadka SC colony.

### FET programme for AR trainees

Organised FET programme for the 113<sup>th</sup> batch FOCARS trainees from September 4 – 22, 2023 in which 6 scientist probationers participated.



Fig.9.7. Coconut seedling distribution by Director

## ICAR KVK-ALAPPUZHA

### I. Major events

#### Inauguration of Mushroom Spawn production Unit and DAESI programme

Mushroom spawn production unit established at KVK-Alappuzha with funding support from State Horticulture Mission (Kerala) was inaugurated by Sri. P. Prasad, Hon. Minister for Agriculture, Govt. of Kerala on March 3, 2023 (Fig. 9.8). Third batch of Diploma in Agricultural Extension Services for Input Dealers (DAESI) programme (2022-23) for Alappuzha district was also inaugurated by the Minister on the occasion. Dr. V. Venkitasubramanian, Director, ICAR-ATARI, Bengaluru in his presidential address highlighted the importance of indigenous foods over the new generation food products and the richness of the country in terms of its diversity and stability in food production. A small exhibition highlighting different aspects of mushroom production and value addition was also arranged in which the Minister interacted with the entrepreneurs.



Fig. 9.8. Exhibition and interaction-MSPU

#### Inauguration of turmeric processing unit

The turmeric processing unit initiated as a part of 'Value chain in turmeric' project implemented by ICAR-KVK-Alappuzha in 10 panchayaths of Mavelikkara and Chengannur blocks of Alappuzha

district with funding from NABARD was inaugurated by Shri. K. Krishnankutty, Hon. Minister for Electricity, Govt. of Kerala in the 'Karshaka Sabha' (Farmers' meet) organized at Chettikulangara on May 9, 2023 (Fig. 9.9). The meeting was presided by Adv. U. Prathibha, MLA, Kayamkulam which was attended by about 150 farmers, many of them being the partner farmers of the project.



Fig. 9.9. VCT Processing unit launching

#### Launching of KVK activities (2023-24)

Sri. P. Prasad, Hon. Minister for Agriculture, Government of Kerala officially launched the action plan activities of KVK for 2023-24 in the operational villages of Muhamma and Thanneermukkom on October 20, 2023 (Fig.9.10).



Fig. 9.10. Address by Minister on Launching of KVK activities

## II. Technology Assessed and Transferred

### On Farm Testing:

Two OFTs were taken up in Muhamma and Thaneermukkam panchayaths during the year.

### Frontline Demonstrations:

Twenty Frontline Demonstrations were taken up in Cherthala block during the year.

## III. Capacity Building Programmes

### Comprehensive Coconut Rejuvenation Campaign (CoCoReCa)

In order to bring back the glory of coconut farming in the Karappuram region to the extent possible, a Comprehensive Coconut Rejuvenation Campaign (CoCoReCa) is implemented by KVK-Alappuzha in Muhamma and Thanneermukkom panchayaths. The inaugural function was held on November 16, 2023.

### Capacity building programmes for tribal farmers and volunteers from Odisha

(a) A capacity building programme on 'Value addition of Jackfruit' for selected master farmers of the NABARD supported NGOs, and Sakha foundation funded by NALCO from Koraput District of Odisha was conducted for 4 days from January 23-26, 2023. NABARD consultant, official of Sakha foundation and 6 trainees from the tribal community attended the programme. A business plan with necessary information on jackfruit processing machineries was prepared and provided to the team.

(b) Three days skill training programme on value addition of Jack fruit, Banana, Ginger and Tamarind was conducted for selected tribal volunteers and staff of ORMAS, an NGO functioning with the support of District Administration of Koraput of Odisha from June 26 - 28, 2023 at the Agro-Processing training cum Incubation Centre (APTIC) of KVK-Alappuzha. A total number of 20 participants attended the programme in which 30 different products of jackfruit, banana, ginger and tamarind were demonstrated to the participants through hands-on-training.

### Capacity building programmes for Extension officials

**District level Extension officials – Scientist interface:** A technology interface meeting of the extension officials and Scientists of Alappuzha

district was organized by KVK – Alappuzha on September 21, 2023 in collaboration with ATMA, Dept. of Agriculture and Kerala Agricultural University (KAU). A total of 55 persons including Project Director and Staff of ATMA, Asst. Directors of Agriculture, and Agricultural Officers of Alappuzha district and resource persons from ICAR-CPCRI, KAU and KVK-Alappuzha attended the programme.

### Training programme for Extension officials:

A capacity building programme for the extension officials of Alappuzha district was organized by KVK – Alappuzha on December 12, 2023 in collaboration with ATMA and Dept. of Agriculture (Fig. 9.11). A total of 50 officials including Project Director and Staff of ATMA, Asst. Directors of Agriculture, and Agricultural Officers of Alappuzha district and resource persons from ICAR-CPCRI and KVK-Alappuzha participated in the day-long sessions.



Fig. 9.11. Capacity building programme extension officials

### Workshop on Agri-Clinic and Agri-Business Centre (ACABC)

KVK in association with NABARD conducted a workshop on Agri-Clinic and Agri-Business Centre (ACABC) programme of MANAGE, Hyderabad on November 17, 2023 in order to create awareness among the targeted group of diploma holders and graduates in Agriculture on the potential of the programme and scope of technical enterprises by them (Fig. 9.12). Forty youth, who had completed VHSE (Agriculture), Diploma in Agriculture, and B Sc. (Agriculture) attended the programme and interacted with the resource persons.



Fig. 9.12. Interaction in the ACABC workshop

**On and Off-campus training programmes:** One hundred and twenty three training programmes were organized during 2023 for a total of 3401 participants.

#### IV. Other Extension Activities

A total of nine radio and two TV programmes by officials from KVK were broadcast during the period.

##### Other programmes

- i. Help line service (calls attended and responded) – 519
- ii. Agro-clinic – 344
- iii. Method demonstration- 25
- iv. Messages to farmers- 74
- v. Farmers' visit to KVK- 143
- vi. Soil & water testing -120
- vii. News paper coverage – 19
- viii. Soil testing campaign – 20 nos

#### V. External funded projects

##### National Innovations in Climate Resilient Agriculture (NICRA) - Phase II

Technology Demonstrations under the project, “National Innovations in Climate Resilient Agriculture” (NICRA) Phase II is being implemented at Edathua village of Kuttanad Taluk. Various technologies like climate resilient practices for paddy in Kuttanad region, modified rain shelter for year round vegetable production, short duration cassava variety- Sree Jaya for escaping flood during monsoon season, large scale composting of aquatic weeds using EM solution and use for vegetable cultivation, soil health card for better nutrient and soil health management, recycling of organic residues for energy generation and crop production using portable biogas plants, modified season cultivation of HY ginger for vegetable purpose, improved goat shelters to withstand water logging/flood, climate resilient cages for poultry rearing, stress, disease and nutritional management in dairy animals are being demonstrated in the farmer's fields. A ‘Village Climate Risk Management Committee’ (VCRMC) is formed with participation of members from the LSG for overseeing and institutionalizing the programme on long term basis in the village.

##### Onattukara Spices Farmer Producer Company Ltd. (OSFPCL)

KVK promotes the Farmer Producer Organization – Onattukara Spices Farmer Producer Company Limited registered in December 2016 with grants from NABARD. The company facilitates cultivation, procurement, processing, and marketing of the major spices viz., Turmeric, Ginger, Pepper and Garcinea in six panchayaths of Bharanikkavu block so as to enhance the net income of the shareholder farmers. At present it has 400 share holders and share capital of 4.5 lakhs. A 'Rural mart cum Farmer Facilitation Centre' with additional funding support from NABARD is also established which started functioning in 2021.

##### Agro Processing Training cum Incubation Centre (APTIC)

To strengthen the capacity of entrepreneurs in processing and value addition of coconut, jackfruit, and seasonal fruits and vegetables, Agriculture Development and Farmers Welfare Department, Govt of Kerala has sanctioned an Agro-Processing Training cum Incubation Centre (APTIC) to KVK-Alappuzha with a sanctioned budget of Rs.73 lakhs. The incubation centre is facilitating the development of technology based and knowledge driven agri business ventures. The centre helps ventures during the start-up period by providing an integrated package of technology, work space, access to specialized equipments and pilot plant. The machineries like water immersion retort, pasta production extruder, hot air oven, deep freezer, fruit pulper, RO water unit, sautiner, pulveriser, band sealing machine, dough mixing machine, weighing balances, mini rice huller, pulveriser, and coconut processing machineries like desheller machine, testa removing machine, and wet pulveriser are installed in the centre. Hundreds of incubatees and farmers visit, learn skills and utilize the facilities for preparation of products every year (Fig.9.13).



Fig. 9.13. Trainees from Odisha to APTIC

### Value chain in turmeric (VCT)

In order to promote the value chain activities of turmeric from production to processing and marketing, the project sanctioned under the FSPF of NABARD with a funding of Rs. 18.0 lakhs was implemented from 2020 to 2023. The project had 300 partner farmers from 10 panchayaths of Mavelikkara and Chengannur Blocks of Alappuzha district who were empowered for scientific cultivation and value addition of turmeric and organized as self sustaining groups.

### Mushroom Spawn Production Unit

With a funding support Rs.15 lakhs from the State Horticulture Mission, a mushroom spawn production unit was established in the KVK for meeting the demand for spawn from the mushroom entrepreneurs trained by the KVK and from adjoining districts.

### Natural Farming Project (NFP)

This national network project funded by ICAR through ATARI was initiated in 2022-23. The project activities are initiated in Cheppad panchayath where demonstration fields were identified and crops grown in partner farmers' fields. Training and awareness programmes are being conducted in different panchayaths of the district.

### Field days of Frontline Demonstrations conducted

#### (I) Salinity management and soil moisture conservation for vegetable cultivation in coastal sandy soils of Alappuzha

Field Day was conducted on April 6, 2023 at Madampil in Kandallloor panchayath. A total of forty eight participants attended the programme.

#### (ii) Integrated management of weedy rice in Kuttanad region

The demonstration was conducted in Muttar

panchayath on April 5, 2023.

#### (iii) Fodder cowpea for yield improvement of dairy cattle

In this demonstration, fodder cowpea seeds of CO (FC)-8 were provided to 20 farmers for cultivation and feeding @ 6 kg per day along with Hybrid Napier grass. A combined field day was conducted on April 6, 2023 in which the Milk society presidents, and partner farmers participated.

#### (iv) Integrated management of leaf blotch and rhizome rot diseases in turmeric

Technologies were demonstrated to 50 partner farmers of Bharanikavu panchayath during field day on April 13, 2023

### VI. Revolving fund activities of KVK

Different inputs *viz.*, vegetable seeds and seedlings, planting materials, bio-agents, Methyl euginol, cue lure and yellow sticky traps, layer chicks, mushroom spawn, mother spawn, multi nutrient mixture for banana and vegetables, azolla, processed products, publications etc. were made available to the farmers of the district (as resource centre) through revolving fund activities. A custom hiring centre is also functioning to provide farm implements like tractor mounted rotavator and transplanter to farmers on hire for land preparation, paddy transplanting etc.



Fig. 9.14. FLD on Upland paddy

## VII. PUBLICATIONS

### Research articles

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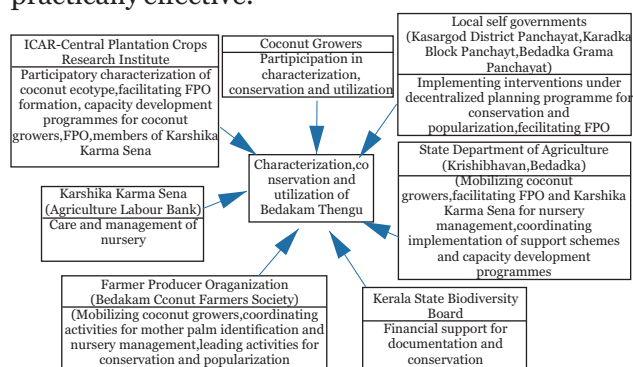
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Nihad, K., Abdul Haris, A. and Anithakumari, P. 2023. E- manual on Horticultural techniques

## VIII. TECHNOLOGY ASSESSED AND TRANSFERRED

### Participatory coconut varietal ecotype conservation model

An extension methodology to facilitate *in situ* conservation of coconut ecotypes was field validated at Bedadakka village in Kasaragod district. Farmer participatory assessment facilitated by ICAR-CPCRI resulted in the identification of the *Bedakam thengu*, a popular cultivar amongst the farmers in Kasaragod district of Kerala, distributed in the area with perceived desirable features. The ecotype endemic to the geographical area was further characterized for conservation and utilization. ICAR-CPCRI provided technical support for characterization and conservation of the coconut ecotype. Bedakam Coconut Growers' Society, a FPO was formed and registered with more than 2000 coconut farmers with the main objective of conservation and popularization of the ecotype. Convergence of interventions by different stakeholders was facilitated to make the efforts practically effective.



Convergence of stakeholders for conservation and utilization of *Bedakam thengu*



Launching of seedling distribution programme at Bedadakam

The activity for the documentation of mother palms of *Bedakam thengu* in farmers gardens and conducting capacity development programmes was supported by Kerala State Biodiversity Board. The local elected bodies viz., Kasaragod district panchayat, Karadka block panchayat and Bedakam grama panchayat implemented schemes under decentralized planning programme to support conservation and utilization of *Bedakam thengu*. Selected members of the 'Karshika Karma Sena' functioning in the grama panchayat facilitated by Department of Agriculture through the local Krishi Bhavan with the objective of providing labour support to farmers for taking up agricultural activities were trained in important cultivation components like mother palm selection, seed nut collection, sowing and other nursery management practices. The FPO entrusted the team of 'Karshika Karma Sena' to manage the nursery to produce the seedlings of *Bedakam thengu*. The seedlings of *Bedakam thengu* thus produced in the nursery located at Pandikandam were distributed to farmers.

### Modified ground pollination technique

The ground pollination technique developed by Devakumar *et al.*, 2018 was later modified by Thomas *et al.* The refinements include using flexible PVC pipe (6 mm diameter) instead of 8 mm diameter clear PVC tube. The vertical air blower powered by a 12 volt battery used for pumping the pollen-talc mixture was replaced with 5 litre hand pump pressure sprayer. A 6 mm micro sprinkler attached to 6 mm adapter was fixed onto the top end of the PVC pipe inserted inside the pollination bag to aid uniform dusting of pollen-talc mixture onto receptive female flowers inside the pollination bag. The technique was tested and validated at three locations of ICAR-CPCRI (Kasaragod, Kayamkulam and Kidu) and two AICRP Centres (Aliyarnagar-Tamil Nadu and Ratnagiri-Maharashtra). The cost of one ground pollination kit was rupees four hundred which does not include the cost of hand sprayer. This technique has potential application in undertaking controlled pollination in other palms also, like, oil palm and date palm.

Observations on harvested seed nuts recorded 11-12 months after pollination revealed that modified ground pollination resulted in a fruit set of 24.10 % during 2019-20, 26.70% during 2020-21 and 24.0% during 2021-22. The lowest fruit setting (18.70%) was reported from Kayamkulam (Kerala) for T X T seed nut production during 2020-21 and the highest fruit setting (32.10%) was reported from Aliyarnagar (Tamil Nadu) for D X T seed nut production during 2021-22.



Demonstration of ground pollinating device

### Detection of adulteration to VCO

Ensuring the authenticity of virgin coconut oil (VCO) is crucial to safeguard the interests of consumers. This technology facilitates the differentiation of VCO from coconut oil (CO), palm oil (PO), and liquid paraffin (mineral oil) by utilizing spectrophotometric measurement of biochemical parameters. Key biochemical attributes such as total phenolic content, iodine value, saponification value, and fatty acids like lauric acid played a significant role in discerning the various oil samples across different levels of adulteration. The classification of samples was

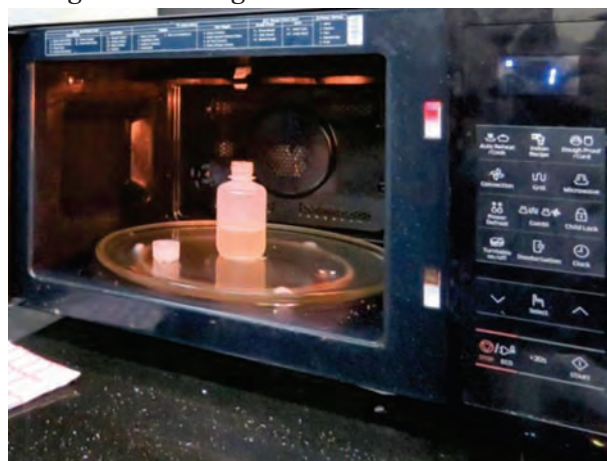


Virgin coconut oil from ICAR-CPCRI

successful even with an adulteration level as minimal as 10%. Through principal component analysis, two components were identified that effectively distinguished various adulterated oil samples. Additionally, multiple regression analysis yielded predictive equation models for adulteration quantitation.

### Microwave processing parameters for preservation of *Kalparasa*<sup>®</sup>

Coconut inflorescence sap, traditionally called as *neera*, is tapped by using the device coco-sap chiller, developed by ICAR-CPCRI, is branded *Kalparasa*<sup>®</sup>. The shelf life of the sap after extraction is 2–3 h in atmospheric condition. For enhanced shelf life, the effect of microwave power level (600 and 900 W) and exposure time (30, 60, 90, and 120 s) were investigated. It was deduced that a combination of high power (900 W) and exposure time (120 s) maintains the physico-chemical quality of *Kalparasa* up to 16 days of storage under refrigerated conditions.



Preservation protocol for *Kalparasa* using microwave oven

## IX. AWARDS AND RECOGNITIONS

### Awards

Dr. T. Sivakumar, SMS, received the Best Technical Staff Award from ICAR-CPCRI during the Foundation Day Celebration held in Kasaragod on January 5, 2023. The award, which includes a cash prize and certificate, was bestowed upon him for his meritorious extension work in supporting the farming community. His contributions include various activities such as OFTs, FFS, technology demonstrations, trainings, publications, etc.

Dr. T.N. Madhu, received the Young Scientist Award at the 5<sup>th</sup> National Conference on "Recent Advances in Agricultural and Industrial Entomology and Environmental Sciences and their Impact on Food Environmental Security", held on September 29-30, 2023, in Chennai.

### Best Research Paper Awards

Dr. R. Pandiselvam, Dr. M.R. Manikantan and Dr. P. P. Shameena Beegum, were honored with the "Best Paper Award" in the Food Technology category from the Journal of Food Science and Technology. This recognition was conferred for their research paper titled "Reaction kinetics of physico-chemical attributes in coconut inflorescence sap during fermentation" at the 29th ICFoST held in Trivandrum on January 5, 2023.

Dr. R.T.P. Pandian Scientist (Plant Pathology) received Best Oral Presentation award for the paper "First report of *Colletotrichum kahawae* subsp. *ciggaro* causing leaf spot disease on arecanut, *Areca catechu* in India" at Platinum Jubilee Conference on "Plant and Soil Health Management: Issues and Challenges" by Indian Phytopathological Society (IPS) at University of Mysuru, Karnataka held from February 2-4, 2023.

Dr. R. Pandiselvam, Dr. M.R. Manikantan and Dr. A.C. Mathew received Second Prize for the oral presentation entitled "Design, Development and Evaluation of Tender Coconut Punching Machine." A research paper presented in 57<sup>th</sup> Annual Convention of Indian Society of Agricultural Engineers & International Symposium on Engineering Interventions for Making Millets a Global Food" held at University of Agricultural

Sciences, Raichur, India during November 6-8, 2023.

Dr. M. Suchithra, Dr. B. Suma, Dr. J.S. Minimol, Dr. Deepu Mathew and Dr. A.V. Santhosh Kumar were conferred Best Poster Presentation Award for their research "Transcriptomic analysis of cocoa (*Theobroma cacao* L.) genotypes for drought tolerance" at International Seminar on Exotic and Underutilized Horticultural Crops: Priorities and Emerging Trends held during October 17-19, 2023, at ICAR-IIHR, Bengaluru, India.

Dr. V.H. Prathibha, Senior Scientist (Plant Pathology) received Best Poster Presentation award for the paper "Occurrence of fruit rot in soursop caused by *Lasiodiplodia theobromae* in India" at International seminar on Exotic and underutilized Horticultural crops: Priorities and Emerging trends held during October 17-19, 2023 at ICAR-IIHR, Bengaluru, India.

Dr. V.H. Prathibha, Dr. N.R. Nagaraja, Dr. M.K. Rajesh, Dr. Daliyamol, Dr. M. Sujithra, Ms. K.P. Thejasri, Ms. C. Sneha and Dr. Vinayaka Hegde, were conferred the Best Poster Presentation Award for the research paper entitled "Rapid, cost effective and efficient technique for screening arecanut against *Phytophthora meadii*" at the International Conference on Plant Health Management ICPHM 2023 - Innovation & Sustainability held during November 15-18, 2023 at Hyderabad, Telangana.

Dr. Jilu V. Sajan, Scientist (Agril Entomology) received Best Paper Award for the paper titled "New record of collateral hosts of coreid bug, *Paradasynus rostratus* Distant (Hemiptera: Coreidae) infesting coconut" in Third International Scientific Conference on Environmental Research: Issues, Challenges and Strategies for Sustainable Development organized by Eurasian Academy of Environmental Science during December 1-2, 2023 at Karwar, Karnataka.

Dr. Rajkumar, Senior Scientist (Nematology) received the Dr. Muthuswamy Anandraj Award for Best Original Research Paper entitled "Ecofriendly management of slow decline in black

pepper under arecanut – A success story” at PLACROSYM XXV (National Symposium on Plantation Crops) organized jointly by Indian Society for Plantation Crops and ICAR- IOPR, Pedavegi, A.P. from December 12-14, 2023.

### Recognitions

- Dr. M.R. Manikantan was elected as NAAS Fellow 2024 in the field of Agril. Engg. & Tech.
- Dr. Shameena Beegum was awarded Associateship of Indian Academy of Horticultural Sciences, New Delhi.
- Dr. R. Pandiselvam featured in world top 2% scientist list published by Stanford



Dr. N. R. Nagaraja being facilitated at Kaniyoor Savanoor, Karnataka

- University, USA and Elsevier.
- Dr. N. R. Nagaraja, Senior Scientist (Plant Breeding) was honoured and felicitated by Nelluru Kemraje Primary Agriculture Co-operative Society on March 9, 2023 at Elimale, Sullia Tk., Dakshina Kannada Dt., Karnataka.
- Dr. N. R. Nagaraja, Senior Scientist (Plant Breeding) was honoured and felicitated by Kaniyoor Savanoor Farmer Producer Company Limited, Dakshina Kannada Dt., Karnataka and Charvaka Milk Producers Cooperative Society on August 10, 2023 at Charvaka Milk Producers Society.



Dr. N. R. Nagaraja being honoured at Savanoor, Karnataka

## X. TRAINING AND CAPACITY BUILDING

### Scientific

| Name & designation                                                                                                                  | Programme                                                                                                                                    | Place & date                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Dr. T. N. Madhu<br>Scientist (Agril. Entomology)                                                                                    | Winter school on "Implication and Impact Assessment of Climate Change on Pests, natural enemies and Pollinators in Horticultural Ecosystems" | ICAR –IIHR, Bangalore during February 7-27, 2023                                                                                       |
| Dr. M. Chaithra<br>Scientist (Plant Pathology)                                                                                      | Workshop on "Significance of Transmission Electron Microscopy and Scanning Electron Microscopy in Disease Diagnostics"                       | ICAR-IIHR, Bangalore during February 27-28, 2023                                                                                       |
| Dr. Sandip Shil,<br>Senior Scientist (Agril. Statistics)                                                                            | Online training on Data Visualization using 'R'                                                                                              | ICAR-NAARM, Hyderabad during March 1-8, 2023                                                                                           |
| Mr. Y. Diwakar<br>Scientist,                                                                                                        | International Workshop on "Phenotyping for Drought Tolerance"                                                                                | Jointly organized by ICAR-NRC for Banana, Tiruchirappalli and Alliance of Bioversity International during March 7-11, 2023.            |
| Dr. Sandip Shil,<br>Senior Scientist (Agril. Statistics)                                                                            | Online National Workshop on "Big Data Analytics in Agriculture"                                                                              | ICAR-NAARM, Hyderabad, during March 9-10, 2023.                                                                                        |
| Dr. R. Sudha, Dr. Jeena Mathew, Dr. K. Nihad, Dr. M. Neema, Senior Scientists<br>Dr. Aparna Veluru and Dr. M. Suchithra, Scientists | Online Training Programme on Multivariate Data Analysis                                                                                      | ICAR-NAARM, Hyderabad during March 20-27, 2023                                                                                         |
| Dr. S. Elaine Apshara,<br>Head (A) & Principal Scientist (Hort)                                                                     | One day Online Workshop on "Prevention, Prohibition and Redressal of Sexual Harassment of Women at Work Place"                               | Institute of Secretariat Training and Management (ISTM), GoI, New Delhi on May 1, 2023                                                 |
| Dr. Jeena Mathew,<br>Senior Scientist (Soil Science)                                                                                | Online training programme on precision farming                                                                                               | ICAR-ATARI, Chandra Shekar Azad University of Agriculture and Technology, Kanpur and ICRISAT, Hyderabad during May 15, - June 13, 2023 |
| Dr Daliyamol, Scientist (Plant Pathology), Dr. T. Sivakumar<br>Subject Matter Specialist (Agril. Entomology)                        | Training on "Remote Pilot training under small rotorcraft category"                                                                          | PBC, Pune during June 23 to 26, 2023.                                                                                                  |
| Dr. S. Elaine Apshara,<br>Head (A) & Principal Scientist (Hort)                                                                     | Online 'Workshop for Liaison Officers (SC/ST)                                                                                                | The Institute of Secretariat Training and Management (ISTM), GoI, New Delhi during August 17-18, 2023.                                 |
| Dr. R. Thava Prakasa Pandian,<br>Scientist (Plant Pathology)                                                                        | Online Training on "Omics Data Analysis: Genome to Proteome"                                                                                 | ICAR- IASRI, New Delhi during October 9-18, 2023                                                                                       |
| Dr. Surekha, Scientist                                                                                                              | ANGRU-IDP-NAHEP sponsored National Training on 'Natural Farming and Strategies for Soil and Crop Sustainability'                             | ANGRAU, Regional Agricultural Res. Station, Tirupati, A.P., during November 1-11, 2023                                                 |
| Dr. A. Abdul Harris,<br>Principal Scientist                                                                                         | Training course "Irrigation Water Management- Micro Irrigation and Advanced Irrigation Techniques"                                           | KSCSTE-CWRDM, Kozhikode during November 20-23, 2023                                                                                    |

|                                                                                                                              |                                                                   |                                                              |
|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------|
| Dr M. Chaithra,<br>Scientist (Plant Pathology)                                                                               | Professional Attachment Training                                  | ICAR-IIHR Bangalore during<br>October 21 – November 21, 2023 |
| Dr. S. Paulraj<br>Senior Scientist (Agril.<br>Microbiology)<br>Dr. S.V. Ramesh<br>Senior Scientist (Agril.<br>Biotechnology) | Laboratory Assessors Training<br>Course as per ISO/IEC 17025:2017 | NABL at ICAR-NIVEDI during<br>February 6-10, 2023            |

### Technical

| Name & designation                                    | Programme                                                                                                                 | Place & date                                          |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Dr. K.S. Muralikrishna,<br>Senior Technical Assistant | Online training program in Hindi<br>on “Introduction to Statistical and<br>Machine Learning Techniques in<br>Agriculture” | ICAR-IASRI, New Delhi,<br>during June 6-12, 2023      |
| Dr. B.J. Nirmal Kumar,<br>Technical Assistant         | Online Training on “Omics Data<br>Analysis: Genome to Proteome”                                                           | ICAR- IASRI, New Delhi<br>during October 09–18, 2023  |
| Dr. B.J. Nirmal Kumar,<br>Technical Assistant         | Online Training on “Metagenomics<br>Data Analysis”                                                                        | ICAR-IASRI, New Delhi<br>during December 11-13, 2023. |

### Administrative

| Name & designation                                                                                                                                                                                                        | Programme                                                                                                | Place & date                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Smt. A. R. Arathi,<br>Stenographer,<br>Smt. A.J. Mary, UDC,<br>Smt. K. Preethi, Assistant,<br>Shri Aswin Reghunath, UDC,<br>Sh. Vivek Singh, Stenographer<br>Grade III,<br>Smt. T. Deepa, UDC,<br>Sri. Arun N.K. Raj, LDC | Online Pre examination<br>training programme for Limited<br>Departmental audit & Accounts<br>Examination | ICAR- NIASM, Pune during<br>July 20 – October 6, 2023 |
| Shri A. Neil Vincer,<br>AAO (Estate)                                                                                                                                                                                      | Organization of training programme<br>during the Vigilance Awareness<br>Campaign                         | ICAR-NAARM, Hyderabad<br>during November 1-3, 2023    |
| Shri Pradeep Kumar Vasu,<br>AAO (Estt.),<br>Shri Narayana Naik, Assistant<br>Smt. K. Preethi, Assistant,                                                                                                                  | Online Training Programme on<br>Reservation in Service for SC/ST/<br>OBC/Ex.SM/PWD's                     | ISTM, New Delhi during<br>November 13-16, 2023        |

## XI. WORKSHOPS, SEMINARS, SUMMER INSTITUTES, FARMERS DAYS ORGANIZED

### **'Kalpa Vajra' (2022-2023) Valedictory cum futurium programme**

ICAR-CPCRI, Regional Station, Kayamkulam, Kerala celebrated Platinum Jubilee 'Kalpa Vajra' (2022-2023) Valedictory cum futurium programme during May 13-14, 2023. Dr. A. K. Singh, Deputy Director General (Horticultural Sciences), ICAR, New Delhi was the chief guest at the function. Directors of ICAR-CTCRI (Thiruvananthapuram), ICAR-IIOPR (Pedavegi), ICAR-CIFT (Kochi), Official representatives from Kerala Agricultural University, Coconut Development Board and Coir board participated in the function. Dr. K. B. Hebbar (Director, ICAR-CPCRI) spoke about the institute's research achievements and the impact it had on the farming community. In his inaugural address, Dr. A. K. Singh mentioned about the need to digitize the agriculture sector and the potential of widespread use of artificial intelligence and sensor-based technologies.

Dr. G. Baiju (Director, ICAR-CTCRI) highlighted the importance of growing tuber crops as intercrops in plantation crops to enable farmers to gain continuous income. Dr. K. Suresh, (Director, ICAR-IIOPR) appreciated the Regional Station for its research achievements and farmer participatory approach.

Dr. V. Mini (Head, Onattukara Regional Agricultural Research Station of Kerala Agricultural University) recalled the contributions of previous generations of researchers in the Institute's 75 years of research accomplishments. Dr. Hanumanthe Gowda (Chief Coconut Development Officer, CDB) mentioned the availability of planting materials and plans to ensure sustainable income to coconut farmers.

Dr. P. Anithakumari (Acting Head, ICAR-CPCRI, RS, Kayamkulam) explained about the 75 years of research achievements of the institute and the credibility achieved among the farmers. Dr. A. Joseph Rajkumar (Principal Scientist) delivered vote of thanks.

Planting of coconut varieties developed from the station in the 'Kalpa Vajra' block, inauguration of

Palmyrah palm genetic evaluation and demonstration of Modified ground pollination kit for coconut were also performed during the function. Dr. A. K. Singh exchanged MoU for micronutrient mixtures 'Kalpa Poshak' and 'Kalpa Vardhini' with FPO, launched millet products and released various publications of the Station. He also announced the 'Kalpa Vajra' award to honor the best coconut farmers in root(wilt) affected tract.

Mr. D. Kuppuramu (Chairman, Coir Board) inaugurated Farmers Seminar. Dr. E. Jayashree (Principal Scientist, ICAR-IISR, Calicut) delivered a lecture on 'Processing and value addition of Spice crops'. Dr. M. R. Manikantan, Dr. M. S. Sajeew, Dr. K. N. Shiva and Dr. U. K. Priya handled sessions on 'Processing and value addition of coconut, tuber crops, banana, arecanut and cocoa'. An extensive exhibition was arranged and around one thousand farmers attended the programme.

### **DST-SERB Kaaryashala on processing and value addition**



Dr. A. K. Singh, Deputy Director General (Hort. Science) inaugurating the valedictory function at kayamkulam



D. D. G (Hort. Science) and Director(CPCRI) along with staffs at CPCRI regional station Kaayamkulam

Organized DST-SERB sponsored Kaaryashala (High End Workshop) on “Opportunities and Challenges in Processing and Value-Addition of Plantation Crops” during September 2-8, 2023 for 30 PG/Ph.D. Students from different SAU's. Also organized 4 weeks Skill Development Training Programme for B.Tech (Agrl. Engg.) students of College of Agricultural Engineering, ANGRAU, Madakasira, AP. during September 4-30, 2023.

### Foundation Day of the ICAR-CPCRI

The 107 Foundation Day of the ICAR-Central Plantation Crops Research Institute was celebrated on January 5, 2023. Dr. A. K. Singh, Deputy Director General (Hort. Science), ICAR, New Delhi, delivered the Foundation Day address virtually. He called upon the scientists to adapt to the changes happening in the sector in order to succeed. He emphasised the need for increased planting material production. Regarding the exports, the growth rates are promising, and we need to further strengthen our efforts to have better global market access.

Dr. V. B. Patel, ADG (Fruits and Plantation Crops), highlighted the rising global demand for coconut water, which is growing at double digit annual growth rates. The Dr. K. V. Ahmed Bavappa memorial lecture was delivered by Dr. Nirmal Babu, former Director, ICAR-IISR, Kozhikode. Dr. Dinesh, Director, ICAR-IISR, Kozhikode and Dr. M. N. Sheela, Director (Acting), ICAR-CTCRI, Thiruvananthapuram and Dr. M. V. Prasad, Director (Acting) ICAR-IIOPR, Pedavegi, offered felicitations online.

Institute awards for the employees were given away by Dr. Dinakara Adiga, Director, ICAR-DCR, Puttur: The Interdisciplinary Award was given to the team comprising Dr. Vinayaka Hegde, Dr. C. Thamban, Dr. Rajesh M. K., Dr. M. Sujithra, Dr.



Inauguration of Foundation Day programme at Kasaragod

Rajkumar, Dr. Daliyamol and Dr. Prathibha V. H. The best technical staff award was given to Dr. T. Sivakumar, SMS, KVK, Kayamkulam.

Dr. K. Muralidharan, Director (Acting), presided over the function. The foundation day programme was followed by a business entrepreneurial "Kalpa – Green Chat" programme.

### 38 Annual Workshop of AICRP on PHET

ICAR-CPCRI, organized 38 Annual Workshop of the All India Coordinated Research Project on Post Harvest Engineering and Technology (AICRP on PHET) during February 20-22, 2023 at Kasaragod. Prof. Prasad Krishna, Director, NIT-Calicut was the chief guest of the inaugural session. The session was chaired by Dr. S. N. Jha, DDG (Agrl. Engg.) and other dignitaries on the dais were Dr. K. Narsaiah, ADG (Process Engg), Dr. K. B. Hebbar, Director, ICAR - CPCRI, Dr. Nachiket Kotwaliwale, Director, ICAR-CIPHET, and Dr. Vishwakarma R. K., Project Coordinator of AICRP on PHET. About 100 delegates from 31 different centres of AICRP on PHET, ICAR-CIPHET and ICAR Head-quarters have participated in the workshop. Two new technical bulletins in Tamil on 'Virgin coconut oil' and 'Coconut chips' by ICAR-CPCRI, Kasaragod centre were released by the chief guest and DDG (Agrl. Engg.). Four memorandums of agreement for transfer of AICRP on PHET, ICAR-CPCRI developed technologies 'Virgin coconut oil', 'coconut chips', 'foam mat dried coconut milk powder' and 'Kalparasa tapping and value addition' were exchanged with three entrepreneurs. A Business and Plenary session was held at the end of the technical sessions on February 22, 2023 which was chaired Dr. S. N. Jha, DDG (Agrl. Engg.) by co-chaired by Dr. K.B. Hebbar, Director, ICAR-CPCRI along with Dr. K. Narsaiah, ADG (Process Engg).



Release of publication by the dignitaries during the function

## Multi-institutional Meeting on Arecanut and Human Health

A Multi-institutional meeting on 'Arecanut and Human Health' was jointly organized by ICAR-Central Plantation Crops Research Institute (ICAR-CPCRI), Kasaragod and Directorate of Arecanut and Spices Development (DASD), Kozhikode at ICAR-CPCRI, Kasaragod on November 14, 2023. The purpose of the event was to formulate research project on the role of arecanut on human health, involving researchers from reputed national organizations, including AIIMS, CSIR-CCMB, CSIR-CIMAP, CSIR-CFTRI, CSIR-NBRI, IISc, KMC, NITTE University, and SDM University, among others.

Dr. Prabhat Kumar, Horticulture Commissioner of the Department of Agriculture and Farmers Welfare, Ministry of Agriculture and Farmers Welfare, New Delhi, emphasized impartiality in assessing the impact of arecanut on human health during his presidential address. He also pointed out inconsistencies in scientific literature related to arecanut and human health, emphasizing the need for a comprehensive research proposal to bring out the facts about arecanut and its effects on human health.



Prof. Sathyamoorthy (SDM University, Dharwad) addressing the delegates

In his welcome address, Dr. K. Balachandra Hebbar, Director ICAR-CPCRI, highlighted the importance of arecanut and research progress in the sector, emphasizing its significance in the livelihood of the farming community in the Indian context. Dr. Homey Cheriyan, Director, DASD, provided insights into the economic scenario of arecanut and its products in the country. Prof. Sathyamoorthy from SDM University, Dharwad, moderated discussions under various scientific sessions, namely, clinical epidemiology and public health, metabolites and metabolomics, neurosciences, and cellular and molecular sciences. The meeting concluded with the

formulation of researchable questions aimed at investigating the effect of arecanut on human health and other modalities, with plans to prepare and submit a research proposal in a multi-institutional network and collaborative mode.

## National Workshop on Science History of India

National Science Day-2023 was celebrated at ICAR-CPCRI, Regional Station, Kayamkulam by organizing National Workshop on 'Science History of India' on February 28, 2023 with the financial support from Kerala State Council for Science, Technology and Environment. In his presidential address, Dr. K. B. Hebbar, Director, ICAR-CPCRI, Kasaragod highlighted on how history leads deeper insights in to science by citing the professional contribution of Prof. J. C. Bose retrieved from history. Prof. Rajan Gurukkal, Former Vice Chancellor, M.G. University & Vice-Chairman, Kerala State Higher Education Council inaugurated the workshop online. Key note address on 'History of Plantation Crops in India' was delivered by Prof. P. K. Michael Tharakan, Former Vice Chancellor, Kannur University & Chairperson, Kerala Council for Historical Research (KCHR).

Publications from this Institute on 'Proceedings of Mentoring Workshop for Strengthening FPO's and 'Compilation on Soil Health Management for Sustained Productivity' were released on this occasion. A Science Day Exhibition was arranged as part of the workshop. More than 120 registered faculties, research scholars and students from South India participated and made the workshop a memorable event of understanding science laced with history.



A glimpse from National Science Day Celebration at Kayamkulam

### Seminar on 'Smart Farming in 5G Era'

A seminar on 'Smart Farming in 5G Era' was organized by the TRAI, Bengaluru at CPCRI Kasaragod in collaboration with the institute on March 1, 2023. Dr. K. Muralidharan, Director in charge, ICAR-CPCRI inaugurated the seminar. While inaugurating he expressed his feelings that the availability of multitude of network options available in the country makes it easy for 5G based smart farming. The other dignitaries present during the inaugural session were Sri. Chandrachud, AGM, TRAI and Sri. Manish Raghav, Sr. Officer, TRAI and Dr. K. P. Chandran, Principal Scientist (Statistics). More than 150 delegates, including agricultural scientists, data networking technicians, students and technocrats participated in the seminar.

### Kisan Diwas and Farmers Training on Coconut Cultivation

Kisan Diwas 2023 was celebrated jointly by ICAR - KVK & ICAR-CPCRI, Kasaragod by organizing training programme to the Scheduled Caste beneficiaries on "Coconut cultivation during December 21-23, 2023 at ICAR - CPCRI, Kasaragod under SCSP programme. The programme commenced with the welcome address by Dr. T.S. Manojkumar, Principal Scientist and Head, KVK, Kasaragod. Dr. K. B. Hebbar, Director, ICAR-CPCRI, Kasaragod gave the Inaugural Address. The programme was presided over by Dr. Vinayak Hegde, Head, Crop Protection Division.

Around 442 coconut hybrid seedlings of Kera Sankara (T x D) were distributed to 85 SC farmers / farm women of villages of Madhur, Badiadka, Puthige, Mogral Puttur Panchayaths as part of the training programme. During the occasion two farmers and two farm women were felicitated for their valuable contribution to farming activities. Dr. Rajkumar, Senior Scientist & Nodal officer of SCSP at CPCRI and Dr. Saritha Hegde, SMS, KVK, Kasaragod, coordinated the training cum distribution of coconut seedlings.

Swachhta Pledge was taken by the participants during the occasion. Technical sessions on "Scientific cultivation of coconut", "diagnosis of disease & its management", "Pest Management & EPN for bio-management of vegetable pests in coconut system" was conducted by Dr. Surekha, Scientist (Agronomy), Dr. V. H. Prathibha, Senior Scientist (Plant pathology) and Dr. Rajkumar, Senior Scientist (Nematology) of ICAR - CPCRI, Kasaragod respectively. Training ended with vote of thanks by Dr Benjamin Mathew, SMS, Horticulture, KVK, Kasaragod.



Coconut Seedling distribution by Director during Kisan Diwas

## XII. PARTICIPATION IN CONFERENCES, MEETINGS, WORKSHOPS, SYMPOSIA, WEBINAR ETC. IN INDIA AND ABROAD

### Participations Abroad

| Name & Designation                                                  | Title                                                                                                                                                                         | Place & Date                                                                      |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Dr. K.B. Hebbar, Director                                           | International Seminar on “Harnessing Coconut Potential for Offsetting Carbon Emission: Integrating Science and Economy for a Sustainable Future”                              | Manado<br>North Sulawesi,<br>Indonesia.<br>October 12-13, 2023,                   |
| Dr. K.B. Hebbar, Director and Dr. Niral V., Head (Crop Improvement) | Fist International Scientific Conference on Conservation, valorization and exchange of resources and germplasm for economic development of the coconut industry (I-CONVERGED) | SEARCA Conference Hall in Los Baños, Laguna, Philippines.<br>November 27-30, 2023 |

### Participations Within India

| Name & Designation                                                                                                                                                                                                                   | Title                                                                                       | Place & date                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Dr. M. Shareefa, Senior Scientist<br>Dr. Maya Lekshmi, Technical Officer                                                                                                                                                             | 2nd DAE-BRNS Workshop on ‘Cryogenic Facility Management’                                    | Tata Institute of Fundamental Research, Colaba, Mumbai.<br>January 4 -7, 2023      |
| Dr. Arun Kumar Sit, Principal Scientist                                                                                                                                                                                              | Conclave of Farmer’s Producer Company of Jalpaiguri                                         | NABARD, Jalpaiguri Unit.<br>January 1, 2023                                        |
| Dr. M.R. Manikantan, Principal Scientist and Dr. R Pandiselvam, Scientist                                                                                                                                                            | 29th Indian Convention of Food Scientists and Technologists                                 | Al-Saj Convention Centre, Trivandrum.<br>January 5-7, 2023                         |
| Dr. Vinayaka Hegde, Head (Crop Protection) and Dr. R. Thava Prakasa Pandian, Scientist                                                                                                                                               | Plant and Soil Health Management: Issues and Challenges                                     | University of Mysuru.<br>February 2 – 4, 2023                                      |
| Dr. S. Elain Apshara, Principal Scientist (Hort.)                                                                                                                                                                                    | Progressive Horticulture Conclave (PHC 2023)                                                | GB Pant University of Agri. & Tech., Pantnagar, Uttarakhand.<br>February 3-5, 2023 |
| Dr. Regi J. Thomas, Principal Scientist                                                                                                                                                                                              | Scientific Advisory Committee meeting                                                       | ICAR-KVK, Pullad, Pathanamthitta.<br>February 8, 2023                              |
| Dr. Rajkumar, Senior Scientist                                                                                                                                                                                                       | Capacity building workshop on 'What makes incubator successful' organized by Start-Up India | IIT, Madras.<br>February 16-17, 2023                                               |
| Dr. P. Muralidharan, Head, KVK,<br>Dr A. Joseph Rajkumar, Principal Scientists<br>Dr. Jilu V. Sajan, Scientist<br>Shri M.S. Rajeev, Mrs. Jissy George,<br>Dr. T. Sivakumar, Mrs. G. Lekha,<br>Dr. S. Ravi, Dr. K. Sajnanath, SMS-KVK | National Workshop on Science History of India                                               | ICAR-CPCRI, RS, Regional Station, Kayamkulam.<br>February 28, 2023                 |
| Dr. T.N. Madhu, Scientist                                                                                                                                                                                                            | National Conference on Cocoa Development in India – Challenges and opportunities            | KSNUAS, Shivamogga<br>March 3, 2023                                                |

| Name & Designation                                                                                                              | Title                                                                                                                                                              | Place & date                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Dr. Y. Diwakar, Scientist                                                                                                       | International Workshop on 'Phenotyping for drought tolerance' jointly organized by ICAR-NRC for Banana and Alliance of Bioversity International                    | ICAR-National Research Center for Banana, Tiruchirappalli, TN. March 7-11, 2023                                  |
| Dr. Sandip Shil, Senior Scientist                                                                                               | National level Workshop on Big Data Analytics in Agriculture (Online Mode)                                                                                         | ICAR-NAARM, Hyderabad. March 09-10, 2023                                                                         |
| Dr. S. Indhuja, Scientist                                                                                                       | National workshop on 'Geobiotics supplementation through millet beverage and fermented millet curd'                                                                | CSIR-CFTRI, Mysuru. April 12-13, 2023                                                                            |
| Dr. V. Niral, Principal Scientist                                                                                               | Virtual workshop for Pacific region farmers and agricultural officers on 'In situ coconut conservation                                                             | Pacific Community (SPC), Suva, Fiji. April 24-25, 2023                                                           |
| Dr. Regi J. Thomas, Principal Scientist                                                                                         | Online Meeting of Central Sub-Committee on Crop Standards, Notification and Release of Varieties for Horticultural Crops                                           | Krishi Bhawan, New Delhi. May 4, 2023                                                                            |
| Dr. C. Thamban, Principal Scientist                                                                                             | Agriculture in Kerala-Challenges and future - 5 <sup>th</sup> International Congress on Kerala Studies                                                             | AKG Centre for Research and Studies, Thiruvananthapuram. May 20-22, 2023                                         |
| Dr. S. Elain Apshara, Principal Scientist, Dr. E.K. Saneera, Dr. M. Chaithra, Scientists                                        | CARDS 2023 - CRC Annual research and development symposium: 'Improving cocoa productivity and quality while managing risks'. (Online)                              | Cocoa Research Centre (CRC), University of the West Indies, St. Augustine Campus, Trinidad. May 24-25, 2023      |
| Dr. A. Joseph Rajkumar, Principal Scientist<br>Dr. Merin Babu,<br>Dr. M. Shareefa,<br>Senior Scientists                         | International Conference on Kerala Studies                                                                                                                         | Devamatha School, Thrissur. May 21, 2023                                                                         |
| Dr. K. Muralidharan, Head (A) (Social Sciences), Dr. Rajkumar, Senior Scientist and Shri Bhavishya, Scientist                   | 30 <sup>th</sup> Swadeshi Science Congress                                                                                                                         | National Institute of Technology (NIT), Calicut. May 25-27, 2023                                                 |
| Dr. K. Muralidharan, Head (A), (Social Science), Dr. S.V. Ramesh, Dr. R. Sudha, Senior Scientists and Shri Bhavishya, Scientist | 15 <sup>th</sup> Swadesh Prem Jagriti Sangosthi- 2023- Global Conference on 'Precision horticulture for improved livelihood, nutrition and environmental services' | Jalgaon, Maharashtra. May 28-31, 2023                                                                            |
| Dr. S. Elain Apshara, Principal Scientist, Dr. M. Chaithra, Dr. E. K. Saneera, Scientists                                       | National Conference on 'Cocoa development in India- Challenges and opportunities'                                                                                  | Keladi Shivappa Nayak University of Agricultural and Horticultural Sciences. Shivamogga, Karnataka. June 3, 2023 |
| Dr. M R Manikantan, Principal Scientist                                                                                         | International exhibition for G20 Ministers                                                                                                                         | Hotel Novotel, Hyderabad. June 15-17, 2023                                                                       |

| Name & Designation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Title                                                                                                                                                             | Place & date                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Dr. P. Muralidharan, Head, KVK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Agri Seminar as part of the "International Congress on Kerala Studies"                                                                                            | At Thrissur organized by AKG Centre for Research and Development, Trivandrum. May 21, 2023. |
| Dr. T. Sivakumar SMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Brainstorming on 'Cassava mealy bugs management'                                                                                                                  | ICAR-CTCRI, Thiruvananthapuram. May 23, 2023                                                |
| Dr. Anitha Kumari, Head (A), Regional Station, Kayamkulam, Dr. P. Muralidharan, Head, KVK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Annual Zonal Workshop of KVKs of Zone-XI                                                                                                                          | ICAR-ATARI and UAS, Bengaluru. June 20-21, 2023                                             |
| Dr. Anitha Kumari, Head (A), Regional Station, Kayamkulam, Dr. P. Muralidharan, Head, KVK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | National Seminar of the Indian Society of Extension Education                                                                                                     | ISEE and UAS, Bengaluru. June 22-24, 2023                                                   |
| Dr A. Joseph Rajkumar, Principal Scientist, Dr K. M. Anes Senior Scientist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Annual review meeting of AICRP on Biological Control                                                                                                              | College of Agriculture, Pune. July 20-21, 2023                                              |
| Dr. Vinayaka Hegde, Head (Crop Protection)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | National Symposium on Plant Health Management: Current Trends and Novel Mitigation Strategies                                                                     | ICAR-CTCRI Thiruvananthapuram. September 11-12, 2023                                        |
| Dr. K.B. Hebbar, Director, Dr. B. Augustine Jerard, Project Coordinator (Palms), Dr. P. Subramanian, Head (Crop Production), Dr. Vinayaka Hegde, Head (Crop Protection), Dr. Murali Gopal, Head (PB&PHT), Dr. V. Niral, Head (Crop Improvement), Dr. K. Muralidharan, Head (A) (Social Sciences), Dr. M. K. Rajesh, Head, Regional Station, Vittal, Dr. P. Anithakumari, Head (A), Regional Station, Kayamkulam, Dr. Ravi Bhat, SIC, PME, Dr. K.P. Chandran, Dr. Regi Jacob Thomas, Dr. A. Joseph Rajkumar, Dr. S. Elain Apshara, Dr. Alpana Das, Principal Scientists, Dr. M. Sujithra, Dr. V.H. Prathibha, Dr. Merin Babu, Dr. N.R. Nagaraja, Senior Scientists, Dr. S. Sumitha, Dr. Anok Uchoi, Dr. L.S. Singh, Scientists. | 32 <sup>nd</sup> Annual Group Meeting of AICRP on (Palms)                                                                                                         | HRS, Kahikuchi, AAU, Jorhat, Assam. September 13-15, 2023                                   |
| Dr. K Muralidharan, Head (A), Social Sciences, Dr. M. R. Manikantan, Principal Scientist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sensitization workshop of ICAR-ABI's                                                                                                                              | NASC Complex, New Delhi. September 21-22, 2023                                              |
| Dr. T. N. Madhu Scientist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5th National conference on "Recent Advances in Agricultural and Industrial Entomology and Environmental sciences and their impact on food environmental security" | Chennai September 29-30, 2023                                                               |

| Name & Designation                                                                                                                                                                                                                                                                                                                                                                                                                     | Title                                                                                                                                                                                           | Place & date                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Dr. M.R. Manikantan, Principal Scientist                                                                                                                                                                                                                                                                                                                                                                                               | International Workshop on Food Loss and Waste Prevention                                                                                                                                        | NASC Complex, New Delhi.<br>September 30 – October 1, 2023                        |
| Dr. P. Muralidharan, Head, KVK<br>Dr. M. Shareefa, Senior Scientist<br>Mrs. T.N. Ranjini, Scientist                                                                                                                                                                                                                                                                                                                                    | XVIth Agricultural Science Congress                                                                                                                                                             | NAAS and ICAR-CMFRI at Kochi. October 10-13, 2023.                                |
| Dr. P. Muralidharan, Head, KVK<br>Mr. M.S. Rajeev, SMS                                                                                                                                                                                                                                                                                                                                                                                 | Zonal workshop on PPV&FR Act                                                                                                                                                                    | PPV&FR Authority, ICAR-ATARI, Bengaluru and KVK-Kottayam. October 14, 2023.       |
| Dr. Vinayaka Hegde, Head (Crop Protection)<br>Dr. V. H. Prathibha, Dr. K. M. Anes<br>Dr. S.V. Ramesh, Senior Scientists,<br>Dr. M.Suchithra, Dr. Panjavarnam, Scientists                                                                                                                                                                                                                                                               | International Seminar on Exotic and Underutilized Horticultural Crops: Priorities and Emerging Trends                                                                                           | ICAR-IIHR, Bengaluru<br>October 17-19, 2023,                                      |
| Dr. M.R. Manikantan, Principal Scientist                                                                                                                                                                                                                                                                                                                                                                                               | World Food India International Exhibition                                                                                                                                                       | Pragati Maidan, New Delhi.<br>November 3-5, 2023                                  |
| Dr. K.B. Hebbar, Director ,Dr. B. Augustine Jerard, Project Coordinator (Palms), Dr. V. Niral Head (Crop Improvement), Dr. P. Subramanian Head (Crop Production), Dr. Vinayaka Hegde Head (Crop Protection), Dr. Murali Gopal Head (PB&PHT), Dr. K. Muralidharan Head (A), Social Sciences, Dr. M.K. Rajesh, Head, Regional Station, Vittal, Dr. Ravi Bhat, Principal Scientist, Dr. N.R. Nagaraja, Dr. S.V. Ramesh, Senior Scientists | Multi-institutional meeting on arecanut and human health                                                                                                                                        | ICAR-CPCRI, Kasaragod.<br>November 14, 2023                                       |
| Dr V. H. Prathibha, Dr M. Sujithra, Senior Scientists, Dr Daliyamol, Scientist                                                                                                                                                                                                                                                                                                                                                         | International conference on plant health management<br>ICPHM 2023-Innovation and sustainability                                                                                                 | Hyderabad<br>November 15-18, 2023                                                 |
| Dr. Regi Jacob Thomas, Head, Regional Station, Kayamkulam, Dr. A. Abdul Haris, Dr. C. Thamban, Dr. A. Joseph Rajkumar, Principal Scientists, Dr. S.V. Ramesh, Dr V. H. Prathiba, Dr M. Sujithra, Dr Rajkumar, Dr. K. Nihad, Dr. Jeena Mathew Dr. R. Sudha, Dr. P. S. Prathibha, Dr. S. Neenu, Sr. Scientists, Dr. Surekha, Dr. Y. Diwakar, Dr. G. Panjavarnam, Dr. Jilu V. Sajan, Dr S. Indhuja, Scientists                            | Fist International Scientific Conference on Conservation, valorization and exchange of resources and germplasm for economic development of the coconut industry (I-CONVERGED) (Attended online) | SEARCA Conference Hall in Los Baños, Laguna, Philippines.<br>November 27-30, 2023 |

| Name & Designation                                                                                                                                                                                                                                                                                                                                                                    | Title                                                                                                                                                      | Place & date                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Dr Merin Babu, Senior Scientist                                                                                                                                                                                                                                                                                                                                                       | International Conference on “Emerging Trends and Innovations in Biotechnology’ 2023                                                                        | Mar Athanasios College for Advanced Studies Tiruvalla (MACFAST). November 30 - December 3, 2023. |
| Dr. A. Abdul Haris, Principal Scientist, Dr. K. Nihad, Dr. S. Neenu, Dr P. S. Prathibha, Sr. Scientists, Dr. Jilu V. Sajan, Scientist                                                                                                                                                                                                                                                 | Third International Scientific Conference on Environmental Research: Issues, Challenges and Strategies for Sustainable Development and Livelihood Security | SGM College and CISSA, Karwar, Karnataka. December 1-2, 2023                                     |
| Dr. Vinayaka Hegde, Head (Crop Protection)                                                                                                                                                                                                                                                                                                                                            | VIROCON-2023; Virology Conference on Advancement in Global Virus Research Towards One Health                                                               | ICAR-NRCB Tiruchirappalli, Tamil Nadu. December 1-3, 2023                                        |
| Dr A. Joseph Rajkumar, Principal Scientist                                                                                                                                                                                                                                                                                                                                            | National Seminar on Advances in Molecular Taxonomy and Phylogenetics                                                                                       | Government College, Nattakom, Kottayam. December 12, 2023                                        |
| Dr. V. Niral, Head (Crop Improvement), Dr. Regi Jacob Thomas, Head, Regional Station, Kayamkulam, Dr. M.K. Rajesh, Head, Regional Station, Vittal, Dr. C. Thamban, Dr. Ravi Bhat, Dr. S. Elain Apshara, Dr. K. P. Chandran, Dr. S. Jayasekhar, Principal Scientists, Dr. Sandip Shil and Dr P. S. Prathibha, Dr Rajkumar, Dr. S. Neenu, Senior Scientists Dr. M. Suchithra, Scientist | 25th Symposium on Plantation Crops (PLACROSYM 25)                                                                                                          | ICAR-IIOPR, Pedavegi. December 12-14, 2023                                                       |
| Dr. K. Sajnanath, SMS                                                                                                                                                                                                                                                                                                                                                                 | Workshop on Natural Farming for KVK Scientists’                                                                                                            | ICAR-KVK-Shivamogga. December 19-23, 2023                                                        |

### XIII. LINKAGES AND COLLABERATIONS

#### International

|                                                           |                                                                                                                                                                               |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| International Coconut Community (ICC), Jakarta, Indonesia | Cooperation between coconut growing countries Coconut genetic resources network, International Coconut Gene Bank for South Asia& Middle East and socio-economic collaboration |
| International Coconut Genetic Resource Network (COGENT)   | Biotechnological activities related to coconut genetic resources                                                                                                              |

#### National

##### ICAR Institutes

|                                                                    |                                                                                                                                                     |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| ICAR-Central Institute of Fisheries Technologies (CIFT), Kochi     | Food processing R&D collaboration                                                                                                                   |
| ICAR-Central Tuber Crop Research Institute, Thiruvananthapuram     | Cassava and coconut based value added products, intercropping of tuber crops in coconut gardens                                                     |
| ICAR-CIARI, Port Blair                                             | Coconut genetic resources collection, conservation and utilization                                                                                  |
| ICAR-CIPHET, Ludhiana                                              | Agricultural pre- and post-harvest machinery                                                                                                        |
| ICAR-CRIDA, Hyderabad                                              | Climate change network and NICRA                                                                                                                    |
| ICAR-Directorate of Cashew Research, Puttur, Karnataka             | Nematological and entomological programmes, cropping systems, value addition training.                                                              |
| ICAR-DMR, Solan                                                    | Agricultural pre- and post-harvest technology development                                                                                           |
| ICAR-Indian Institute of Horticultural Research, Bengaluru         | Phytoplasma disease related studies, varietal screening, cropping systems, agricultural tools and machinery and horticultural IP related activities |
| ICAR-Indian Institute of Oil Palm Research (IIOPR), Pedavegi       | Phytoplasma disease related studies and other common activities under plantation crops sector, tissue culture and biotechnological investigations   |
| ICAR-Indian Institute Spices Research, Kozhikode                   | Cropping system studies, Phytophthora diseases in plantation crops                                                                                  |
| ICAR-National Bureau of Plant Genetic Resources (NBPGR), New Delhi | Germplasm registration and exchange of PGPR, Cryo-preservation of germplasm                                                                         |
| ICAR-NBAIM, Mau                                                    | Microbial research network R&D                                                                                                                      |
| ICAR-NBAIR, Bengaluru                                              | Biological control R&D                                                                                                                              |
| ICAR-NRC for Orchids, Pakyong                                      | Technology Mission for the development of North Eastern states, Intercropping of orchids in coconut/ arecanut multispecies based cropping system    |
| ICAR-Sugarcane Breeding Institute, Coimbatore                      | Cytological studies of in vitro cultures of coconut and arecanut and Food processing R&D                                                            |

##### Others

|                                                                                                        |                                                |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Agricultural Technology Management Agency (ATMA)                                                       | ToT activities                                 |
| All India Radio (AIR), Kannur, All India Radio (AIR), Thiruvananthapuram, Doordarshan (Prasar Bharati) | Transfer of technology programme through media |
| Bannari Institute of Technology, Sathyamangalam, Tamil Nadu                                            | Food technology R & D collaboration            |
| Bidhan Chandra Krishi Vishwavidyalaya, Mohanpur, Nadia, West Bengal                                    | Collaborating centre under AICRP               |
| CAMPCO, Mangalore                                                                                      | Arecanut/ cocoa research and development       |

|                                                                                                        |                                                                                                       |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Central University of Kerala, Kasaragod, Kerala                                                        | R & D collaboration in Biotechnology                                                                  |
| Coconut Development Board (CDB), Kochi                                                                 | Research and development in coconut                                                                   |
| CSIR-NIIST, Trivandrum                                                                                 | Technology programmes                                                                                 |
| DBT, New Delhi                                                                                         | Advancements in Biotechnology and Bioinformatics                                                      |
| Department of Agricultural Development and Farmers Welfare, Govt. of Kerala                            | ToT activities, Plating material production                                                           |
| Directorate of Arecanut and Spices Development (DASD), Kozhikode                                       | Research and development in arecanut                                                                  |
| Directorate of Cashew and Cocoa Research (DCCD), Kochi                                                 | Research and development in cocoa                                                                     |
| District Industries Centre, Kasaragod                                                                  | Capacity building and entrepreneurship programmes                                                     |
| District Panchayath, Alappuzha                                                                         | ToT activities                                                                                        |
| District Panchayath, Kasaragod                                                                         | ICAR-CPCRI, Kasaragod & KVK, Kasaragod ToT activities                                                 |
| DIT, New Delhi                                                                                         | Bioinformatics programmes                                                                             |
| DST, New Delhi                                                                                         | Molecular biology research and women empowerment programmes                                           |
| General Aeronautics Ltd., Bangaluru                                                                    | Unmanned Aerial Vehicle (UAV- Drone) for palm spraying                                                |
| IIPM, Bengaluru                                                                                        | Technology programmes in plantations management                                                       |
| K.S. Hegde Medical Academy, NITTE – Deemed to be University, Deralakatte, Mangalore                    | Deciphering the biochemical machinery underlying <i>in vitro</i> regeneration in coconut and arecanut |
| KCAET, KAU, Tavanur                                                                                    | Technology programmes                                                                                 |
| Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences, Shivamogga               | NSC of arecanut and academic programmes                                                               |
| Kerala Agricultural University                                                                         | R & D collaboration                                                                                   |
| Kerala State Council for Science Technology & Environment (KSCSTE), Thiruvananthapuram                 | R & D collaboration                                                                                   |
| Kerala State Planning Board                                                                            | R & D collaboration                                                                                   |
| KVASU, Wayanad                                                                                         | Technology programmes                                                                                 |
| National Bank for Agriculture and Rural Development (NABARD), Mumbai                                   | Developmental programmes/ demonstrating model coconut clusters in root (wilt) affected areas          |
| National Innovation Foundation (NIF), Gandhinagar, Gujarat                                             | Evaluation of innovator's technology                                                                  |
| National Institute of Food Technology, Entrepreneurship and Management (NIFTEM), Thanjavur, Tamil Nadu | R & D collaboration in PHT                                                                            |
| Nitte University, Mangalore                                                                            | Research and frontier diagnostic collaboration                                                        |
| Onattukara Regional Agricultural Research Station (ORARS), Kerala Agricultural University              | KVK, Alappuzha for NICRA activities                                                                   |
| PPV & FRA, New Delhi                                                                                   | DUS Centre on coconut, arecanut and cocoa                                                             |
| Rashtriya Krishi Vikas Yojana (RKVY)                                                                   | R & D collaboration                                                                                   |
| Tamil Nadu Agricultural University, Coimbatore                                                         | AICRP Centre collaboration                                                                            |
| Tamil Nadu Veterinary and Animal Sciences University, Chennai                                          | AICRP Centre collaboration                                                                            |
| University of Agricultural Sciences, Bangalore                                                         | AICRP Centre collaboration                                                                            |
| Vegetable and Fruit Promotion Council, Kerala                                                          | R & D collaboration                                                                                   |

## Linkage and Collaboration with educational institutes

Alvas College, Moodbidri  
Amrita Viswavidyapeetham, Kollam  
B A M College, Thuruthicad  
BCM College, Kottayam  
CHRIST (Deemed University), Bangaluru  
College of Indegenous Food Technology, Konni  
Dr. YSR Horticulture University, Andhra Pradesh  
Govt. College, Kasaragod  
Hindustan College of Arts & Science, Coimbatore  
Hindustan College of Engineering & Technology, Coimbatore  
Kannur University  
Kelappaji College of Agricultural Engineering and Technology, Tavanur, Mapalluram  
Khansa Women's College, Kumbala, Kasaragod  
KUFOS, Kochi  
Lovely Professional University, Punjab  
M E S M K Mackar Pillay College for Adnanced Studied, Aluva  
Mangalore University, Mangalore  
Mar Athanesious College of Advanced Studies, Thiruvalla  
Marthoma College of Science and Technology, Kollam  
Marthoma College, Thiruvalla  
National Center For Aquatic Animal Health, CUSAT, Kochi  
Nehru Arts and Science College (Autonomous), Coimbatore  
NIFTEM, Haryana  
NMAM Institute of Technology, Nitte, Karkala, Karnataka  
PA First Grade Coollege, Mangalore  
Pazhassi raja College, Pulpally, Wayanad  
PGP College of Arts & Science, Namakkal  
Sacred Heart College Thevara, Kochin  
SAFI Institure of Advanced Study, Vazhayur, Kozhikod  
St. Agnes College Autonomous, Mangalore  
St. Aloysius College (Autonomous), Mangalore  
St. George College, Aruvithura, Kottayam  
St. Mary's College of Wimen, Thiruvalla  
TNAU, Coimbatore  
UAS, Raichur  
University of Calicut, School of Health Science  
Uttar Banga Krishi Vidyalaya, West Bengal  
Vimala College, Thrissur  
Vivekananda College of Engineering & Tech., Puttur, DK, Karnataka

## XIV. RESEARCH PROJECTS

There were 37 institute projects, 40 externally funded projects and two consultancy projects under operation during the period.

### Institute Projects

| Project No. | Name of Project                                                                              | Project Leader/s       | Associate/s                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1000761028  | Genetic resources management in coconut, arecanut and cocoa                                  | V. Niral               | S. Elain Apshara, N.R. Nagaraja, K. Samsudeen, A.K. Sit, L.S. Singh, Alpana Das, R. Sudha, Y. Diwakar, T.N. Ranjini, M. Suchithra, M. Sujithra, V. H. Prathibha, K. B. Hebbar, S.V. Ramesh, P. Subramanian, P. P. Shameena Beegum, M. R. Manikantan, R. Pandiselvam, C. Thamban, T.N. Madhu, E.K. Saneera, R. Thava Prakasa Pandian, M. Chaithra, Anok Uchoi, M. Chaithra, B.A. Jerard and Babli Mog, ICAR-DCR, Puttur |
| 1000761029  | Genetic investigations and breeding in coconut, arecanut and cocoa                           | Regi Jacob Thomas      | K. Samsudeen, V. Niral, R. Sudha, T.N. Ranjini, Y. Diwakar, M. Shareefa, Merin Babu, A. Josephraj Kumar, Sandip Shil, N.R. Nagaraja, A.K. Sit, L.S. Singh, S. Elain Apshara, M. Suchithra, E. K. Saneera, T. N. Madhu, Alpana Das and S. Sumitha.                                                                                                                                                                      |
| 1000761101  | Studies on phenology of coconut genotypes in different climatic zones                        | T. N. Ranjini          | K. Samsudeen, R. Sudha, A.K. Sit, L.S. Singh, Y. Diwakar, and V. Niral                                                                                                                                                                                                                                                                                                                                                 |
| 1000761102  | Studies on impact of high temperature stress on reproductive phase in coconut                | R. Sudha               | Ravi Bhat, K. Samsudeen, S. V. Ramesh, K. B. Hebbar, T. N. Ranjini and V. Niral                                                                                                                                                                                                                                                                                                                                        |
| 1000761103  | Development of palmyrah descriptor and basic studies on growth and development               | S. Sumitha             | B. A. Jerard, S. Elain Apshara, M.R. Manikantan, A. Abdul Haris and Scientist from AICRP on Palms Killikulam, Pandirimamidi and Kondamallepally                                                                                                                                                                                                                                                                        |
| 1000761104  | Phenotyping and gene expression studies for low moisture stress tolerance in cocoa genotypes | M. Suchithra           | S. Elain Apshara and S. V. Ramesh                                                                                                                                                                                                                                                                                                                                                                                      |
| 1000761030  | Biotechnological applications in palms and cocoa                                             | M.K. Rajesh            | V. Aparna, Murali Gopal, Alpana Das, V. Niral, K. Samsudeen, R. Sudha, S.V. Ramesh, S. Elain Apshara, N.R. Nagaraja, V. H. Prathibha, M. Sujithra, T.N. Ranjini, M. Suchithra, K.P. Chandran, Y. Diwakar, Bhavishya, G. Panjavarnam, A. Josephraj Kumar, Merin Babu, R. Thava Prakasa Pandian, Rajkumar and S. Paulraj                                                                                                 |
| 1000761031  | Development of tissue culture techniques in coconut                                          | M. Neema /M. K. Rajesh | V. Aparna, M. Shareefa, Regi Jacob Thomas, V. Niral, K. Samsudeen, Y. Diwakar, Alpana Das, A. Abdul Haris, G. Panjavarnam, K. Nihad and Daliyamol                                                                                                                                                                                                                                                                      |
| 1000761032  | Development of double-stranded RNA based food bait for the suppression of red palm weevil    | M.K. Rajesh            | A. Josephraj Kumar, S.V. Ramesh, M. Sujithra, V. Aparna                                                                                                                                                                                                                                                                                                                                                                |

| Project No. | Name of Project                                                                                                          | Project Leader/s         | Associate/s                                                                                                                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1000763057  | Cropping/ farming approaches for improving soil health and system productivity in coconut, arecanut and cocoa            | P. Subramanian           | Ravi Bhat, V. Selvamani, Surekha, G. Panjavarnam, Alka Gupta, U. K. Priya, Bhavishya, A. Abdul Haris, K. Nihad, Arun Kumar Sit, S. Neenu, AnokUchoi, S. Paulraj, Indhuja S, Jeena Mathew, S. Jayasekhar, C. Thamban and M. Chaitra                                       |
| 1000763058  | Enhancing nutrient and water use efficiency for sustained productivity in coconut, arecanut and cocoa                    | V. Selvamani             | P. Subramanian, Surekha, K. Nihad, Ravi Bhat, S. Neenu, A. Abdul Haris, Jeena Mathew, U. K. Priya, Bhavishya, G. Panjavarnam, S. Indhuja, Arun Kumar Sit, S. Paul Raj, AnokUchoi, Merin Babu, P. Anitha Kumari, R. Thava Prakasa Pandian, Jilu V. Sajan and S. V. Ramesh |
| 1000763055  | Bioresources management in coconut, arecanut and cocoa                                                                   | Alka Gupta               | P. Anithakumari, Murali Gopal, P. Subramanian, Ravi Bhat, Surekha, S. Elain Apshara, Rajkumar, Sandip Shil, S. Neenu, V. Selvamani, U.K. Priya, S. Indhuja, A. Abdul Harris, K. Nihad, Jeena Mathew, Merin Babu, R. Pandiselvam and S.V. Ramesh                          |
| 1000763103  | Bioprospecting of phyllosphere bacteria in coconut and cocoa                                                             | S. Paulraj               | M.K. Rajesh, S. V. Ramesh, R. Thava Prakasa Pandian and M. Sujithra                                                                                                                                                                                                      |
| 1000763106  | Studies on carbon sequestration potential of coconut based fruit cropping system – A strategy to mitigate climate change | G. Panjavarnam           | P. Subramanian, Murali Gopal and S. Neenu                                                                                                                                                                                                                                |
| 1000765039  | Integrated approaches for management of fungal diseases of palms and cocoa                                               | Vinayaka Hegde           | V. H. Prathibha, Daliyamol, R. Thava Prakasa Pandian, M. Chaithra, M. K. Rajesh, Bhavishya, T. N. Madhu, M. Chaithra and Rajkumar                                                                                                                                        |
| 1000765040  | Diagnostics and management of root (wilt) disease (RWD) in coconut and yellow leaf disease (YLD) in arecanut             | Vinayaka Hegde           | K.B. Hebbar, Murali Gopal, A. Josephraj Kumar, M.K. Rajesh, S. V. Ramesh, Merin Babu, Daliyamol, S. Indhuja, R. Thava Prakash Pandian, M. Chaithra, Bhavishya, Jilu V. Sajan, E. K. Saneera, T. N. Madhu, , and M. Chaithra                                              |
| 1000765101  | Elucidating host pathogenic response against fruit rot disease in arecanut                                               | V. H. Prathibha          | N. R. Nagaraja, M. K. Rajesh, R. Thava Prakasa Pandian, Daliyamol and Vinayaka Hegde                                                                                                                                                                                     |
| 1000765102  | Epidemiology and management of <i>Colletotrichum</i> diseases in palms and cocoa                                         | R. Thava Prakasa Pandian | Vinayaka Hegde, Merin Babu, V. H. Prathibha, M. Chaithra, Daliyamol, M. Chaithra, M. K. Rajesh, S. V. Ramesh, Bhavishya, T. N. Madhu, N. R. Nagaraja, Y. Diwakar, and K. P. Chandran                                                                                     |
| 1000765108  | Biological management of <i>Phytophthora</i> diseases in cocoa with indigenous microbes                                  | M. Chaithra              | R. Thava Prakasa Pandian, S. Elain Apshara, Vinayaka Hegde, Bhavishya, T. N. Madhu, E. K. Saneera, and M. Suchithra                                                                                                                                                      |
| 1000765041  | Integrated management of pests and nematodes in palms and cocoa                                                          | A. Josephraj-kumar       | P. S. Prathibha, Rajkumar, M. Sujithra, Jilu V. Sajan, E. K. Saneera, T. N. Madhu, Merin Babu, K. M. Anes, M. Chaithra, R. Thava Prakasa Pandian, Daliyamol, Bhavishya and M. Chaitra                                                                                    |

| Project No. | Name of Project                                                                                                          | Project Leader/s              | Associate/s                                                                                                                                                                           |
|-------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1000765104  | Management of sucking pests using microbial volatile and essential oils in palms and cocoa                               | T. N. Madhu                   | E. K. Saneera, R. Thava Prakasa Pandian, A. Josephraj Kumar, Jilu V. Sajan, M. Sujithra, P. S. Prathibha, Murali Gopal, Y. Diwakar, and M. Chaithra                                   |
| 1000765105  | Characterization of mite complex in arecanut and development of botanical based formulations for their management.       | E. K. Saneera                 | T. N. Madhu, A. Josephraj Kumar, S. V. Ramesh, M. K. Rajesh, N. R. Nagaraja, P. S. Prathibha, M. Chaitra and M. Suchithra                                                             |
| 1000765106  | Biological control of coconut coreid bug, <i>Paradasynusrostratus</i> Distant using egg parasitoid, <i>Anastatus</i> sp. | Jilu V. Sajan                 | A. Josephraj Kumar, P. S. Prathibha, Merin Babu, K. M. Anes and T. N. Madhu                                                                                                           |
| 1000765107  | Studies on host range expansion of red palm weevil and IPM in arecanut                                                   | P. S. Prathibha               | E. K. Saneera, T. N. Madhu, and M. Chaithra                                                                                                                                           |
| 1000766014  | Physiological and biochemical investigations of yield and quality in palms and cocoa                                     | K. B. Hebbar/<br>Murali Gopal | S.V. Ramesh, S. Elain Apshara, S. Neenu, A. K. Sit, N. R. Nagaraja, L. S. Singh, Anok Uchoi, R. Sudha, Bhavishya, Y. Diwakar, R. Pandiselvam, and M. R. Manikantan                    |
| 1000767018  | Mechanization, processing, product diversification and nutraceutical properties                                          | M.R. Manikantan               | R. Pandiselvam, P. P. Shameena Beegum, S. V. Ramesh, Murali Gopal and S. Paul Raj                                                                                                     |
| 1000767027  | Development of process technology for minimal processing of mature coconut kernel and its value added products           | P. P. Shameena Beegum         | K. B. Suresha, from UAS, Bangalore, M. R. Manikantan, R. Pandiselvam and Murali Gopal                                                                                                 |
| 1000767028  | Development and evaluation of rotary dryer cum flavor coating machine for production of coconut chips                    | M. R. Manikantan              | R. Pandiselvam and P. P. Shameena Beegum                                                                                                                                              |
| 1000767029  | Development and characterization of biodegradable plate from cocoa pod husk                                              | R. Pandiselvam                | M. R. Manikantan, S. Elain Apshara and Anjineyulu Kothakota (CSIR-NIIST)                                                                                                              |
| 1000769020  | Technology transfer and co-learning action research approaches                                                           | C. Thamban                    | K. Muralidharan, P. Anithakumari, K. P. Chandran, S. Jayasekhar, P. S. Prathibha, Daliyamol, A. Bhavishya, R. Thava Prakash Pandian, Alpana Das, A. K. Sit, Sandip Shil and K.M. Anes |
| 1000769013  | Socio-economic dimensions and value chain dynamics in policy perspective                                                 | S. Jayasekhar                 | K. P. Chandran, Sandip Shil, K. Muralidharan and C. Thamban                                                                                                                           |
| 1000769019  | Development of statistical and computational techniques for improving research methodology                               | K. P. Chandran                | K. Muralidharan, Sandip Shil, S. Jayasekhar, R. Thava Prakash Pandian, Bhavishya, T. N. Madhu and E. K. Saneera                                                                       |

| Project No. | Name of Project                                                                                                                                        | Project Leader/s | Associate/s                                                                                                                                                                                                                                                                                           |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1000767110  | Use of simulation models for the production system analysis of palms and cocoa                                                                         | M. Sujithra      | K.B. Hebbar, B. A. Jerard, K. P. Chandran, Sandip Shil, K. Muralidharan, A. Josephraj Kumar, V. Selvamani, Daliyamol, T.N. Ranjini, R. Sudha, Surekha, Bhavishya, T. N. Madhu, Anok Uchoi, E.K. Saneera, Vinayaka Hegde, P. S. Prathibha, V. H. Prathibha, R. Thava Prakasa Pandian and Jilu V. Sajan |
| 1000765103  | Studies on acclimatization potential of <i>Phytophthora palmivora</i> under climate change scenario for prediction of disease risk                     | Daliyamol        | Vinayaka Hegde, V. H. Prathibha, , M. K. Rajesh, , M. Sujithra, V. Selvamani, Sandeep Shil, and K. P. Chandran                                                                                                                                                                                        |
| 1000765109  | Development of standard operating procedures (SoPs) for drone based spraying operations in the management of pest and diseases in coconut and arecanut | Daliyamol        | R. Pandiselvam, Vinayaka Hegde, V. H. Prathibha, R Thava Prakasa Pandian, M. Sujithra, Bhavishya<br>K. Manjunatha, , Scientist (Farm Mach. & Power), ICAR-DCR Puttur                                                                                                                                  |
| 1000765110  | Development of standard operating procedures (SoPs) for drone based spraying of nutrients in coconut and arecanut                                      | Bhavishya        | P. Subramanian, Ravi Bhat, Surekha, Jeena Mathew R. Pandiselvam, R. Thava Prakasa Pandian, Vinayaka Hedge, K. Manjunatha, Scientist (Farm Mach. & Power), ICAR-DCR Puttur                                                                                                                             |
| 1000765111  | Development of an AI based Mobile Application for detection and advisory of diseases in Coconut                                                        | Daliyamol        | Vinayaka Hegde, V. H. Prathibha B. A. Jerard, R. Thava Prakasa Pandian Merin Babu, M. Chaithra, M. Chaithra, Md. Ashraful Haque ICAR-IASRI, Shashi Dahiya ICAR-IASRI, New Delhi                                                                                                                       |

### Externally Funded Projects

| Proj. No.  | Proj. Title                                                                                                           | Project Leader   | Associate (s)                                                                                                                                                      |
|------------|-----------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1050761086 | DUS Centre for coconut (Kasaragod)                                                                                    | V. Niral         | K. Samsudeen                                                                                                                                                       |
| 1050761114 | Development of DUS testing criteria and establishment of genebank for arecanut                                        | N. R. Nagaraja   | A.K. Sit and L.S. Singh                                                                                                                                            |
| 2010760004 | Seed Production in Coconut, Arecanut, Cocoa (Under ICAR Project on Seed Production in Agricultural Crops) (Kasaragod) | K. Samsudeen     | V. Niral, M. K. Rajesh, S. Elain Apshara, N.R. Nagaraja, Regi Jacob Thomas, M. Shareefa, K. M. Anes, Y. Diwakar, R. Sudha, T. N. Ranjini, A. K. Sit and L.S. Singh |
| 1050761150 | Identification of drought tolerant coconut palms in Tamil Nadu and utilization for developing adaptive gene pool      | K. Samsudeen     | V. Niral, P. Subramanian and Ragothuman, CDB                                                                                                                       |
| 1050761151 | 'Evaluation of Farmer's 'MM 20' Betel vine variety                                                                    | A. K. Sit        | V. Niral                                                                                                                                                           |
| 1050761156 | Cocoa seed gardens/ clonal orchards – Establishment and enrichment                                                    | S. Elain Apshara | M. Chaithra                                                                                                                                                        |

| Proj. No.  | Proj. Title                                                                                                                                    | Project Leader  | Associate (s)                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------|
| 1050761127 | Commercial production of arecanut – tissue culture of planting material of yellow leaf disease resistant palms and dwarf hybrids               | M. Neema        | M.K. Rajesh and R. Thava Prakasa Pandian               |
| 1050761140 | Evaluation of stage-specific modulation of specific miRNAs during zygotic and somatic embryogenesis in coconut and their functional validation | M.K. Rajesh     | K. Samsudeen and S.V. Ramesh                           |
| 1050761142 | Refinement of <i>in vitro</i> inflorescence culture of coconut for mass multiplication of true-to-type planting material                       | M. Shareefa     | Anitha Karun, Regi Jacob Thomas, M.K. Rajesh           |
| 1050761148 | Establishment of a molecular marker based accreditation laboratory for coconut hybrids                                                         | M.K. Rajesh     | Regi Jacob Thomas and K. Samsudeen                     |
| 1050761157 | Establishment of centre of excellence of biotechnology for plantation crops breeding                                                           | N.R. Nagaraja   | S. Elain Apshara                                       |
| 1050761138 | Participatory Demonstration Plots of Cinnamon intercropping in coconut                                                                         | Ravi Bhat       | P. Subramanian, Surekha                                |
| 1050761143 | Impact of water/ soil moisture conservation to enhancing production of coconut under rainfed and deficit irrigated farms                       | P. Subramanian  | A. C. Mathew, C. Thamban and Ravi Bhat                 |
| 1050761146 | Establishment of advanced facilities for measuring physiological processes                                                                     | K. Nihad        | --                                                     |
| 1050761154 | Technology dissemination through demonstration – Horticulture based crop cafeteria for nutritional and livelihood security of marginal farmers | K. Nihad        | A. Abdul Haris and U. K. Priya                         |
| 1050761155 | Demonstration of Kalpa Vardhini as a component of integrated nutrient management for sustained soil and palm health of coconut                 | Jeena Mathew    | A. Abdul Harris and S. Indhuja                         |
| 1050761159 | <i>In situ</i> structure for recording real-time observation on coconut crown                                                                  | Ravi Bhat       | V. Niral, Vinayaka Hegde, R. Pandiselvam, V. Selvamani |
| 1050761139 | Establishment of FLD on arecanut fruit rot diseases management using mandipropamid fungicide                                                   | V. H. Prathibha | Vinayaka Hegde, R. Thava Prakasa Pandian and Rajkumar  |
| 1050761149 | Diversity analysis of <i>Ganoderma</i> sp. infecting coconut and their eco-friendly management                                                 | Daliyamol       | Vinayaka Hegde, V. H. Prathibha, Merin Babu            |

| Proj. No.  | Proj. Title                                                                                                                                                   | Project Leader  | Associate (s)                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------|
| 1050761160 | Field survey for assessing the recovery of arecanut from leaf spot disease under different management conditions in major arecanut growing areas in Karnataka | Vinayaka Hegde  | V. H. Prathibha, R. Thava Prakasa Pandian, K. P. Chandran, Bhavishya and Gangadhara Naik, KSNUASH Shivamogga  |
| 1050761161 | Establishment of state of the art laboratory for diagnosis of YLD                                                                                             | Vinayaka Hegde  | M. K. Rajesh, Merin Babu, R. Thava Prakasa Pandian, Bhavishya and M. Chaithra                                 |
| 1050761141 | Formulation development of entomopathogenic fungus, <i>Simplicium</i> spp. and its utilization in coconut invasive whiteflies                                 | M. Sujithra     | V. H. Prathibha                                                                                               |
| 1050761152 | Frontline demonstration of biointensive integrated pest management strategies against coconut white grub <i>Leucopholisconiophora</i> Burmeister              | P. S. Prathibha | Jilu V. Sajan, K. M. Anes, , C. Thamban and A. K. Sit                                                         |
| 1050761153 | Demonstration of Effective and Eco-friendly management of white grubs using Entomopathogenic nematodes in arecanut                                            | Rajkumar        |                                                                                                               |
| 1050761166 | Bio-suppression of red palm weevil <i>Rhyncophorusferrugineus</i> Oliv. in coconut using entomopathogenic nematodes                                           | K.M. Anes       |                                                                                                               |
| 1050761147 | Development of biodegradable plate from tender coconut husk                                                                                                   | R. Pandiselvam  | M.R. Manikantan and Anjineyulu Kothakotha, CSIR-NIIST, Thiruvananthapuram                                     |
| 1050761158 | Design, fabrication and field demonstration of portable electrical <i>in-situ</i> cocosap concentrator                                                        | K. B. Hebbar    | R. Pandiselvam                                                                                                |
| 1050761117 | Participatory technology integration to empower and ensure livelihood security of farmers in Alappuzha district(Farmer FIRST Programme)                       | P. Anithakumari | A. Joseph Rajkumar, K. Nihad, M. Shareefa, Jeena Mathew, S. Indhuja, Merin Babu, K. M. Anes and S. Jayasekhar |
| 2010760007 | Intellectual property management and transfer/ commercialization of agricultural technology scheme                                                            | K. Muralidharan | M.R. Manikantan and A.C. Mathew                                                                               |
| 1050761110 | Establishment of Agri-Business Incubation (ABI) Center at ICAR-CPCRI, Kasaragod                                                                               | K. Muralidharan | A. C. Mathew, M. R. Manikantan, R. Pandiselvam, S. Jayasekhar, Murali Gopal, P. P. Shameena Beegum            |
| 1050761135 | Establishing Demonstration Plots on Arecanut Based Multispecies Cropping System (Nutmeg & Cocoa)                                                              | N.R. Nagaraja   | Rajkumar and U.K. Priya                                                                                       |
| 1050761136 | Establishing Demonstration Plots on Arecanut Dwarf Hybrids                                                                                                    | N.R. Nagaraja   | Rajkumar                                                                                                      |

| Proj. No.  | Proj. Title                                                                                                                | Project Leader           | Associate (s)                                                                                                                                                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1050761137 | Farmer Producer Organization with ICAR-CPCRI as Producer Organization Promoting Institution (POPI)                         | P. Anithakumari          | A. Joseph rajkumar, Merin Babu, M. Shareefa, Jeena Mathew, K. Nihad, S. Indhuja                                                                                                                                                                                                                        |
| 1050761145 | Demonstration of integrated management of inflorescence dieback disease in arecanut                                        | R. Thava Prakasa Pandian | Bhavishya, M. Chaithra, T. N. Madhu, Rajkumar, V. H. Pratibha                                                                                                                                                                                                                                          |
| 2022760008 | Profile analysis of farmer producer companies in India                                                                     | C. Thampan (CC-PI)       | Anok Uchoi                                                                                                                                                                                                                                                                                             |
| 1050761162 | Global Value chain analysis of plantation crops of India with special emphasis on food safety standards                    | S. Jayasekhar            | K. Venkateswaran, IIPM, Bengaluru                                                                                                                                                                                                                                                                      |
| 1050761168 | Value chain analysis of arecanut in India                                                                                  | S. Jayasekhar            |                                                                                                                                                                                                                                                                                                        |
| 1050761163 | Study with respect to conversion factor standardisation of coconut                                                         | K.P. Chandran            | K. Muralidharan, S. Jayasekhar, B. A. Jerard, S. Sumitha, P. P. Shameena Beegum, K. M. Anes, K. C. Kiran Kumar (AICRP Palms, Arsikere), C. Sudha Laxmi (AICRP Palms, Aliyarnagar), R. Arunkumar (AICRP Palms, Veppankulam), A. Kireeti (AICRP Palms, Ambjipeta), K. V. Malshe (AICRP Palms, Ratnagiri) |
| 1050761165 | Coconut based enterprises in Lakshadweep Island: Present status, outlook and development of a revamped strategic framework | C. Thampan               | P. N. Anand (KVK, Lakshadweep)                                                                                                                                                                                                                                                                         |
| 1050761167 | Natural farming in coconut based farming system – An analysis of farmers experiences and impact                            | C. Thampan               | P. Subramanian, Ravi Bhat, K. P. Chandran, S. Jayasekhar, S. Neenu, S. Paulraj, Surekha                                                                                                                                                                                                                |

## XV. RESEARCH AND ORGANISATIONAL MANAGEMENT

### 25<sup>th</sup> RAC Meeting

The XXV meeting of the Research Advisory Committee was held at ICAR-CPCRI, Kasaragod on June 9, 2023. Dr. V. A. Parthasarathy FNABS, Former Director, IISR, Kozhikode, Chairman, Dr. Nirmal Babu, Former Director, ICAR-IISR, Kozhikode, Dr. George V. Thomas, Former Director, ICAR-CPCRI, Kasaragod, Dr. V. B Patel, ADG (Fruits and Plantation Crops), ICAR, New Delhi, Dr. H.B. Singh, Former Professor & Head, Plant Pathology, 13/21 Vikas Nagar, P.O. Vikas Nagar, Lucknow-226022 (U.P.), Dr. P. S Negi, Chief Scientist, Fruits and Vegetables Technology, CSIR-CFTRI, Mysuru, Dr. R. N. Padaria, Joint Director (Extension), ICAR-IARI, Pusa, New Delhi, Mr. Krishna Kumar, Managing Director, CAMPCO Ltd., Mangaluru, Shri. Suresh Kumar Shetty, Progressive Farmer, Kumbala, Kasaragod, Shri. Sadananda Shetty, Progressive Farmer, Paivalike, Manjeshwara, Dr. K. Balachandra Hebbar, Director, ICAR-CPCRI, Kasaragod, Members and Dr. Vinayaka Hegde, Head, Div. Crop Protection, ICAR-CPCRI, Kasaragod Member Secretary were present. Dr. V.B. Patel and Dr. R.N. Padaria attended the meeting online through videoconferencing.

All the Heads of the Divisions, the Project Coordinator (Palms), Heads of Regional Stations, Heads of KVKs, programme leaders and other scientists from the headquarters, Regional Station, Vittal and Research Centre, Kidu attended the meeting. Scientists from Kayamkulam, Mohitnagar and Kahikuchi also attended the meeting through videoconferencing.

Dr. K.B. Hebbar, the Director, presented the Institute's achievements and highlighted various aspects including the development of standard operating procedures for drone-based spray of agro-chemicals, non-invasive detection of red palm weevil in palms, progress and roadblocks in tissue culture-based planting material production, technology commercialization and revenue generation. The need for a centralized laboratory facility, one scientist one product, and one technical, one skill was emphasized to achieve the action points as enumerated under eight thematic

areas during the Directors' presentation.

Two publications, 'One CPCRI Multiple Innovations' and 'Ready Reckoner: Coconut, Arecanut and cocoa (English and Malayalam)' were released.



Release of publications during the RAC meeting



The RAC team along with project leaders during the field visit

Later research achievements under the eight mega-project leaders presented the salient achievements of the respective project. Based on discussions RAC recommended future line of research.

### 51<sup>st</sup> Institute Research Committee Meeting

The 51st Annual Institute Research Committee Meeting of the ICAR-CPCRI was held at ICAR-CPCRI Kasaragod during July 10–14, 2023. Progress of 67 different research projects and achievements under crop improvement, biotechnology, crop production, integrated disease management, integrated pest management, physiology & biochemistry, value chain management and social sciences including transfer of technologies were presented by the respective Principal Investigators.

During the meeting, under the chairmanship of Dr. K. B. Hebbar, Director, all the ongoing research

projects, including externally funded projects, under seven research programmes were discussed in detail. Recommendations of RAC and QRT were duly incorporated, while the inputs from farmers and experts from the extension/developmental sectors were given due consideration while finalizing the programme. During the deliberations, 26 new projects were approved for the ensuing year.

Plenary Session of the 51st institute research committee meeting of the ICAR-CPCRI, Kasaragod was held on July 14, 2023. The session was chaired by Dr. K. B. Hebbar, Director, ICAR-CPCRI, Kasaragod. Dr. Hanumanthe Gowda, CCDO, CDB, Kochi, was the chief guest and Shri. Krishna Kumar, MD, CAMPCO, Mangaluru was the guest of honor. Heads of the divisions, Heads of Regional Stations and Research centres and distinguished scientists of ICAR-CPCRI, IMC members, Mr. Sadananda Shetty and Shri. Murali, progressive farmer, Mr. Sathyanarayana Udupa, Kundapur and Shri. Varadharajan, Coimbatore, entrepreneurs, Mrs. Jyothi, DDA (NHM) and Mrs. Anitha, DDA, have also attended the session.

Based on the recommendations of the House and the Research Advisory, activities were planned according to the identified researchable areas mentioned below and technical programme under

each project for the year 2023-24 were formulated.

### Institute Management Committee

The IMC had met twice during the year on February 23, 2023 and September 22, 2023 at CPCRI, Kasaragod under the chairmanship of Dr. K. B. Hebbar, Director.

The IMC of the Institute has the following members:

1. Dr. K. Balachandra Hebbar, Director, ICAR-CPCRI, Kasaragod
2. Dr. V.B. Patel, Assistant Director General (F&PC), ICAR, New Delhi
3. Shri. K. Sadananda Shetty
4. Shri. Suresh Kumar Shetty
5. Chief Administrative Officer, ICAR-CPCRI, Kasaragod

During the first meeting Dr. V. Niral, Principal Scientist, Dr. M.R. Manikantan, Principal Scientist and Shri K.R. Nithianandan, FAO I/c, were the special invitees. During the second meeting, Dr. Vikramaditya Pandey represented the ICAR and Shri Ram Avtar Parashar was the special invitee. The IMC has overviewed the institutional management aspects and financial management aspects and has taken appropriate decisions for implementation.



IRC meeting at CPCRI Kasaragod

## XVI. INTELLECTUAL PROPERTY AND TECHNOLOGY MANAGEMENT

During the year, the Institute Technology Management Committee met five times to fix technology transfer fees, approving registration of incubatees of Kalpa ABI and selection of best entrepreneurs from different states who are successful in using the Institute technologies. New technologies recommended for commercialization are: arecanut variety Shatamangala; Kalpa Eco Pot (for nurseries); 'Kalpa EPN, *Heterorhabditis indica* (CPCRI- HI1) Aqua suspension; and

Motorized Tender Coconut Punching Machine. A total of 54 MoAs were signed for technology transfer to realize a revenue of Rs.22,30,000/-. Institute provided 26 consultancy/contractual services and earned Rs.77,700/-. The Dream Big-Kalpa, Institute-Industry Interface meeting was conducted on August 2, 2023 in which 60 people participated. Awareness programmes on IP management were conducted for students from different universities visited the Institute.

### Technologies certified by ICAR

| Sl. No. | Technology                                                                                                  | Developers                                                                                                                                               |
|---------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | A set up to study the physiological impact of high temperature stress on the progamic phase of coconut.     | Dr. K.B. Hebbar, Dr. S.V. Ramesh, Dr. M. Sujithra Dr. V. Niral and Shri G. S. Hareesh                                                                    |
| 2       | Controlled hydroponics to study the physiological and biochemical basis of salinity tolerance in coconut    | Dr. K.B. Hebbar, Dr. S.V. Ramesh and Dr. V. Selvamani                                                                                                    |
| 3       | Kalpa Vardhini, nutrient mixture for adult coconut palms                                                    | Dr. Jeena Mathew, Dr. A. Abdul Haris, Dr. Ravi Bhat and Dr. P. Anithakumari                                                                              |
| 4       | Foam mat drying of coconut milk                                                                             | Dr. M.R. Manikantan, Dr. P. P. Shameena Beegum, Dr. R. Pandiselvam and Dr. S. Jayashekhar                                                                |
| 5       | Jaggery infused osmo-dehydrated coconut chips                                                               | Dr. M.R. Manikantan Dr. M. Pravitha, Dr. P. P. Shameena Beegum and Dr. R. Pandiselvam                                                                    |
| 6       | Design and development of portable pyrolysis system for the production of biochar from tender coconut husk  | Dr. A.C. Mathew, Dr. R. Pandiselvam, Dr. M.R. Manikantan Dr. S.V. Ramesh and Dr. S. Neenu                                                                |
| 7       | Heliconia "Iris"-A potential intercrop for shaded plantations                                               | Dr. K. Nihad, Dr. V. Krishna Kumar, Dr. A. Abdul Haris and Dr. Ravi Bhat                                                                                 |
| 8       | Technology on mass production of <i>Trichoderma harzianum</i> (CPTD28) using Arecanut leaf sheath           | Dr. V.H. Prathibha, Dr. Daliyamol and Dr. Vinayaka Hegde                                                                                                 |
| 9       | Neera honey infused extrudates                                                                              | Dr. R. Pandiselvam, Dr. M.R. Manikantan, Dr. P. P. Shameena Beegum Dr. S.V. Ramesh, Dr. S. Neenu, Dr. Murali Gopal Dr. A. C. Mathew and Dr. K. B. Hebbar |
| 10      | A preservation process of microwave treatment of coconut inflorescence Sap (Kalparasa®)                     | Dr. R. Pandiselvam, Dr. K.B. Hebbar, Dr. M .R. Manikantan Dr. S.V. Ramesh and Dr. P. P. Shameena Beegum                                                  |
| 11      | Thermal treatment of tender coconut water using transient computer simulation                               | Dr. R. Pandiselvam, Dr. M.R. Manikantan Dr. S.V. Ramesh and Dr. P.P. Shameena Beegum                                                                     |
| 12      | Facile coconut inflorescence sap mediated synthesis of silver nanoparticles and its anti-microbial property | Dr. M. K. Rajesh, Dr. K. S. Muralikrishna, Dr. Anitha Karun and Dr. K. B. Hebbar                                                                         |

|    |                                                                                                       |                                                                                                                                                                                                                                       |
|----|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | Single-nucleotide polymorphism (SNP) markers through genotyping-by-sequencing (GBS) approach          | Dr. M. K. Rajesh, Dr. Sunil Shivaji Gangurde, Dr. M.K. Pandey, Dr. V. Niral Dr. R. Sudha, Dr. B.A. Jerard, Dr. Ganesh N. Kadke Dr. A.A. Sabana, Dr. K.S. Muralikrishna, Dr. K. Samsudeen Dr. Anitha Karun and Dr. T.S. Keshava Prasad |
| 14 | Developed noncoding RNA derived simple sequence repeat markers in coconut ( <i>Cocos nucifera</i> L.) | Dr. M. K. Rajesh, Dr. A. A. Sabana, Dr. Ginny Antony, Dr. K. P. Gangaraj, Dr. V. Niral, Dr. R. Sudha and Dr. B. A. Jerard                                                                                                             |
| 15 | Kalpa EPN, <i>Steinernema carpocapsae</i> (CPCRI-SC1) aqua suspension technology                      | Dr. Rajkumar, Dr. M. Sujithra, Dr. M. K. Rajesh, Dr. V. H. Pratibha Dr. Surekha, Dr. N. R. Nagraja, Dr. A. Joseph Rajkumar and Dr. Vinayaka Hegde                                                                                     |
| 16 | Trait identification and quality standards of testa oils derived from diverse coconut genotypes       | Dr. S.V. Ramesh, Dr. K.B. Hebbar, Dr. P. P. Shameena Beegum, Dr. R. Pandiselvam Dr. V. Niral, Dr. M.R. Manikantan, Smt. Sugatha Padmanabhan, Dr. S. Neenu, Dr. Sandip Shil, Dr. A. N. Lokesh and Dr. K.S. Shivashankara               |
| 17 | Quality standards of variants of virgin coconut oil                                                   | Dr. S.V. Ramesh, Dr. R. Pandiselvam, Dr. P. P. Shameena Beegum, Dr. S. Neenu, Dr. Sandip Shil, Dr. M.R. Manikantan, Dr. A.C. Mathew and Dr. K.B. Hebbar                                                                               |
| 18 | Coconut infused dark chocolates                                                                       | Dr. P. P. Shameena Beegum, Dr. M.R. Manikantan, Dr. R. Pandiselvam, Dr. S.V. Ramesh Dr. S. Neenu, Dr. Alka Gupta and Dr. K. B. Hebbar                                                                                                 |

### Consultancy/ Analytical Services

About forty six sets of arecanut samples received from the different stakeholders Customs, Department of Horticulture and Food Processing of Assam etc were analysed for its microbial quality as per FSSAI guidelines.

| Sl. No. | Name of Technology/ Know-How/ Service Provided                           | Name of Contracting Party                                                                | Date                     | Revenue Earned (₹) |
|---------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------|--------------------|
| 1       | Investigation report of dried arecanut 1 No. samples                     | Deputy Commissioner CUSTOMS, ICD, Kanpur                                                 | 10.01.2023               | 1180.00            |
| 2       | Analysis of Bone meal ENKJ 02/22-23                                      | Agriculture Officer, Krishi Bhavan, Enmakaje, P.O. Perla, Kasaragod - 671552             | 19.01.2023               | 3750.00            |
| 3       | Investigation report of dried arecanut 2 Nos. samples                    | District Agri. Officer, Karbi Anglong, Diphu, Assam                                      | 02.02.2023               | 2360.00            |
| 4       | Investigation report of dried arecanut 1 No. samples                     | District Agri. Officer, Morigaon, , Assam                                                | 02.02.2023               | 1180.00            |
| 5       | Visit by Dr. P. Subramanian, Dr. A. Joseph Rajkumar and Dr. V. Selvamani | M/s. DeeJay Farm DeeJay Coconut Farm Pvt. Ltd. (Balli Farm-GOA) Tiloi, Balli, Goa-403703 | 02.02.2023 to 21.02.2023 | 11800.00           |
| 6       | Investigation report of dried arecanut 3 Nos. samples                    | District Agri. Officer, Dhubri, Assam                                                    | 06.02.2023               | 3540.00            |
| 7       | Investigation report of dried arecanut 1 No. samples                     | District Agri. Officer, Kamrup, Ulubari, Assam                                           | 06.03.2023               | 1180.00            |
| 8       | Investigation report of dried arecanut 2 Nos. samples                    | Superintendent of Police, Chirang (BTAD), Kajalgaon, Assam                               | 22.03.2023               | 2360.00            |

|    |                                                       |                                                                                                                                              |            |           |
|----|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| 9  | Investigation report of dried arecanut 2 Nos. samples | District Agri. Officer, Dhubri                                                                                                               | 02.05.2023 | 2360.00   |
| 10 | Analysis of soil fertility of umikkari and soil       | Mr. Deepakumar M, Sreelakam, Vidyanagar, Near Civil Station, Kasaragod                                                                       | 16-05-2023 | 3750.00   |
| 11 | Investigation report of dried arecanut 1 No. samples  | District Agri. Officer, Dhubri                                                                                                               | 25.05.2023 | 1180.00   |
| 12 | Analysis of soil fertility                            | Mr. Jomon Mathew, Kuzhikkattu, Chullikkara                                                                                                   | 02.06.2023 | 7500.00   |
| 13 | Analysis of Bio ash Coffee Ash                        | Mr. K.M. Ali, Chattanchal, Thekkel, P.O. Thekkel, Kasaragod 671541.                                                                          | 27.06.2023 | 2500.00   |
| 14 | Analysis of dried arecanut samples                    | The Asi, Dipen Bhyan, GRPS, Guwahati- 01                                                                                                     | 06.07.2023 | 1180.00   |
| 15 | Analysis of agri lime Agri waste                      | Mr. Surendran K, Partner, VAMCO Industries, Thekkil Post, Kasaragod.                                                                         | 17.07.2023 | 2500.00   |
| 16 | Analysis of calcium hydroxide in Lime                 | Arecanut Research and Development Foundation (R) (ARDF), Varanashitowers, Mission-Street, Mangalore, D.K. – 575001, Karnataka                | 08.08.2023 | 500.00    |
| 17 | Analysis of soil fertility of 4 samples               | Ganesha V, Mohtisham Complexes Pvt. Ltd., 6 <sup>th</sup> & 7 <sup>th</sup> floor, Empire Building, M.G. Road, Mangalore – 575003, Karnataka | 08.09.2023 | 10000     |
| 18 | Investigation report of dried arecanut 2 Nos. samples | The Sub -Divisional Agri. Officer, Govt. of Assam, Pathsala                                                                                  | 11.07.2023 | 2360.00   |
| 19 | Investigation report of dried arecanut 1 No. samples  | District Agri. Officer, Kamrup, Ulubari, Assam                                                                                               | 11.07.2023 | 1180.00   |
| 20 | -do-                                                  | District Agri. Officer, Kamrup, Ulubari, Assam                                                                                               | 21.07.2023 | 1180.00   |
| 21 | -do-                                                  | District Agri. Officer, Kamrup, Ulubari, Assam                                                                                               | 24.07.2023 | 1180.00   |
| 22 | -do-                                                  | Superintendent of Police, Govt. of Assam, Nalbari                                                                                            | 24.07.2023 | 1180.00   |
| 23 | -do-                                                  | District Agri. Officer, Kamrup, Ulubari, Assam                                                                                               | 09.08.2023 | 9440.00   |
| 24 | -do-                                                  | District Agri. Officer, Kamrup, Ulubari, Assam                                                                                               | 09.08.2023 | 1180.00   |
| 25 | -do-                                                  | District Agri. Officer, Kamrup, Ulubari, Assam                                                                                               | 30.08.2023 | 1180.00   |
|    |                                                       | Total                                                                                                                                        |            | 77,700.00 |

## Commercialization of Technology

During 2023, total 24 technologies were commercialised by the Institute to 44 entrepreneurs through licensing and an amount of Rs. 22.30 lakhs have been collected as technology transfer fees.

| Sl. No. | Name of Technology Commercialized                                                                | Date of MOA | Value (In INR) | To Whom Commercialized                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------|-------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Shatamangala                                                                                     | 02.09.2023  | 2,00,000       | Mr. Gangadhara Alva, Integrated Rural Agri Nursery, Badiadka – 671 551, Kasaragod Dt., Kerala.                                      |
| 2       | -do-                                                                                             | 02.09.2023  | 2,00,000       | Mrs. Vanitha Yadukumar, Yashaswi Nursery, Kemminje – 574 202, Puttur, Dakshina Kannada Dt., Karnataka.                              |
| 3       | Varieties – Kalpa Samrudhi and Chandra Sankara and Aqua formulation of EPN Kalpa EPN (CPCRI-SC1) | 17.07.2023  | 1,15,000       | Mr. Ram Laxman Polabhai and Mr. Jotva Bharat Hajabhai, Mahakali Coconut Farm, Veraval, Dist. Gir-Somnath-362265, Gujarat.           |
| 4       | Kalpa EPN, <i>Heterorhabditis indica</i> (CPCRI- HI1) Aqua suspension – Nematode bio-agent       | 18.08.2023  | 25,000         | Mr Deepak Bhat, Citron Field Agro, Kanchigadde village, Kodnagadde Post, Sirsi Taluk, Uttara Kannada, Karnataka – 581336.           |
| 5       | Aqua suspension of EPN, Kalpa EPN (CPCRI- SC1)                                                   | 02.09.2023  | 10,000         | ICAR Krishi Vigyan Kendra, Kanhirangad (PO), Taliparamba, Kannur – 670142, Kerala.                                                  |
| 6       | -do-                                                                                             | 19.01.2023  | 5,000          | Mr. Krishna Kumar, Pollachi Farm, Belathur, Talavady, Sathyamangalam, Erode - Dist, Tamil Nadu, PIN-638461.                         |
| 7       | Snowball Tender Nut Machine                                                                      | 25.01.2023  | 2,500          | Mr. Jyothish K.M., Karithakarakuzhiyil (house), Akachery P.O., Menachody, Kannur – 670650, Kerala.                                  |
| 8       | -do-                                                                                             | 28.03.2023  | 2,500          | Mr Akash Kumar Shakya, M/s. Fastest Service, Bhawapur, Allahabad - 211003, Uttara Pradesh.                                          |
| 9       | -do-                                                                                             | 22.09.2023  | 2,500          | Mr. Kiran D.B., Managing Director, M/s Nutriwadi Natural Foods International, Lekshmi Nivas, Mundur (PO), Thrissur – 680541, Kerala |
| 10      | -do-                                                                                             | 04.10.2023  | 2,500          | M/s Coconut Delight, Hanumakonda, Warangal Urban, Telangana - 506001.                                                               |
| 11      | -do-                                                                                             | 31.10.2023  | 2,500          | Mr Vasamsetti Lakshmi Prsanneswara Rao, NedunuruPedapalem, Dr. B.R. Ambedkar KonaseemaDist, Andhra Pradesh - 533241                 |
| 12      | Technical knowhow of production of virgin coconut oil                                            | 13.02.2023  | 40,000         | Mr. Rajakumar M., Kalluvila, Neyyattinkara P.O., Thiruvananthapuram - 695121, Kerala.                                               |
| 13      | -do-                                                                                             | 15.03.2023  | 40,000         | M/s Organo resource management, Kizhsery building, Near Municipal Office, Perinthalmanna, Malappuram Dist, Kerala.                  |

|    |                                                        |            |        |                                                                                                                                                                                  |
|----|--------------------------------------------------------|------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14 | -do-                                                   | 03.05.2023 | 40,000 | Mrs. Renjini P Nambiar,<br>Kaneeladukkam, Karivedakam P O,<br>Kasaragod, Kerala-671541.                                                                                          |
| 15 | -do-                                                   | 15.11.2023 | 40,000 | M/s Hotballoon Foods Pvt. Ltd.,<br>Panangangara, Puzhakkattiri (P.O.),<br>Pin – 679321, Malappuram Dist,<br>Kerala                                                               |
| 16 | Coconut Chips                                          | 20.02.2023 | 25,000 | The General Manager, Eramala<br>Service Co-Operative Bank Ltd.,<br>Orkkatteri P.O., Vadakara, 673501,<br>Kozhikode, Kerala.                                                      |
| 17 | -do-                                                   | 17.04.2023 | 25,000 | Mr. N.P. Subramaniam,<br>M/s. Protaman Foods,<br>Parameshwaranpalayam,<br>Coimbatore - 641109, Tamil Nadu.                                                                       |
| 18 | -do-                                                   | 22.05.2023 | 25,000 | Lakshadweep Khadi and Village<br>Industries Board, Kavaratti,<br>Kavaratti P.O. – 682555, U.T. of<br>Lakshadweep.                                                                |
| 19 | Technical Knowhow of<br>production of Coconut<br>Chips | 26.07.2023 | 25,000 | Mr. Peter Paul Leonard Irudayaraj,<br>Kaspapuram, Padavettammankoil<br>Street, Chennai- 600126,<br>Tamilnadu.                                                                    |
| 20 | -do-                                                   | 02.09.2023 | 25,000 | Dr. Venugopal Nainala, Managing<br>Director, Narial Bharat Nutra<br>Producer Company Limited,<br>Mamidikuduru, Dr BR Ambedkar<br>Konaseema district, Andhra<br>Pradesh – 533247. |
| 21 | -do-                                                   | 15.11.2023 | 25,000 | M/s Hotballoon Foods Pvt. Ltd.,<br>Panangangara, Puzhakkattiri (P.O.),<br>Pin – 679321, Malappuram Dist,<br>Kerala                                                               |
| 22 | Preservation of<br>Carbonated Tender<br>Coconut Water  | 15.03.2023 | 25,000 | M/s Organo resource management,<br>Kizhsery building, Near Municipal<br>Office, Perinthalmanna,<br>Malappuram Dist, Kerala.                                                      |
| 23 | -do-                                                   | 22.05.2023 | 25,000 | Lakshadweep Khadi and Village<br>Industries Board, Kavaratti,<br>Kavaratti P.O. – 682555, U.T. of<br>Lakshadweep.                                                                |
| 24 | -do-                                                   | 22.06.2023 | 25,000 | M/s Maniyur Food Park<br>LLP, Cherandathur P.O., Thiruvallur<br>Via, Kozhikode Dist - 673541,<br>Kerala.                                                                         |
| 25 | -do-                                                   | 20.07.2023 | 25,000 | Mr Abdul Rasheed C, Proprietor,<br>E4 Enterprises, Kalady P.O,<br>Malappuram, Kerala – 679582.                                                                                   |
| 26 | -do-                                                   | 02.09.2023 | 25,000 | Mr. Abdul Sawad T.A., Managing<br>Director of Redhive ODM<br>LLP, Apsara Tiger Garden<br>Business Centre, KPR Rao Road,<br>Kasaragod-671121, Kerala.                             |
| 27 | -do-                                                   | 14.09.2023 | 25,000 | M/s Malabar Farmer<br>Producer Company Limited.,<br>Kunnathupalam, P.O.<br>Guruvayoorappan College,<br>Kozhikode, PIN – 673014, Kerala                                           |

|    |                                                                                           |            |          |                                                                                                                                     |
|----|-------------------------------------------------------------------------------------------|------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|
| 28 | -do-                                                                                      | 07.12.2023 | 25,000   | Mr. Najeed Muhammed, United Agro Foods & Beverages, Kinfra Textile Centre, Pallivayal (PO), Taliparamba – 670142, Kerala.           |
| 29 | Foam mat dried coconut milk powder                                                        | 20.02.2023 | 10,000   | The General Manager, Eramala Service Co-Operative Bank Ltd., Orkkatteri P.O., Vadakara, 673501, Kozhikode, Kerala.                  |
| 30 | -do-                                                                                      | 08.03.2023 | 10,000   | Mr. Ameen Ashraf T.P., Maataste Food Products LLP, KINFRA, Nadukani, Kannur, PIN- 686691, Kerala.                                   |
| 31 | -do-                                                                                      | 15.03.2023 | 10,000   | M/s Organo resource management, Kizhsery building, Near Municipal Office, Perinthalmanna, Malappuram Dist, Kerala.                  |
| 32 | -do-                                                                                      | 22.05.2023 | 10,000   | Lakshadweep Khadi and Village Industries Board, Kavaratti, Kavaratti P.O. – 682555, U.T. of Lakshadweep.                            |
| 33 | -do-                                                                                      | 25.07.2023 | 10,000   | Mrs. Manjusha. T, Dakshin Food Products, Chelannur, Eruvallur P.O, Kozhikode-673616, Kerala.                                        |
| 34 | Collection of fresh and hygienic Kalparasa and production of natural coconut sugar        | 20.02.2023 | 1,00,000 | Global Coconut Farmers Producer Company Limited, 39, Chettipalayam Road, Palladam-641664, Tirupur District, Tamil Nadu.             |
| 35 | -do-                                                                                      | 01.05.2023 | 1,00,000 | Yadiyuru Farmer Producer Company Ltd, Yadiyuru, Kunigal, Tumkur District – 572142, Karnataka.                                       |
| 36 | -do-                                                                                      | 02.09.2023 | 1,00,000 | YGP Coconut Farmers Producer Company Ltd, Avaragere (Village), Chitradurga Road, Davangere, Karnataka-577003.                       |
| 37 | -do-                                                                                      | 02.11.2023 | 1,00,000 | M/s SVM Organic Palm Products, Bhattiprolu, BapatlaDist, PIN – 522256, Andhra Pradesh.                                              |
| 38 | Preservation protocol for trimmed tender coconut                                          | 22.09.2023 | 15,000   | Mr. Kiran D.B., Managing Director, M/s Nutriwadi Natural Foods International, Lekshmi Nivas, Mundur (PO), Thrissur – 680541, Kerala |
| 39 | -do-                                                                                      | 04.10.2023 | 15,000   | M/s Coconut Delight, Hanumakonda, Warangal Urban, Telangana - 506001.                                                               |
| 40 | Technology for mass production of <i>Trichoderma harzianum</i> using arecanut leaf sheath | 27.02.2023 | 5,000    | Raitajanya Farmer Producer Company Limited, Alangar, Moodubidire – 574227, Dakshina Kannada (dist), Karnataka.                      |
| 41 | -do-                                                                                      | 10.11.2023 | 5,000    | Ms Navaneetha Gowda K.N., Ivarnadu post, Sullia taluk, 574239, Dakshina Kannada (dist), Karnataka                                   |
| 42 | Trichoderma Coir Pith Cake                                                                | 20.09.2023 | 5,000    | ICAR Krishi Vigyan Kendra, KCAET Campus, Tavanur, Malappuram – 679573, Kerala.                                                      |

|    |                                                                             |            |                          |                                                                                                                                        |
|----|-----------------------------------------------------------------------------|------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 43 | -do-                                                                        | 07.12.2023 | 5,000                    | Krishi Vigyan Kendra, ICAR-IISR, Peruvannamuzhi P.O., Kozhikode-673528, Kerala                                                         |
| 44 | Desiccated coconut incubation                                               | 18.05.2023 | 2,000                    | Mr Abdul Sawad T A, Managing Director of Redhive ODM LLP, Apsara Tiger Garden Business Centre, KPR Rao Road, Kasaragod-671121, Kerala. |
| 45 | Preservation of coconut gratings                                            | 12.09.2023 | 10,000                   | Mrs. Sunitha D.V., Director, M/s Cocopalms Pixel Pvt. Ltd., Rangapura, Tumkur, Karnataka - 572106.                                     |
| 46 | Knowhow for production of bean to bite chocolate                            | 17.08.2023 | 10,000                   | Mr. Sivasubramaniam Umapathy, Umapathy Poultry Farms, Venkitapuram Post, Palladam (TK), Tirupur, Tamil Nadu 641664.                    |
| 47 | -do-                                                                        | 15.11.2023 | 10,000                   | M/s Hotballoon Foods Pvt. Ltd., Panangangara, Puzhakkattiri (P.O.), Pin – 679321, Malappuram Dist, Kerala                              |
| 48 | Kalpa Vardhini                                                              | 05.06.2023 | 10,000                   | ICAR Krishi Vigyan Kendra, Mitraniketan, Vellanad, Thiruvananthapuram-695543, Kerala.                                                  |
| 49 | -do-                                                                        | 21.08.2023 | 10,000                   | Green Kollam Farmers Producer Company Limited, Curzon Nagar 114, Kollam – 691013                                                       |
| 50 | Kalpa Poshak                                                                | 22.05.2023 | 10,000                   | Odanadu Farmer Producer Company Ltd., Alappuzha, Kerala.                                                                               |
| 51 | Kalpa Soil Care - Coir pith composting (urea free) technology               | 30.06.2023 | 25,000                   | M/s Stars of Pantnagar Pvt. Ltd., Vikas Khand Gomti Nagar, Lucknow, Uttar Pradesh – 226010.                                            |
| 52 | Kalpa Eco-pots                                                              | 18.08.2023 | 15,000                   | Mr. Gangadhara Alva, Integrated Rural Agri Nursery, Badiadka – 671 551, Kasaragod Dt., Kerala.                                         |
| 53 | Modified ground pollination technique for hybrid seed production in coconut | 03.11.2023 | 25,000                   | M/s Silpa Clonal Nursery, Pendalanka (village), Bapatla Dist, PIN – 522257, Andhra Pradesh.                                            |
| 54 | Matured coconut water based value added products                            | 13.02.2023 | Complementary with VCO   | Mr. Rajakumar M., Kalluvila, Neyyattinkara P.O., Thiruvananthapuram - 695121, Kerala.                                                  |
| 55 | -do-                                                                        | 20.02.2023 | Complementary with chips | The General Manager, Eramala Service Co-Operative Bank Ltd., Orkkatteri P.O., Vadakara, 673501, Kozhikode, Kerala.                     |
| 56 | -do-                                                                        | 15.03.2023 | Complementary with VCO   | M/s Organo resource management, Kizhsery building, Near Municipal Office, Perinthalmanna, Malappuram Dist, Kerala.                     |
| 57 | -do-                                                                        | 17.04.2023 | Complementary with VCO   | Mr. N.P. Subramaniam, M/s. Protaman Foods, Parameshwaranpalayam, Coimbatore - 641109, Tamil Nadu.                                      |
| 58 | -do-                                                                        | 03.05.2023 | Complementary with VCO   | Mrs. Renjini P Nambiar, Kaneeladukkam, Karivedakam P O, Kasaragod, Kerala-671541.                                                      |
| 59 | -do-                                                                        | 22.05.2023 | Complementary with chips | Lakshadweep Khadi and Village Industries Board, Kavaratti, Kavaratti P.O. – 682555, U.T. of Lakshadweep.                               |

|    |      |                  |                          |                                                                                                                                                                   |
|----|------|------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 60 | -do- | 26.07.2023       | Complementary with chips | Mr. Peter Paul Leonard Irudayaraj, Kaspapuram, Padavettammankoil Street, Chennai- 600126, Tamilnadu.                                                              |
| 61 | -do- | 02.09.2023       | Complementary with chips | Dr. Venugopal Nainala, Managing Director, Narial Bharat Nutra Producer Company Limited, Mamidikuduru, Dr BR Ambedkar Konaseema district, Andhra Pradesh – 533247. |
| 62 | -do- | 15.11.2023       | Complementary with chips | M/s Hotballoon Foods Pvt. Ltd., Panangangara, Puzhakkattiri (P.O.), Pin – 679321, Malappuram Dist, Kerala                                                         |
| 63 | -do- | 15.11.2023       | Complementary with VCO   | M/s Hotballoon Foods Pvt. Ltd., Panangangara, Puzhakkattiri (P.O.), Pin – 679321, Malappuram Dist, Kerala                                                         |
|    |      | Total Amount Rs. | 22,30,000                |                                                                                                                                                                   |

### Sales of technology products including planting materials

| Sl. No. | Item                                                                         | Quantity (nos.) | Amount (Rs.)        |
|---------|------------------------------------------------------------------------------|-----------------|---------------------|
| 1       | Coconut seedlings (hybrids)                                                  | 12,531          | 35,08,680.00        |
| 2       | Coconut seedlings (tall varieties)                                           | 14,664          | 17,59,680.00        |
| 3       | Coconut seedlings (dwarf varieties)                                          | 4,293           | 9,01,530.00         |
| 4       | Coconut seed nuts                                                            | 2,015           | 1,61,520.00         |
| 5       | Coconut leaf vermicompost                                                    | 668             | 10,230.00           |
| 6       | Earthworms                                                                   | 28,300          | 20,376.00           |
| 7       | <i>Trichoderma</i>                                                           | 25Kg            | 2,500.00            |
| 8       | Trichoderma enhanced coir pith cake                                          | 100             | 500.00              |
| 9       | Polybag seedlings                                                            | 1,180           | 2,20,680.00         |
| 10      | Extension publications                                                       | 28              | 2,456.00            |
| 11      | Arecanut seedlings                                                           | 12,144          | 5,46,480.00         |
| 12      | Areca seednuts                                                               | 51,040          | 6,12,480.00         |
| 13      | Areca seednuts                                                               | 2kg             | 180.00              |
| 14      | Cocoa seedlings                                                              | 30,460 nos.     | 3,04,600.00         |
| 15      | Cocoa pods                                                                   | 113             | 3,955.00            |
| 16      | Cocoa grafts                                                                 | 950             | 28,500.00           |
| 17      | Other planting materials                                                     | 2,607           | 95,800.00           |
| 18      | Coconuts (commodity sale)                                                    | 50,918 nos.     | 5,33,673.00         |
| 19      | Arecanuts (commodity sale)                                                   | 4,572 kg        | 2,05,609.00         |
| 20      | Pepper (commodity sale)                                                      | 524 kg          | 60,260.00           |
| 21      | Cashew Auction (Trees)                                                       | 370             | 7,376.00            |
| 22      | Kalpa EPN (CPCRI-SC1) aqua suspension                                        | 125             | 14,750.00           |
| 23      | Kalpa EPN (CPCRI-HI1 CPCRI-SC1) infected <i>Galleria</i> cadaver formulation | 190             | 1,121.00            |
| 24      | Other farm produce                                                           | 1,00,096.25     | 63,749.30           |
| 25      | Others                                                                       |                 | 14,500.00           |
|         | <b>Total</b>                                                                 |                 | <b>90,81,185.30</b> |

## XVII. PERSONNEL

### SCIENTIFIC STAFF

| Sl. No.                             | Name                            | Designation                                                                       |
|-------------------------------------|---------------------------------|-----------------------------------------------------------------------------------|
| <b>Headquarters, Kasaragod</b>      |                                 |                                                                                   |
| 1                                   | Dr. K.B. Hebbar                 | Director                                                                          |
| 2                                   | Dr. B. Augustine Jerard         | PC(Palms) (w.e.f. 08.05.2023)                                                     |
| 3                                   | Dr. Vinayaka Hegde              | HoD (Crop Protection) (w.e.f. 14.07.2023)                                         |
| 4                                   | Dr. P. Subramanian              | HoD (Crop Production) (w.e.f. 14.07.2023)                                         |
| 5                                   | Dr. Murali Gopal                | HoD (PB& PHT) (w.e.f. 14.07.2023)                                                 |
| 6                                   | Dr. (Mrs.) V. Niral             | HoD (Crop Improvement) (w.e.f. 14.07.2023)                                        |
| 7                                   | Dr. K. Ponnuswamy               | HoD (Social Science) (w.e.f. 22.12.2023)                                          |
| 8                                   | Dr. K. Muralidharan             | AHoD (Social Science) (up to 21.12.2023) /Principal Scientist (Agril. Statistics) |
| 9                                   | Dr. H.P. Maheswarappa           | Principal Scientist (Agronomy) (on deputation)                                    |
| 10                                  | Dr. Ravi Bhat                   | AHoD (Crop Production) (up to 13.07.2023) /Principal Scientist (Agronomy)         |
| 11                                  | Dr. C. Thamban                  | Principal Scientist (Agril. Extension)                                            |
| 12                                  | Dr. (Mrs.) Alka Gupta           | Principal Scientist (Agril. Microbiology)                                         |
| 13                                  | Dr. A.C. Mathew                 | Principal Scientist (SWC) (Engg.) (Retired on 31.07.2023)                         |
| 14                                  | Dr. K. Samsudeen                | Principal Scientist (Economic Botany)                                             |
| 15                                  | Dr. M.K. Rajesh                 | Principal Scientist (Agril. Biotechnology) (up to 13.07.2023)                     |
| 16                                  | Dr. M.R. Manikantan             | Principal Scientist (Agril. Structure & Process Engg.)                            |
| 17                                  | Dr. K.P. Chandran               | Principal Scientist (Agril. Statistics)                                           |
| 18                                  | Dr. S. Jayasekhar               | Principal Scientist (Agril. Economics)                                            |
| 19                                  | Dr. (Ms) V. Selvamani           | Senior Scientist (Soil Science)                                                   |
| 20                                  | Dr. R. Sudha                    | Senior Scientist (Fruit Science)                                                  |
| 21                                  | Dr. (Mrs.) S. Neenu             | Senior Scientist (Soil Science)                                                   |
| 22                                  | Dr. S. Paulraj                  | Senior Scientist (Agril. Microbiology)                                            |
| 23                                  | Dr. S.V. Ramesh                 | Senior Scientist (Agril. Biotechnology)                                           |
| 24                                  | Dr. Rajkumar                    | Senior Scientist (Nematology)                                                     |
| 25                                  | Dr. (Mrs.) V. H. Prathibha      | Senior Scientist (Plant Pathology)                                                |
| 26                                  | Dr. (Mrs.) M. Sujithra          | Senior Scientist (Agril. Entomology)                                              |
| 27                                  | Dr. (Mrs.) P. S. Pratibha       | Senior Scientist (Agril. Entomology)                                              |
| 28                                  | Dr. (Mrs.) Surekha              | Scientist (Agronomy)                                                              |
| 29                                  | Dr. Neema Mohammed              | Scientist (SPM&AP) (Transferred on 12.05.2023)                                    |
| 30                                  | Dr. (Mrs.) P.P. Shameena Beegum | Scientist (SPM&AP)                                                                |
| 31                                  | Dr. (Mrs.) Daliya Mol           | Scientist (Plant Pathology)                                                       |
| 32                                  | Dr. R. Pandiselvam              | Scientist (Agril. Process & Food Engg.)                                           |
| 33                                  | Mrs. T.N. Ranjini               | Scientist (SPM&AP)                                                                |
| 34                                  | Dr. (Mrs.) S. Sumitha           | Scientist (SPM&AP)                                                                |
| 35                                  | Dr. (Mrs.) Aparna Veluru        | Scientist (SPM&AP)                                                                |
| 36                                  | Dr. (Mrs.) G. Panjavarlam       | Scientist (Fruit Science)                                                         |
| 37                                  | Mrs. Bandela Sravanthi          | Scientist (SP&M&AP) (Transferred on 04.05.2023)                                   |
| <b>KVK, Kasaragod</b>               |                                 |                                                                                   |
| 38                                  | Dr. T.S. Manojkumar             | Principal Scientist (Agril. Structure & Processing Engg.) & Head, KVK             |
| <b>Regional Station, Kayamkulam</b> |                                 |                                                                                   |
| 39                                  | Dr. Regi Jacob Thomas           | Head, Principal Scientist(Horticulture) (w.e.f. 10.11.2023)                       |
| 40                                  | Dr. (Mrs.) P. Anithakumari      | Acting Head (up to 09.11.2023) / Principal Scientist (Ag. Extension)              |
| 41                                  | Dr. A. Abdul Haris              | Principal Scientist (Agronomy)                                                    |
| 42                                  | Dr. A. Joseph Rajkumar          | Principal Scientist (Agril. Entomology)                                           |
| 43                                  | Dr. (Mrs.) M. Shareefa          | Senior Scientist (Horticulture)                                                   |
| 44                                  | Dr. (Mrs.) Merin Babu           | Senior Scientist (Plant Pathology)                                                |
| 45                                  | Dr. (Mrs.) K. Nihad             | Senior Scientist (Horticulture)                                                   |
| 46                                  | Dr. (Mrs.) Jeena Mathew         | Senior Scientist (Soil Science)                                                   |
| 47                                  | Dr. K.M. Anes                   | Senior Scientist (Agril. Nematology)                                              |
| 48                                  | Dr. S. Indhuja                  | Scientist (Agril. Microbiology)                                                   |

| Sl. No.                            | Name                           | Designation                                                          |
|------------------------------------|--------------------------------|----------------------------------------------------------------------|
| 49                                 | Dr. (Mrs.) Jilu V. Sajan       | Scientist (Agril. Entomology)                                        |
| 50                                 | Dr. (Mrs.) U.K. Priya          | Scientist (Soil Science)                                             |
| <b>KVK, Alappuzha</b>              |                                |                                                                      |
| 51                                 | Dr. P. Muralidharan            | Principal Scientist (Soil Science) & Head, KVK                       |
| <b>Regional Station, Vittal</b>    |                                |                                                                      |
| 52                                 | Dr. M.K. Rajesh                | Head, Principal Scientist (Plant Biotechnology.) (w.e.f. 14.07.2023) |
| 53                                 | Dr. S. Elain Apshara           | Acting Head (up to 13.07.2023) / Principal Scientist (Fruit Science) |
| 54                                 | Dr. N.R. Nagaraja              | Senior Scientist (Plant Breeding)                                    |
| 55                                 | Shri Bhavishya                 | Scientist (SPM &AP)                                                  |
| 56                                 | Dr. R. Thava Prakash Pandian   | Scientist (Plant Pathology)                                          |
| 57                                 | Dr. T.N. Madhu                 | Scientist (Agril. Entomology)                                        |
| 58                                 | Dr. M. Chaithra                | Scientist (Plant Pathology)                                          |
| 59                                 | Dr. (Mrs.) M. Suchithra        | Scientist (SPM &AP)                                                  |
| 60                                 | Dr. (Mrs.) E.K. Saneera        | Scientist (Agril. Entomology)                                        |
| <b>Research Center, Mohitnagar</b> |                                |                                                                      |
| 61                                 | Dr. Arunkumar Sit              | Principal Scientist (Hort.) & Scientist-In-charge                    |
| 62                                 | Dr. Sandip Shil                | Senior Scientist (Agril. Statistics)                                 |
| <b>Research Center, Kahikuchi</b>  |                                |                                                                      |
| 63                                 | Dr. (Mrs.) Alpana Das          | Principal Scientist (Agril. Biotechnology) & Scientist-In-charge     |
| 64                                 | Dr. Leichombam Singhajit Singh | Scientist (SPM &AP)                                                  |
| 65                                 | Dr. Anok Uchoi                 | Scientist (SPM &AP)                                                  |
| 66                                 | Dr. (Ms.) M. Chaithra          | Scientist (Plant Pathology) (w.e.f. 11.07.2023)                      |
| <b>Research Center, Kidu</b>       |                                |                                                                      |
| 67                                 | Shri Y. Diwakar                | Scientist (SPM&AP) and Scientist-In-Charge                           |

## TECHNICAL STAFF

| Sl. No.                        | Name                     | Designation                                                  |
|--------------------------------|--------------------------|--------------------------------------------------------------|
| <b>Headquarters, Kasaragod</b> |                          |                                                              |
| 1                              | Shri H. Muralikrishna    | Chief Technical Officer                                      |
| 2                              | Smt. Sugatha Padmanabhan | Chief Technical Officer (Retired on 31.07.23)                |
| 3                              | Shri K. Shyam Prasad     | Chief Technical Officer                                      |
| 4                              | Shri G.S. Hareesh        | Senior Technical Officer (Instrumentation Engg.)             |
| 5                              | Shri K. Ajith Kumar      | Senior Technical Officer (Civil Engg.) (w.e.f. 05.06.2023)   |
| 6                              | Shri V.K. Gopalakrishnan | Technical Officer                                            |
| 7                              | Shri S. Manohara         | Technical Officer (Vehicle)                                  |
| 8                              | Shri K. Krishnan Nair    | Technical Officer (Field /Farm)                              |
| 9                              | Shri A.K. Ramadas        | Technical Officer (Vehicle) (Retired on 28.02.2023)          |
| 10                             | Shri K. Devaraj          | Technical Officer (Jr. Eng.) (on study leave)                |
| 14                             | Dr. K.S. Muralikrishna   | Senior Technical Assistant (Laboratory)                      |
| 15                             | Shri A.V. Satheeshkumar  | Senior Technical Assistant (Vehicle)                         |
| 16                             | Shri P. Santhosh Kumar   | Senior Technical Assistant (Field /Farm) (w.e.f. 12.06.2023) |
| 17                             | Shri K. Panduranga       | Senior Technical Assistant                                   |
| 18                             | Shri K. Bhavani Shankar  | Technical Assistant                                          |
| 20                             | Shri S. Sunil            | Technical Assistant (Electrical Engineering)                 |
| 22                             | Shri G. Arunji           | Technical Assistant (LIB)                                    |
| 23                             | Shri P.P. Anoop Kumar    | Technical Assistant (Field/Farm)                             |
| 19                             | Shri A. Divakara         | Senior Technician (Retired on 28.02.2023)                    |
| 11                             | Shri K.J. Sebastian      | Senior Technician (Retired on 30.04.2023)                    |
| 12                             | Shri K. Raghavan         | Senior Technician (Retired on 30.05.2023)                    |
| 13                             | Shri A.R. Padmanabhan    | Senior Technician (Retired on 30.05.2023)                    |
| 21                             | Shri N. Dinesh Kumar     | Senior Technician (Field /Farm)                              |
| 24                             | Smt. E.P. Ashamol        | Technician (Field/Farm)                                      |
| 25                             | Shri P.S. Suvith         | Technician (Field/Farm)                                      |
| 26                             | Shri R. Ajith Kumar      | Technician (Field/Farm)                                      |
| 27                             | Shri M. Krishnan         | Technician                                                   |
| 28                             | Smt. Chithralekha Kodoth | Technician                                                   |

| Sl. No.                             | Name                             | Designation                                                             |
|-------------------------------------|----------------------------------|-------------------------------------------------------------------------|
| 29                                  | Shri B. Sundara                  | Technician                                                              |
| 30                                  | Shri B. Chandrasaha              | Technician                                                              |
| 31                                  | Shri V.T. Rameshan               | Technician                                                              |
| 32                                  | Shri K. Sukumaran                | Technician                                                              |
| <b>KVK, Kasaragod</b>               |                                  |                                                                         |
| 33                                  | Dr. (Mrs.) Saritha Hegde         | Chief Technical Officer (SMS-Home Science)                              |
| 34                                  | Dr. (Mrs.) Neelofar Illias Kutty | Chief Technical Officer (Programme Assistant) (Home Science)            |
| 35                                  | Smt. M.P. Jayasree               | Assistant Chief Technical Officer (SMS) (Agril. Extn.) (On study leave) |
| 36                                  | Shri K. Manikandan               | Senior Technical Officer (Programme Assistant)                          |
| 37                                  | Dr. Benjamin Mathew              | Senior Technical Officer (SMS – Horticulture)                           |
| 38                                  | Shri K.P. Lagesh                 | Technician (Vehicle)                                                    |
| <b>Regional Station, Kayamkulam</b> |                                  |                                                                         |
| 39                                  | Shri C.G. Narayanan Nampoothiri  | Assistant Chief Technical Officer (Field/Farm)                          |
| 40                                  | Shri B. Anilkumar                | Assistant Chief Technical Officer (Field Farm)                          |
| 41                                  | Dr. (Mrs.) Maya Lekshmi          | Technical Officer(F/F)                                                  |
| 42                                  | Shri Sunny Thomas                | Technical Officer(F/F)                                                  |
| 43                                  | Shri V.P. Joy                    | Senior Technical Assistant (Field/Farm) (up to 31.12.2023)              |
| 44                                  | Smt. Asha K. Chandran            | Technical Assistant (F/F)                                               |
| 45                                  | Smt. Poonam Khatra               | Technical Assistant (F/F) (From 02.01.2023)                             |
| 46                                  | Shri Premjith Antony             | Technician (Field Farm)                                                 |
| <b>KVK, Alappuzha</b>               |                                  |                                                                         |
| 47                                  | Smt. Jissy George                | Chief Technical Officer (SMS –Home Science)                             |
| 48                                  | Smt. G. Lekha                    | Chief Technical Officer (SMS-Plant Pathology)                           |
| 49                                  | Shri M.S. Rajeev                 | Chief Technical Officer (SMS-Agronomy)                                  |
| 50                                  | Shri S. Ravi                     | Chief Technical Officer (SMS-Animal Husbandry)                          |
| 51                                  | Dr. T. Sivakumar                 | Chief Technical Officer (SMS-Agricultural Entomology)                   |
| 52                                  | Dr. K. Sajnanath                 | Assistant Chief Technical Officer (SMS-Soil Science)                    |
| 53                                  | Shri K.M. Ansary                 | Senior Technical Officer (Computer)                                     |
| 54                                  | Smt. P.V. Bijila                 | Senior Technical Officer (Horticulture)                                 |
| 55                                  | Shri Dayanandan Unnithan         | Technical Officer (Vehicle)                                             |
| 56                                  | Shri B.J. Sajin                  | Technician                                                              |
| <b>Regional Station, Vittal</b>     |                                  |                                                                         |
| 57                                  | Shri K. Ajith Kumar              | Senior Technical Officer (Civil Engg.) (up to 04.06.2023)               |
| 58                                  | Shri C. Purandhara               | Technical Officer (Retired on 31.08.2023)                               |
| 59                                  | Shri Abdul Aziz                  | Technical Officer (Retired on 31.07.2023)                               |
| 60                                  | Shri M. Narayana Naik            | Technical Officer (F/F) (w.e.f. 27.09.2023)                             |
| 61                                  | Shri B. Tharanath Naik           | Senior Technical Assistant (Vehicle)                                    |
| 62                                  | Shri P. Santhosh Kumar           | Senior Technical Assistant (Field /Farm) (up to 11.06.2023)             |
| 63                                  | Shri B.J. Nirmal Kumar           | Technical Assistant (F/F)                                               |
| 64                                  | Shri Bisun Bhaskar               | Technical Assistant (Lab)                                               |
| 65                                  | Shri D. Isubu                    | Technician                                                              |
| 66                                  | Shri Dharmapala                  | Technician                                                              |
| 67                                  | Shri Mohana                      | Technician                                                              |
| <b>Research Center, Mohitnagar</b>  |                                  |                                                                         |
| 68                                  | Shri Avarjyothi Gosh             | Assistant Chief Technical Officer                                       |
| 69                                  | Shri Pratap Kumar Sarkar         | Technical Officer                                                       |
| 70                                  | Shri Jagadish Roy                | Technical Officer (Vehicle)                                             |
| 71                                  | Shri Prakash Burman              | Senior Technician                                                       |
| 72                                  | Shri Kartick Chandra Biswas      | Technician                                                              |

| Sl. No.                           | Name                    | Designation                               |
|-----------------------------------|-------------------------|-------------------------------------------|
| <b>Research Center, Kahikuchi</b> |                         |                                           |
| 73                                | Shri Bikash Chowdhari   | Chief Technical Officer                   |
| 74                                | Shri Gopinath Malakar   | Technical Officer (Vehicle)               |
| <b>Research Center, Kidu</b>      |                         |                                           |
| 75                                | Shri M. Manamohan       | Technical Officer (Retired on 30.04.2023) |
| 76                                | Shri M. Narayana Naik   | Technical Officer(F/F) (up to 26.09.2023) |
| 77                                | Shri A.S. Gopalakrishna | Technical Officer                         |
| 78                                | Shri V. Kamal Kumar     | Technical Assistant                       |
| 79                                | Shri V. Chennappa       | Technician                                |
| 80                                | Shri S. Chennappa       | Technician                                |
| 81                                | Shri V. Jathappa Gowda  | Technician                                |

## ADMINISTRATIVE STAFF

| Sl. No.                             | Name                        | Designation                               |
|-------------------------------------|-----------------------------|-------------------------------------------|
| <b>Headquarters, Kasaragod</b>      |                             |                                           |
| 1                                   | Shri Ram Avtar Parashar     | CF&AO (w.e.f. 15.05.2023)                 |
| 2                                   | Smt. Jessymol Antony        | FAO (up to 17.02.2023)                    |
| 3                                   | Shri K.R. Nityanandan       | AAO (up to 28.06.2023)                    |
| 4                                   | Shri P. Krishna Kumar       | AO                                        |
| 5                                   | Shri Pradeep Kumar Vasu     | AAO                                       |
| 6                                   | Shri M. Ravindran           | AAO (w.e.f. 12.05.2023)                   |
| 7                                   | Shri A. Neil Vincer         | AAO                                       |
| 8                                   | Shri Narayani K.            | Principal Private Secretary               |
| 9                                   | Smt. Girija Chandran        | Private Secretary (Retired on 28.02.2023) |
| 10                                  | Smt. Sulochana Nair         | Private Secretary                         |
| 11                                  | Smt. A.R. Arathi            | Stenographer Gr. III                      |
| 12                                  | Shri P.M. Thomas            | Assistant                                 |
| 13                                  | Smt. K.T.K. Sheenakumari    | Assistant                                 |
| 14                                  | Shri P. Narayana Naik       | Assistant                                 |
| 15                                  | Smt. Rupa Manikandan        | Assistant                                 |
| 16                                  | Shri Paulson Sam George     | Assistant                                 |
| 17                                  | Smt. K. Preethi             | Assistant                                 |
| 18                                  | Shri Aswin Regunath         | UDC                                       |
| 19                                  | Shri T.K. Gangadharan       | UDC                                       |
| 20                                  | Shri Mohammed Haneefa       | UDC (w.e.f. 26.09.2023)                   |
| 21                                  | Shri N. Udayakumar          | UDC                                       |
| 22                                  | Smt. A.J. Mary              | UDC                                       |
| 23                                  | Shri P.K. Pramod Kumar      | UDC                                       |
| 24                                  | Shri V. Jayarajan           | LDC                                       |
| 25                                  | Shri Ratan Singh            | LDC (on deputation)                       |
| 26                                  | Shri Dinesh                 | LDC (on deputation)                       |
| 27                                  | Shri K.P. Ibrahim           | LDC (on deputation)                       |
| 28                                  | Shri Sathya Bratha Moharana | LDC (on deputation)                       |
| <b>KVK, Kasaragod</b>               |                             |                                           |
| 29                                  | Smt. K. Jayashree           | Assistant                                 |
| <b>Regional Station, Kayamkulam</b> |                             |                                           |
| 30                                  | Shri K.K. Sasi              | AFAO (up to 18.07.2023)                   |
| 31                                  | Smt. Prasanna Sarngan       | Private Secretary (w. e. f. 09.11.2023)   |
| 32                                  | Shri K. Haridasan           | Assistant                                 |
| 33                                  | Shri K. Venu Gopal          | Assistant                                 |
| 34                                  | Smt. V. Madhavikutty        | Assistant (up to 31.05.2023)              |
| 35                                  | Smt. K. Sreelatha           | Assistant                                 |
| 36                                  | Smt. T. Deepa               | UDC                                       |
| 37                                  | Shri N.K. Arun Raj          | LDC                                       |
| <b>KVK, Alappuzha</b>               |                             |                                           |
| 38                                  | Smt. K.R. Regitha           | PA                                        |
| <b>Regional Station, Vittal</b>     |                             |                                           |
| 39                                  | Shri Vivek Singh            | Stenographer Gr. III                      |
| 40                                  | Shri Mohammed Haneefa       | UDC (up to 25.09.2203)                    |

| Sl. No.                            | Name                | Designation |
|------------------------------------|---------------------|-------------|
| 41                                 | Shri Chandu Naik    | LDC         |
| 42                                 | Shri Laxmi Narayana | UDC         |
| 43                                 | Shri K.N. Sajeev    | LDC         |
| 44                                 | Shri B. Choma       | LDC         |
| <b>Research Centre, Mohitnagar</b> |                     |             |

| Sl. No.                           | Name              | Designation                   |
|-----------------------------------|-------------------|-------------------------------|
| 45                                | Shri Subash Paul  | Assistant                     |
| <b>Research Centre, Kahikuchi</b> |                   |                               |
| 46                                | Shri Deepak Meena | LDC (on deputation)           |
| 47                                | Shri Umesh Kumar  | LDC (on deputation)           |
| 48                                | Shri T.J. Saji    | Assistant (w.e.f. 18.08.2023) |
| <b>Research Center, Kidu</b>      |                   |                               |
| 49                                | Shri M. Ravindran | AAO (up to 11.05.2023)        |
| 50                                | Shri M. Durgasha  | LDC                           |

## SKILLED SUPPORT STAFF

| Sl. No.                             | Name                              |
|-------------------------------------|-----------------------------------|
| <b>Headquarters, Kasaragod</b>      |                                   |
| 1                                   | Shri K.G. Sureshababu             |
| 2                                   | Shri V.S. Pakeeran                |
| 3                                   | Smt. V. Thambai                   |
| 4                                   | Smt. G. Kamala (up to 31.12.2023) |
| 5                                   | Shri K. Jayaprakash               |
| 6                                   | Shri V. Krishnakutty              |
| 7                                   | Shri P.P. Prabhakaran             |
| 8                                   | Shri B. Ramachandran              |
| 9                                   | Shri B. Sanjeeva Patali           |
| 10                                  | Smt. N.V. Sasikala                |
| 11                                  | Shri Laxshmana Naik               |
| 12                                  | Smt. Lalitha Bai                  |
| 13                                  | Shri M. Velayudhan                |
| 14                                  | Shri N. Bhaskaran                 |
| 15                                  | Shri K. Sureshan                  |
| 16                                  | Shri A. Madhu                     |
| 17                                  | Shri K.A. Madhavan                |
| 18                                  | Shri Vanamalini                   |
| 19                                  | Shri E.M. Aneesh                  |
| 20                                  | Shri Praveen Raj                  |
| 21                                  | Shri R. Ashok Kumar               |
| 22                                  | Shri Sarath Kumar                 |
| 23                                  | Shri Kripesh                      |
| 24                                  | Smt. N. Rohini                    |
| <b>KVK, Kasaragod</b>               |                                   |
| 25                                  | Shri P. Chaniya Naik              |
| <b>Regional Station, Kayamkulam</b> |                                   |
| 26                                  | Shri M. E. Sivan                  |
| 27                                  | Shri K. Soman                     |
| 28                                  | Shri K. Omanakuttan               |
| 29                                  | Shri K. Ravi                      |
| 30                                  | Shri C. Sukumaran                 |
| 31                                  | Smt. K. Valsala                   |
| 32                                  | Shri C. Sundaran                  |

| Sl. No.                            | Name                                  |
|------------------------------------|---------------------------------------|
| 33                                 | Shri A.T. Harikuttan                  |
| 34                                 | Shri S. Rajesh                        |
| 35                                 | Smt. N. Suma                          |
| 36                                 | Smt. L. Leena                         |
| 37                                 | Smt. Arathi R. Pillai                 |
| <b>KVK, Alappuzha</b>              |                                       |
| 38                                 | Shri R. Rajesh                        |
| 39                                 | Smt. A.S. Aswathy                     |
| <b>Regional Station, Vittal</b>    |                                       |
| 40                                 | Shri Ibrahim                          |
| 41                                 | Shri K. Somappa                       |
| 42                                 | Shri M. Ananda                        |
| <b>Research Center, Mohitnagar</b> |                                       |
| 43                                 | Shri Sailen Seal                      |
| 44                                 | Shri Krishna Kumar Mandal             |
| 45                                 | Shri Nripendra Chandra Roy            |
| 46                                 | Shri Mahadev Misra                    |
| 47                                 | Shri Sushanta Burman                  |
| <b>Research Center, Kahikuchi</b>  |                                       |
| 48                                 | Shri Sathish Baishya                  |
| 49                                 | Shri Pankaj Das                       |
| 50                                 | Shri Thanka Bahadur Thapa             |
| <b>Research Center, Kidu</b>       |                                       |
| 51                                 | Smt. D. Bhavani (up to 31.03.2023)    |
| 52                                 | Shri Dasappa Gowda (up to 31.05.2023) |
| 53                                 | Smt. T. Susheela (up to 31.07.2023)   |
| 54                                 | Smt. N. Bhavani                       |
| 55                                 | Shri Padmayya Gowda                   |
| 56                                 | Smt. S. Rukmini                       |
| 57                                 | Smt. S. Chandravathi                  |
| 58                                 | Shri S. Bhojappa                      |
| 59                                 | Smt. Komalangi                        |
| 60                                 | Shri S. Neelappa                      |
| 61                                 | Shri S. Regappa                       |
| 62                                 | Smt. Meenakshi B.                     |

## XVIII. DISTINGUISHED VISITORS

| Sl.No. | Visitors                                                                                                                                                                                                                                                                                                                                                                                                       | Place and Date                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1      | Dr. Dinakara Adiga, Director, ICAR-DCR, Puttur,                                                                                                                                                                                                                                                                                                                                                                | ICAR-CPCRI, Kasaragod<br>January 5, 2023                |
| 2      | Dr. S.N. Jha, DDG (Agril. Engg.), Prof. Prasad Krishna, Director, NIT, Kozhikode, Dr. K. Narsaiah, ADG (Process Engg.), Dr. Nachiket Kotwaliwale, Director, ICAR-CIPHET, and Dr. Vishwakarma R.K., Project Coordinator of AICRP on PHET.                                                                                                                                                                       | ICAR-CPCRI, Kasaragod<br>February 20-22, 2023           |
| 3      | Sri. P. Prasad, Hon. Minister of Agriculture, Govt. of Kerala                                                                                                                                                                                                                                                                                                                                                  | ICAR-CPCRI, RS, Kayamkulam<br>March 3, 2023             |
| 4      | Dr. G. Kadirvel, Director, ATARI Zone VI, Guwahati                                                                                                                                                                                                                                                                                                                                                             | ICAR-CPCRI, RC, Kahikuchi<br>April 26, 2023             |
| 5      | Sri. K. Krishnankutty, Hon. Minister for Electricity, Govt. of Kerala                                                                                                                                                                                                                                                                                                                                          | ICAR-CPCRI, RS, Kayamkulam<br>May 9, 2023               |
| 6      | Dr. A. K. Singh, Deputy Director General (Horticultural Sciences), ICAR, New Delhi, Dr. Baiju G., Director, ICAR-CTCRI,                                                                                                                                                                                                                                                                                        | ICAR-CPCRI, RS, Kayamkulam<br>May 13-14, 2023           |
| 7      | Heads of 11 KVKs of Karnataka on inter-KVK exposure visit                                                                                                                                                                                                                                                                                                                                                      | ICAR-CPCRI, RS, Kayamkulam<br>August 17, 2023           |
| 8      | Sushri Shobha Karandlaje, Honourable Minister of State for Agriculture and Farmers Welfare, Government of India, Sri N.A. Nellikkunnu, MLA, Kasaragod, Dr. V. B. Patel, Assistant Director of General (Fruits and Plantation Crops), Horticulture Science Division, ICAR, New Delhi,                                                                                                                           | ICAR-CPCRI, Kasaragod<br>September 2, 2023              |
| 9      | Dr. V.B. Patel, ADG (F&PC), ICAR, New Delhi, Dr. N. K. Mohan, Hon'ble Member, Agricultural Commission, Govt. of Assam, Dr. K. Suresh, Director, ICAR-IIOPR, Pedavegi; Dr. R.K. Mathur, Director, ICAR-IIOR, Hyderabad; Dr. V. Venkatasubramanian, Director, ICAR-ATARI, Bangalore; Dr. G. Kathirvel, Director, ICAR-ATARI, Guwahati; Dr. KUK. Namboothiri and Dr. P. Chowdappa, former directors of ICAR-CPCRI | ICAR-CPCRI, RC, Kahikuchi<br>September 13-15, 2023      |
| 10     | Mr. M.S. Arunkumar, MLA, Mavelikkara                                                                                                                                                                                                                                                                                                                                                                           | KVK, Alappuzha, RS,<br>Kayamkulam<br>September 27, 2023 |
| 11     | Dr. Prabhat Kumar, Hort. Commissioner, DAFW, Ministry of Agri. & Farmer's Welfare Homey Cheriyan, Director, DASD, Kozhikode                                                                                                                                                                                                                                                                                    | ICAR-CPCRI, Kasaragod<br>November 14, 2023              |
| 12     | Sri. T.K. Premkumar, DDM, NABARD                                                                                                                                                                                                                                                                                                                                                                               | KVK, Alappuzha, RS,<br>Kayamkulam<br>November 17, 2023  |

## XIX. MERA GAON - MERA GAURAV

During the year, ICAR-CPCRI, Kasaragod and its regional stations and research centres have implemented the MGMG initiative in collaboration with various stakeholders, viz. Department of Agriculture/ Horticulture, KVKs, LSG Institutions, input dealers, progressive farmers, SHGs, etc. A total of 56 villages were covered by 18 teams comprising a total of 55 scientists. Summary of activities is provided in the following table.

| Particulars                   | Total Numbers | Name of CPCRI institutes/ RS/RC        |
|-------------------------------|---------------|----------------------------------------|
| No. of Scientists involved    | 55            | ICAR – CPCRI, Kasaragod,               |
| No. of teams                  | 18            | ICAR-CPCRI, RSs, Kayamkula and Vittal, |
| No. of villages covered       | 56            | ICAR-CPCRI, RCs Mohitnagar,            |
| No. of visits to the villages | 60            | Kahikuchi and Kidu                     |

A total of 20 interface programmes conducted, benefitting 215 farmers. Topics covered were the following:

- Arecanut leaf spot disease management,
- Coconut bud rot management,
- coconut cultivation practices,
- Awareness programme on coconut value addition products,
- Soil health management in mandate crops,
- Groundnut and millet as intercrops in coconut,
- Coconut Rhinoceros beetle management,
- Coconut cultivation practices,
- Awareness programme on coconut value added products, and
- Soil health management in coconut and intercrops

A total of 13 technology demonstrations were conducted on the following topics:

- Nylon net technique for management of rhinoceros beetle in juvenile palms,
- EPN based integrated root grub management in coconut and arecanut,
- Borax and Zinc sulphate application in arecanut for management of nut splitting,
- Phyto-sanitation for management of arecanut leaf spot disease,
- Soil sampling in coconut and arecanut,
- arecanut fruit rot management using mandipropamid,
- integrated management of arecanut inflorescence dieback,
- EPN capsules for red palm weevil (RPW)

management in coconut,

- Integrated Root (Wilt) Disease Management in coconut gardens, and
- Soil sampling & INM in coconut.

Ten training programme conducted benefitting 155 farmers. Topics covered were the following:

- Arecanut leaf spot management,
- Integrated soil health management in coconut,
- Arecanut root grub management using EPN,
- Eco-friendly management of lepidopteron caterpillar in vegetables intercropped in coconut using EPN aqua suspension,
- Integrated coconut bud rot management,
- Integrated soil health management in coconut,
- Integrated plant protection approach in coconut,
- Seedling production techniques for vegetable and coconut,
- Community Coconut Nursery management in root (wilt) disease affected areas, and
- Organic manure preparation techniques.

Above interventions and diagnostic field visits could solve the farmers problems such as, integrated nutrient management; nutrient disorders in coconut; root grub and bud rot management in coconut; root (wilt) and leaf rot disease; arecanut leaf spot disease; red palm weevil in coconut; pest infestation in millets as coconut intercrop; and moisture conservation measures in coconut.

As a part of these activities, agencies such as Krishi Bhavan, FPOs, Krishi Vigyan Kendra, water shed development of NABARD, Extension Training Centre, Vorkody, farmers clubs, educational institutions in the locations with whom functional linkages were strengthened.

A total of 14 extension publications numbering about 200 were distributed with list of topics as given below:

- Tengu (Kannada) technical bulletin,
- Coconut (English) technical bulletin,
- Root grub management in arecanut using EPN (Kannada) Extension folder,
- Coconut rhinoceros beetle management (Malayalam) extension folder, and
- Red palm weevil management in coconut (Kannada) extension folder.



Grama Sabha at Madhur panchayath



Training on arecanut root grub management at Shivapur village, Udupi



Workshop on arecanut in Dakshina Kannada



Sanitation and cleanliness drive in Mangad village



Training programme on papaya cultivation in coconut homesteads in Badiadka (29/11/2023)



Training programme on coconut cultivation at ICAR – CPCRI, Kasaragod

## XX. SWACHH BHARAT ABHIYAN

As a part of Swachh Bharat Abhiyan, regular cleaning activities have been carried out in office premises and laboratories regularly including some of the special programmes conducted at headquarters and other units of the Institute.

### Swachhta Hi Sewa Campaign

Swachhata special campaign 3.0 observed from September 25 to October 2, 2020 to commemorate the Birth Anniversary of Mahatma Gandhi. National highways, schools, tourist spots, MGMG villages were identified for swachhata outdoor campaign and total pending references, record management includes reviewing of old files and weeding out of files, e-files, scrap disposal activities were scheduled for swachhata indoor activities. Swachhata Diwas was celebrated at ICAR-CPCRI on October 2, 2023 and swachhata campaign 3.0 started with taking swachhata pledge. 2<sup>nd</sup> October to 6<sup>th</sup> November, Swachhata campaign conducted at Kendriya Vidyalaya No.1, National high way 31A (Mohitnagar), Baniapara Junior High School of Mohitnagar, Azheekkal beach and light house of Arattupuzha Panchayath, Alappuzha district other than institute guest houses and labs. At Kasaragod, a swachhata awareness programme was organized in which Ms. Sneha, Health Inspector, Mogral Puttur Panchayat delivered a lecture on 'Efficient waste management' and Mrs. Prameela, chairman for management of plastic waste, delivered a talk on 'Plastic waste management'. They highlighted the activity of 'Haritha Karma Sena' for collection of plastic wastes, e-wastes, bottles and other wastes. Under scrap disposal activity during the Special Campaign 3.0, a total of 21 tonnes of scrap materials were auctioned through GeM and generated Rs. 6,12,000/-.

### Swachhta Pakhwada

The campaign activity was conducted during December 16-31, 2023. During this fortnight, various activities such as digitization of office records, sanitation drives at Institute premises,

MGMG villages, laboratory waste management, method demonstration on converting kitchen waste to wealth through composting, training on 'Recycling of biodegradable wastes in coconut based farming system (CBFS): Production of *Trichoderma* enriched neem cake', Cleaning the premises of local areas, Selecting model residential quarters with regard to waste management, visit to community waste disposal sites/ compost pits and application of *Metarhizium* in compost pits, campaign on cleaning of sewerage and water lines and an awareness campaign at Madhur village were organized.

Kisan Diwas was celebrated on December 23, 2023 at Kasaragod where coconut hybrid seedlings were distributed to SC farmers/ farm women of villages of Madhur, Badiadka, Puthige, Mogral Puttur panchayaths and progressive farmers and farm women were felicitated for their valuable contribution to coconut and vegetable farming activities. At Kayamkulam, Dr. V. Krishnakumar, Retired Head, ICAR-CPCRI Regional Station, Kayamkulam delivered a talk on 'Recycling of bioresources for sustainable production in coconut-based farming system' and best performing SC women farmers were honoured during the function. Climbing machines, coconut seedlings, vermicompost and growbags were distributed to SCSP farmers during the programme. At Kahikuchi, Mr. Sanjay Kumar Rabha, General Member of Rabha Hasong Tribal Community (autonomous council), Rani, Kampur district, Guwahati was invited and he delivered a lecture on need of environmentally friendly approaches at NE region and shared their success stories.

On valedictory function, Mr. K. Balakrishnan, District Co-ordinator, Haritha Keralam Mission, Kasaragod delivered a lecture on 'Solid and liquid waste management'.



Cleaning activities as a part of Swachh Bharat Abhiyan at CPCRI Kasargod



Campaign at Madhur village



Valedictory session of Swachhta Pakhwada at CPCRI Kasargod



Coconut hybrid seedlings distribution to SC farmers



Proposed to make green, plastic free campus



Disposal of scrape and unserviceable items



110-KVA Solar Installation at CPCRI Kasaragod

## XXI. WOMEN'S CELL ACTIVITIES

The Women Cell under the Chairmanship of Dr. V. Niral, Principal Scientist, organized various events for the welfare of women staff of the Institute.

An awareness programme on “Eco- friendly personal hygiene for women” was conducted at ICAR-CPCRI Kasargod on August 21, 2023 for the benefit of the women staff. Dr. V. Niral, Chairperson Women's Cell, in her welcome address emphasized the need for awareness among women for the eco friendly personal hygiene and related health issues. Miss Lavina, Marine Biologist/Researcher was the Chief guest and delivered a speech on eco friendly personal hygiene for women and Dr Shruthi A.G Director Shruthi IVF fertility and cosmetics Gynaecology Centre also delivered speech on health related issues.

International women's day was celebrated at ICAR-CPCRI on March 8, 2023 with Ms. Oli Aman Jodha, the first woman farrier in India and young entrepreneur as the chief guest.

Oli is a 16 year old girl, hailing from Thirunelli of Wayand district of Kerala, and at the age of nine she went to Nepal and learnt the art of fixing horse shoes and continues to be the only woman farrier in the country. Ms. Oli in her speech stressed that the basic education on gender equality and gender neutrality should be initiated at every home. Further, she emphasized that mutual respect is the basis of any kind of relationship. She summoned the students to be independent, self sufficient and

not to sacrifice their genuine desires for other's sake. She stressed that children need to bridge the generation gap that exists among children and parents by educating the parent and should always ensure that their actions do not cause trouble to others. She shared her views on nature, honeybees, and role of honey bees in sustaining our ecosystem and biodiversity etc.

Three quarterly meetings were conducted to discuss and review the activities women cell under the Chairmanship of Dr. V. Niral and the executive members. The members of the Women Cell of ICAR-CPCRI, Kasaragod bid farewell to Mrs. Sugatha Padmanabhan, CTO and Mrs. G. Kamala, SSS two of our long serving colleagues on the occasion of their superannuation from ICAR service.



Ms. Oli Aman Jodha, inaugurating the International Women's day function

## XXII. MAJOR EVENTS AND OTHER INFORMATION

### World Coconut Day Celebration and National Teachers' Day - 2023

The World Coconut Day 2023 was Inaugurated by Sushri Shobha Karandlaje, Honourable Minister of State for Agriculture and Farmers Welfare, Government of India, at ICAR-CPCRI Kasaragod. The programme was organised jointly by ICAR-CPCRI, Kasaragod and Coconut Development Board, Kochi. The minister urged the entrepreneurs to double their exports by marketing value-added products to customers not only in local bazar but worldwide bazars. She said, "We are number one in coconut production and also 8th biggest world exporter of agricultural commodities such as horticultural crops, spices and food grains".

The minister visited various research laboratories of the institute including tissue culture & the agro-processing unit laboratory. She has also witnessed drone-based spraying on coconut plantations, ground pollination device demonstration, products of organic farming technology, integrated



Hon'ble minister inaugurating World Coconut Day at Kasaragod

pest and disease management practices adopted by the institute.

Sushri Shobha Karandlaje released QR coded coconut seedlings. She has also planted a coconut sapling in memory of her visit to the institute. She lauded the institute for their achievements and told that these should be reflected in the sector.

Sri N.A. Nellikkunnu, MLA, who presided over the function, highlighted the present crisis faced by the coconut farmers. He wanted to bring production

cost to Rs 10 from the present Rs. 12 by adopting technologies. He wanted the departments to encourage the youth to take up farming.

Dr. V.B. Patel, Assistant Director General (F&PC), ICAR, New Delhi felicitating the function. He said that the policymakers from the government, scientists from the institute, entrepreneurial personalities and the farmers with their combined efforts in the coconut sector, will find significant developments in the long term.

Sri Renukumar, Vice Chairman, CDB, and Sri P.R. Muralidharan, President, BAMCO, offered felicitations. Earlier, Dr. K.B. Hebbar, Director, CPCRI, in his welcome address, mentioned about the restructuring of institute research to the national priorities. He also described the need for climate-resilient research foreseeing the climate change effect on the crops. He also highlighted the technologies developed at the institute to give emphasis to value addition and entrepreneurship development.

Dr. Hanumanthe Gowda, Chief Coconut Development Officer, Coconut Development Board, Kochi, dwelt upon the various schemes implemented by the Board for farmers.

Seven publications were released during the programme, which include, 1) Cultivation practices of coconut - Northeast India, 2) Advisory on integrated management of leaf spot disease in arecanut in English and Malayalam, 3) Ready Reckoner on Cocoa, Coconut and Arecanut (bilingual in English & Kannada), 4) Coconut Products by Coconut Development Board in English & Hindi and 5) Schemes for Prosperity by Coconut Development Board.



Release of publications during the programme

An app on QR code development for seedlings was also executed by the Hon'ble Minister.

During the programme, the following 15 farmer entrepreneurs were honoured by the minister: Shri P.A. Elango, Chief Executive Officer & Shri Prabhu P.K., Executive Promotions, M/s Global Coconut Farmers Producer Company Limited, T.N.; Shri N.U. Raam Mohan, Managing Director, M/s Umapathy Coconut Hybrid Centre, T.N.; Shri Vinodan T.K., General Manager, Shri Vinayan E.V., Chairman, M/s Thrissur Coconut Producer Company, Kerala; Shri Sarvajjan T.V., Director, M/s Eramala Coconut Complex, Kerala; Shri Sathyanarayana Udupa, Chairman and MD, M/s Udupi Kalparasa Coconut and All Spices Producer Company Ltd, Karnataka; Shri Muralidhar, Advisor of the company; Shri Kusumadhar, S. K., Chairman, M/s South Canara Coconut Farmers Producer Company Limited, Karnataka; Shri Chethan A., Chief Executive Officer, Shri Dinesh Prayag, General Manager, M/s Krishna Plantations Pvt. Ltd., Goa; Shri Nagesh Prayag, Marketing Manager, Shri Ram Laxman Polabhai, Owner, Shri Ram Rajabhai Rambhai, Co. Owner, Shri Jotva Jagdish Lakshman Bhai, Chief Executive Officer, M/s Mahakali Coconut farm, Gujarat.

Entrepreneurs obtained licenses for six technologies on the day from the minister. They were Mr. Gangadhara Alva, Integrated Rural Agri Nursery, Badiadka – 671 551, Kerala on Shatamangala; Mrs. Vanitha Yadukumar, Yashaswi Nursery, Korungu, Kemminje – 574 202, Puttur, Karnataka State on Shatamangala; Mr Abdul Sawad T A, Managing Director of REDHIVE ODM LLP, Kasaragod-671121, Kerala, on Preservation of carbonated tender coconut water; YGP Coconut Farmers Producer Company Ltd., Chitradurga Road, Davangere, Karnataka-577003 on Collection of fresh and hygienic Kalparasa and production of natural coconut sugar; Programme co-ordinator, KVK, Kannur on Aqua formulation of EPN Kalpa EPN (CPCRI- SC1); and Dr Venugopal Nainala, Managing Director, Narial Bharat Nutra Producer Company Limited, Mamdikuduru Village, Konaseema district, A.P. on Coconut chips.

More than 500 farmers from Kerala, Tamil Nadu, Karnataka, Andhra Pradesh, Goa and Gujarat attended the celebration.

There were various technical sessions on coconut production during the programme in different sessions. A DST sponsored 'SERB Accelerate Vigyan Karyashala' on coconut value addition was also inaugurated by the Hon'ble Minister.

ICAR-CPCRI Regional Station, Kayamkulam in collaboration with Coconut Development Board, Kochi observed World Coconut Day celebrations and National Teachers' Day on 05-09-2023 with various activities including farmers' seminar, farmer-scientist interaction, felicitation to retired teachers from Kerala Agricultural University (KAU) and competitions for staff members. The programme was inaugurated by Dr. C.K. Peethambaran, Retired Director of Research, KAU. Dr. P. Anithakumari, Acting Head, ICAR-CPCRI, RS, Kayamkulam was the chairperson of the session. Dr. A. Abdul Haris, Principal Scientist, led the interactive session with the farmers.

Dr. C.K. Peethambaran, Dr. D. Alexander (Retired Director of Research, KAU), Dr. Sverup John (Retired Dean, Faculty of Agriculture, KAU), Dr. Shylaja S. (Retired Professor (Agricultural Extension), Dr. Arthur Jacob (Retired Associated Director of Research (Southern Region), Dr. A.V. Mathew (Retired Associate Director of Research, RARS, Kumarakom) acted as the panelists during the technical session in farmers seminar. The retired teachers from KAU who nourished three generations of agriculture graduates were honored. '*Kalpasthupam*', reminiscence for commemorating the legacy of 75 years of service to coconut farming community was unveiled during the event. Sixty farmers, people representatives and officials of Department of Agriculture Development & Farmers Welfare from Thanneermukkom Panchayath participated in the programme.



Inauguration of World Coconut Day at CPCRI RS Kayamkulam by Dr. C. K. Peethambaran, retired Director of Research KAU.



Staff of RS Kayamkulam along with dignitaries

### World Environmental Day - 'Lifestyle for environment'

As a part of the Environment Day, a series of programmes were conducted on environmental conservation. On 5<sup>th</sup> June 2023 pledge was taken in the headquarters, regional stations and research centres to safeguard environment. A series of programmes including trainings, interface programmes, seminars, exhibitions and demonstrations were carried out for farmers and students. To highlight, a training programme on 'Climate resilient coconut cultivation practices', at ICAR-CPCRI, RS, Kayamkulam, a seminar on 'Climate Resilient and Smart Agriculture' at CPCRI, Kasaragod, and exhibition on 'Climate smart technologies in agriculture' and 25 such programmes were conducted by regional stations and research centres.

A tree planting programme was held as a part of the celebrations in the main office campus, residential campus and around the Kendriya Vidyalaya school ground. Fruit tree saplings of guava, lemon, berries, etc. were planted. In the main office, Dr. C. George Thomas, planted a sapling and Dr. K.B. Hebbar, Director and Dr. M.K. Rajesh, Principal Scientist planted saplings in the memory of former Directors, Dr. K.V.A. Bavappa, and Dr. M.K. Nair, respectively.

### World Soil Day

ICAR-CPCRI, Kasaragod, celebrated World Soil Day on 5<sup>th</sup> December, 2023. Dr. P.R. Suresh (former Dean, College of Agriculture, Padannakkad) inaugurated the programme. Dr. K.B. Hebbar, Director, ICAR-CPCRI, Kasaragod presided over the function.

Dr. Ramesha B., Professor & Head, Extension Training Centre, Vorkady, Manjeshwar, delivered felicitations. Dr. Suresh distributed soil health cards to the farmers. In his inaugural address, he emphasised the importance of soil and water for the future generation.



World Soil day celebration at CPCRI Kasaragod

Dr. P. Subramanian, Head, Crop Production Division of CPCRI welcomed the gathering and briefed about the Soil Day event.

A technical session on the theme of the year 2023, 'Soil and Water: A Source for Life' was handled by Dr. P.R. Suresh, Dr. Ramesha B., Dr. Thamban, C., Principal Scientist, Dr. P. Subramanian, Head, Crop Production, Dr. Neenu, S., Sr Scientist, Soil Science and Dr. V. Selvamani, Sr. Scientist, Soil Science.

About thirty farmers of Kasaragod district actively participated in the technical session. Students of KV -2 CPCRI, Kasaragod took part in poster competitions held as a part of World Soil Day 2023.

A workshop was also conducted at ICAR-CPCRI, Regional Station, Kayamkulam on 5<sup>th</sup> December 2023, as part of the celebrations. Dr. A. Abdul Haris, Principal Scientist delivered a lecture on the topic 'Soil and water in synergy for sustainable plant health'. The lecture session was followed by a quiz for enriching the knowledge base of the farmers on aspects pertaining to soil and water relations in plant growth. Dr. Jeena Mathew, Senior Scientist interacted with farmers during the quiz programme covering various aspects of soil health and plant nutrition. The programme was attended by fifty farmers from the district.

### Kalpa Quiz - Interactive program for knowledge dissemination

A special program for knowledge sharing and promoting interests in coconut information KALPA QUIZ program was conducted during 2023 involving eight panchayats in which 172 farmers, women and youth participated in the quiz. More than 1000 farmers attended and cash awards were given to the first three teams and

Indian Coconut Journal to all the contestants as a means of knowledge dissemination through extension literature and popular articles. The program was financially supported by Coconut Development Board (CDB), Kochi. Scientists of ICAR-CPCRI were the quiz masters and linkage established with Department of Agriculture, Kerala for the program. Dr Sreevalsan J Menon, Professor KAU and renowned musician was the chief guest in the finale.



Grand Finale of Kalpa Quiz- exclusive program on Coconut technologies and knowledge

### Kalpa - Coconut quiz competition

ICAR-CPCRI, RS, Kayamkulam organized the Mega Finale of Kalpa quiz competition on 16<sup>th</sup> March 2023, as a part of Kalpa Vajra Celebrations. The Coconut Development Board sponsored cash Prizes for the Mega Finale winners. The competition was organized at the Panchayath level, and five to eleven groups of three members each participated from seven panchayaths of Muthukulam Block and Kayamkulam municipality. The Scientists of the station organized panchayath level competitions from 21<sup>st</sup> February 2023 to 24<sup>th</sup> February 2023. The participated Panchayaths and quiz masters were Pathiyoor (Dr. Indhuja. S.), Krishnapuram (Dr. Merin Babu), Devikulangara (Dr. Jilu V. Sajan), Arattupuzha (Dr. Anes K. M.), Muthukulam (Dr. Shareefa M.), Kayamkulam (Dr. Nihad K.), Kandallloor (Dr. Jeena Mathew) and Cheppad (Dr. Priya U. K.). A total of 163 participants, including 65 students, farmers, people's representatives, SHG members, FPO members, etc., participated in the quiz competitions. All the participants received one-year subscription of Indian Naliker Journal and certificates of participation. The first and second prize winners at the Panchayath level received Bio-primed polybag coconut seedlings, sponsored by Puthupally Village Co-operative

Bank Ltd. No.1900.

On 16<sup>th</sup> March 2023, the first prize winners from each Panchayath participated in the Mega Finale, which was conducted in the Auditorium of Puthuppally Service Cooperative Bank Ltd. No. 1900 at Devikulangara Panchayath. The quiz master for the Mega Finale was Dr. Sivakumar T. (SMS, KVK, Alappuzha).

The valedictory function was presided over by Sri. Pavanathan, President, Devikulangara Panchayath and inaugurated by Smt. Ambujakshi Teacher (Muthukulam Block Panchayath President). Dr. Sreevalsan J. Menon (Professor, KAU and Musician) was the Chief Guest of the program. Adv. Gopinathan K. President, SCB delivered the keynote address. More than 125 participants including people's representatives, teachers, farmers, students, etc. attended the function.

All finalists were felicitated by Muthukulam Block Panchayath. The winners of the Mega Finale at the Block level were awarded cash prizes of Rs. 25,000/-, Rs. 15,000/-, and Rs. 10,000/- for 1<sup>st</sup>, 2<sup>nd</sup> and 3<sup>rd</sup> positions, respectively, which were sponsored by CDB. All participated students received a special gift sponsored by ICAR-CPCRI. The whole program was organized as an innovative technology dissemination of coconut.

### Millet promotion in coconut gardens

In 2023 demonstrated finger millet, little millet, proso millet and sorghum as intercrops in 100 acres of coconut gardens of FFP implementation locations. The interventions included field level training programs, MGNREGS convergence for cultivation of millets under land consolidation approach, field visits, newspaper and media publicity, visits to plots from various agencies and students. The potential yield was achieved in case of proso, little and finger millets. Sorghum was a big draw among visitors and linkage programs for millet popularization under took with food safety department, Department of Agriculture, Amritha Institute, Colleges, KMML and LSGDs. A food festival with 32 millet products out of the harvest attracted almost 1200 participants. The Hon. Minister of Agriculture, Kerala, Mr. P. Prasad, Hon. MP Alappuzha Sri. Arif, Hon' MLA, Kayamkulam Adv. U. Prathibha, Kerala Haritha

Mission Director Dr. N. Seema, Block and Panchayath members visited the cultivation of millet in coconut gardens and the model was appreciated to be replicated in the millet mission activities of the district.

### World Intellectual Property Day—'Women and IP: Accelerating Innovation and Creativity

ICAR- CPCRI, RC, Kahikuchi celebrated 'World Intellectual Property Day, on 26 April, 2023. The programme was graced by Dr. K. Balachandra Hebbar, Director, ICAR-CPCRI, Kasaragod and Dr. G. Kadirvel, Director, ATARI Zone VI, Guwahati. There was a talk on the theme “Women and IP: Accelerating Innovation and Creativity” by Dr. Alpana Das, Principal Scientist and SIC, ICAR-CPCRI, RC, Kahikuchi. In the auspicious occasion, 'MoA for Enhancing Adoption of Scientific Production and Processing Technologies of Arecanut in NE' was signed in between ICAR CPCRI, Kasaragod and ICAR-ATARI, Guwahati, Assam.

### Library and Information Services



Dr. Alpana Das delivering a talk on women and IP

The library web page (<https://cpcri.icar.gov.in/page/library>) provides an overall view of activities, mainly the Online Public Access Catalogue (OPAC), Institute Digital Repository and services provided by the library such as Consortium of E-resources in Agriculture (CeRA)/Krishikosh/Krishiprabha/Agricat.

The institute digital repository, holding a literature collection to the tune of 7835 is very user friendly with nine communities and provides full text

| Station    | Database/ Journal subscription |                  |                 | Books | Back volumes | Other publications |
|------------|--------------------------------|------------------|-----------------|-------|--------------|--------------------|
|            | Online database                | Foreign Journals | Indian Journals |       |              |                    |
| Kasaragod  | 1                              | 1                | 11              | 11816 | 13364        | 3650               |
| Kayamkulam | -                              | -                | -               | 3584  | 3255         | 3976               |
| Vittal     | -                              | -                | -               | 5312  | 5801         | 3052               |
| Mohitnagar | -                              | -                | -               | 537   | -            | 15                 |
| Kahikuchi  | -                              | -                | -               | 134   | -            | -                  |

Current Library Holdings 2023

access to its resources through the intranet. The usage of the digital repository came to around 450 hits per month.

### Administrative Reforms

Prior to the cadre restructuring of administrative posts, the sanctioned strength of Asst. Administrative Officers (AAO) was eight, out of which five posts of AAOs were allocated to ICAR-CPCRI headquarters, one each to the ICAR-CPCRI Regional Station, Kayamkulam and Vittal and one to the Research Centre, Kidu. Consequent upon the revision of the cadre strength of AAO from 8 posts to 3 posts, the institute was not in a position to post AAOs to the regional stations or research centers. In view of the above and in light of the implementation of eOffice, the charge of AAOs could be discharged from the headquarters.

Further, the existing system of maintaining separate accounts for headquarters and regional stations and consolidating the annual accounts of the institute was cumbersome. Hence, it was decided that from 2023–24, only a single account shall be maintained at headquarters and irrespective of stations, centers, KVKs, schemes, etc. and all payments shall be made from CPCRI headquarters only.

In addition, one or two channels of file movement in eOffice were done away with to speed up the file movements as well as the disposal of administrative work.

## XXIII. BUDGET AND EXPENDITURE

The Budget and Expenditure for the period from April 1, 2023 to March 31, 2024

(Figures in Rupees Lakhs)

| Budget Head                             | Plan        |                |
|-----------------------------------------|-------------|----------------|
|                                         | Budget      | Expenditure    |
| <b>Revenue</b>                          |             |                |
| Estt. Charges                           | 2949.52     | 2949.46        |
| OTA                                     |             |                |
| Pension                                 | 3929.66     | 3929.42        |
| TA                                      | 50          | 49.70          |
| Research & Operational expenses         | 450         | 447.90         |
| <b>Works: Repair &amp; Maintenance</b>  |             |                |
| Office Buildings                        | 3           | 2.70           |
| Residential Buildings                   | 9           | 9              |
| Minor Works                             | 30          | 31.63          |
| Other Administrative Charges            | 630         | 630.51         |
| <b>Total</b>                            | <b>1172</b> | <b>1171.44</b> |
| Miscellaneous Expenses (including HRD)  | 63          | 63.56          |
| Tribal Sub Plan - General               | 75.00       | 75.00          |
| Scheduled Cast/ Scheduled Tribe-General | 50.00       | 50.00          |
| <b>Total</b>                            | <b>188</b>  | <b>188.56</b>  |
| <b>Total</b>                            | <b>1360</b> | <b>1360.00</b> |
| <b>Capital</b>                          |             |                |
|                                         | 100.00      |                |
| Equipments                              |             | 85.99          |
| Information Technology                  |             | 2.36           |
| Library                                 |             | 0.55           |
| Furniture & Fixtures                    |             | 0.56           |
| Livestock                               |             |                |
| Works                                   |             | 9.81           |
| Minor Work                              |             | 0.73           |
| Tribal Sub Plan - Capital               |             |                |
| NEH                                     | 120.00      | 120.00         |
| <b>Total</b>                            | <b>220</b>  | <b>220</b>     |
| <b>TOTAL</b>                            | <b>1580</b> | <b>1580</b>    |

### Other Projects

|                                     | Opening Balance | Receipts | Expenditure | Refund  |
|-------------------------------------|-----------------|----------|-------------|---------|
| Other Plan Schemes                  | 17.19           | 862.67   | 861.94      | 0.04363 |
| Deposit Schemes (Externally funded) | 321.11          | 421.93   | 335.31      | NIL     |
| <b>KVK, Kasaragod</b>               | (-)6.30         | 167.22   | 165.98      | NIL     |
| <b>KVK, Alappuzha</b>               | 0.00324         | 259.62   | 258.37      | NIL     |

### Revenue receipts

| Head                                                | Target | Achievement     |
|-----------------------------------------------------|--------|-----------------|
| Income from sales/ services (Schedule-8)            | 412.86 | 17348894        |
| Fee/Subscription (Schedule-10)                      |        | 758000          |
| Income from Royalty, Publication etc. (Schedule-12) |        | 2346            |
| Other Income (Schedule-14)                          |        | 34531169        |
| STD Interest                                        |        | 0               |
| Recoveries on Loans & Advances                      |        | 206136          |
| <b>TOTAL</b>                                        |        | <b>52843545</b> |

## XXIV. WEATHER DATA

### Headquarters, Kasaragod

| Month     | Temp. Max.<br>(°C) | Temp. Min<br>(°C) | RH (%) (FN) | RH(%) (AN) | Rainfall (mm) | Rainy Days<br>(No.) |
|-----------|--------------------|-------------------|-------------|------------|---------------|---------------------|
| January   | 32.8               | 21.7              | 82.3        | 63.9       | 0             | 0                   |
| February  | 34.2               | 22.7              | 86.7        | 61.7       | 0             | 0                   |
| March     | 34.7               | 22.8              | 78.8        | 54.2       | 6.2           | 1                   |
| April     | 34.7               | 25.2              | 79.3        | 65.4       | 0             | 0                   |
| May       | 34.58              | 25.5              | 78.5        | 64.8       | 123.9         | 10                  |
| June      | 31.1               | 24.6              | 87.0        | 74.5       | 507.8         | 25                  |
| July      | 27.9               | 23.4              | 91.3        | 85.1       | 1373.6        | 31                  |
| August    | 29.9               | 24.1              | 85.1        | 75.2       | 142.2         | 14                  |
| September | 29.6               | 23.6              | 90.3        | 80.5       | 537.7         | 25                  |
| October   | 31.3               | 24.0              | 87.5        | 73.0       | 171.8         | 12                  |
| November  | 32.2               | 24.2              | 88.8        | 82.8       | 224.6         | 10                  |
| December  | 33.2               | 23.7              | 71.0        | 62.1       | 0             | 0                   |

### Regional Station, Kayamkulam

| Month     | Temp.<br>Max (°C) | Temp.<br>Min (°C) | RH (%)<br>(FN) | RH(%)<br>(AN) | Wind<br>Velocity<br>(km/h) | Sunshine<br>(h) | Evaporation<br>(litre) | Rainfall<br>(mm) | Rainy<br>Days<br>(No.) |
|-----------|-------------------|-------------------|----------------|---------------|----------------------------|-----------------|------------------------|------------------|------------------------|
| January   | 32.4              | 20.6              | 93             | 61            | 1.1                        | 8.2             | 3.8                    | 12.4             | 1                      |
| February  | 33.4              | 20.9              | 92             | 56            | 1.6                        | 8.8             | 3.9                    | 00.6             | 0                      |
| March     | 34.4              | 23.9              | 91             | 57            | 1.9                        | 9.2             | 4.0                    | 47.0             | 3                      |
| April     | 33.8              | 25.2              | 93             | 64            | 1.6                        | 8.2             | 3.9                    | 103.0            | 8                      |
| May       | 33.2              | 25.6              | 94             | 71            | 1.4                        | 8.6             | 3.9                    | 234.0            | 13                     |
| June      | 31.1              | 22.6              | 95             | 75            | 0.9                        | 6.0             | 3.6                    | 357.0            | 20                     |
| July      | 30.4              | 24.0              | 94             | 74            | 0.7                        | 7.3             | 3.7                    | 408.3            | 14                     |
| August    | 32.1              | 24.2              | 93             | 65            | 0.3                        | 8.7             | 4.0                    | 104.7            | 3                      |
| September | 30.5              | 23.9              | 95             | 74            | 0.2                        | 4.8             | 3.5                    | 556.2            | 21                     |
| October   | 31.6              | 24.2              | 95             | 71            | 0.0                        | 6.0             | 3.6                    | 658.6            | 19                     |
| November  | 32.4              | 24.3              | 94             | 69            | 0.3                        | 6.4             | 3.7                    | 299.7            | 11                     |
| December  | 32.9              | 23.5              | 94             | 68            | 0.3                        | 6.3             | 3.7                    | 155.3            | 07                     |

### Research Centre, Kidu

| Month     | Temp.       |          | RH % |      | Wind<br>velocity<br>(Km/h) | Sunshine<br>(h) | Evaporation<br>(mm) | Rainfall<br>(mm) | Rainy<br>days |
|-----------|-------------|----------|------|------|----------------------------|-----------------|---------------------|------------------|---------------|
|           | Max<br>(*C) | Min (*C) | FN   | AN   |                            |                 |                     |                  |               |
| January   | 35.3        | 18.4     | 88.3 | 42.1 | 0.03                       | 286.3           | 151.4               | 0.0              | 0             |
| February  | 38.0        | 23.5     | 89.3 | 35.9 | 0.12                       | 257.3           | 158.6               | 0.0              | 0             |
| March     | 40.1        | 28.1     | 90.6 | 36.4 | 0.33                       | 247.9           | 193.3               | 0.0              | 0             |
| April     | 40.3        | 31.9     | 85.7 | 42.4 | 0.66                       | 245.6           | 219.2               | 63.0             | 2             |
| May       | 39.9        | 29.9     | 94.4 | 50.5 | 0.31                       | 198.0           | 205.0               | 260.3            | 6             |
| June      | 35.5        | 25.6     | 95.1 | 69.5 | 0.20                       | 140.9           | 142.1               | 326.2            | 14            |
| July      | 29.8        | 25.2     | 98.5 | 88.8 | 0.19                       | 24.5            | 59.6                | 1730.2           | 30            |
| August    | 31.8        | 25.2     | 94.8 | 70.9 | 0.02                       | 108.6           | 103.1               | 417.0            | 14            |
| September | 33.9        | 26.3     | 96.8 | 79.5 | 0.02                       | 64.3            | 89.8                | 516.7            | 20            |
| October   | 34.0        | 29.3     | 86.1 | 68.7 | 0.02                       | 152.1           | 108.5               | 429.9            | 16            |
| November  | 35.4        | 30.7     | 83.2 | 63.8 | 0.00                       | 127.9           | 111.3               | 62.2             | 5             |
| December  | 36.4        | 31.2     | 77.0 | 56.2 | 0.03                       | 123.1           | 103.3               | 21.5             | 2             |

## XXV. राजभाषा कार्यान्वयन

भाकृअनुप- केंद्रीय रोपण फसल अनुसंधान संस्थान में 14 सितम्बर 2023 से 28 सितम्बर 2023 तक हिंदी पखवाड़ा का आयोजन किया गया। इसका उद्घाटन समारोह अनुसंधान केंद्र, काहिकुची में 14 सितम्बर 2023 को पूर्वाह्न 9.30 बजे ऑनलाइन माध्यम में संपन्न हुआ। आई सी ऐ आर गीत के उपरांत श्री अश्विन रघुनाथ ने सर्वशक्तिमान का प्रार्थना गीत गाया & डॉ. अल्पना दास, प्रमुख एवं प्रधान वैज्ञानिक, अनुसंधान केंद्र, काहिकुची ने अतिथियों का स्वागत किया & मुख्य अतिथि डॉ. आर.के. माथुर, निदेशक, भारतीय तिलहन अनुसंधान संस्थान, हैदराबाद थे और उन्होंने हिन्दी के महत्व की व्याख्या की। उद्घाटन एवं अध्यक्षता डॉ. वी.बी. पटेल, सहायक महानिदेशक (बागवानी विज्ञान), भा.कृ.अनु.प., नई दिल्ली ने किया। उन्होंने हिन्दी की आवश्यकता और उसको बढ़ावा देने की जरूरत को समझाया। श्री पी. कृष्ण कुमार, प्रशासनिक अधिकारी ने धन्यवाद ज्ञापन किया। केंद्रीय रोपण फसल अनुसंधान संस्थान के सभी अधिकारियों एवं कर्मचारियों ने विभिन्न केंद्रों से इस समारोह में हाइब्रिड मोड में भाग लिया।

हिंदी पखवाड़े के अवसर पर कर्मचारियों के लिए शब्द निर्माण, अनुवाद, स्मरण परीक्षा, हिंदी कविता पाठ, अंताक्षरी, निबंध लेखन, मौके पर बोलें, सुलेख एवं तस्वीर देखें और लिखें जैसे प्रतियोगिताओं का आयोजन भी किया गया।

हिंदी पखवाड़ा समारोह का समापन 29 सितंबर 2022 दोपहर 3.30 बजे डॉ. के बालचंद्र हेब्बार, निदेशक महोदय की अध्यक्षता में संपन्न हुआ। अपने अध्यक्षीय भाषण में जितना हो सके, उतना काम हिन्दी में करने के लिए उन्होंने अनुरोध किया। डॉ. सीमा चंद्रन, सहायक प्राध्यापक, केरल केंद्रीय विश्वविद्यालय, कासरगोड

कार्यक्रम का मुख्य अतिथि रही। एकत्रित दर्शकों को सम्बोधित करते हुए, उन्होंने बताया कि हिंदी भाषा के सीखने और प्रचार का प्रयत्न भरपूर किया जा रहा है। इसी दिशा निर्देशों से काम करने से और सरल शब्दों का प्रयोग से इस भाषा को जन जन की भाषा बनाया जा सकता है। समापन समारोह में विविध प्रतियोगिताओं के विजेताओं को मुख्य अतिथि ने नकद पुरस्कार प्रदान किए। कार्यक्रम के अंतर्गत श्री एन. उदयकुमार और श्री मणिकंठन ने अपने मधुर गीतों से सभी का मनोरंजन किया। स्वागत भाषण श्री राम अवतार पराशर, प्रमुख वित्त एवं लेखा अधिकारी ने किया और धन्यवाद ज्ञापन श्री कृष्णकुमार पी प्रशासनिक अधिकारी ने किया।

क्षेत्रीय स्टेशन, कायमकुलम में 19 से 25 सितंबर 2023 तक हिंदी सप्ताह मनाया गया। हिंदी गीत, हिंदी श्रुतलेख, हिंदी टंकण, हिंदी अनुवाद, हिंदी हस्तलिपि, टिप्पणी और हिंदी मसौदा लेखन सहित विभिन्न प्रतियोगिता प्रतियोगिताओं का आयोजन किया गया। स्टेशन के प्रत्येक कर्मचारी ने प्रतियोगिताओं में भाग लिया। 25 सितंबर, 2023 को समापन समारोह का आयोजन डॉ. पी. अनिताकुमारी (कार्यकारी प्रमुख) के अध्यक्षता में किया गया। मुख्य अतिथि श्रीमती राजप्रिया, हिंदी शिक्षक, कायमकुलम बॉयज हायर सेकेंडरी स्कूल से थीं। श्री राम अवतार पराशर, मुख्य वित्त एवं लेखा अधिकारी, और श्री ए. नील विन्सर, सहायक प्रशासनिक अधिकारी द्वारा अभिनंदन भाषण दिया गया।

दिनांक 26.04.2023 को भाकृअनुप — के.रो.फ.अ.सं., कासरगोड में 'कंठस्थ' प्रशिक्षण कार्यक्रम भी ऑनलाइन जरिए आयोजित किया गया था।



उद्घाटन कार्यक्रम



राजभाषा प्रशिक्षण



क्षेत्रीय स्टेशन, कायमकुलम में समापन समारोह



समापन समारोह



हर कदम, हर डगर  
किसानों का हमसाफर  
भारतीय कृषि अनुसंधान परिषद

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